ASVAB Math Workbook

1,000 Practice Problems with Step-by-Step Solutions

Arithmetic Reasoning & Mathematics Knowledge

Key concepts and worked examples for every topic
A diagnostic test with a personal study plan
750 drill problems in 27 chapters
4 full-length math practice tests
Every answer checked by computer algebra

Ivan Sikuten

ASVAB Math Workbook: 1,000 Practice Problems with Step-by-Step Solutions

First edition, 2026

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Test formats, question counts, time limits, and qualifying scores are set by the Department of Defense and the individual services and can change. Always confirm current details with your recruiter or at officialasvab.com. The author makes no guarantee of any particular test score.

The practice problems in this book were generated and their answers verified with the SymPy computer algebra system; every answer was computed exactly and then confirmed by a second, independent calculation.

Contents

How to Use This Book . . . 1
The ASVAB Math Subtests . . . 2
Test-Day Strategies . . . 3
Diagnostic Test . . . 5

I Number Skills **11**
1 Whole Numbers & Order of Operations . . . 12
2 Negative Numbers & Absolute Value . . . 18
3 Factors, Multiples, Primes, GCF & LCM . . . 24
4 Fractions & Mixed Numbers . . . 30
5 Decimals . . . 36
6 Percents . . . 42
7 Ratios & Proportions . . . 47
8 Exponents & Scientific Notation . . . 53
9 Square Roots & Radicals . . . 59

II Algebra **65**
10 Algebraic Expressions . . . 66
11 Linear Equations . . . 72
12 Inequalities . . . 78
13 Systems of Equations . . . 85
14 Polynomials & Exponent Rules . . . 91
15 Factoring & Quadratic Equations . . . 97

III Geometry **103**
16 Lines & Angles . . . 104
17 Triangles & the Pythagorean Theorem . . . 111
18 Perimeter & Area . . . 118
19 Circles . . . 125
20 Volume & Surface Area . . . 132
21 Coordinate Geometry . . . 139

IV Word Problems **145**
22 Money: Shopping, Discounts, Tax & Wages . . . 146
23 Interest, Commission & Percent Change . . . 156
24 Rate, Distance & Time . . . 164
25 Work Rates & Mixtures . . . 172
26 Averages, Statistics & Probability . . . 180

27 Measurement, Units & Multi-Step Problems 188

V Practice Tests **196**

Practice Test 1 197
Practice Test 2 209
Practice Test 3 220
Practice Test 4 232
Formula Sheet 245
Quick Reference Tables 246
Glossary of Math Terms 247
Progress Tracker 249

How to Use This Book

The math on the ASVAB is mostly middle-school and early high-school math—percents, fractions, equations, areas, word problems—asked quickly and without a calculator. What raises scores is *practice*: seeing each kind of question often enough that you recognize it instantly and know the first step.

This workbook gives you that practice: **1,000 problems**, each with a complete step-by-step solution.

The plan

1. **Take the Diagnostic Test** (page 5). Thirty questions, about 33 minutes, no calculator. The answer key tells you which chapter each question comes from.
2. **Study the chapters you need.** Start with the topics you missed on the diagnostic. Each chapter has three parts:
   - *Key Concepts*—the rules, formulas, a worked example, a no-calculator tip, and the traps test writers love.
   - *Practice Set*—warm-up problems first, then test-level problems, then challenge problems.
   - *Answers & Explanations*—a quick answer key plus a full solution for every problem.
3. **Take the four Practice Tests** under real time limits once you have worked through the chapters. Each has an Arithmetic Reasoning section (30 questions, 36 minutes) and a Mathematics Knowledge section (25 questions, 24 minutes), the format of the paper ASVAB.
4. **Review every miss.** The answer keys of the tests list the chapter for every question, so you always know where to go back.

Reading the solutions

Every solution is broken into numbered steps that you can follow with pencil and paper. Many solutions also include:

- **TIP**—a shortcut, an estimate, or a way to check your answer.
- **WHY NOT**—what went wrong if you picked one of the tempting wrong answers. The wrong choices in this book are not random numbers; each comes from a real, common mistake (dividing by the wrong number, forgetting to convert minutes to hours, adding percents that should be multiplied). If your answer matches one of them, the explanation tells you exactly which mistake you made.

Good habits

- **No calculator.** The ASVAB does not allow one, so don't practice with one. Every problem here is designed to be done by hand.
- **Write your work down.** Scratch paper is provided at the test; use it here too.
- **Time yourself.** Each practice set shows a target time. Don't worry about speed at first—accuracy comes first, speed follows.
- **Redo your misses a week later** and log them in the Progress Tracker at the back of the book.

The ASVAB Math Subtests

The Armed Services Vocational Aptitude Battery (ASVAB) is the entrance test for the U.S. Army, Navy, Air Force, Marine Corps, Space Force, and Coast Guard. It consists of several subtests. Two of them are math:

Arithmetic Reasoning (AR)

Word problems. Each question describes a real-life situation—a purchase, a trip, a paycheck, a recipe, a work schedule—and you have to decide which operations to use. Typical topics: percents and discounts, ratios, rates and distance, averages, interest, work, measurement, and multi-step reasoning.

Mathematics Knowledge (MK)

Direct math questions: number properties, fractions, exponents and roots, algebra (expressions, equations, inequalities, factoring), and geometry (angles, triangles, area, circles, volume, coordinates). Knowing the vocabulary and formulas is half the battle.

Format and timing

The ASVAB is given in two main formats. On the **computer-adaptive** version (CAT-ASVAB), the difficulty of each question depends on how you answered the previous ones, and you cannot go back to change an answer. On the **paper-and-pencil** version, everyone answers the same questions and you can move around within a subtest. Typical question counts and time limits:[1]

	Arithmetic Reasoning	Mathematics Knowledge
CAT-ASVAB (computer)	15 questions, 55 minutes	15 questions, 31 minutes
Paper and pencil	30 questions, 36 minutes	25 questions, 24 minutes

Every question is multiple choice with four answer choices. **Calculators are not allowed**; scratch paper is provided. Formulas are *not* given, so memorize the ones on the Formula Sheet at the back of this book.

Why math matters so much

Your Armed Forces Qualification Test (AFQT) score—the score that decides whether you can enlist at all—is computed from four subtests: Arithmetic Reasoning, Mathematics Knowledge, Word Knowledge, and Paragraph Comprehension. The two math subtests make up roughly **half** of the AFQT. The AFQT is reported as a percentile from 1 to 99, and each service sets its own minimum.

Arithmetic Reasoning and Mathematics Knowledge also feed many of the *line scores* (composite scores) that qualify you for specific jobs, such as technical, electronics, and administrative specialties. A few extra correct math answers can open up a wider choice of careers and enlistment bonuses.

[1] These figures can change; confirm them with your recruiter or at officialasvab.com.

Test-Day Strategies

1. Read the last sentence first

In a word problem, the last sentence tells you what to find. Read it first, then read the whole problem looking for the numbers you need. Many wrong answer choices are the right answer to a *different* question (the discount instead of the sale price, the number who passed instead of the number who failed).

2. Estimate before you calculate

A rough answer tells you which choices are impossible. If a $48 item is 25% off, the sale price must be a bit more than $35—so $12 and $60 are out before you do any exact arithmetic.

3. Work backward from the choices

When a question asks "which number…" or gives an equation, you can test the answer choices. Start with a middle value; if it is too big, the answer is smaller, and vice versa. For "which value satisfies" questions this is often the fastest method.

4. Pick numbers

If the problem uses variables or percents of an unknown amount, choose an easy number (100 for percents, a common multiple for fractions) and see which answer choice gives the same result.

5. Watch the units

Minutes vs. hours, inches vs. feet, cents vs. dollars, radius vs. diameter. Convert everything to the units the question asks for *before* you calculate.

6. Manage your time

On the paper test you have a little over a minute per question. If a question takes more than two minutes, make your best guess, mark it, and move on. Never leave a question blank: a blank is always wrong, a guess might be right, and on the computer test unanswered questions at the end of a subtest lower your score.

7. Check that your answer makes sense

A sale price cannot be higher than the original price. A probability cannot be more than 1. An average must lie between the smallest and largest values. A quick reasonableness check catches many careless errors.

No-Calculator Tip Learn by heart: the squares from 1^2 to 15^2, the fraction–decimal–percent equivalents ($\frac{1}{8} = 0.125 = 12.5\%$, $\frac{1}{3} \approx 33.3\%$, ...), and the formulas on the Formula Sheet. Instant recall of these saves minutes on test day.

Mental Math Toolkit

Because the ASVAB bans calculators, a handful of shortcuts pay off on almost every page of this book. Practice them until they are automatic.

Trick	Example
Multiply by 5	Multiply by 10, then halve: $48 \times 5 = 480 \div 2 = 240$.
Multiply by 25	Multiply by 100, then divide by 4: $36 \times 25 = 3{,}600 \div 4 = 900$.
Multiply by 9 or 11	$47 \times 9 = 470 - 47 = 423$; $34 \times 11 = 340 + 34 = 374$.
Break a number apart	$7 \times 98 = 7 \times 100 - 7 \times 2 = 700 - 14 = 686$.
Square a number ending in 5	35^2: $3 \times 4 = 12$, then write 25: 1,225.
Divide by 5	Double, then divide by 10: $340 \div 5 = 680 \div 10 = 68$.
Divide by a decimal	Move both decimal points: $4.8 \div 0.06 = 480 \div 6 = 80$.
Build percents from 10%	15% of 60: 10% = 6, 5% = 3, total 9.
Swap the percent	8% of 50 = 50% of 8 = 4.
Compare fractions	Cross-multiply: $\frac{5}{8}$ vs. $\frac{3}{5}$: $5 \times 5 = 25 > 8 \times 3 = 24$, so $\frac{5}{8}$ is larger.
Estimate first	$398 \times 51 \approx 400 \times 50 = 20{,}000$ rules out answers near 2,000 or 200,000.

Diagnostic Test

This test samples every chapter of the book. Take it before you begin studying, without a calculator, and try to finish in about 33 minutes. Don't guess wildly—if you have no idea how to start a problem, leave it blank so that the result shows what you really need to study.

Score: _____ / 30 Target time: 33 minutes

1. Round 878,359 to the nearest thousand.

A 877,000 **B** 878,000 **C** 878,400 **D** 879,000

2. What number is halfway between 7 and −15 on a number line?

A −4 **B** 4 **C** 11 **D** 22

3. What is the greatest common factor of 54, 66, and 30?

A 3 **B** 6 **C** 30 **D** 2,970

4. Lance Corporal Jensen's platoon is on an 8-mile ruck march. So far the platoon has covered $3\frac{3}{4}$ miles. How many miles are left?

A $4\frac{1}{4}$ miles **B** $4\frac{3}{4}$ miles
C $5\frac{1}{4}$ miles **D** $5\frac{3}{4}$ miles

5. Yuki works at a warehouse and earns $19.50 per hour for the first 40 hours each week. Every hour over 40 is overtime and pays 1.5 times the regular rate. Last week she worked 47 hours. How much did she earn?

A $916.50 **B** $984.75
C $994.75 **D** $1,374.75

6. Last year the value of a used car was $10,900. This year it is $7,630. What is the percent decrease?

A 30% **B** 60% **C** 70% **D** 3,270%

7. Solve for x: $\frac{45}{25} = \frac{x}{30}$

A 39 **B** 50 **C** 54 **D** 1,350

8. Which of the following is equal to $\frac{9^7 \times 9^8}{9^3}$?

A 9^5 **B** 9^{53} **C** 9^{18} **D** 9^{12}

9. What is $8\sqrt{3} - 6\sqrt{3}$?

A $2\sqrt{3}$ **B** $5\sqrt{3}$ **C** $14\sqrt{3}$ **D** $48\sqrt{3}$

10. If n stands for a number, which expression represents "the sum of 12 times a number and 2"?

A $12 + 2n$ **B** $12(n + 2)$
C $12n + 2$ **D** $14n$

11. Solve for x: $2x + 17 = 27$.

A $-\frac{7}{2}$ **B** 5 **C** 10 **D** 22

12. Which of the following is the solution of $-4x + 12 \le -20$?

A $x \ge 8$ **B** $x \ge -8$ **C** $x \le 8$ **D** $x \ge 2$

13. A jar holds 47 coins, all quarters and dimes. The coins are worth $11 in all. How many quarters are in the jar?

A 5 **B** 38 **C** 42 **D** 44

14. Subtract $2x^2 - 2x - 5$ from $7x^2 - x + 6$.

A $9x^2 - 3x + 1$ **B** $5x^2 - 3x + 1$
C $5x^2 + x + 11$ **D** $-5x^2 - x - 11$

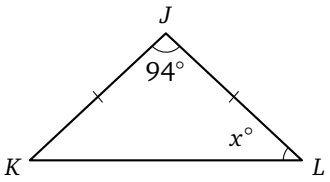
15. Which expression is equal to $x^2 - 18x + 81$?

A $(x - 9)^2$ **B** $(x - 81)(x - 1)$
C $(x + 9)^2$ **D** $(x - 9)(x + 9)$

16. Tasha designs a regular polygon for a garden path. Each interior angle is 156°. How many sides does her polygon have?

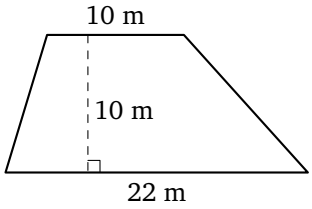
A 13 **B** 15 **C** 17 **D** 24

17. In isosceles triangle JKL, the vertex angle $\angle J$ is 94°. What is the value of x, the measure of a base angle?



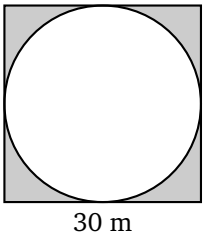
A 33 **B** 38 **C** 43 **D** 86

18. What is the area of a trapezoid with bases of 10 meters and 22 meters and a height of 10 meters?



A 100 m^2 **B** 156 m^2 **C** 158 m^2 **D** 160 m^2

19. In the figure, a circle fits exactly inside a square with sides of 30 meters, touching all four sides. What is the area of the shaded region?



A $(900 - 225\pi)$ m^2 **B** 225π m^2
C $(900 - 30\pi)$ m^2 **D** $(900 - 15\pi)$ m^2

20. How many cubes that measure 3 inches on each side are needed to fill a box that is 9 inches long, 6 inches wide, and 9 inches high?

A 6 **B** 18 **C** 162 **D** 486

21. How far apart are the points $(7, -11)$ and $(-5, -2)$?

A 15 **B** 21 **C** 108 **D** 225

22. Petty Officer Ruiz's take-home pay is $2,820 a month. He puts $\frac{1}{10}$ of it into savings and sends $\frac{1}{3}$ of it home to his family. How much is left each month for everything else?

A $1,222 **B** $1,598 **C** $1,880 **D** $2,538

23. An online store lists a $1,160 laptop at 10% off. Members get another 15% off the sale price at checkout. What is the final price of the laptop?

A $870.00 **B** $887.40
C $986.00 **D** $1,044.00

24. A bus travels 130 miles on the highway at 65 miles per hour, then 30 miles through town at 20 miles per hour. How many hours does the whole trip take?

A 2 hours **B** 2.5 hours
C 3 hours **D** 3.5 hours

25. A convoy leaves the motor pool at 11:00 a.m. and travels 125 miles to a training area at 50 miles per hour. At what time does it arrive?

A 12:45 p.m. **B** 1:00 p.m.
C 1:30 p.m. **D** 1:50 p.m.

26. A gardener has 18 gallons of a 75% fertilizer solution. How many gallons of water must be added to make a 25% solution?

A 6 gallons **B** 36 gallons
C 46 gallons **D** 54 gallons

27. A gardener mixes 12 gallons of a 25% fertilizer solution with 8 gallons of a 30% fertilizer solution. What percent of the mixture is fertilizer?

A 27% **B** 27.5% **C** 28% **D** 55%

28. Keisha packed 7 shirts, 2 pairs of pants, and 3 pairs of shoes for a trip. How many different outfits of one shirt, one pair of pants, and one pair of shoes can she make?

A 12 **B** 14 **C** 42 **D** 343

29. A supply convoy departs at 0555 and arrives at 1715. How long is the trip?

A 11 hours 20 minutes
B 12 hours
C 12 hours 20 minutes
D 12 hours 40 minutes

30. A water tank holds 360 gallons. Soldiers use $\frac{1}{5}$ of the water in the morning and $\frac{1}{3}$ of what is left in the afternoon. How many gallons are used in the afternoon?

A 66 **B** 72 **C** 76 **D** 96

Answers and Your Study Plan

Check your answers against the table. Each row shows the chapter the question comes from. **Circle the chapter number of every question you missed or left blank**—those chapters are your study plan. Work through them first, then the rest of the book in order.

#	Ans.	Ch.	Topic
1	B	1	Whole Numbers & Order of Operations
2	A	2	Negative Numbers & Absolute Value
3	B	3	Factors, Multiples, Primes, GCF & LCM
4	A	4	Fractions & Mixed Numbers
5	B	5	Decimals
6	A	6	Percents
7	C	7	Ratios & Proportions
8	D	8	Exponents & Scientific Notation
9	A	9	Square Roots & Radicals
10	C	10	Algebraic Expressions
11	B	11	Linear Equations
12	A	12	Inequalities
13	C	13	Systems of Equations
14	C	14	Polynomials & Exponent Rules
15	A	15	Factoring & Quadratic Equations
16	B	16	Lines & Angles
17	C	17	Triangles & the Pythagorean Theorem
18	D	18	Perimeter & Area
19	A	19	Circles
20	B	20	Volume & Surface Area
21	A	21	Coordinate Geometry
22	B	22	Money: Shopping, Discounts, Tax & Wages
23	B	23	Interest, Commission & Percent Change
24	D	24	Rate, Distance & Time
25	C	24	Rate, Distance & Time
26	B	25	Work Rates & Mixtures
27	A	25	Work Rates & Mixtures
28	C	26	Averages, Statistics & Probability
29	A	27	Measurement, Units & Multi-Step Problems
30	D	27	Measurement, Units & Multi-Step Problems

How did you do?

26–30 correct	Strong. Skim the Key Concepts, do the challenge problems, then the practice tests.
18–25 correct	Solid base. Study your circled chapters fully, then the rest.
10–17 correct	Work through every chapter in order; take your time with the warm-ups.
0–9 correct	Start at Chapter 1. Every chapter builds on the ones before it.

Explanations

1. **B** 878,000

1 The thousands digit is 8. Look at the digit just to its right (the hundreds digit): 3.

2 Because 3 is less than 5, keep the thousands digit 8 as it is.

3 Change every digit after the thousands place to zero: 878,000.

WHY NOT (C) rounds to the nearest hundred instead. (D) rounds up even though the next digit, 3, is less than 5.

2. **A** -4

1 The halfway point is the average of the two numbers: $(-15 + 7) \div 2$.

2 $-15 + 7 = -8$, and $-8 \div 2 = -4$.

3 Check: -4 is 11 units from each number.

WHY NOT (B) has the right size but the wrong sign. (C) is half the distance between the numbers, not the point halfway between them. (D) is the distance between the numbers.

3. **B** 6

1 Write each number as a product of primes: $30 = 2 \times 3 \times 5$, $54 = 2 \times 3^3$, $66 = 2 \times 3 \times 11$.

2 Take each prime that all three numbers share, with its lowest power: $2 \times 3 = 6$.

WHY NOT (A) is a common factor, but not the greatest one. (C) is the smallest number, but it is not a factor of 54. (D) is the least common multiple, not the greatest common factor.

4. **A** $4\frac{1}{4}$ miles

1 Subtract: $8 - 3\frac{3}{4}$.

2 Regroup 8 as $7\frac{4}{4}$.

3 $7\frac{4}{4} - 3\frac{3}{4} = 4\frac{1}{4}$ miles.

WHY NOT (C) regroups but forgets to take 1 from the whole number. (D) subtracts the whole numbers and keeps the fraction.

5. **B** \$984.75

1 Regular pay: $40 \times 19.50 = 780.00$.

2 Overtime rate: $1.5 \times 19.50 = 29.25$ per hour.

3 Overtime pay: $7 \times 29.25 = 204.75$.

4 Total: $780.00 + 204.75 = 984.75$, so \$984.75.

WHY NOT (A) pays the overtime hours at the regular rate. (D) pays every hour at the overtime rate.

6. **A** 30%

1 Find the amount of decrease: 10,900 – 7,630 = 3,270 dollars.

2 Divide by the *original* amount: $\frac{3{,}270}{10{,}900} = 0.3$.

3 Convert to a percent: 0.3 = 30% decrease.

TIP Hard division? Test the choices instead: 30% of \$10,900 is \$3,270, exactly the change.

WHY NOT (C) gives the new amount as a percent of the old amount. (D) gives the dollar change instead of the percent change.

7. **C** 54

1 Cross-multiply: $25 \times x = 30 \times 45$, so $25x = 1{,}350$.

2 Divide both sides by 25: $x = 1{,}350 \div 25 = 54$.

TIP Simplify $\frac{45}{25}$ to $\frac{9}{5}$ first to work with smaller numbers: $5x = 30 \times 9 = 270$.

WHY NOT (B) uses a constant difference: 30 is 5 more than 25, so it adds 5 to 45 instead of scaling. (D) multiplies 30 by 45 but forgets to divide by 25.

8. **D** 9^{12}

1 Numerator: same base, so add the exponents. $9^7 \times 9^8 = 9^{15}$.

2 Divide: subtract the exponents. $\frac{9^{15}}{9^3} = 9^{15-3} = 9^{12}$.

WHY NOT (A) divides the exponents instead of subtracting. (B) multiplies the exponents in the numerator instead of adding them. (C) adds the exponent in the denominator instead of subtracting it.

9. **A** $2\sqrt{3}$

1 Both terms have $\sqrt{3}$, so they are like radicals: combine the numbers in front, just like $8x - 6x$.

2 $8 - 6 = 2$, so the answer is $2\sqrt{3}$. The $\sqrt{3}$ stays the same.

WHY NOT (C) adds instead of subtracting. (D) multiplies the numbers in front instead of combining them.

10. **C** $12n + 2$

1 The two amounts being added are "12 times a number" ($12n$) and 2.

2 With n for the number, the expression is $12n + 2$.

WHY NOT (A) attaches the variable to the wrong number. (B) multiplies the whole sum $n + 2$ by 12; only the number should be multiplied by 12.

11. **B** 5

1 Undo the addition: subtract 17 from both sides. $2x = 27 - 17 = 10$.

2 Undo the multiplication: divide both sides by 2. $x = \frac{10}{2} = 5$.

3 Check: $2(5) + 17 = 10 + 17 = 27$. ✓

WHY NOT (A) divides only 27 by 2, not the 17: $x + 17 = \frac{27}{2}$. (C) forgets to divide by 2. (D) moves 17 to the other side without changing its sign.

12. **A** $x \geq 8$

1 Subtract 12 from both sides (adding or subtracting never changes the direction): $-4x \leq -20 - 12$, so $-4x \leq -32$.

2 Divide both sides by -4. Dividing by a *negative* number reverses the inequality symbol: $\leq$ becomes $\geq$: $x \geq \frac{-32}{-4}$, so $x \geq 8$.

3 Check with a number in the answer, $x = 9$: $-4(9) + 12 = -24$, and $-24 \leq -20$ is true. ✓

WHY NOT (B) gets the number's sign wrong (positive vs. negative) when dividing by -4. (C) forgets to reverse the inequality symbol when dividing by -4. (D) moves 12 across without changing its sign.

13. **C** 42

1 Work in cents so there are no decimals: a quarter is 25 cents, a dime is 10 cents, and \$11 is 1,100 cents.

2 Let q be the number of quarters; then there are $47 - q$ dimes. Total value: $25q + 10(47 - q) = 1{,}100$.

3 Distribute: $25q + 470 - 10q = 1{,}100$, so $15q = 1{,}100 - 470 = 630$.

4 Divide by 15: $q = 42$.

WHY NOT (A) gives the number of dimes. (D) divides the total value by the value of one coin.

14. **C** $5x^2 + x + 11$

1 "Subtract $2x^2 - 2x - 5$ from $7x^2 - x + 6$" means $(7x^2 - x + 6) - (2x^2 - 2x - 5)$.

2 Distribute the minus sign to *every* term of the second polynomial: $-(2x^2 - 2x - 5) = -2x^2 + 2x + 5$.

3 Combine like terms: $(7 - 2)x^2 + (-1 - (-2))x + (6 - (-5)) = 5x^2 + x + 11$.

WHY NOT (A) adds the polynomials instead of subtracting. (B) changes the sign of only the first term of the second polynomial. (D) subtracts in the wrong order: "subtract A from B" means $B - A$.

15. **A** $(x - 9)^2$

1 The first and last terms are perfect squares: $x^2 = (x)^2$ and $81 = 9^2$.

2 The middle term is $-2(x)(9) = -18x$, so it is a perfect square: $x^2 - 18x + 81 = (x - 9)^2$.

WHY NOT (B) uses 81 and 1, which multiply correctly but give the wrong middle term. (C) gets the sign wrong; the middle term tells you the sign. (D) is a difference of squares, which has no middle term.

16. **B** 15

1 Each exterior angle is $180° - 156° = 24°$.

2 The exterior angles add up to 360°, so the number of sides is $360 \div 24 = 15$.

3 Check: $(15 - 2) \times 180° \div 15 = 156°$. ✓

WHY NOT (D) gives the exterior angle instead of the number of sides.

17. **C** 43

1 In an isosceles triangle the two base angles (opposite the equal sides) are equal. Call each one x.

2 The angles add up to 180: $x + x + 94 = 180$, so $2x = 180 - 94 = 86$.

3 Divide by 2: $x = 86 \div 2 = 43$.

WHY NOT (D) forgets to split what is left between the two equal base angles.

18. **D** 160 m^2

1 Area of a trapezoid $= \frac{1}{2} \times (b_1 + b_2) \times h$: average the two parallel bases, then multiply by the height.

2 Add the bases: $22 + 10 = 32$.

3 $\frac{1}{2} \times 32 \times 10 = 16 \times 10 = 160$, so the area is 160 square meters.

WHY NOT (A) uses only the shorter base.

19. **A** $(900 - 225\pi)$ m^2

1 The circle touches all four sides, so its diameter equals the side of the square: $d = 30$, so $r = 15$.

2 Square: $30^2 = 900$. Circle: $\pi \times 15^2 = 225\pi$.

3 Shaded region = square − circle $= 900 - 225\pi$ square meters.

TIP Sense check: 225π is a little more than 675, so the shaded corners are only a small part of the square's 900, just as the figure shows.

WHY NOT (B) finds the area of the circle, not the shaded corners. (C) subtracts the circumference instead of the area of the circle. (D) forgets to square the radius.

20. **B** 18

1 Count how many fit along each edge: $9 \div 3 = 3$, $6 \div 3 = 2$, and $9 \div 3 = 3$.

2 Multiply: $3 \times 2 \times 3 = 18$.

TIP Check with volumes: the container holds $9 \times 6 \times 9 = 486$ cubic inches, and one cube is $3^3 = 27$, so $486 \div 27 = 18$.

WHY NOT (A) counts only the bottom layer. (C) divides the volume by the edge length instead of by the volume of one cube. (D) finds the volume of the container but forgets to divide.

21. **A** 15

1 Horizontal change: $|-5 - 7| = 12$. Vertical change: $|-2 - (-11)| = 9$.

2 These are the legs of a right triangle, and the distance is the hypotenuse: $d = \sqrt{12^2 + 9^2} = \sqrt{144 + 81} = \sqrt{225} = 15$.

TIP The legs 9 and 12 belong to the Pythagorean triple 9-12-15.

WHY NOT (B) adds the horizontal and vertical distances instead of using the Pythagorean theorem. (C) multiplies the horizontal and vertical distances. (D) forgets to take the square root.

22. **B** $1,598

1 Amount saved: $\frac{1}{10} \times \$2{,}820 = \$282$.

2 Amount sent home: $\frac{1}{3} \times \$2{,}820 = \$940$.

3 What is left: $\$2{,}820 - \$282 - \$940 = \$1{,}598$.

WHY NOT (A) is the total of the two amounts, not what is left. (C) subtracts only the amount sent home. (D) subtracts only the amount saved.

23. **B** $887.40

1 First discount: $\$1{,}160 - 0.1 \times \$1{,}160 = \$1{,}160 - \$116 = \$1{,}044$.

2 The second discount is taken from the *new* price: $\$1{,}044 - 0.15 \times \$1{,}044 = \$1{,}044 - \$156.60 = \$887.40$.

TIP In one line: $\$1{,}160 \times 0.9 \times 0.85 = \$887.40$.

WHY NOT (A) adds the discounts to get 25% off. (C) applies only the second discount. (D) stops after the first discount.

24. **D** 3.5 hours

1 First part: $130 \div 65 = 2$ hours.

2 Second part: $30 \div 20 = 1.5$ hours.

3 Total: $2 + 1.5 = 3.5$ hours.

WHY NOT (A) finds the time for the first part only.

25. **C** 1:30 p.m.

1 Travel time: $125 \div 50 = 2\frac{1}{2}$ hours.

2 Change the fraction of an hour to minutes: $\frac{1}{2} \times 60 = 30$ minutes, so the trip takes 2 hours 30 minutes.

3 Add to the start time: 11:00 a.m. plus 2 hours 30 minutes is 1:30 p.m.

WHY NOT (B) leaves out the part of an hour. (D) reads 2.5 hours as 2 hours 50 minutes.

26. **B** 36 gallons

1 Pure fertilizer now: $0.75 \times 18 = 13.5$ gallons. Adding water does not change this amount.

2 In the new mixture these 13.5 gallons must be 25% of the total: $13.5 \div 0.25 = 54$ gallons in all.

3 Water to add: $54 - 18 = 36$ gallons.

WHY NOT (A) makes two slips: flips the percents ($\frac{25}{75}$ instead of $\frac{75}{25}$) and stops at the total instead of finding the water added. (D) is the final amount of the mixture, not the water added.

27. **A** 27%

1 Pure fertilizer in each: $0.25 \times 12 = 3$ and $0.3 \times 8 = 2.4$ gallons.

2 Totals: $3 + 2.4 = 5.4$ gallons of fertilizer in $12 + 8 = 20$ gallons of mixture.

3 Percent: $\frac{5.4}{20} = 0.27 = 27\%$.

TIP Check: 27% is between 25% and 30%, closer to the percent of the larger amount.

WHY NOT (B) averages the two percents without weighting by the amounts. (C) weights each percent by the wrong amount. (D) adds the two percents.

28. **C** 42

1 Each choice can be paired with every other choice, so multiply the number of options at each step (the counting principle).

2 $7 \times 2 \times 3 = 42$.

WHY NOT (A) adds the choices instead of multiplying them. (B) leaves out one of the choices.

29. **A** 11 hours 20 minutes

1 15 minutes minus 55 minutes won't go, so borrow 1 hour: 1715 becomes 16 hours 75 minutes.

2 Subtract: $16 - 5 = 11$ hours and $75 - 55 = 20$ minutes.

3 The time is 11 hours 20 minutes.

WHY NOT (B) subtracts 1715 – 0555 like ordinary numbers and reads 1160 as 11 hours 60 minutes (that is, 12 hours). (C) borrows the hour but forgets to take it away from the hours. (D) subtracts the smaller number of minutes from the larger instead of borrowing.

30. **D** 96

1 First step: $\frac{1}{5} \times 360 = 72$. Remainder: $360 - 72 = 288$.

2 Second step: $\frac{1}{3}$ of the *remainder*: $\frac{1}{3} \times 288 = 96$.

WHY NOT (B) gives the water used in the morning, not the water used in the afternoon.

PART I

Number Skills

Whole numbers, signed numbers, factors, fractions, decimals, percents, ratios, exponents, and roots—the toolkit behind every other question.

1 Whole Numbers & Order of Operations
2 Negative Numbers & Absolute Value
3 Factors, Multiples, Primes, GCF & LCM
4 Fractions & Mixed Numbers
5 Decimals
6 Percents
7 Ratios & Proportions
8 Exponents & Scientific Notation
9 Square Roots & Radicals

CHAPTER 1

Whole Numbers & Order of Operations

Tested on: Mathematics Knowledge • Arithmetic Reasoning

Key Concepts

Every other chapter in this book rests on whole-number arithmetic. With no calculator on the ASVAB, you need three things: a feel for place value, one firm rule for the order of operations, and a habit of reading word problems for what they *really* ask.

Place value and rounding

In 4,725,316 the digits name, from the left: millions (4), hundred-thousands (7), ten-thousands (2), thousands (5), hundreds (3), tens (1), ones (6). The *value* of the 7 is 700,000.
To round, look at the digit just to the *right* of the rounding place: 5 or more → round up; 4 or less → keep the digit. Then change every digit after the rounding place to 0. 38,462 → 38,000 (nearest thousand) and 38,500 (nearest hundred).

Order of operations (PEMDAS)

1. **P**arentheses (innermost first)
2. **E**xponents ($4^2 = 4 \times 4 = 16$)
3. **M**ultiplication and **D**ivision, a tie: work *left to right*
4. **A**ddition and **S**ubtraction, a tie: work *left to right*

Example: $20 - 12 \div 4 \times 2 = 20 - 3 \times 2 = 20 - 6 = 14$.

Properties that make mental math easy

Commutative (order)	$a + b = b + a, \quad a \times b = b \times a$
Associative (grouping)	$(a + b) + c = a + (b + c)$
Distributive	$a \times (b + c) = a \times b + a \times c$
Identity	$a + 0 = a, \quad a \times 1 = a$

The distributive property turns hard products into easy ones: $7 \times 98 = 7 \times 100 - 7 \times 2 = 700 - 14 = 686$.

WORKED EXAMPLE

A convoy must move 250 soldiers. Each truck carries 24 soldiers. How many trucks are needed?
Solution. $250 \div 24 = 10$ remainder 10. Ten trucks carry 240 soldiers, and the other 10 still need a ride, so you need **11** trucks. When a question asks how many containers, vehicles, or trips are *needed*, round *up*.

No-Calculator Tip Estimate by rounding each number to its first digit, then count zeros: $398 \times 51 \approx 400 \times 50$; $4 \times 5 = 20$, and the three zeros give 20,000. Estimating first also catches answers that are off by a factor of 10.

Common Traps

- Adding before multiplying: $6 + 4 \times 3 = 18$, not 30.
- Multiplying before an earlier division: $24 \div 4 \times 2 = 12$, not 3.
- Rounding *down* the number of buses, boxes, or trips needed.
- Giving the remainder when the question asks for the quotient (or the reverse).

Practice Set 1

Score: _____ / 25 Target time: 27 minutes

Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

1. Simplify: $(7 + 6) \times 2$

A 16 **B** 19 **C** 26 **D** 36

2. Which digit is in the ten-thousands place in 54,315?

A 3 **B** 4 **C** 5 **D** 7

3. Evaluate: $(6 + 11) \times 6$

A 72 **B** 102 **C** 112 **D** 122

4. Round 8,184 to the nearest hundred.

A 8,000 **B** 8,100 **C** 8,180 **D** 8,200

5. Which equation shows the distributive property?

A $(5 + 6) + 11 = 5 + (6 + 11)$
B $7 \times (6 - 3) = 7 \times 6 - 7 \times 3$
C $74 + 0 = 74$
D $2 + 5 = 5 + 2$

6. What is 375×6?

A 1,650 **B** 1,820 **C** 2,240 **D** 2,250

7. Victor buys 2 rolls of tape at \$3 each and 2 gallons of paint at \$32 each. What is the total cost?

A \$35 **B** \$38 **C** \$70 **D** \$140

8. Corporal Lopez divides 445 rounds of ammunition equally among 5 soldiers. How many rounds does each soldier get?

A 8 **B** 87 **C** 89 **D** 440

9. Evaluate: $20 + 5 \times 5 - 25 \div 5$

A 20 **B** 40 **C** 60 **D** 80

10. Round 21,907 to the nearest hundred.

A 21,900 **B** 21,910 **C** 22,000 **D** 22,200

11. What is the value of $(3 + 4) \times 3 - 15$?

A 0 **B** 5 **C** 6 **D** 12

12. What is 12×304?

A 3,604 **B** 3,648 **C** 4,148 **D** 4,648

13. Which of the following is the closest estimate of 297×83?

A 24,000 **B** 27,000
C 240,000 **D** 2,400,000

14. Divide: $840 \div 8$

A 15 **B** 105 **C** 150 **D** 1,050

15. Multiply: 74×63

A 666 **B** 4,662 **C** 4,762 **D** 44,622

16. A wedding has 237 guests. Each table seats 8 guests. What is the least number of tables needed so that every guest has a seat?

A 5 **B** 29 **C** 30 **D** 31

17. A supply clerk packs 247 MREs into cases of 12. After filling as many full cases as possible, how many MREs are left over?

A 3 **B** 5 **C** 6 **D** 7

18. Carlos buys 8 packs of socks at \$7 each and 6 hats at \$17 each and pays with two \$100 bills. How much change should Carlos get back?

A \$42 **B** \$127 **C** \$158 **D** \$176

19. Evaluate: $6 \times (3 + 3)^2 \div 4$

A 18 **B** 49 **C** 54 **D** 324

20. A total of 38 people are going to a family reunion. Each van can carry 12 people. Each van costs $95 to rent for the day. What is the total cost of the vans needed?

A $190 **B** $285 **C** $380 **D** $475

21. Evaluate: $3 \times (3 + 3)^2 \div 2$

A 18 **B** 44 **C** 49 **D** 54

22. A school is taking 311 students on a field trip. Each bus holds 40 students. If they use the least number of buses possible, how many more students could the last bus hold?

A 7 **B** 8 **C** 9 **D** 31

23. Evaluate: $41 - 6 \times (6 + 3) \div 3 + 2$

A 8 **B** 10 **C** 21 **D** 25

24. Tyrone buys a laptop for $1,280 and 2 software programs for $50 each. He pays $300 up front and splits the rest into 4 equal monthly payments. How much is each monthly payment?

A $270 **B** $345 **C** $420 **D** $1,080

25. Each of 4 trucks carries 372 cases of water to a training area. The water is shared equally among 12 companies. How many cases does each company get?

A 31 **B** 122 **C** 123 **D** 124

Answers & Explanations 1

1 C	**6** D	**11** C	**16** C	**21** D
2 C	**7** C	**12** B	**17** D	**22** C
3 B	**8** C	**13** A	**18** A	**23** D
4 D	**9** B	**14** B	**19** C	**24** A
5 B	**10** A	**15** B	**20** C	**25** D

1. **C** 26

1 Parentheses first: $7 + 6 = 13$, which leaves 13×2.

2 Multiply: $13 \times 2 = 26$.

WHY NOT (B) ignores the parentheses.

2. **C** 5

1 Read the digits from the right, naming each place: 5 (ones), 1 (tens), 3 (hundreds), 4 (thousands), 5 (ten-thousands).

2 So the ten-thousands digit is 5.

WHY NOT (A) is the digit in the hundreds place. (B) is the digit in the thousands place.

3. **B** 102

1 Parentheses first: $6 + 11 = 17$, which leaves 17×6.

2 Multiply: $17 \times 6 = 102$.

WHY NOT (A) ignores the parentheses.

4. **D** 8,200

1 The hundreds digit is 1. Look at the digit just to its right (the tens digit): 8.

2 Because 8 is 5 or more, round up: the hundreds digit 1 becomes 2.

3 Change every digit after the hundreds place to zero: 8,200.

WHY NOT (A) rounds to the nearest thousand instead. (B) rounds down even though the next digit, 8, is 5 or more. (C) rounds to the nearest ten instead.

5. **B** $7 \times (6 - 3) = 7 \times 6 - 7 \times 3$

1 The distributive property says a number times a sum (or difference) equals the sum (or difference) of the separate products: $a \times (b + c) = a \times b + a \times c$.

2 Only $7 \times (6 - 3) = 7 \times 6 - 7 \times 3$ fits that pattern.

WHY NOT (A) shows the associative property: only the grouping changes. (C) shows the identity property: adding 0 or multiplying by 1 changes nothing. (D) shows the commutative property: only the order changes.

6. **D** 2,250

1 Break 375 into place values: $300 + 70 + 5$.

2 Multiply each part by 6: $300 \times 6 = 1{,}800$, $70 \times 6 = 420$, $5 \times 6 = 30$.

3 Add the parts: $1{,}800 + 420 + 30 = 2{,}250$.

WHY NOT (B) forgets to add the carried digits.

7. **C** $70

1 Find the cost of each kind of item: $2 \times 3 = 6$ and $2 \times 32 = 64$.

2 Add: $6 + 64 = 70$ dollars.

WHY NOT (A) adds one of each price and ignores the quantities. (B) forgets to multiply the second price by 2. (D) multiplies the total number of items by the sum of the prices.

8. **C** 89

1 Sharing equally means dividing the total by the number of shares.

2 $445 \div 5 = 89$. Check: $5 \times 89 = 445$.

WHY NOT (A) stops dividing too soon and drops the last digit of the quotient. (D) subtracts instead of dividing.

9. **B** 40

1 Multiply (multiplication and division go left to right): $5 \times 5 = 25$, which leaves $20 + 25 - 25 \div 5$.

2 Divide before you add or subtract: $25 \div 5 = 5$, which leaves $20 + 25 - 5$.

3 Add (addition and subtraction go left to right): $20 + 25 = 45$, which leaves $45 - 5$.

4 Subtract: $45 - 5 = 40$.

WHY NOT (A) works straight from left to right instead of multiplying and dividing first.

10. **A** 21,900

1 The hundreds digit is 9. Look at the digit just to its right (the tens digit): 0.

2 Because 0 is less than 5, keep the hundreds digit 9 as it is.

3 Change every digit after the hundreds place to zero: 21,900.

WHY NOT (B) rounds to the nearest ten instead. (C) rounds up even though the next digit, 0, is less than 5.

11. **C** 6

1 Parentheses first: $3+4=7$, which leaves $7\times3-15$.

2 Multiply before you add or subtract: $7\times3=21$, which leaves $21-15$.

3 Subtract: $21-15=6$.

WHY NOT (A) ignores the parentheses.

12. **B** 3,648

1 Write 304 as $300+4$, an easy number plus a little.

2 Distribute the 12: $12\times300+12\times4=3{,}600+48$.

3 Add: $3{,}600+48=3{,}648$.

WHY NOT (A) adds 4 instead of 12×4.

13. **A** 24,000

1 Round each number to its first digit: 297 is about 300, and 83 is about 80.

2 Multiply the leading digits: $3\times8=24$.

3 Attach the three zeros (two from 300, one from 80): 24,000.

WHY NOT (B) rounds 83 to 90 instead of 80. (C) adds an extra zero. (D) adds two extra zeros.

14. **B** 105

1 Start with 8: $8\div8=1$.

2 Bring down the 4: 8 does not go into 4, so write 0 in the quotient.

3 Bring down the 0 to make 40: $40\div8=5$.

4 The quotient is 105. Check: $8\times105=840$.

WHY NOT (A) skips the zero in the quotient. (C) puts the zero in the wrong place. (D) adds an extra zero.

15. **B** 4,662

1 Split 63 into $60+3$ and multiply 74 by each part.

2 $74\times3=222$ and $74\times60=4{,}440$.

3 Add the partial products: $222+4{,}440=4{,}662$.

WHY NOT (A) forgets the placeholder zero, multiplying by 6 instead of 60. (D) puts two placeholder zeros in the second row instead of one.

16. **C** 30

1 Divide: $237\div8=29$ remainder 5.

2 The 29 full tables hold $29\times8=232$ guests, so 5 guests are still left.

3 They need one more table: $29+1=30$ tables.

WHY NOT (A) is the number of guests left over after filling 29 tables. (B) rounds down, which leaves 5 guests with no table.

17. **D** 7

1 Divide: $247\div12$. Since $20\times12=240$ and $21\times12=252$ is too many, 20 full cases can be filled.

2 Left over: $247-240=7$ MREs.

WHY NOT (B) is how many more MREs it would take to fill another case.

18. **A** $42

1 Find the cost of each kind of item: $8\times7=56$ and $6\times17=102$.

2 Total cost: $56+102=158$ dollars.

3 Change: $200-158=42$ dollars.

WHY NOT (B) forgets to multiply the second price by 6. (C) is the total cost, not the change. (D) subtracts one of each price and ignores the quantities.

19. **C** 54

1 Parentheses first: $3+3=6$, which leaves $6\times(6)^2\div4$.

2 Exponent next: $6^2=6\times6=36$, which leaves $6\times36\div4$.

3 Multiply (multiplication and division go left to right): $6\times36=216$, which leaves $216\div4$.

4 Divide: $216\div4=54$.

WHY NOT (A) doubles instead of squaring. (D) multiplies before applying the exponent.

20. **C** $380

1 Divide: $38\div12=3$ remainder 2.

2 The 2 leftover people need one more van, so 4 vans are needed.

3 Cost: $4\times95=380$ dollars.

WHY NOT (B) rounds the number of vans down.

21. **D** 54

1 Parentheses first: $3+3=6$, which leaves $3\times(6)^2\div2$.

2 Exponent next: $6^2=6\times6=36$, which leaves $3\times36\div2$.

3 Multiply (multiplication and division go left to right): $3\times36=108$, which leaves $108\div2$.

4 Divide: $108\div2=54$.

WHY NOT (A) doubles instead of squaring.

22. **C** 9

1 Divide: $311\div40=7$ remainder 31. So 8 buses are needed.

2 The last bus holds only the 31 leftover students.

3 Open spaces in it: $40-31=9$.

WHY NOT (A) is the number of full buses. (B) is the number of buses needed. (D) is the number of students in the last bus, not the open spaces.

23. **D** 25

1 Parentheses first: $6 + 3 = 9$, which leaves $41 - 6 \times 9 \div 3 + 2$.

2 Multiply (multiplication and division go left to right): $6 \times 9 = 54$, which leaves $41 - 54 \div 3 + 2$.

3 Divide before you add or subtract: $54 \div 3 = 18$, which leaves $41 - 18 + 2$.

4 Subtract (addition and subtraction go left to right): $41 - 18 = 23$, which leaves $23 + 2$.

5 Add: $23 + 2 = 25$.

WHY NOT (A) ignores the parentheses. (C) adds before subtracting; addition and subtraction go left to right.

24. **A** $270

1 Total cost: $1{,}280 + 2 \times 50 = 1{,}280 + 100 = 1{,}380$ dollars.

2 Subtract the up-front payment: $1{,}380 - 300 = 1{,}080$ dollars.

3 Split into 4 equal payments: $1{,}080 \div 4 = 270$ dollars.

WHY NOT (B) forgets to subtract the up-front payment. (C) adds the up-front payment instead of subtracting it. (D) is the amount still owed, not each monthly payment.

25. **D** 124

1 Total cases: $4 \times 372 = 1{,}488$.

2 Share among 12 companies: $1{,}488 \div 12 = 124$ cases each.

WHY NOT (A) shares only one truck's load.

CHAPTER 2

Negative Numbers & Absolute Value

Tested on: Mathematics Knowledge • Arithmetic Reasoning

Key Concepts

Negative numbers show up whenever something goes below a starting point: temperatures below zero, depths below sea level, an overdrawn bank account. A handful of sign rules covers every problem in this chapter.

The number line and absolute value

Numbers increase to the right. Every negative number is less than 0, and the farther left, the smaller: $-9 < -4 < 0 < 3$.
The *absolute value* $|x|$ is the distance from 0, so it is never negative: $|-7| = 7$ and $|7| = 7$. The distance between two numbers is the absolute value of their difference: from -3 to 5 is $|5 - (-3)| = 8$.

Adding and subtracting

- **Same signs:** add the absolute values, keep the sign: $-6 + (-9) = -15$.
- **Different signs:** subtract the absolute values, keep the sign of the number with the larger absolute value: $-11 + 4 = -7$.
- **Subtracting** means adding the opposite: $5 - 8 = 5 + (-8) = -3$ and $-7 - (-12) = -7 + 12 = 5$.

Multiplying and dividing

Same signs give a positive answer; different signs give a negative answer: $(-6)(-7) = 42$, $-48 \div 6 = -8$. With several factors, count the negatives: an even number of them gives a positive product, an odd number a negative one. Watch exponents: $(-3)^2 = (-3)(-3) = 9$, but $-3^2 = -(3 \times 3) = -9$.

WORKED EXAMPLE

At 5 a.m. the temperature was $-8°$F. By noon it was 13°F. How many degrees did the temperature rise?
Solution. Rise = new − old = $13 - (-8) = 13 + 8 = 21$ degrees. On a number line: 8 degrees up to 0, then 13 more.

No-Calculator Tip For a change or a distance, think "how far apart on the number line." If the two numbers are on opposite sides of 0, *add* their absolute values (-8 to 13: $8 + 13 = 21$); if they are on the same side, *subtract* them.

Common Traps

- Reading $-7 - (-12)$ as $-7 - 12$. Two negatives in a row make a plus.
- Thinking -12 is greater than -5 because $12 > 5$.

- Squaring before the sign: $-4^2 = -16$, while $(-4)^2 = 16$.
- Keeping a negative sign inside absolute value bars: $|3 - 9| = 6$, not -6.

Practice Set 2

Score: _____ / 25 Target time: 26 minutes

Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

1. What is $-15 - 9$?

A -27 **B** -25 **C** -24 **D** 24

2. What is $6 \div (-2)$?

A -13 **B** -5 **C** -3 **D** 3

3. What is $7 - 8$?

A -5 **B** -1 **C** 1 **D** 15

4. What is $(-5)(-8)$?

A -40 **B** -13 **C** 40 **D** 44

5. Evaluate: $|-7| - |15|$

A -22 **B** -8 **C** 8 **D** 22

6. How many units apart are 2 and -9 on a number line?

A -11 **B** 7 **C** 11 **D** 14

7. Of -14, -10, -19, and -1, which number is the greatest?

A -1 **B** -19 **C** -14 **D** -10

8. At noon the temperature at a farm in North Dakota was 11°F. By midnight it had dropped 30 degrees. What was the temperature at midnight?

A -41°F **B** -19°F **C** 19°F **D** 41°F

9. Simplify: $-13 - (-17)$

A 4 **B** 7 **C** 9 **D** 30

10. A Navy helicopter hovers at an altitude of 1,300 feet. Directly below it, a submarine is at -400 feet (below sea level). What is the vertical distance between the helicopter and the submarine?

A 900 ft **B** 1,700 ft **C** 1,750 ft **D** 1,800 ft

11. Evaluate: $-5 - (-5) - (-2)$

A -12 **B** -2 **C** -1 **D** 2

12. The elevation of a ridge on a training range is 2,050 feet. The elevation of a dry lake bed is -220 feet (below sea level). How much higher is the ridge on a training range than the dry lake bed?

A 1,830 ft **B** 2,270 ft **C** 2,280 ft **D** 2,320 ft

13. Evaluate: $(-42 \div 7) \times 2$

A -13 **B** -12 **C** -3 **D** 12

14. Evaluate: $|-2 - 12| - |8|$

A -22 **B** -6 **C** 5 **D** 6

15. Which list shows the numbers 9, -15, -5, -11, 8 in order from least to greatest?

A $-15, -11, -5, 8, 9$ **B** $9, 8, -5, -11, -15$
C $-5, 8, 9, -11, -15$ **D** $-5, -11, -15, 8, 9$

16. On a number line, point M is halfway between -11 and -7. What number is at M?

A -14 **B** -9 **C** 2 **D** 9

17. At 6 p.m. the temperature at a ski resort was 2°F. It then fell 5 degrees every hour for 4 hours. What was the temperature at the end of that time?

A -23°F **B** -20°F **C** -19°F **D** -18°F

18. Private Lopez has \$170 in a checking account. The private pays \$85 for groceries and \$197 for a car repair. Because the balance went below \$0, the bank also charges a \$30 overdraft fee. What is the account balance now?

A $-\$482$ **B** $-\$147$ **C** $-\$142$ **D** $-\$112$

19. If $x = -9$ and $y = -4$, what is $-x^2 - y$?

A -77 **B** 22 **C** 77 **D** 85

20. Evaluate: $(-9 - 7) \times (-4) - 12$

A 52 **B** 54 **C** 76 **D** 256

21. What is the value of $-x^2 - y$ when $x = -9$ and $y = -5$?

A -86 **B** -84 **C** -76 **D** 86

22. What is the value of $-3 \times (3 - 9) + (-16) \div (-2)$?

A -10 **B** 26 **C** 30 **D** 36

23. What is the value of $|x+y| - |x| - |y|$ when $x = -5$ and $y = 2$?

A -8 **B** -7 **C** -4 **D** 0

24. A submarine is at -600 feet (below sea level). It rises 25 feet per minute for 7 minutes and then dives 450 feet. What is its new position?

A $-1{,}225$ ft **B** $-1{,}025$ ft
C -875 ft **D** 25 ft

25. Specialist Haddad has \$80 in a checking account. The specialist pays \$50 for a concert ticket and \$126 for car insurance. Because the balance went below \$0, the bank also charges a \$30 overdraft fee. How much must Specialist Haddad deposit to bring the balance back up to \$0?

A \$96 **B** \$116 **C** \$121 **D** \$126

Answers & Explanations 2

1 C	**6** C	**11** D	**16** B	**21** C
2 C	**7** A	**12** B	**17** D	**22** B
3 B	**8** B	**13** B	**18** C	**23** C
4 C	**9** A	**14** D	**19** A	**24** C
5 B	**10** B	**15** A	**20** A	**25** D

1. **C** -24

1 Rewrite each subtraction as adding the opposite: $-15 - 9 = -15 + (-9)$.

2 Same signs: add the absolute values ($15 + 9 = 24$) and keep the sign: $-15 + (-9) = -24$.

WHY NOT (D) has the right size but the wrong sign.

2. **C** -3

1 Divide the absolute values: $6 \div 2 = 3$.

2 The signs are different, so the answer is negative: $6 \div (-2) = -3$.

WHY NOT (D) uses the wrong sign: different signs give a negative answer.

3. **B** -1

1 Rewrite each subtraction as adding the opposite: $7 - 8 = 7 + (-8)$.

2 Different signs: subtract the absolute values ($8 - 7 = 1$) and keep the sign of -8, which has the larger absolute value: $7 + (-8) = -1$.

WHY NOT (C) has the right size but the wrong sign. (D) adds the numbers instead of subtracting.

4. **C** 40

1 Multiply the absolute values: $5 \times 8 = 40$.

2 The signs are the same, so the answer is positive: $(-5)(-8) = 40$.

WHY NOT (A) uses the wrong sign: same signs give a positive answer. (B) adds the numbers instead of multiplying.

5. **B** -8

1 Find each absolute value: $|-7| = 7$ and $|15| = 15$.

2 Subtract: $7 - 15 = -8$.

WHY NOT (A) subtracts the numbers inside the bars and keeps their signs. (C) has the right size but the wrong sign. (D) adds the absolute values instead of subtracting.

6. **C** 11

1 Distance is the larger number minus the smaller number (or the absolute value of the difference).

2 $2 - (-9) = 2 + 9 = 11$, or count 9 units to 0 and 2 more.

WHY NOT (A) subtracts in the wrong order; a distance is never negative. (B) subtracts the absolute values; the points are on opposite sides of 0, so the distances add.

7. **A** -1

1 All four numbers are negative. On a number line, the greatest one is the one farthest to the right, which is the one closest to 0.

2 -1 is closest to 0, so it is the greatest: $-19 < -14 < -10 < -1$.

WHY NOT (B) has the largest absolute value, but it is the farthest left of 0. (C) is farther left of 0 than -1. (D) is farther left of 0 than -1.

8. **B** $-19°$F

1 A drop means subtract: $11 - 30$.

2 $11 - 30 = 11 + (-30)$. Different signs: subtract the absolute values ($30 - 11 = 19$) and keep the sign of -30, which has the larger absolute value: $11 + (-30) = -19$. The temperature was $-19°$F.

WHY NOT (C) subtracts the smaller number from the larger and drops the negative sign. (D) adds the drop instead of subtracting it.

9. **A** 4

1 Rewrite each subtraction as adding the opposite: $-13 - (-17) = -13 + 17$.

2 Different signs: subtract the absolute values ($17 - 13 = 4$) and keep the sign of 17, which has the larger absolute value: $-13 + 17 = 4$.

WHY NOT (D) ignores the signs and adds all the numbers.

10. **B** 1,700 ft

1 Distance = higher position − lower position: 1,300 − (−400).

2 Subtracting a negative means adding: 1,300 + 400 = 1,700 feet.

TIP Sea level is 0: 1,300 feet above it plus 400 feet below it.

WHY NOT (A) subtracts the depth instead of adding it.

11. **D** 2

1 Rewrite each subtraction as adding the opposite: $-5-(-5)-(-2) = -5+5+2$.

2 Opposites add to zero: $-5+5=0$.

3 $0+2=2$.

WHY NOT (A) subtracts 5 instead of adding it; subtracting a negative means adding. (B) has the right size but the wrong sign.

12. **B** 2,270 ft

1 Difference = higher − lower: 2,050 − (−220).

2 Subtracting a negative means adding: 2,050 + 220 = 2,270 feet.

WHY NOT (A) subtracts the depth instead of adding it.

13. **B** −12

1 Parentheses first: $-42 \div 7 = -6$ (different signs, negative).

2 Then multiply: $(-6) \times 2 = -12$ (different signs, negative).

WHY NOT (C) divides −42 by 7×2 instead of doing the division in parentheses first. (D) gets the sign wrong in one of the steps.

14. **D** 6

1 Work inside the bars first: $-2-12=-14$, so $|-2-12| = |-14| = 14$.

2 Also $|8| = 8$.

3 Subtract: $14-8=6$.

WHY NOT (A) treats $|-2-12|$ as −14 instead of 14. (B) has the right size but the wrong sign.

15. **A** $-15, -11, -5, 8, 9$

1 Negative numbers come first. Among them, the one farthest from 0 is the least: $-15 < -11 < -5$.

2 Then the positive numbers: $8 < 9$.

3 So the order is $-15, -11, -5, 8, 9$.

WHY NOT (B) lists the numbers from greatest to least. (C) orders the numbers by their distance from 0, ignoring the signs. (D) puts the negative numbers in the wrong order; for negatives, the larger absolute value is the smaller number.

16. **B** −9

1 The halfway point is the average of the two numbers: $(-11+(-7)) \div 2$.

2 $-11+(-7) = -18$, and $-18 \div 2 = -9$.

3 Check: −9 is 2 units from each number.

WHY NOT (C) is half the distance between the numbers, not the point halfway between them. (D) has the right size but the wrong sign.

17. **D** −18°F

1 Total drop: $5 \times 4 = 20$ degrees.

2 Subtract it from the starting temperature: $2-20 = -18$.

3 The temperature was −18°F.

WHY NOT (B) forgets the starting temperature.

18. **C** −$142

1 Start with $170 and subtract both payments: $170-85-197=-112$.

2 Subtract the fee: $-112-30=-142$.

3 The balance is −$142: the account is overdrawn by $142.

WHY NOT (A) treats the starting balance as money owed instead of money in the account. (D) forgets the overdraft fee.

19. **A** −77

1 Substitute, keeping negatives in parentheses: $-(-9)^2-(-4)$.

2 Square first: $(-9)^2 = 81$, so $-(-9)^2 = -81$.

3 Subtracting −4 means adding 4: $-81+4=-77$.

WHY NOT (B) multiplies x by 2 instead of squaring it: $-2(-9) = 18$. (C) makes both mistakes: treats $-x^2$ as positive and subtracts 4 instead of adding it. (D) treats $-x^2$ as $(-x)^2$, which is positive.

20. **A** 52

1 Parentheses first: $-9-7=-16$.

2 Multiply: $(-16)\times(-4) = 64$ (same signs, positive).

3 Subtract: $64-12=52$.

WHY NOT (C) adds 12 instead of subtracting it. (D) subtracts before multiplying.

21. **C** −76

1 Substitute, keeping negatives in parentheses: $-(-9)^2-(-5)$.

2 Square first: $(-9)^2 = 81$, so $-(-9)^2 = -81$.

3 Subtracting −5 means adding 5: $-81+5=-76$.

WHY NOT (A) subtracts 5 instead of subtracting −5; subtracting a negative means adding. (D) treats $-x^2$ as $(-x)^2$, which is positive.

22. **B** 26

1 Parentheses first: $3 - 9 = -6$.

2 Multiply: $-3 \times (-6) = 18$ (same signs, positive).

3 Divide: $(-16) \div (-2) = 8$ (same signs, positive).

4 Add: $18 + 8 = 26$.

WHY NOT (A) gets the sign of $-3 \times (-6)$ wrong.

23. **C** -4

1 Substitute: $|-5 + 2| - |-5| - |2|$.

2 Inside the first bars: $-5 + 2 = -3$, and $|-3| = 3$. Also $|-5| = 5$ and $|2| = 2$.

3 Subtract from left to right: $3 - 5 - 2 = -4$.

WHY NOT (D) keeps the signs instead of taking absolute values.

24. **C** -875 ft

1 The rise: $25 \times 7 = 175$ feet up.

2 After rising: $-600 + 175 = -425$ feet.

3 After diving: $-425 - 450 = -875$ feet.

WHY NOT (A) treats the rise as going down. (B) forgets to multiply the rate by the number of minutes. (D) treats the dive as a rise.

25. **D** \$126

1 Balance after the payments and the fee: $80 - 50 - 126 - 30 = -126$.

2 Deposit needed = target − balance: $0 - (-126) = 0 + 126 = 126$.

3 Specialist Haddad must deposit \$126.

WHY NOT (A) forgets the overdraft fee.

CHAPTER 3

Factors, Multiples, Primes, GCF & LCM

Tested on: Mathematics Knowledge • Arithmetic Reasoning

Key Concepts

Factors and multiples are the building blocks of fractions (Chapter 4) and of many "when will it happen again?" word problems. Keep the vocabulary straight and the rest is careful listing.

Vocabulary

- A **factor** of n divides n evenly: the factors of 12 are $1, 2, 3, 4, 6, 12$.
- A **multiple** of n is n times a whole number: $12, 24, 36, \ldots$
- A **prime** number has exactly two factors, 1 and itself: $2, 3, 5, 7, 11, 13, 17, 19, 23, 29, \ldots$ The number 1 is *not* prime, and 2 is the only even prime.
- The **reciprocal** of $\frac{a}{b}$ is $\frac{b}{a}$; a number times its reciprocal is 1.

Prime factorization, GCF and LCM

Split a number into primes with a factor tree: $360 = 2^3 \times 3^2 \times 5$. For $36 = 2^2 \times 3^2$ and $48 = 2^4 \times 3$:

- **GCF** (greatest common factor): shared primes, *lowest* powers: $2^2 \times 3 = 12$.
- **LCM** (least common multiple): every prime, *highest* powers: $2^4 \times 3^2 = 144$.

Divisibility rules

by 2: last digit even
by 3: digit sum divisible by 3
by 4: last two digits divisible by 4
by 5: last digit 0 or 5
by 6: divisible by 2 *and* 3
by 9: digit sum divisible by 9

WORKED EXAMPLE

One bus leaves the depot every 12 minutes and another every 18 minutes. Both leave at 8:00 a.m. When do they next leave together?
Solution. You need the LCM of 12 and 18. Multiples of 18: $18, 36, \ldots$; 36 is also a multiple of 12. They leave together again after 36 minutes, at 8:36 a.m.

No-Calculator Tip Word-problem signal: "greatest number of identical kits," "largest equal groups," or "longest equal pieces" means **GCF**. "Next time together," "least number of packs so the counts match" means **LCM**.

Common Traps

- Mixing up GCF and LCM; the GCF is never larger than the smaller number, and the LCM is never smaller than the larger number.

- Multiplying two numbers and calling it the LCM ($12 \times 18 = 216$, but the LCM is 36).
- Calling 1 prime, or calling numbers like $51 = 3 \times 17$ and $91 = 7 \times 13$ prime.

Practice Set 3

Score: _____ / 25 Target time: 27 minutes

Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

1. Which one of 3, 21, 27, or 57 is a prime number?

A 3 **B** 21 **C** 27 **D** 57

2. Which of the following is a factor of 129?

A 15 **B** 25 **C** 43 **D** 258

3. What is the GCF of 32 and 12?

A 4 **B** 5 **C** 12 **D** 96

4. What is the least common multiple of 15 and 24?

A 3 **B** 60 **C** 120 **D** 360

5. What is the GCF of 60 and 48?

A 3 **B** 12 **C** 48 **D** 240

6. What is the LCM of 15 and 25?

A 5 **B** 37 **C** 65 **D** 75

7. Which number is a factor of 78?

A 11 **B** 16 **C** 39 **D** 156

8. Which one of 9, 23, 27, or 51 is a prime number?

A 9 **B** 23 **C** 27 **D** 51

9. What is the prime factorization of 420?

A $2^3 \times 3 \times 5 \times 7$ **B** $2^2 \times 3 \times 5 \times 7$
C $2 \times 3 \times 7 \times 10$ **D** $2 \times 3 \times 5 \times 7$

10. Which of the numbers 144, 524, 558, and 780 is divisible by both 4 and 9?

A 144 **B** 524 **C** 558 **D** 780

11. Two warning lights blink together. One blinks every 12 seconds and the other every 18 seconds. After how many seconds will they next blink at the same time?

A 36 seconds **B** 41 seconds
C 72 seconds **D** 216 seconds

12. A youth club has 96 bottles of water and 36 granola bars. Every care package must be identical, with nothing left over. What is the greatest number of care packages that can be made?

A 3 **B** 11 **C** 12 **D** 288

13. Specialist Chen checks the generator every 6 hours and refuels the vehicles every 9 hours. Both tasks were done at 0600. After how many hours will both tasks next fall at the same time?

A 3 hours **B** 15 hours
C 18 hours **D** 54 hours

14. A church group has 36 protein bars and 27 energy drinks. Every snack pack must be identical, with nothing left over. What is the greatest number of snack packs that can be made?

A 9 **B** 27 **C** 63 **D** 108

15. Which one of 336, 465, 740, or 924 is divisible by 3 *and* by 5?

A 336 **B** 465 **C** 740 **D** 924

16. What is the prime factorization of 1,056?

A $4 \times 2^3 \times 3 \times 11$ **B** $2 \times 3 \times 11$
C $2^4 \times 3 \times 11$ **D** $2^5 \times 3 \times 11$

17. What is the greatest common factor of 45, 40, and 35?

A 4 **B** 5 **C** 35 **D** 2,520

18. What is the smallest number that is a multiple of both 6 and 26?

A 2 **B** 39 **C** 68 **D** 78

19. A gardener wants to plant 152 tomato plants in equal rows with none left over. Each row must have at least 2 plants, and there must be at least 2 rows. How many different row lengths are possible?

A 2 **B** 4 **C** 5 **D** 6

20. Emily is buying juice boxes and granola bars. Juice boxes come in packs of 4, and granola bars come in packs of 6. She wants exactly the same number of juice boxes and granola bars. What is the least number of packs of juice boxes she can buy?

A 2 **B** 3 **C** 6 **D** 12

21. A training class has 18 soldiers from Alpha Company and 30 from Bravo Company. The instructor wants to form teams so that every team has the same number of Alpha soldiers and the same number of Bravo soldiers, with no one left out. If the greatest possible number of teams is formed, how many Alpha soldiers are on each team?

A 3 **B** 5 **C** 6 **D** 9

22. Ana is buying cups and lids. Cups come in packs of 18, and lids come in packs of 8. She wants exactly the same number of cups and lids. What is the least number of packs of lids she can buy?

A 4 **B** 9 **C** 18 **D** 72

23. A carpenter has two boards, 72 inches and 54 inches long. She wants to cut both boards into pieces that are all the same length, with no wood left over. If she cuts the longest possible pieces, how many pieces will she have in all?

A 7 **B** 14 **C** 18 **D** 126

24. How many positive factors does 154 have?

A 4 **B** 6 **C** 8 **D** 9

25. What is the smallest positive number that is divisible by each of 8, 10, 4, and 9?

A 40 **B** 72 **C** 180 **D** 360

Answers & Explanations 3

1 A	**6** D	**11** A	**16** D	**21** A
2 C	**7** C	**12** C	**17** B	**22** B
3 A	**8** B	**13** C	**18** D	**23** A
4 C	**9** B	**14** A	**19** D	**24** C
5 B	**10** A	**15** B	**20** B	**25** D

1. **A** 3

1 A prime number has exactly two factors: 1 and itself.

2 Each of the other choices has another factor: $21 = 3 \times 7$; $27 = 3 \times 9$; $57 = 3 \times 19$.

3 That leaves 3, which is prime: its only factors are 1 and 3.

WHY NOT (B) is not prime: $21 = 3 \times 7$. (C) is not prime: $27 = 3 \times 9$. (D) is not prime: $57 = 3 \times 19$.

2. **C** 43

1 A factor of 129 divides it with no remainder.

2 $129 \div 43 = 3$, so 43 is a factor. The choice 258 is bigger than 129, so it can only be a multiple.

WHY NOT (A) does not divide 129 evenly. (B) does not divide 129 evenly. (D) is a multiple of 129, not a factor.

3. **A** 4

1 List the factors of 12: 1, 2, 3, 4, 6, 12.

2 List the factors of 32: 1, 2, 4, 8, 16, 32.

3 The greatest number in both lists is 4.

WHY NOT (C) is the smallest number, but it is not a factor of 32. (D) is the least common multiple, not the greatest common factor.

4. **C** 120

1 List multiples of the larger number, 24: 24, 48, 72, 96, 120.

2 The first one that is also a multiple of 15 is 120 ($15 \times 8 = 120$).

WHY NOT (A) is the greatest common factor, not the least common multiple. (D) multiplies the numbers; that is a common multiple, but not the least one.

5. **B** 12

1 List the factors of 48: 1, 2, 3, 4, 6, 8, 12, 16, 24, 48.

2 List the factors of 60: 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60.

3 The greatest number in both lists is 12.

WHY NOT (A) is a common factor, but not the greatest one. (C) is the smallest number, but it is not a factor of 60. (D) is the least common multiple, not the greatest common factor.

6. **D** 75

1 List multiples of the larger number, 25: 25, 50, 75.

2 The first one that is also a multiple of 15 is 75 ($15 \times 5 = 75$).

WHY NOT (A) is the greatest common factor, not the least common multiple.

7. **C** 39

1 A factor of 78 divides it with no remainder.

2 $78 \div 39 = 2$, so 39 is a factor. The choice 156 is bigger than 78, so it can only be a multiple.

WHY NOT (A) does not divide 78 evenly. (B) does not divide 78 evenly. (D) is a multiple of 78, not a factor.

8. **B** 23

1 A prime number has exactly two factors: 1 and itself.

2 Each of the other choices has another factor: $9 = 3 \times 3$; $27 = 3 \times 9$; $51 = 3 \times 17$.

3 That leaves 23. It is not divisible by 2 or 3. Since $5 \times 5 = 25$ is already more than 23, no larger factor needs testing, so 23 is prime.

WHY NOT (A) is not prime: $9 = 3 \times 3$. (C) is not prime: $27 = 3 \times 9$. (D) is not prime: $51 = 3 \times 17$.

9. **B** $2^2 \times 3 \times 5 \times 7$

1 Divide by 2 as many times as you can: $420 \rightarrow 210 \rightarrow 105$ (2 factors of 2).

2 Divide by 3 once: $105 \rightarrow 35$ (1 factor of 3).

3 Divide by 5 once: $35 \rightarrow 7$ (1 factor of 5), and the last factor is 7.

4 So $420 = 2^2 \times 3 \times 5 \times 7$.

WHY NOT (A) multiplies out to 840, not 420. (C) uses 10, which is not prime. (D) multiplies out to 210, not 420.

10. **A** 144

1 Divisible by 4: the number formed by its last two digits is divisible by 4. Divisible by 9: its digit sum is divisible by 9.

2 The number 144 passes both tests (last two digits 44; digit sum 9): $144 \div 36 = 4$.

3 Each of the other choices fails one of the tests.

WHY NOT (B) is divisible by 4, but not by 9: its digit sum, 11, is not divisible by 9. (C) is divisible by 9, but not by 4: its last two digits, 58, are not divisible by 4. (D) is divisible by 4, but not by 9: its digit sum, 15, is not divisible by 9.

11. **A** 36 seconds

1 The events line up again after a common multiple of 12 and 18; the first time is the least common multiple.

2 Multiples of 18: 18, 36. The first one that is also a multiple of 12 is 36.

WHY NOT (C) is the second time they line up, not the first. (D) multiplies the two numbers; that is a common multiple, but not the first one.

12. **C** 12

1 Every care package gets the same number of bottles of water and the same number of granola bars, so the number of care packages must divide both 96 and 36. The greatest such number is the GCF.

2 Write each number as a product of primes: $96 = 2^5 \times 3$, $36 = 2^2 \times 3^2$.

3 Take each prime that both numbers share, with its lowest power: $2^2 \times 3 = 12$.

WHY NOT (A) is a common factor, but not the greatest one. (D) is the least common multiple, not the greatest common factor.

13. **C** 18 hours

1 The events line up again after a common multiple of 6 and 9; the first time is the least common multiple.

2 Multiples of 9: 9, 18. The first one that is also a multiple of 6 is 18.

WHY NOT (A) is the greatest common factor, not the least common multiple. (B) adds the two numbers. (D) multiplies the two numbers; that is a common multiple, but not the first one.

14. **A** 9

1 Every snack pack gets the same number of protein bars and the same number of energy drinks, so the number of snack packs must divide both 36 and 27. The greatest such number is the GCF.

2 Write each number as a product of primes: $36 = 2^2 \times 3^2$, $27 = 3^3$.

3 Take each prime that both numbers share, with its lowest power: $3^2 = 9$.

WHY NOT (B) does not divide 36 evenly. (C) adds the two numbers. (D) is the least common multiple, not the greatest common factor.

15. **B** 465

1 Divisible by 3: its digit sum is divisible by 3. Divisible by 5: its last digit is 0 or 5.

2 The number 465 passes both tests (digit sum 15; ends in 5): $465 \div 15 = 31$.

3 Each of the other choices fails one of the tests.

WHY NOT (A) is divisible by 3, but not by 5: it does not end in 0 or 5. (C) is divisible by 5, but not by 3: its digit sum, 11, is not divisible by 3. (D) is divisible by 3, but not by 5: it does not end in 0 or 5.

16. **D** $2^5 \times 3 \times 11$

1 Divide by 2 as many times as you can: $1{,}056 \rightarrow 528 \rightarrow 264 \rightarrow 132 \rightarrow 66 \rightarrow 33$ (5 factors of 2).

2 Divide by 3 once: $33 \rightarrow 11$ (1 factor of 3).

3 Divide by 11 once: $11 \rightarrow 1$ (1 factor of 11).

4 So $1{,}056 = 2^5 \times 3 \times 11$.

WHY NOT (A) uses 4, which is not prime. (B) lists each prime only once; that product is only 66. (C) multiplies out to 528, not 1,056.

17. **B** 5

1 Write each number as a product of primes: $35 = 5 \times 7$, $40 = 2^3 \times 5$, $45 = 3^2 \times 5$.

2 Take each prime that all three numbers share, with its lowest power: 5.

WHY NOT (C) is the smallest number, but it is not a factor of 40. (D) is the least common multiple, not the greatest common factor.

18. **D** 78

1 Write each number as a product of primes: $6 = 2 \times 3$, $26 = 2 \times 13$.

2 Take every prime that appears, with its highest power: $2 \times 3 \times 13 = 78$.

WHY NOT (A) is the greatest common factor, not the least common multiple.

19. **D** 6

1 Each possible row length is a factor of 152.

2 Factor pairs of 152: 1×152, 2×76, 4×38, 8×19.

3 Leave out 1 and 152 (a row of 1, or only one row). That leaves 2, 4, 8, 19, 38, 76: 6 possibilities.

WHY NOT (A) counts only the different prime factors (2 and 19).

20. **B** 3

1 The number of each item must be a common multiple of 4 and 6; the least one is the LCM.

2 Multiples of 6: 6, 12. The first that is also a multiple of 4 is 12, so she needs 12 of each.

3 Packs of juice boxes: $12 \div 4 = 3$.

WHY NOT (A) is the number of packs of granola bars. (C) buys one pack of juice boxes for each item in a pack of granola bars; that works, but it is not the least. (D) is the number of juice boxes, not the number of packs.

21. **A** 3

1 First find the greatest number of teams: it must divide both 18 and 30, so it is the GCF.

2 Write each number as a product of primes: $18 = 2 \times 3^2$, $30 = 2 \times 3 \times 5$.

3 Take each prime that both numbers share, with its lowest power: $2 \times 3 = 6$.

4 There are 6 teams, so each gets $18 \div 6 = 3$ Alpha soldiers.

WHY NOT (B) is the number of Bravo soldiers in each. (C) is the number of teams, not the number of Alpha soldiers in each. (D) splits into only 2 groups, a common factor that is not the greatest.

22. **B** 9

1 The number of each item must be a common multiple of 18 and 8; the least one is the LCM.

2 Multiples of 18: 18, 36, 54, 72. The first that is also a multiple of 8 is 72, so she needs 72 of each.

3 Packs of lids: $72 \div 8 = 9$.

WHY NOT (A) is the number of packs of cups. (C) buys one pack of lids for each item in a pack of cups; that works, but it is not the least. (D) is the number of lids, not the number of packs.

23. **A** 7

1 First find the longest piece: it must divide both lengths, so it is the GCF.

2 Write each number as a product of primes: $72 = 2^3 \times 3^2$, $54 = 2 \times 3^3$.

3 Take each prime that both numbers share, with its lowest power: $2 \times 3^2 = 18$.

4 Pieces: $72 \div 18 = 4$ and $54 \div 18 = 3$, so $4 + 3 = 7$ pieces in all.

WHY NOT (B) cuts pieces 9 inches long, a common factor that is not the greatest. (C) is the length of each piece, not the number of pieces. (D) adds the lengths of the boards.

24. **C** 8

1 List the factor pairs, starting from 1: 1×154, 2×77, 7×22, 11×14.

2 The factors are 1, 2, 7, 11, 14, 22, 77, 154.

3 Count them: 8 factors.

TIP Shortcut: $154 = 2 \times 7 \times 11$; add 1 to each exponent and multiply: $(1 + 1) \times (1 + 1) \times (1 + 1) = 8$.

WHY NOT (A) counts the factor pairs instead of the factors. (B) leaves out 1 and 154.

25. **D** 360

1 Write each number as a product of primes: $4 = 2^2$, $8 = 2^3$, $9 = 3^2$, $10 = 2 \times 5$.

2 Take every prime that appears, with its highest power: $2^3 \times 3^2 \times 5 = 360$.

3 Check: 360 divided by each of the numbers gives a whole number.

WHY NOT (A) is not divisible by 9. (B) is not divisible by 10. (C) is not divisible by 8.

CHAPTER 4

Fractions & Mixed Numbers

Tested on: Mathematics Knowledge • Arithmetic Reasoning

Key Concepts

Fractions are everywhere on the ASVAB: in recipes, distances, shares of a budget, and parts of a group. Every fraction skill comes down to a few moves done carefully, one step at a time.

Simplifying and comparing

Divide the top and bottom by their greatest common factor: $\frac{18}{24} = \frac{18\div6}{24\div6} = \frac{3}{4}$. Multiplying the top and bottom by the same number gives an *equivalent* fraction: $\frac{3}{4} = \frac{15}{20}$. To compare, rewrite over a common denominator: $\frac{2}{3} = \frac{8}{12}$ and $\frac{3}{4} = \frac{9}{12}$, so $\frac{3}{4}$ is larger.

The four operations

- **Add/subtract:** common denominator first, then combine the numerators: $\frac{1}{4} + \frac{2}{3} = \frac{3}{12} + \frac{8}{12} = \frac{11}{12}$.
- **Multiply:** top times top, bottom times bottom (cancel first): $\frac{4}{9} \times \frac{3}{8} = \frac{1}{3} \times \frac{1}{2} = \frac{1}{6}$.
- **Divide:** keep the first, flip the *second*, multiply: $\frac{2}{3} \div \frac{4}{5} = \frac{2}{3} \times \frac{5}{4} = \frac{5}{6}$.

Mixed numbers

$3\frac{2}{5} = \frac{3\times5+2}{5} = \frac{17}{5}$. To add or subtract, work with the whole numbers and the fractions; when the top fraction is too small, *regroup*: $5\frac{1}{4} = 4\frac{5}{4}$. To multiply or divide, change to improper fractions first.

WORKED EXAMPLE

Maria spent $\frac{1}{3}$ of her \$1,200 paycheck on rent and then $\frac{1}{4}$ of what was left on groceries. How much money does she have now?
Solution. Rent: $\frac{1}{3} \times 1{,}200 = 400$, leaving 800. Groceries: $\frac{1}{4}$ of 800 is 200, leaving 800 – 200 = \$600.

No-Calculator Tip To find a fraction of a number, divide by the bottom, then multiply by the top: $\frac{3}{8}$ of 32 is $32 \div 8 = 4$, then $4 \times 3 = 12$.

Common Traps

- Adding tops and bottoms: $\frac{1}{2} + \frac{1}{3}$ is $\frac{5}{6}$, not $\frac{2}{5}$.
- Flipping the wrong fraction when dividing (flip the second one only).
- Taking the second fraction of the *original* amount instead of what was left.
- Forgetting to take 1 from the whole number when you regroup.

Practice Set 4

Score: _____ / 25 Target time: 26 minutes

Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

1. Simplify $\frac{18}{99}$ to lowest terms.

A $\frac{6}{33}$ **B** $\frac{11}{2}$ **C** $\frac{2}{11}$ **D** $\frac{2}{99}$

2. Add: $\frac{4}{5} + \frac{14}{15}$

A $\frac{9}{10}$ **B** $1\frac{11}{15}$ **C** $1\frac{4}{5}$ **D** $1\frac{14}{15}$

3. A parking lot holds 60 vehicles, and $\frac{5}{12}$ of them are pickup trucks. How many vehicles are pickup trucks?

A 5 **B** 10 **C** 24 **D** 25

4. What is $\frac{5}{6} - \frac{1}{2}$?

A $\frac{1}{3}$ **B** $\frac{2}{3}$ **C** 1 **D** $1\frac{1}{3}$

5. A class has 20 students, and $\frac{3}{4}$ of them take the bus to school. How many students take the bus?

A 5 **B** 15 **C** 16 **D** 21

6. What is $\frac{10}{30}$ in simplest form?

A $\frac{2}{6}$ **B** $\frac{2}{3}$ **C** $\frac{1}{3}$ **D** $\frac{1}{30}$

7. If $\frac{5}{8} = \frac{x}{40}$, what is the value of x?

A 5 **B** 10 **C** 24 **D** 25

8. What is $\frac{4}{5} \times 35$?

A 7 **B** 28 **C** $43\frac{3}{4}$ **D** 140

9. Which of the fractions $\frac{9}{10}$, $\frac{1}{6}$, $\frac{2}{3}$, and $\frac{13}{15}$ is the smallest?

A $\frac{1}{6}$ **B** $\frac{13}{15}$ **C** $\frac{9}{10}$ **D** $\frac{2}{3}$

10. What is $\frac{4}{9} \div \frac{1}{2}$?

A $\frac{2}{9}$ **B** $\frac{5}{9}$ **C** $\frac{8}{9}$ **D** $1\frac{1}{8}$

11. Add: $5\frac{5}{6} + 7\frac{1}{2}$

A $12\frac{1}{3}$ **B** $13\frac{1}{3}$ **C** $14\frac{1}{3}$ **D** $43\frac{3}{4}$

12. Specialist Novak's platoon is on an 8-mile ruck march. So far the platoon has covered $3\frac{1}{8}$ miles. How many miles are left?

A $\frac{8}{39}$ miles **B** $4\frac{3}{4}$ miles
C $4\frac{7}{8}$ miles **D** $5\frac{1}{8}$ miles

13. What is $5\frac{3}{5} + 1\frac{9}{10}$?

A $6\frac{1}{2}$ **B** $7\frac{1}{2}$ **C** $8\frac{1}{2}$ **D** $10\frac{16}{25}$

14. Hannah ran $3\frac{3}{4}$ miles on Saturday and $3\frac{2}{3}$ miles on Sunday. How many miles did she run in all?

A $6\frac{5}{12}$ miles **B** $6\frac{5}{7}$ miles
C $7\frac{5}{12}$ miles **D** $8\frac{5}{12}$ miles

15. Divide: $3 \div \frac{3}{4}$

A $\frac{1}{4}$ **B** 2 **C** $2\frac{1}{4}$ **D** 4

16. Which is the largest: $\frac{3}{10}$, $\frac{7}{12}$, $\frac{1}{10}$, or $\frac{14}{15}$?

A $\frac{3}{10}$ **B** $\frac{1}{10}$ **C** $\frac{7}{12}$ **D** $\frac{14}{15}$

17. Subtract: $\frac{1}{4} - \frac{1}{10}$

A $\frac{3}{20}$ **B** $\frac{1}{5}$ **C** $\frac{3}{10}$ **D** $\frac{7}{20}$

18. Multiply: $\frac{4}{15} \times \frac{3}{14}$

A $\frac{2}{35}$ **B** $\frac{101}{210}$ **C** $\frac{6}{7}$ **D** $1\frac{11}{45}$

19. What is $7\frac{2}{5} - 4\frac{3}{5}$?

A $1\frac{4}{5}$ **B** $2\frac{1}{5}$ **C** $2\frac{2}{5}$ **D** $2\frac{4}{5}$

20. What is $4\frac{2}{3} \div 1\frac{2}{3}$?

A $\frac{5}{14}$ **B** $1\frac{2}{5}$ **C** $2\frac{4}{5}$ **D** 5

21. Private Kim received an \$8,100 reenlistment bonus, used $\frac{2}{3}$ of it to pay off a car loan, and then spent $\frac{1}{3}$ of the remaining money on new furniture. How much of the bonus is left?

A \$0 **B** \$1,800 **C** \$2,700 **D** \$6,300

22. Multiply: $4\frac{1}{3} \times 2\frac{5}{8}$

A $1\frac{11}{24}$ **B** $6\frac{23}{24}$ **C** $8\frac{5}{24}$ **D** $11\frac{3}{8}$

23. Private Jensen received a $10,900 reenlistment bonus, used $\frac{4}{5}$ of it to pay off a car loan, and then spent $\frac{2}{5}$ of the remaining money on new furniture. How much of the bonus is left?

A $1,308 **B** $1,309 **C** $2,180 **D** $7,412

24. What is $9\frac{1}{3} - 4\frac{3}{4}$?

A $3\frac{7}{12}$ **B** $4\frac{5}{12}$ **C** $4\frac{7}{12}$ **D** $5\frac{5}{12}$

25. A drill sergeant has 60 recruits, and $\frac{1}{12}$ of them passed the swim test on the first try. Of those, $\frac{3}{5}$ also passed the obstacle course on the first try. How many recruits passed both on the first try?

A 2 **B** 3 **C** 5 **D** 36

Answers & Explanations 4

1 C	**6** C	**11** B	**16** D	**21** B
2 B	**7** D	**12** C	**17** A	**22** D
3 D	**8** B	**13** B	**18** A	**23** A
4 A	**9** A	**14** C	**19** D	**24** C
5 B	**10** C	**15** D	**20** C	**25** B

1. **C** $\frac{2}{11}$

1 Find the greatest common factor of 18 and 99: it is 9.

2 Divide the top and bottom by 9: $\frac{18}{99} = \frac{18\div9}{99\div9} = \frac{2}{11}$.

TIP You can also divide in smaller steps (for example by 3 first); just keep going until the top and bottom share no factor.

WHY NOT (A) is equal to $\frac{18}{99}$, but not in lowest terms: 6 and 33 still share a factor of 3. (B) flips the fraction. (D) divides only the top by 9.

2. **B** $1\frac{11}{15}$

1 15 is a multiple of 5, so use 15 as the common denominator.

2 Rewrite $\frac{4}{5}$ with denominator 15: $\frac{4}{5} = \frac{12}{15}$.

3 Add the numerators: $\frac{12}{15} + \frac{14}{15} = \frac{26}{15} = 1\frac{11}{15}$.

WHY NOT (A) adds the numerators and adds the denominators.

3. **D** 25

1 Find $\frac{5}{12}$ of 60: divide by the bottom, $60 \div 12 = 5$.

2 Multiply by the top: $5 \times 5 = 25$.

WHY NOT (A) finds only $\frac{1}{12}$ of the total.

4. **A** $\frac{1}{3}$

1 6 is a multiple of 2, so use 6 as the common denominator.

2 Rewrite $\frac{1}{2}$ with denominator 6: $\frac{1}{2} = \frac{3}{6}$.

3 Subtract the numerators: $\frac{5}{6} - \frac{3}{6} = \frac{2}{6} = \frac{1}{3}$.

WHY NOT (B) uses the common denominator without rewriting the numerators. (C) subtracts the numerators and subtracts the denominators. (D) adds instead of subtracting.

5. **B** 15

1 Find $\frac{3}{4}$ of 20: divide by the bottom, $20 \div 4 = 5$.

2 Multiply by the top: $5 \times 3 = 15$.

WHY NOT (A) finds only $\frac{1}{4}$ of the total.

6. **C** $\frac{1}{3}$

1 Find the greatest common factor of 10 and 30: it is 10.

2 Divide the top and bottom by 10: $\frac{10}{30} = \frac{10\div10}{30\div10} = \frac{1}{3}$.

TIP You can also divide in smaller steps (for example by 2 first); just keep going until the top and bottom share no factor.

WHY NOT (A) is equal to $\frac{10}{30}$, but not in lowest terms: 2 and 6 still share a factor of 2. (D) divides only the top by 10.

7. **D** 25

1 The bottom changed from 8 to 40: $40 \div 8 = 5$, so it was multiplied by 5.

2 Multiply the top by the same number: $5 \times 5 = 25$.

WHY NOT (A) stops after finding $40 \div 8 = 5$.

8. **B** 28

1 Divide by the bottom: $35 \div 5 = 7$.

2 Multiply by the top: $7 \times 4 = 28$.

WHY NOT (A) finds only $\frac{1}{5}$ of 35. (C) divides by the fraction instead of multiplying. (D) multiplies by 4 but forgets to divide by 5.

9. **A** $\frac{1}{6}$

1 Rewrite each fraction with the common denominator 30: $\frac{9}{10} = \frac{27}{30}$, $\frac{1}{6} = \frac{5}{30}$, $\frac{2}{3} = \frac{20}{30}$, $\frac{13}{15} = \frac{26}{30}$.

2 The smallest numerator is 5, so $\frac{1}{6}$ is the smallest.

WHY NOT (B) equals $\frac{26}{30}$, which is more than $\frac{5}{30}$. (C) equals $\frac{27}{30}$, which is more than $\frac{5}{30}$. (D) equals $\frac{20}{30}$, which is more than $\frac{5}{30}$.

10. **C** $\frac{8}{9}$

1 Keep the first fraction, change $\div$ to $\times$, and flip the second fraction: $\frac{4}{9} \times \frac{2}{1}$.

2 Multiply top times top and bottom times bottom: $\frac{4}{9} \times 2 = \frac{8}{9}$.

WHY NOT (A) multiplies without flipping the second fraction. (D) divides in the wrong order (flips the answer).

11. **B** $13\frac{1}{3}$

1 Add the whole numbers: $5 + 7 = 12$.

2 Add the fractions with the common denominator 6: $\frac{5}{6} + \frac{3}{6} = \frac{8}{6} = 1\frac{1}{3}$.

3 Combine: $12 + 1\frac{1}{3} = 13\frac{1}{3}$.

WHY NOT (A) drops the extra whole number from $\frac{8}{6}$. (D) multiplies instead of adding.

12. **C** $4\frac{7}{8}$ miles

1 Subtract: $8 - 3\frac{1}{8}$.

2 Regroup 8 as $7\frac{8}{8}$.

3 $7\frac{8}{8} - 3\frac{1}{8} = 4\frac{7}{8}$ miles.

WHY NOT (D) subtracts the whole numbers and keeps the fraction.

13. **B** $7\frac{1}{2}$

1 Add the whole numbers: $5 + 1 = 6$.

2 Add the fractions with the common denominator 10: $\frac{6}{10} + \frac{9}{10} = \frac{15}{10} = 1\frac{1}{2}$.

3 Combine: $6 + 1\frac{1}{2} = 7\frac{1}{2}$.

WHY NOT (A) drops the extra whole number from $\frac{15}{10}$. (D) multiplies instead of adding.

14. **C** $7\frac{5}{12}$ miles

1 Add the whole numbers: $3 + 3 = 6$.

2 Add the fractions: $\frac{9}{12} + \frac{8}{12} = \frac{17}{12} = 1\frac{5}{12}$.

3 Total: $6 + 1\frac{5}{12} = 7\frac{5}{12}$ miles.

WHY NOT (A) drops the extra whole number from $\frac{17}{12}$. (B) adds the numerators and adds the denominators of the fractions.

15. **D** 4

1 Dividing by $\frac{3}{4}$ means multiplying by its reciprocal, $\frac{4}{3}$.

2 $3 \times \frac{4}{3} = \frac{12}{3} = 4$.

WHY NOT (A) divides in the wrong order. (C) multiplies instead of dividing.

16. **D** $\frac{14}{15}$

1 Rewrite each fraction with the common denominator 60: $\frac{3}{10} = \frac{18}{60}$, $\frac{7}{12} = \frac{35}{60}$, $\frac{1}{10} = \frac{6}{60}$, $\frac{14}{15} = \frac{56}{60}$.

2 The largest numerator is 56, so $\frac{14}{15}$ is the largest.

WHY NOT (A) equals $\frac{18}{60}$, which is less than $\frac{56}{60}$. (B) equals $\frac{6}{60}$, which is less than $\frac{56}{60}$. (C) equals $\frac{35}{60}$, which is less than $\frac{56}{60}$.

17. **A** $\frac{3}{20}$

1 The least common denominator of 4 and 10 is 20.

2 Rewrite: $\frac{1}{4} = \frac{5}{20}$ and $\frac{1}{10} = \frac{2}{20}$.

3 Subtract the numerators: $\frac{5}{20} - \frac{2}{20} = \frac{3}{20}$.

WHY NOT (D) adds instead of subtracting.

18. **A** $\frac{2}{35}$

1 Cancel before multiplying: 4 and 14 share 2 and 3 and 15 share 3, so the problem becomes $\frac{2}{5} \times \frac{1}{7}$.

2 Multiply top times top and bottom times bottom: $\frac{2}{5} \times \frac{1}{7} = \frac{2}{35}$.

WHY NOT (B) adds instead of multiplying. (C) multiplies the numerators but keeps only one denominator. (D) flips the second fraction; that is division, not multiplication.

19. **D** $2\frac{4}{5}$

1 Rewrite the fractions with the common denominator 5: $7\frac{2}{5} - 4\frac{3}{5}$.

2 Since $\frac{2}{5}$ is less than $\frac{3}{5}$, regroup: take 1 from 7, so $7\frac{2}{5} = 6\frac{7}{5}$.

3 Subtract: whole numbers $6 - 4 = 2$, fractions $\frac{7}{5} - \frac{3}{5} = \frac{4}{5}$. The answer is $2\frac{4}{5}$.

WHY NOT (B) lowers the whole number by 1 but never adds that 1 to the fraction, then subtracts the smaller fraction from the larger.

20. **C** $2\frac{4}{5}$

1 Change each mixed number to an improper fraction: $4\frac{2}{3} = \frac{4\times3+2}{3} = \frac{14}{3}$ and $1\frac{2}{3} = \frac{5}{3}$.

2 Dividing means multiplying by the reciprocal: $\frac{14}{3} \div \frac{5}{3} = \frac{14}{3} \times \frac{3}{5}$.

3 Cancel before multiplying: 3 and 3 share 3, so the problem becomes $14 \times \frac{1}{5}$.

4 Multiply top times top and bottom times bottom: $14 \times \frac{1}{5} = \frac{14}{5} = 2\frac{4}{5}$.

WHY NOT (A) divides in the wrong order. (D) divides the whole numbers and the fractions separately.

21. **B** \$1,800

1 The car loan: $\frac{2}{3} \times 8{,}100 = 5{,}400$, which leaves $8{,}100 - 5{,}400 = 2{,}700$ dollars.

2 The furniture: $\frac{1}{3}$ of what was left is $\frac{1}{3} \times 2{,}700 = 900$.

3 What remains: $2{,}700 - 900 = 1{,}800$ dollars.

TIP Shortcut: keep $\frac{1}{3}$ and then $\frac{2}{3}$ of that: $\frac{1}{3} \times \frac{2}{3} \times 8{,}100 = 1{,}800$.

WHY NOT (A) takes $\frac{1}{3}$ of the original amount instead of what was left. (C) stops after the car loan. (D) multiplies the two fractions and subtracts that from the original amount.

22. **D** $11\frac{3}{8}$

1 Change each mixed number to an improper fraction: $4\frac{1}{3} = \frac{4\times3+1}{3} = \frac{13}{3}$ and $2\frac{5}{8} = \frac{21}{8}$.

2 Cancel before multiplying: 21 and 3 share 3, so the problem becomes $13 \times \frac{7}{8}$.

3 Multiply top times top and bottom times bottom: $13 \times \frac{7}{8} = \frac{91}{8} = 11\frac{3}{8}$.

WHY NOT (A) adds the whole number to the numerator instead of multiplying it by the denominator. (B) adds instead of multiplying. (C) multiplies the whole numbers and the fractions separately.

23. **A** \$1,308

1 The car loan: $\frac{4}{5} \times 10{,}900 = 8{,}720$, which leaves $10{,}900 - 8{,}720 = 2{,}180$ dollars.

2 The furniture: $\frac{2}{5}$ of what was left is $\frac{2}{5} \times 2{,}180 = 872$.

3 What remains: $2{,}180 - 872 = 1{,}308$ dollars.

TIP Shortcut: keep $\frac{1}{5}$ and then $\frac{3}{5}$ of that: $\frac{1}{5} \times \frac{3}{5} \times 10{,}900 = 1{,}308$.

WHY NOT (C) stops after the car loan. (D) multiplies the two fractions and subtracts that from the original amount.

24. **C** $4\frac{7}{12}$

1 Rewrite the fractions with the common denominator 12: $9\frac{4}{12} - 4\frac{9}{12}$.

2 Since $\frac{4}{12}$ is less than $\frac{9}{12}$, regroup: take 1 from 9, so $9\frac{4}{12} = 8\frac{16}{12}$.

3 Subtract: whole numbers $8 - 4 = 4$, fractions $\frac{16}{12} - \frac{9}{12} = \frac{7}{12}$. The answer is $4\frac{7}{12}$.

WHY NOT (B) lowers the whole number by 1 but never adds that 1 to the fraction, then subtracts the smaller fraction from the larger. (D) subtracts the smaller fraction from the larger instead of regrouping.

25. **B** 3

1 First group: $\frac{1}{12} \times 60 = 5$.

2 Then take $\frac{3}{5}$ of that group: $\frac{3}{5} \times 5 = 3$.

WHY NOT (A) is the part of the smaller group that is *not* counted. (C) is the size of the first group only. (D) takes $\frac{3}{5}$ of the whole group instead of the smaller group.

CHAPTER 5

Decimals

Tested on: Mathematics Knowledge • Arithmetic Reasoning

Key Concepts

Decimals are fractions with denominators of 10, 100, 1,000, and so on. Money is the everyday example, and most decimal word problems on the ASVAB are about prices, pay, and fuel. The whole chapter rests on keeping track of the decimal point.

Place value

In 3.472: 4 tenths, 7 hundredths, 2 thousandths. Compare decimals place by place from the left: $0.5 > 0.45 > 0.405 > 0.054$ (look at the tenths first). Extra zeros on the right change nothing: $0.5 = 0.50$. To **round**, look at the next digit to the right: 5 or more rounds up. $7.486 \approx 7.49$ (nearest hundredth) ≈ 7.5 (nearest tenth).

The four operations

- **Add/subtract:** line up the decimal points; fill empty places with zeros: $15.00 - 3.84 = 11.16$.
- **Multiply:** multiply as whole numbers, then count the decimal places in both factors: 0.6×0.04: $6 \times 4 = 24$, three places, 0.024.
- **Divide by a decimal:** move the point in *both* numbers until the divisor is whole: $7.2 \div 0.08 = 720 \div 8 = 90$.

Fractions and decimals

Fraction → decimal: divide the top by the bottom ($\frac{3}{8} = 3 \div 8 = 0.375$). Decimal → fraction: read the place value and simplify ($0.45 = \frac{45}{100} = \frac{9}{20}$). Worth memorizing: $\frac{1}{2} = 0.5$, $\frac{1}{4} = 0.25$, $\frac{3}{4} = 0.75$, $\frac{1}{5} = 0.2$, $\frac{1}{8} = 0.125$.

WORKED EXAMPLE

Gas costs \$3.49 per gallon. How much do 12 gallons cost?
Solution. $3.49 \times 12 = 3.49 \times 10 + 3.49 \times 2 = 34.90 + 6.98 = \$41.88$.

No-Calculator Tip Estimate first to place the decimal point: 3.49×12 is about $3.5 \times 12 = 42$, so the answer must be about \$42, not \$4.19 or \$418.80.

Common Traps

- Lining up the last digits instead of the decimal points.
- Thinking a longer decimal is larger: $0.405 < 0.5$.
- Moving the decimal point in only one number when dividing.

- Forgetting to borrow across zeros: 20.00 – 7.36 = 12.64.

Practice Set 5

Score: _____ / 25 Target time: 27 minutes

Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

1. Add: 2.7 + 14.42

A 14.69 **B** 17.04 **C** 17.12 **D** 171.2

2. Round 40.8 to the nearest whole number.

A 39 **B** 40 **C** 41 **D** 43

3. Subtract: 27.26 – 25.843

A 1.417 **B** 2.623 **C** 2.834 **D** 53.103

4. Round 5.268 to the nearest hundredth.

A 5.27 **B** 5.28 **C** 5.29 **D** 5.3

5. What is 0.3×0.2?

A 0.006 **B** 0.02 **C** 0.04 **D** 0.06

6. What is $14.4 \div 2$?

A 0.72 **B** 7.2 **C** 28.8 **D** 72

7. At the base exchange, energy drinks cost $2.47 each. How much do 9 energy drinks cost?

A $22.23 **B** $22.24 **C** $23.23 **D** $222.30

8. Ethan buys a phone charger for $9.39 and a pair of socks for $6.14. He pays with a $20 bill. How much change should he get back?

A $3.97 **B** $4.47 **C** $10.61 **D** $15.53

9. Divide: $5.4 \div 0.06$

A 0.9 **B** 9 **C** 90 **D** 900

10. Convert 0.45 to a fraction in simplest form.

A $\frac{9}{20}$ **B** $\frac{1}{2}$ **C** $\frac{20}{9}$ **D** $\frac{9}{2}$

11. Which of the numbers 0.085, 0.508, 0.0805, and 0.8 is the greatest?

A 0.8 **B** 0.085 **C** 0.508 **D** 0.0805

12. Which fraction is equal to 0.65?

A $\frac{13}{200}$ **B** $\frac{13}{40}$ **C** $\frac{1}{2}$ **D** $\frac{13}{20}$

13. Which of the numbers 0.97, 0.09, 0.709, and 0.9 is the least?

A 0.9 **B** 0.09 **C** 0.97 **D** 0.709

14. What is $4.4 \div 0.2$?

A 0.88 **B** 2.2 **C** 16 **D** 22

15. Subtract: 50 – 6.53

A 43.47 **B** 43.48 **C** 44.47 **D** 56.53

16. Multiply: 3.8×1.5

A 0.57 **B** 5.7 **C** 57 **D** 570

17. Corporal Haddad buys 2 notebooks at $1.25 each and 3 bottles of water at $1.70 each. What is the total cost?

A $2.95 **B** $4.20 **C** $7.60 **D** $8.50

18. An 8-pack of soda costs $2.24. What is the price per can?

A $0.23 **B** $0.27 **C** $0.28 **D** $2.80

19. Store A sells a pack of 4 batteries for $4.00. Store B sells a pack of 10 batteries for $12.70. What is the lower price per battery?

A $0.27 **B** $1.00 **C** $1.27 **D** $4.00

20. Tyrone works at a hardware store and earns $19.50 per hour for the first 40 hours each week. Every hour over 40 is overtime and pays 1.5 times the regular rate. Last week he worked 42 hours. How much did he earn?

A $819.00 **B** $838.50
C $839.50 **D** $1,228.50

21. Store A sells a bag of 2 pounds of rice for $3.74. Store B sells a bag of 5 pounds of rice for $8.55. What is the lower price per pound?

A $1.71 **B** $1.87 **C** $3.74 **D** $17.10

22. Derek works at a restaurant and earns $17.50 per hour for the first 40 hours each week. Every hour over 40 is overtime and pays 1.5 times the regular rate. Last week he worked 43 hours. How much did he earn?

A $700.00 **B** $752.50 **C** $757.00 **D** $778.75

23. A motorcycle traveled 120.9 miles on 3.1 gallons of fuel. How many miles per gallon did it get?

A 39 miles per gallon
B 41 miles per gallon
C 117.8 miles per gallon
D 390 miles per gallon

24. Sofia buys 4.5 pounds of cheese at $6.40 per pound and 4 pounds of shrimp at $7.90 per pound. How much does she spend?

A $22.80 **B** $54.40 **C** $60.40 **D** $319.60

25. Malik buys 2 drinks at $1.58 each and 3 bags of chips at $1.60 each. He pays with a $20 bill. How much change should he get back?

A $7.96 **B** $12.03 **C** $12.04 **D** $16.84

Answers & Explanations 5

1 C	**6** B	**11** A	**16** B	**21** A
2 C	**7** A	**12** D	**17** C	**22** D
3 A	**8** B	**13** B	**18** C	**23** A
4 A	**9** C	**14** D	**19** B	**24** C
5 D	**10** A	**15** A	**20** B	**25** C

1. **C** 17.12

1 Line up the decimal points and fill in zeros so both numbers have 2 decimal places: 2.70 + 14.42.

2 Add column by column, keeping the point in place: 2.70 + 14.42 = 17.12.

WHY NOT (A) lines up the last digits instead of the decimal points. (D) misplaces the decimal point.

2. **C** 41

1 The ones digit is 0. The digit to its right (the tenths digit) is 8.

2 Since 8 is 5 or more, round up and drop the digits after it: 41.

WHY NOT (B) drops the extra digits without rounding up (the next digit, 8, is 5 or more).

3. **A** 1.417

1 Line up the decimal points and fill in zeros so both numbers have 3 decimal places: 27.260 – 25.843.

2 Subtract column by column, keeping the point in place: 27.260 – 25.843 = 1.417.

WHY NOT (B) subtracts the smaller digit from the larger in each column instead of borrowing. (D) adds instead of subtracting.

4. **A** 5.27

1 The hundredths digit is 6. The digit to its right (the thousandths digit) is 8.

2 Since 8 is 5 or more, round up and drop the digits after it: 5.27.

WHY NOT (D) rounds to the nearest tenth instead.

5. **D** 0.06

1 Ignore the decimal points and multiply: $3 \times 2 = 6$.

2 Count the decimal places: 1 in 0.3 and 1 in 0.2, 2 in all.

3 Put 2 decimal places in the product: 0.06.

WHY NOT (A) puts one decimal place too many in the answer.

6. **B** 7.2

1 Divide as with whole numbers and put the point in the answer directly above the point in 14.4.

2 $14.4 \div 2 = 7.2$. Check: $2 \times 7.2 = 14.4$.

WHY NOT (A) puts the decimal point one place too far left. (C) multiplies instead of dividing. (D) drops the decimal point.

7. **A** $22.23

1 Multiply the price by the number: 2.47×9.

2 Split it up: $2 \times 9 = 18$ and $0.47 \times 9 = 4.23$.

3 Add: 18 + 4.23 = 22.23, so the cost is $22.23.

TIP Estimate: $2.47 is about $2.50, and $2.5 \times 9 = 22.5$, so the answer is near $22.50.

WHY NOT (D) misplaces the decimal point.

8. **B** $4.47

1 Total: 9.39 + 6.14 = 15.53.

2 Change: 20.00 – 15.53 = 4.47, so $4.47.

TIP Count up from $15.53: to the next dollar, then to $20.

WHY NOT (C) subtracts only the first item. (D) is the total cost, not the change.

9. **C** 90

1 Make the divisor a whole number: move the decimal point 2 places to the right in *both* numbers.

2 $5.4 \div 0.06$ becomes $540 \div 6$.

3 Divide: $540 \div 6 = 90$. Check: $0.06 \times 90 = 5.4$.

WHY NOT (A) moves the decimal point in the divisor but not in the number being divided. (B) moves the decimal point in the number being divided one place too few. (D) moves the decimal point in the number being divided one place too many.

10. **A** $\frac{9}{20}$

1 The last digit is in the hundredths place, so $0.45 = \frac{45}{100}$.

2 Simplify by dividing the top and bottom by 5: $\frac{45}{100} = \frac{9}{20}$.

WHY NOT (C) flips the fraction. (D) reads the last digit as tenths instead of hundredths.

11. **A** 0.8

1 Give every number the same number of decimal places by adding zeros: 0.0850, 0.5080, 0.0805, 0.8000.

2 Now compare them like whole numbers. In order from greatest to least: 0.8 > 0.508 > 0.085 > 0.0805.

3 The greatest is 0.8.

WHY NOT (B) is less than 0.8: in the tenths place it has 0, while 0.8 has 8. (C) is less than 0.8: in the tenths place it has 5, while 0.8 has 8. (D) has more digits, but it is less than 0.8: in the tenths place it has 0, while 0.8 has 8.

12. **D** $\frac{13}{20}$

1 The last digit is in the hundredths place, so 0.65 = $\frac{65}{100}$.

2 Simplify by dividing the top and bottom by 5: $\frac{65}{100} = \frac{13}{20}$.

WHY NOT (A) reads the last digit as thousandths instead of hundredths.

13. **B** 0.09

1 Give every number the same number of decimal places by adding zeros: 0.970, 0.090, 0.709, 0.900.

2 Now compare them like whole numbers. In order from least to greatest: 0.09 < 0.709 < 0.9 < 0.97.

3 The least is 0.09.

WHY NOT (A) is greater than 0.09: in the tenths place it has 9, while 0.09 has 0. (C) is greater than 0.09: in the tenths place it has 9, while 0.09 has 0. (D) is greater than 0.09: in the tenths place it has 7, while 0.09 has 0.

14. **D** 22

1 Make the divisor a whole number: move the decimal point 1 place to the right in *both* numbers.

2 4.4 ÷ 0.2 becomes 44 ÷ 2.

3 Divide: 44 ÷ 2 = 22. Check: 0.2 × 22 = 4.4.

WHY NOT (A) multiplies instead of dividing. (B) moves the decimal point in the divisor but not in the number being divided.

15. **A** 43.47

1 Line up the decimal points and fill in zeros so both numbers have 2 decimal places: 50.00 – 6.53.

2 Subtract column by column, keeping the point in place: 50.00 – 6.53 = 43.47.

WHY NOT (C) borrows but forgets to reduce the ones digit it borrowed from. (D) subtracts the smaller digit from the larger in each column instead of borrowing.

16. **B** 5.7

1 Ignore the decimal points and multiply: 38 × 15 = 570.

2 Count the decimal places: 1 in 3.8 and 1 in 1.5, 2 in all.

3 Put 2 decimal places in the product: 5.70, which is 5.7.

WHY NOT (A) puts one decimal place too many in the answer. (C) counts only the decimal place in 3.8 and ignores the one in 1.5. (D) ignores the decimal points completely.

17. **C** $7.60

1 First item: 2 × 1.25 = 2.50.

2 Second item: 3 × 1.70 = 5.10.

3 Total: 2.50 + 5.10 = 7.60, so $7.60.

WHY NOT (A) adds one of each price and ignores the quantities. (B) forgets to multiply the second price by 3. (D) charges every item at the higher price.

18. **C** $0.28

1 Price per can = total price ÷ number of cans.

2 2.24 ÷ 8 = 0.28. Check: 8 × 0.28 = 2.24.

WHY NOT (D) puts the decimal point one place too far right.

19. **B** $1.00

1 Store A: 4.00 ÷ 4 = 1.00 per battery.

2 Store B: 12.70 ÷ 10 = 1.27 per battery.

3 The lower price is at Store A: $1.00 per battery.

WHY NOT (A) is the difference between the prices per battery. (C) is the higher of the two prices per battery. (D) is the cheaper pack price, not the price per battery.

20. **B** $838.50

1 Regular pay: 40 × 19.50 = 780.00.

2 Overtime rate: 1.5 × 19.50 = 29.25 per hour.

3 Overtime pay: 2 × 29.25 = 58.50.

4 Total: 780.00 + 58.50 = 838.50, so $838.50.

WHY NOT (A) pays the overtime hours at the regular rate. (D) pays every hour at the overtime rate.

21. **A** $1.71

1 Store A: 3.74 ÷ 2 = 1.87 per pound.

2 Store B: 8.55 ÷ 5 = 1.71 per pound.

3 The lower price is at Store B: $1.71 per pound.

WHY NOT (B) is the higher of the two prices per pound. (C) is the cheaper bag price, not the price per pound. (D) misplaces the decimal point.

22. **D** $778.75

1 Regular pay: $40 \times 17.50 = 700.00$.

2 Overtime rate: $1.5 \times 17.50 = 26.25$ per hour.

3 Overtime pay: $3 \times 26.25 = 78.75$.

4 Total: $700.00 + 78.75 = 778.75$, so $778.75.

WHY NOT (A) leaves out the overtime pay. (B) pays the overtime hours at the regular rate. (C) adds 1.5 dollars to the rate instead of multiplying by 1.5.

23. **A** 39 miles per gallon

1 Miles per gallon = miles ÷ gallons: $120.9 \div 3.1$.

2 Move the decimal point one place to the right in both numbers: $1{,}209 \div 31$.

3 Divide: $31 \times 30 = 930$, and $1{,}209 - 930 = 279$, which is 31×9. So $1{,}209 \div 31 = 30 + 9 = 39$ miles per gallon.

WHY NOT (C) subtracts instead of dividing. (D) moves the decimal point in the miles but not in the gallons.

24. **C** $60.40

1 First item: $4.5 \times 6.40 = 28.80$.

2 Second item: $4 \times 7.90 = 31.60$.

3 Total: $28.80 + 31.60 = 60.40$, so $60.40.

TIP For 4.5×6.40, think 4×6.40 plus half of $6.40.

WHY NOT (A) adds the weights and prices instead of multiplying. (B) uses the first price for both items. (D) misplaces the decimal point in 4.5×6.40.

25. **C** $12.04

1 Cost of each kind: $2 \times 1.58 = 3.16$ and $3 \times 1.60 = 4.80$.

2 Total: $3.16 + 4.80 = 7.96$.

3 Change: $20.00 - 7.96 = 12.04$, so $12.04.

TIP Count up from $7.96: to the next dollar, then to $20.

WHY NOT (A) is the total cost, not the change. (D) subtracts only the first item.

CHAPTER 6

Percents

Tested on: Mathematics Knowledge • Arithmetic Reasoning

Key Concepts

Percent means "per hundred." Percent questions appear on both math subtests, and most of them reduce to one sentence: **part = percent × whole**.

Converting between forms

- Percent → decimal: move the decimal point two places *left* (35% = 0.35, 4% = 0.04, 150% = 1.5).
- Decimal → percent: move it two places *right* (0.6 = 60%, 0.075 = 7.5%).
- Fraction → percent: divide, then convert ($\frac{3}{8}$ = 0.375 = 37.5%).

The three percent questions

Every basic percent problem gives you two of the three quantities in

$$\text{part} = \text{percent} \times \text{whole}$$

and asks for the third.

- **Find the part:** 25% of 64 is $0.25 \times 64 = 16$.
- **Find the percent:** 18 is what percent of 72? $\frac{18}{72} = 0.25 = 25\%$.
- **Find the whole:** 24 is 30% of what number? $24 \div 0.30 = 80$.

Percent change

$$\text{percent change} = \frac{\text{new} - \text{original}}{\text{original}} \times 100\%$$

Always divide by the *original* (starting) value.

WORKED EXAMPLE

A jacket costs \$60. Its price goes up 15%. What is the new price?
Solution. The increase is $0.15 \times 60 = 9$ dollars, so the new price is $60 + 9 = \$69$. *Shortcut:* a 15% increase means you pay 115%, and $1.15 \times 60 = 69$.

No-Calculator Tip Benchmark percents make mental math fast: 10% = move the decimal one place left, 5% = half of 10%, 1% = move it two places, 25% = $\frac{1}{4}$, 50% = $\frac{1}{2}$, 75% = $\frac{3}{4}$, 20% = $\frac{1}{5}$. So 35% of 80 is 30% + 5% = 24 + 4 = 28.

Common Traps

- Answering the wrong question: if 35% *passed*, the number who *did not* pass uses 65%.
- Dividing by the new value in a percent-change problem.
- Adding successive percents: up 25% then down 20% is *not* a 5% increase.

Practice Set 6

Score: _____ / 25 Target time: 27 minutes

Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

1. What is 80% of 60?

A 12 **B** 48 **C** 108 **D** 480

2. What is $\frac{1}{4}$ written as a percent?

A 1% **B** 2.5% **C** 5% **D** 25%

3. What is 75% of 280?

A 21 **B** 70 **C** 170 **D** 210

4. Which of the following is equal to 0.075?

A 7.5% **B** 8% **C** 15% **D** 75%

5. A motor pool has 120 vehicles, and 10% of them are scheduled for maintenance this week. How many vehicles are *not* scheduled for maintenance?

A 12 **B** 108 **C** 110 **D** 118

6. Which of the following is equal to 0.78?

A 0.78% **B** 7.8% **C** 58% **D** 78%

7. Of the 380 students at a high school who took the ASVAB, 75% scored above 50 on the AFQT. How many of these students did *not* score above 50?

A 55 **B** 85 **C** 95 **D** 285

8. What is 75% of 396?

A 99 **B** 297 **C** 693 **D** 2,970

9. What percent of 152 is 190?

A 1.25% **B** 12.5% **C** 80% **D** 125%

10. Find the number if 50% of it is 60.

A 120 **B** 160 **C** 240 **D** 3,000

11. The number 53 is what percent of 212?

A 0.25% **B** 2.5% **C** 12.5% **D** 25%

12. If 40% of a number is 186, what is the number?

A 465 **B** 515 **C** 930 **D** 7,440

13. Last year the price of a gallon of paint was \$80. This year it is \$96. What is the percent increase?

A 20% **B** 24% **C** 28% **D** 120%

14. If 60% of a number is 27, what is the number?

A 30 **B** 45 **C** 87 **D** 1,620

15. Last year monthly rent for an apartment was \$950. This year it is \$1,026. What is the percent increase?

A 6% **B** 8% **C** 76% **D** 108%

16. The number 570 is what percent of 380?

A 1.5% **B** 15% **C** 150% **D** 300%

17. Which of the following is equal to 0.8?

A 0.8% **B** 8% **C** 80% **D** 800%

18. What is 80% of 700?

A 560 **B** 710 **C** 1,260 **D** 5,600

19. A store priced a television at \$360. Later it increased the price by 50%, then put the television on sale at 10% off the new price. What is the sale price?

A \$286 **B** \$324 **C** \$486 **D** \$504

20. What is 25% of 10% of 240?

A 5 **B** 6 **C** 24 **D** 84

21. A winter coat was priced at $260. The store raised the price by 25%, and a month later it marked the new price down by 20%. What is the final price?

A $185 **B** $208 **C** $260 **D** $273

22. A warehouse holds 900 crates. Of the crates, 45% contain medical supplies, and $\frac{3}{5}$ of those crates must be kept refrigerated. How many crates must be refrigerated?

A 243 **B** 268 **C** 405 **D** 540

23. A store priced a pair of running shoes at $84. Later it increased the price by 50%, then put the pair of running shoes on sale at 50% off the new price. What is the sale price?

A $63 **B** $84 **C** $93 **D** $126

24. A training battalion has 650 recruits. Of these, 20% are assigned to the morning shift, and $\frac{2}{5}$ of the morning-shift recruits are on kitchen duty. How many recruits are on kitchen duty?

A 32 **B** 37 **C** 52 **D** 130

25. What is 80% of 75% of 200?

A 10 **B** 120 **C** 150 **D** 310

Answers & Explanations 6

1 B	**6** D	**11** D	**16** C	**21** C
2 D	**7** C	**12** A	**17** C	**22** A
3 D	**8** B	**13** A	**18** A	**23** A
4 A	**9** D	**14** B	**19** C	**24** C
5 B	**10** A	**15** B	**20** B	**25** B

1. **B** 48

1 Change the percent to a decimal: 80% = 0.8.

2 Multiply by the number: $0.8 \times 60 = 48$.

WHY NOT (A) is what is left after removing 80%, not 80% of the number. (C) adds 80% to the number. (D) moves the decimal point only one place.

2. **D** 25%

1 Divide the numerator by the denominator: $1 \div 4 = 0.25$.

2 Multiply by 100 to get a percent: 0.25 = 25%.

TIP Or scale the fraction to hundredths: $\frac{1}{4} = \frac{25}{100}$.

WHY NOT (A) just writes the numerator as if it were a percent. (B) moves the decimal point only one place.

3. **D** 210

1 Change the percent to a decimal: 75% = 0.75.

2 Multiply by the number: $0.75 \times 280 = 210$.

WHY NOT (A) moves the decimal point three places. (B) is what is left after removing 75%, not 75% of the number.

4. **A** 7.5%

1 To change a decimal to a percent, multiply by 100 (move the decimal point two places to the right).

2 $0.075 \times 100 = 7.5$, so 0.075 = 7.5%.

WHY NOT (D) moves the decimal point three places.

5. **B** 108

1 The question asks about the *other* group: 100% – 10% = 90%.

2 Find that part: $0.9 \times 120 = 108$.

TIP Or find 10% first (12) and subtract: 120 – 12 = 108.

WHY NOT (A) is the number in the group described, not the number asked for. (C) subtracts the percent as if it were a count.

6. **D** 78%

1 To change a decimal to a percent, multiply by 100 (move the decimal point two places to the right).

2 $0.78 \times 100 = 78$, so 0.78 = 78%.

WHY NOT (A) forgets to move the decimal point at all. (B) moves the decimal point only one place.

7. **C** 95

1 The question asks about the *other* group: 100% – 75% = 25%.

2 Find that part: $0.25 \times 380 = 95$.

TIP Or find 75% first (285) and subtract: 380 – 285 = 95.

WHY NOT (D) is the number in the group described, not the number asked for.

8. **B** 297

1 Change the percent to a decimal: 75% = 0.75.

2 Multiply by the number: $0.75 \times 396 = 297$.

WHY NOT (A) is what is left after removing 75%, not 75% of the number. (C) adds 75% to the number. (D) moves the decimal point only one place.

9. **D** 125%

1 Write the part over the whole: $\frac{190}{152}$.

2 Simplify by dividing the top and bottom by 38: $\frac{190}{152} = \frac{5}{4}$.

3 Rewrite as hundredths: $\frac{5}{4} = \frac{125}{100} = 125\%$.

TIP Or test the choices: 125% of 152 is $1.25 \times 152 = 190$.

WHY NOT (A) forgets to multiply by 100. (B) moves the decimal point only one place. (C) divides the whole by the part instead of the part by the whole.

10. **A** 120

1 Use part = percent × whole: $60 = 0.5 \times w$.

2 Divide both sides by 0.5: $w = 60 \div 0.5 = 120$.

3 Check: 50% of 120 is 60. ✓

WHY NOT (D) multiplies by the percent without converting it.

11. **D** 25%

1 Write the part over the whole: $\frac{53}{212}$.

2 Simplify by dividing the top and bottom by 53: $\frac{53}{212} = \frac{1}{4}$.

3 Rewrite as hundredths: $\frac{1}{4} = \frac{25}{100} = 25\%$.

TIP Or test the choices: 25% of 212 is $0.25 \times 212 = 53$.

WHY NOT (A) forgets to multiply by 100. (B) moves the decimal point only one place.

12. **A** 465

1 Use part = percent × whole: $186 = 0.4 \times w$.

2 Divide both sides by 0.4: $w = 186 \div 0.4 = 465$.

3 Check: 40% of 465 is 186. ✓

WHY NOT (D) multiplies by the percent without converting it.

13. **A** 20%

1 Find the amount of increase: 96 – 80 = 16 dollars.

2 Divide by the *original* amount: $\frac{16}{80} = 0.2$.

3 Convert to a percent: 0.2 = 20% increase.

TIP Hard division? Test the choices instead: 20% of \$80 is \$16, exactly the change.

WHY NOT (D) gives the new amount as a percent of the old amount.

14. **B** 45

1 Use part = percent × whole: $27 = 0.6 \times w$.

2 Divide both sides by 0.6: $w = 27 \div 0.6 = 45$.

3 Check: 60% of 45 is 27. ✓

WHY NOT (C) adds the percent to the part. (D) multiplies by the percent without converting it.

15. **B** 8%

1 Find the amount of increase: 1,026 – 950 = 76 dollars.

2 Divide by the *original* amount: $\frac{76}{950} = 0.08$.

3 Convert to a percent: 0.08 = 8% increase.

TIP Hard division? Test the choices instead: 8% of \$950 is \$76, exactly the change.

WHY NOT (C) gives the dollar change instead of the percent change. (D) gives the new amount as a percent of the old amount.

16. **C** 150%

1 Write the part over the whole: $\frac{570}{380}$.

2 Simplify by dividing the top and bottom by 190: $\frac{570}{380} = \frac{3}{2}$.

3 Rewrite as hundredths: $\frac{3}{2} = \frac{150}{100} = 150\%$.

TIP Or test the choices: 150% of 380 is $1.5 \times 380 = 570$.

WHY NOT (A) forgets to multiply by 100. (B) moves the decimal point only one place.

17. **C** 80%

1 To change a decimal to a percent, multiply by 100 (move the decimal point two places to the right).

2 $0.8 \times 100 = 80$, so 0.8 = 80%.

WHY NOT (A) forgets to move the decimal point at all. (B) moves the decimal point only one place. (D) moves the decimal point three places.

18. **A** 560

1 Change the percent to a decimal: 80% = 0.8.

2 Multiply by the number: $0.8 \times 700 = 560$.

WHY NOT (C) adds 80% to the number. (D) moves the decimal point only one place.

19. **C** \$486

1 Raise the price by 50%: $360 \times 1.5 = 540$.

2 Take 10% off the *new* price: $540 \times 0.9 = 486$.

3 The final price is \$486.

WHY NOT (B) applies only the decrease. (D) combines the changes into a single 40% increase.

20. **B** 6

1 Work from the inside out. First, 10% of 240: $0.1 \times 240 = 24$.

2 Then 25% of that result: $0.25 \times 24 = 6$.

WHY NOT (C) stops after taking 10%. (D) adds the two percents.

21. **C** \$260

1 Raise the price by 25%: $260 \times 1.25 = 325$.

2 Take 20% off the *new* price: $325 \times 0.8 = 260$.

3 The final price is \$260.

WHY NOT (B) applies only the decrease. (D) combines the changes into a single 5% increase.

22. **A** 243

1 Find the first group: $0.45 \times 900 = 405$.

2 Take $\frac{3}{5}$ of that group: $\frac{3}{5} \times 405 = 243$.

WHY NOT (C) is the size of the first group only. (D) takes $\frac{3}{5}$ of the whole total.

23. **A** \$63

1 Raise the price by 50%: $84 \times 1.5 = 126$.

2 Take 50% off the *new* price: $126 \times 0.5 = 63$.

3 The final price is \$63.

WHY NOT (B) assumes the two changes cancel out. (D) stops after the increase.

24. **C** 52

1 Find the first group: $0.2 \times 650 = 130$.

2 Take $\frac{2}{5}$ of that group: $\frac{2}{5} \times 130 = 52$.

WHY NOT (D) is the size of the first group only.

25. **B** 120

1 Work from the inside out. First, 75% of 200: $0.75 \times 200 = 150$.

2 Then 80% of that result: $0.8 \times 150 = 120$.

TIP Percents of percents multiply: $0.8 \times 0.75 = 0.6$, and $0.6 \times 200 = 120$.

WHY NOT (A) subtracts the two percents. (C) stops after taking 75%. (D) adds the two percents.

CHAPTER 7

Ratios & Proportions

Tested on: Mathematics Knowledge • Arithmetic Reasoning

Key Concepts

A *ratio* compares two quantities; a *proportion* says that two ratios are equal. Together they handle recipes, maps, mixtures, unit rates, and every "how many of each" question on the test.

Writing and simplifying ratios

The ratio of a to b can be written $a : b$, "a to b," or $\frac{a}{b}$. **Order matters**: the first quantity named goes first.

- Simplify a ratio like a fraction: divide both terms by their greatest common factor. $18 : 24 = 3 : 4$ (divide both by 6).
- Put both amounts in the *same unit* first: 2 feet to 8 inches is 24 inches to 8 inches, so the ratio is $24 : 8 = 3 : 1$.

Ratios as parts of a whole

If the ratio of boys to girls is $3 : 5$, think of the group as $3 + 5 = 8$ equal *parts*: boys are $\frac{3}{8}$ of the group and girls are $\frac{5}{8}$. With 40 people, one part is $40 \div 8 = 5$, so there are $3 \times 5 = 15$ boys and $5 \times 5 = 25$ girls.

Proportions, unit rates, and direct variation

- In a proportion $\frac{a}{b} = \frac{c}{d}$ the *cross products* are equal: $a \times d = b \times c$. To solve $\frac{x}{12} = \frac{15}{36}$, write $36x = 12 \times 15 = 180$, so $x = 180 \div 36 = 5$.
- A *unit rate* compares to one unit: 300 miles on 12 gallons is $300 \div 12 = 25$ miles per gallon.
- "y varies directly with x" means $y = kx$: the ratio $\frac{y}{x}$ is always the same number k.

WORKED EXAMPLE

A cookie recipe uses 3 cups of flour to make 24 cookies. How many cups of flour are needed to make 40 cookies?

Solution. Keep the same order in both ratios (cups over cookies): $\frac{3}{24} = \frac{x}{40}$. Cross-multiply: $24x = 3 \times 40 = 120$. Divide: $x = 120 \div 24 = 5$ cups.

No-Calculator Tip Simplify before you multiply. In the example, $\frac{3}{24} = \frac{1}{8}$: one cup makes 8 cookies, so 40 cookies need $40 \div 8 = 5$ cups. Smaller numbers mean fewer mistakes.

Common Traps

- Reversing the order: "girls to boys" is $5 : 3$, not $3 : 5$.

- Treating the ratio 3 : 5 as the fraction $\frac{3}{5}$ of the *total*; the total has 3 + 5 = 8 parts.
- Adding instead of scaling: going from 24 to 40 cookies does *not* mean adding 16 cups of flour.
- Comparing amounts in different units (feet with inches, hours with minutes) without converting first.

Practice Set 7

Score: _____ / 25 Target time: 28 minutes

Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

1. In a training class, 40 students passed the first exam and 32 did not. What is the ratio of students who passed to students who did not pass, in simplest form?

A 4 : 9 **B** 5 : 9 **C** 4 : 5 **D** 5 : 4

2. A bag holds only red and green marbles, in the ratio 3 : 7 (red to green). What fraction of the marbles are green?

A $\frac{3}{10}$ **B** $\frac{7}{10}$ **C** $\frac{4}{5}$ **D** $\frac{7}{3}$

3. Solve for x: $\frac{x}{18} = \frac{15}{27}$

A 6 **B** 10 **C** 12 **D** 270

4. A pump removes 225 gallons of water from a flooded basement in 5 minutes. How many gallons per minute does it pump?

A 4.5 **B** 45 **C** 46 **D** 450

5. If $\frac{x}{28} = \frac{40}{35}$, what is the value of x?

A 12 **B** 17 **C** 32 **D** 1,120

6. A supply convoy traveled 266 miles in 7 hours. What was its average speed, in miles per hour?

A 3.8 **B** 37 **C** 38 **D** 380

7. The ratio of boys to girls in a class is 3 : 8. What fraction of the students in the class are girls?

A $\frac{1}{11}$ **B** $\frac{3}{11}$ **C** $\frac{8}{11}$ **D** $\frac{8}{3}$

8. A box holds 30 red pens and 50 blue pens. What is the ratio of red pens to blue pens, in simplest form?

A 3 : 5 **B** 5 : 8 **C** 1 : 1 **D** 5 : 3

9. Written in simplest form, what is the ratio of 3 pounds to 4 ounces?

A 3 : 64 **B** 1 : 12 **C** 3 : 4 **D** 12 : 1

10. Solve for x: $\frac{30}{x} = \frac{35}{49}$

A 21 **B** 42 **C** 44 **D** 1,470

11. Written in simplest form, what is the ratio of 1 foot to 3 inches?

A 1 : 4 **B** 1 : 3 **C** 10 : 3 **D** 4 : 1

12. A company surveyed 65 soldiers about PT. Each soldier chose morning PT or evening PT, and the ratio of morning to evening choices was 8 : 5. How many soldiers chose evening PT?

A 25 **B** 40 **C** 45 **D** 50

13. A school club has 50 members, and every member is a junior or a senior. The ratio of juniors to seniors is 3 : 7. How many seniors are in the club?

A 15 **B** 35 **C** 40 **D** 55

14. A concrete mix uses 2 bags of cement for every 5 bags of sand. How many bags of cement are needed for 10 bags of sand?

A 1 **B** 2 **C** 3 **D** 4

15. A mess hall recipe uses 3 pounds of rice to feed 30 soldiers. How many pounds of rice are needed to feed 230 soldiers?

A 23 **B** 25 **C** 26 **D** 690

16. On a map of an Army training area, 1 inch represents 2 miles. Two checkpoints are $5\frac{1}{2}$ inches apart on the map. How far apart are the checkpoints on the ground?

A 11 miles **B** 12 miles
C 22 miles **D** 110 miles

17. A blueprint uses the scale $\frac{1}{2}$ inch $= 1$ foot. A room is $7\frac{1}{2}$ inches long on the blueprint. How long is the actual room, in feet?

A 3.75 **B** 14 **C** 15 **D** 150

18. If y varies directly with x, and $y = 24$ when $x = 3$, what is the value of y when $x = 12$?

A 6 **B** 8 **C** 33 **D** 96

19. The angles of a triangle are in the ratio 4 : 3 : 5. What is the measure of the smallest angle?

A 15° **B** 25° **C** 45° **D** 60°

20. Three partners split a profit of \$12,000 in the ratio 5 : 4 : 6. How much does the partner with the largest share receive?

A \$3,200 **B** \$4,000 **C** \$4,800 **D** \$6,300

21. In a school club, the ratio of boys to girls is 2 : 11. There are 72 more girls than boys. How many girls are in the club?

A 16 **B** 44 **C** 83 **D** 88

22. A motor pool has only Humvees and trucks, in the ratio 4 : 7 (Humvees to trucks). There are 36 more trucks than Humvees. How many trucks are there?

A 84 **B** 132 **C** 168 **D** 252

23. On a map, 2 inches represent 125 miles. The actual distance between two Air Force bases is 875 miles. How far apart are they on the map?

A 7 inches **B** 9 inches
C 14 inches **D** 16 inches

24. If y varies directly as x, and $y = 12$ when $x = 8$, what is the value of x when $y = 18$?

A 6 **B** 12 **C** 14 **D** 27

25. A chili recipe that makes 12 servings uses $1\frac{1}{2}$ pounds of ground beef. How many pounds of ground beef are needed to make 30 servings?

A $\frac{3}{5}$ **B** $3\frac{3}{4}$ **C** $19\frac{1}{2}$ **D** 45

Answers & Explanations 7

1 D	**6** C	**11** D	**16** A	**21** D
2 B	**7** C	**12** A	**17** C	**22** A
3 B	**8** A	**13** B	**18** D	**23** C
4 B	**9** D	**14** D	**19** C	**24** B
5 C	**10** B	**15** A	**20** C	**25** B

1. **D** 5 : 4

1 Write the amounts in the order asked, students who passed first: 40 : 32.

2 Divide both terms by their greatest common factor, 8: 40 ÷ 8 = 5 and 32 ÷ 8 = 4. The ratio is 5 : 4.

WHY NOT (A) compares the students who did not pass to the total. (B) compares the students who passed to the total instead of to the students who did not pass. (C) reverses the order; the question asks for students who passed to students who did not pass.

2. **B** $\frac{7}{10}$

1 Think in parts: the whole group is 3+7 = 10 equal parts.

2 The green are 7 of those 10 parts, so the fraction is $\frac{7}{10}$.

WHY NOT (A) is the fraction for the other group. (D) uses the ratio 7 : 3 as if it were the fraction of the total.

3. **B** 10

1 Cross-multiply: $27 \times x = 18 \times 15$, so $27x = 270$.

2 Divide both sides by 27: $x = 270 \div 27 = 10$.

TIP Simplify $\frac{15}{27}$ to $\frac{5}{9}$ first to work with smaller numbers: $9x = 18 \times 5 = 90$.

WHY NOT (A) uses a constant difference: 18 is 9 less than 27, so it subtracts 9 from 15 instead of scaling. (D) multiplies 18 by 15 but forgets to divide by 27.

4. **B** 45

1 *Per* means "for each one," so divide the total by the number of units: 225 ÷ 5 = 45.

2 The unit rate is 45 gallons per minute.

TIP Check by multiplying back: 45 × 5 = 225.

WHY NOT (A) misplaces the decimal point when dividing. (D) misplaces the decimal point when dividing.

5. **C** 32

1 Cross-multiply: $35 \times x = 28 \times 40$, so $35x = 1{,}120$.

2 Divide both sides by 35: $x = 1{,}120 \div 35 = 32$.

TIP Simplify $\frac{40}{35}$ to $\frac{8}{7}$ first to work with smaller numbers: $7x = 28 \times 8 = 224$.

WHY NOT (D) multiplies 28 by 40 but forgets to divide by 35.

6. **C** 38

1 *Per* means "for each one," so divide the total by the number of units: 266 ÷ 7 = 38.

2 The unit rate is 38 miles per hour.

TIP Check by multiplying back: 38 × 7 = 266.

WHY NOT (A) misplaces the decimal point when dividing. (D) misplaces the decimal point when dividing.

7. **C** $\frac{8}{11}$

1 Think in parts: the whole group is 3+8 = 11 equal parts.

2 The girls are 8 of those 11 parts, so the fraction is $\frac{8}{11}$.

WHY NOT (A) is the size of just one part, not the whole group's share. (B) is the fraction for the other group. (D) uses the ratio 8 : 3 as if it were the fraction of the total.

8. **A** 3 : 5

1 Write the amounts in the order asked, red pens first: 30 : 50.

2 Divide both terms by their greatest common factor, 10: 30 ÷ 10 = 3 and 50 ÷ 10 = 5. The ratio is 3 : 5.

WHY NOT (B) compares the blue pens to the total. (D) reverses the order; the question asks for red pens to blue pens.

9. **D** 12 : 1

1 Both amounts must be in the same unit. Change pounds to ounces: 3 × 16 = 48 ounces.

2 Now compare ounces to ounces, in the order asked: 48 : 4.

3 Divide both terms by their greatest common factor, 4: 48 ÷ 4 = 12 and 4 ÷ 4 = 1. The ratio is 12 : 1.

WHY NOT (A) multiplies the ounces by 16 instead of the pounds. (B) reverses the order of the ratio. (C) forgets to change pounds to ounces before comparing.

10. **B** 42

1 Cross-multiply: $35 \times x = 30 \times 49$, so $35x = 1{,}470$.

2 Divide both sides by 35: $x = 1{,}470 \div 35 = 42$.

3 Check: $\frac{30}{42}$ and $\frac{35}{49}$ both simplify to $\frac{5}{7}$. ✓

WHY NOT (C) uses a constant difference: 30 is 5 less than 35, so it subtracts 5 from 49 instead of scaling. (D) multiplies 30 by 49 but forgets to divide by 35.

11. **D** 4 : 1

1 Both amounts must be in the same unit. Change feet to inches: $1 \times 12 = 12$ inches.

2 Now compare inches to inches, in the order asked: 12 : 3.

3 Divide both terms by their greatest common factor, 3: $12 \div 3 = 4$ and $3 \div 3 = 1$. The ratio is 4 : 1.

WHY NOT (A) reverses the order of the ratio. (B) forgets to change feet to inches before comparing. (C) uses 10 inches in a foot instead of 12.

12. **A** 25

1 Add the terms of the ratio to get the number of equal parts: $8 + 5 = 13$.

2 Find the size of one part: $65 \div 13 = 5$.

3 The soldiers who chose evening PT make up 5 parts: $5 \times 5 = 25$.

TIP Check: $25 + 40 = 65$, and 40 : 25 simplifies to 8 : 5.

WHY NOT (B) is the number of soldiers who chose morning PT.

13. **B** 35

1 Add the terms of the ratio to get the number of equal parts: $3 + 7 = 10$.

2 Find the size of one part: $50 \div 10 = 5$.

3 The seniors make up 7 parts: $7 \times 5 = 35$.

TIP Check: $35 + 15 = 50$, and 15 : 35 simplifies to 3 : 7.

WHY NOT (A) is the number of juniors.

14. **D** 4

1 Set up a proportion with the same order on both sides (bags of cement over bags of sand): $\frac{2}{5} = \frac{x}{10}$.

2 Cross-multiply: $5x = 2 \times 10 = 20$.

3 Divide by 5: $x = 20 \div 5 = 4$ bags of cement.

WHY NOT (A) matches the 2 bags with 10 bags of sand instead of 5. (B) finds only the extra amount needed and forgets the original 2 bags.

15. **A** 23

1 Set up a proportion with the same order on both sides (pounds over soldiers): $\frac{3}{30} = \frac{x}{230}$.

2 Cross-multiply: $30x = 3 \times 230 = 690$.

3 Divide by 30: $x = 690 \div 30 = 23$ pounds.

TIP Unit rate: 1 pound of rice feeds 10 soldiers, so $230 \div 10 = 23$ pounds.

WHY NOT (D) multiplies by 230 but forgets to divide by 30.

16. **A** 11 miles

1 Each inch on the map stands for 2 miles, so multiply the measurement by 2.

2 Split it up: $5 \times 2 = 10$ and $\frac{1}{2} \times 2 = 1$, so the total is $10 + 1 = 11$ miles.

WHY NOT (D) misplaces the decimal point in the product.

17. **C** 15

1 Each $\frac{1}{2}$ inch stands for 1 foot, so each whole inch stands for 2 feet.

2 Multiply: $7\frac{1}{2} \times 2 = 15$ feet ($7 \times 2 = 14$ and $\frac{1}{2} \times 2 = 1$).

WHY NOT (A) multiplies by $\frac{1}{2}$ instead of dividing by it. (B) ignores the $\frac{1}{2}$ inch. (D) misplaces the decimal point in the product.

18. **D** 96

1 Direct variation means $y = kx$, where k is a constant.

2 Find k by dividing: $k = \frac{y}{x} = \frac{24}{3} = 8$.

3 Use it: $y = 8 \times 12 = 96$.

WHY NOT (A) uses inverse variation (constant product) instead of direct variation. (B) stops after finding the constant k. (C) adds the change in x instead of multiplying by the constant.

19. **C** 45°

1 The three angles of a triangle add up to 180°. Add the terms of the ratio: $4 + 3 + 5 = 12$ equal parts.

2 One part is $180 \div 12 = 15$ degrees.

3 The smallest angle is 3 parts: $3 \times 15 = 45°$.

TIP Check: $60 + 45 + 75 = 180$.

WHY NOT (A) is the size of one part. (D) is the middle angle.

20. **C** $4,800

1 Add the terms of the ratio: $5 + 4 + 6 = 15$ equal parts.

2 One part is $12{,}000 \div 15 = 800$ dollars.

3 The largest share is 6 parts: $6 \times 800 = 4{,}800$ dollars.

TIP Check: $4{,}000 + 3{,}200 + 4{,}800 = 12{,}000$.

WHY NOT (A) is the smallest share. (B) is the middle share.

21. **D** 88

1 In parts, the girls are 11 parts and the boys are 2 parts, so the difference is $11 - 2 = 9$ parts.

2 Those 9 parts equal 72, so one part is $72 \div 9 = 8$.

3 The girls are 11 parts: $11 \times 8 = 88$.

TIP Check: $88 - 16 = 72$, and $16 : 88$ simplifies to $2 : 11$.

WHY NOT (A) is the number of boys.

22. **A** 84

1 In parts, the trucks are 7 parts and the Humvees are 4 parts, so the difference is $7 - 4 = 3$ parts.

2 Those 3 parts equal 36, so one part is $36 \div 3 = 12$.

3 The trucks are 7 parts: $7 \times 12 = 84$.

TIP Check: $84 - 48 = 36$, and $48 : 84$ simplifies to $4 : 7$.

WHY NOT (B) is the total, not one group. (D) treats the difference of 36 as one part, but it is 3 parts.

23. **C** 14 inches

1 Set up a proportion, inches over miles: $\frac{2}{125} = \frac{x}{875}$.

2 Cross-multiply: $125x = 2 \times 875 = 1{,}750$.

3 Divide: $x = 1{,}750 \div 125 = 14$ inches.

TIP Or count how many 125-mile chunks fit: $875 \div 125 = 7$; each chunk is 2 inches on the map, so $7 \times 2 = 14$.

WHY NOT (A) divides 875 by 125 but forgets that the scale uses 2 inches, not 1.

24. **B** 12

1 Direct variation means $y = kx$. Find k by dividing: $k = \frac{y}{x} = \frac{12}{8} = \frac{3}{2}$.

2 Now solve $18 = \frac{3}{2}x$: divide by $\frac{3}{2}$, which is the same as multiplying by $\frac{2}{3}$: $x = 18 \times \frac{2}{3} = 12$.

TIP Or use a proportion: $\frac{12}{8} = \frac{18}{x}$, so $12x = 144$ and $x = 12$.

WHY NOT (C) adds the change in y instead of scaling. (D) multiplies by k instead of dividing by it.

25. **B** $3\frac{3}{4}$

1 Find the scale factor: 30 servings is $\frac{30}{12} = \frac{5}{2}$ times as many as 12 servings.

2 Multiply the amount by that factor: $1\frac{1}{2} \times \frac{5}{2} = \frac{3}{2} \times \frac{5}{2} = \frac{15}{4} = 3\frac{3}{4}$ pounds.

TIP Unit rate: each serving needs $\frac{3}{2} \div 12 = \frac{1}{8}$ pound, and $30 \times \frac{1}{8} = 3\frac{3}{4}$ pounds.

WHY NOT (A) multiplies by $\frac{12}{30}$ instead of $\frac{30}{12}$. (C) adds 18 (the number of extra servings) instead of scaling. (D) multiplies by 30 but forgets to divide by 12.

CHAPTER 8

Exponents & Scientific Notation

Tested on: Mathematics Knowledge

Key Concepts

An *exponent* is shorthand for repeated multiplication: $2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$. *Scientific notation* uses powers of 10 to write very large and very small numbers compactly. Both show up on the test as quick one-step questions.

Powers, signs, and special exponents

- b^n means n factors of b: $3^4 = 3 \times 3 \times 3 \times 3 = 81$.
- The exponent applies only to what it touches: $(-3)^2 = 9$, but $-3^2 = -(3^2) = -9$. An even power of a negative number is positive; an odd power is negative.
- Zero exponent: $b^0 = 1$ (for any $b \neq 0$). Negative exponent = reciprocal: $2^{-3} = \frac{1}{2^3} = \frac{1}{8}$, and $\left(\frac{2}{3}\right)^{-2} = \left(\frac{3}{2}\right)^2 = \frac{9}{4}$.

Rules for the same base

Multiply: add exponents	$a^m \cdot a^n = a^{m+n}$	$2^3 \cdot 2^4 = 2^7$
Divide: subtract exponents	$\frac{a^m}{a^n} = a^{m-n}$	$\frac{5^9}{5^3} = 5^6$
Power of a power: multiply	$(a^m)^n = a^{mn}$	$(3^2)^4 = 3^8$
Power of a fraction	$\left(\frac{p}{q}\right)^n = \frac{p^n}{q^n}$	$\left(\frac{2}{3}\right)^3 = \frac{8}{27}$

Scientific notation

A number is in scientific notation when it is written as $m \times 10^k$ with $1 \leq m < 10$. A positive k makes a big number (move the decimal point k places right); a negative k makes a small number (move it left): $4.5 \times 10^4 = 45{,}000$ and $3.6 \times 10^{-3} = 0.0036$.

WORKED EXAMPLE

Write $(3 \times 10^4)(4 \times 10^5)$ in scientific notation.

Solution. Multiply the first factors and add the exponents: $3 \times 4 = 12$ and $10^4 \times 10^5 = 10^9$, giving 12×10^9. But 12 is not between 1 and 10: $12 = 1.2 \times 10^1$, so the answer is 1.2×10^{10}.

No-Calculator Tip Count the jumps. To write 0.0052 in scientific notation, the decimal point jumps 3 places right to make 5.2, so the exponent is –3: 5.2×10^{-3}. For 52,000 it jumps 4 places left to make 5.2, so the exponent is +4. Small numbers get negative exponents; big numbers get positive ones.

Common Traps

- Multiplying the base by the exponent: 3^4 is 81, not 12.
- Thinking a negative exponent makes a negative number: $2^{-3} = \frac{1}{8}$, not -8.
- Multiplying exponents when you should add them: $2^3 \cdot 2^4 = 2^7$, not 2^{12}.
- Leaving 12×10^9 unadjusted: the first factor must be less than 10.

Practice Set 8

Score: _____ / 25 Target time: 25 minutes

Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

1. What is the value of 9^2?

A 81 **B** 101 **C** 512 **D** 729

2. Evaluate 7×9^0.

A 0 **B** 1 **C** 7 **D** 63

3. Simplify $\frac{9^6}{9^4}$.

A 9^{10} **B** 9^2 **C** 1^2 **D** 9^{-2}

4. Which of the following is equal to $\frac{1}{729}$?

A 9^3 **B** 3^{-9} **C** 9^{-3} **D** -9^3

5. Simplify $(6^4)^5$.

A $5 \cdot 6^4$ **B** 6^{20} **C** 6^9 **D** 6^{25}

6. Evaluate $5^3 + 3^2$.

A 21 **B** 54 **C** 131 **D** 134

7. Write 8.95×10^6 as an ordinary number.

A 8,950,000 **B** 89,500,000
C 895,000,000 **D** 8,950,000,000

8. What is 3,700 written in scientific notation?

A 37×10^2 **B** 3.7×10^3
C 3.7×10^4 **D** 3.7×10^2

9. $(-3)^2$ is equal to which of the following?

A -27 **B** -9 **C** -6 **D** 9

10. What is the value of $(2.5)^3$?

A 15.625 **B** 39.0625 **C** 156.25 **D** 1,562.5

11. What is 0.079 written in scientific notation?

A 7.9×10^{-2} **B** 7.9×10^2
C 7.9×10^{-1} **D** 7.9×10^{-3}

12. Multiply: $(5 \times 10^5)(3 \times 10^9)$. Give the answer in scientific notation.

A 1.5×10^{14} **B** 8×10^{14}
C 1.5×10^{15} **D** 1.5×10^{46}

13. Which of the following shows 934,000,000 in scientific notation?

A 9.34×10^7 **B** 93.4×10^7
C 9.34×10^9 **D** 9.34×10^8

14. What is $(6 \times 10^9)(2 \times 10^4)$ written in scientific notation?

A 1.2×10^{13} **B** 1.2×10^{14}
C 3.2×10^{14} **D** 1.2×10^{37}

15. Evaluate $(0.2)^2$.

A 0.004 **B** 0.008 **C** 0.04 **D** 0.4

16. What is the value of $(-3)^3 + 2^3$?

A -35 **B** -19 **C** -3 **D** 35

17. Which of the following is equal to 5.29×10^7?

A 37.03 **B** 5,290,000
C 52,900,000 **D** 529,000,000

18. Simplify $(4^3)^3 \times 4^2$.

A 16^{11} **B** 4^{18} **C** 4^8 **D** 4^{11}

19. Simplify $(9 \times 10^{11}) \div (5 \times 10^{7})$. Give the answer in scientific notation.

A 1.8×10^{-4} **B** 1.8×10^{2}
C 1.8×10^{3} **D** 1.8×10^{4}

20. Which number below is less than 2×10^{-3}?

A 1.1×10^{-3} **B** 2×10^{3}
C 1.4×10^{-2} **D** 6.5×10^{-3}

21. If $9 \times 3^{3} = 3^{n}$, what is the value of n?

A 5 **B** 6 **C** 12 **D** 27

22. Which of the following numbers is greater than 7.7×10^{-6}?

A 9.1×10^{-8} **B** 9.1×10^{-7}
C 2.4×10^{-5} **D** 7.1×10^{-6}

23. If $4 \times 2^{6} = 2^{n}$, what is the value of n?

A 8 **B** 10 **C** 12 **D** 24

24. Simplify $(5 \times 10^{-4})(8 \times 10^{10})$. Give the answer in scientific notation.

A 4×10^{5} **B** 4×10^{6} **C** 3×10^{7} **D** 4×10^{7}

25. Evaluate $3^{-6} \times 3^{7}$.

A $\frac{1}{3}$ **B** 3 **C** $\frac{15}{4}$ **D** 9

Answers & Explanations 8

1 A	**6** D	**11** A	**16** B	**21** A
2 C	**7** A	**12** C	**17** C	**22** C
3 B	**8** B	**13** D	**18** D	**23** A
4 C	**9** D	**14** B	**19** D	**24** D
5 B	**10** A	**15** C	**20** A	**25** B

1. **A** 81

1 The exponent tells how many times to use the base as a factor: $9^2 = 9 \times 9$.

2 Multiply step by step: $9 \times 9 = 81$.

WHY NOT (C) switches the base and the exponent. (D) uses 3 factors of 9 instead of 2.

2. **C** 7

1 Any nonzero number raised to the zero power equals 1, so $9^0 = 1$.

2 Then $7 \times 1 = 7$.

WHY NOT (A) treats 9^0 as 0. (B) raises the whole product 7×9 to the zero power. (D) treats 9^0 as 9.

3. **B** 9^2

1 When you divide powers with the same base, keep the base and *subtract* the exponents.

2 $\frac{9^6}{9^4} = 9^{6-4} = 9^2$.

WHY NOT (A) adds the exponents instead of subtracting them. (C) divides the bases as well as subtracting the exponents. (D) subtracts the exponents in the wrong order.

4. **C** 9^{-3}

1 Write the denominator as a power: $729 = 9 \times 9 \times 9 = 9^3$.

2 A reciprocal of a power is the same power with a negative exponent: $\frac{1}{9^3} = 9^{-3}$.

WHY NOT (A) is the number itself; the reciprocal needs a negative exponent. (B) switches the base and the exponent. (D) confuses a negative exponent with a negative number.

5. **B** 6^{20}

1 For a power of a power, keep the base and *multiply* the exponents.

2 $(6^4)^5 = 6^{4 \cdot 5} = 6^{20}$.

WHY NOT (A) multiplies by the outside exponent instead of raising to it. (C) adds the exponents instead of multiplying them.

6. **D** 134

1 Exponents come before multiplying, adding, or subtracting, so find each power first.

2 $5^3 = 5 \times 5 \times 5 = 125$ and $3^2 = 3 \times 3 = 9$.

3 Then $125 + 9 = 134$.

WHY NOT (A) multiplies each base by its exponent. (C) evaluates 5^3 correctly but multiplies 3 by 2.

7. **A** 8,950,000

1 Multiplying by 10^6 moves the decimal point 6 places to the right.

2 8.95 becomes 8,950,000 (fill the empty places with zeros).

WHY NOT (B) moves the decimal point one place too many.

8. **B** 3.7×10^3

1 Move the decimal point to get a number that is at least 1 but less than 10: 3.7.

2 The decimal point moved 3 places to the left, which made the number smaller, so multiply by 10^3 to keep the same value.

3 So $3{,}700 = 3.7 \times 10^3$.

WHY NOT (A) is equal to 3,700 but is not scientific notation: the first factor must be less than 10. (C) counts one place too many. (D) counts one place too few.

9. **D** 9

1 The parentheses make -3 the base: $(-3)^2 = (-3) \times (-3)$.

2 Count the negative signs: there are 2, and an even number of negative factors gives a positive product.

3 Since $3^2 = 9$, the answer is 9.

WHY NOT (A) uses 3 factors instead of 2. (B) gets the sign wrong: an even number of negative factors gives a positive product. (C) multiplies the base by the exponent.

10. **A** 15.625

1 Multiply the digits as whole numbers first: 25^3 = 15,625.

2 Count decimal places: 2.5 has 1 decimal place, and it is used as a factor 3 times, so the answer has $1 \times 3 = 3$ decimal places: 15.625.

TIP Estimate: 2.5 is between 2 and 3, so its cube is between $2^3 = 8$ and $3^3 = 27$.

WHY NOT (B) uses 4 factors instead of 3. (C) puts too few decimal places in the answer (it needs 3).

11. **A** 7.9×10^{-2}

1 Move the decimal point to get a number that is at least 1 but less than 10: 7.9.

2 The decimal point moved 2 places to the right, which made the number larger, so multiply by 10^{-2} to keep the same value. (Numbers less than 1 always get a negative exponent.)

3 So $0.079 = 7.9 \times 10^{-2}$.

WHY NOT (B) uses a positive exponent, but numbers less than 1 have negative exponents. (C) counts one place too few. (D) counts one place too many.

12. **C** 1.5×10^{15}

1 Multiply the first factors: $5 \times 3 = 15$.

2 Multiply the powers of 10 by adding exponents: $10^5 \times 10^9 = 10^{14}$.

3 15 is not less than 10, so write $15 = 1.5 \times 10^1$: the answer is 1.5×10^{15}.

WHY NOT (A) rewrites 15 as 1.5 but forgets to add 1 to the exponent. (B) adds the first factors instead of multiplying them. (D) multiplies the exponents instead of adding them.

13. **D** 9.34×10^8

1 Move the decimal point to get a number that is at least 1 but less than 10: 9.34.

2 The decimal point moved 8 places to the left, which made the number smaller, so multiply by 10^8 to keep the same value.

3 So $934{,}000{,}000 = 9.34 \times 10^8$.

WHY NOT (A) counts one place too few. (B) is equal to 934,000,000 but is not scientific notation: the first factor must be less than 10. (C) counts one place too many.

14. **B** 1.2×10^{14}

1 Multiply the first factors: $6 \times 2 = 12$.

2 Multiply the powers of 10 by adding exponents: $10^9 \times 10^4 = 10^{13}$.

3 12 is not less than 10, so write $12 = 1.2 \times 10^1$: the answer is 1.2×10^{14}.

WHY NOT (A) rewrites 12 as 1.2 but forgets to add 1 to the exponent. (D) multiplies the exponents instead of adding them.

15. **C** 0.04

1 Multiply the digits as whole numbers first: $2^2 = 4$.

2 Count decimal places: 0.2 has 1 decimal place, and it is used as a factor 2 times, so the answer has $1 \times 2 = 2$ decimal places: 0.04.

TIP Estimate: 0.2 is less than 1, so its square must be *smaller* than 0.2.

WHY NOT (A) puts too many decimal places in the answer. (B) uses 3 factors instead of 2. (D) multiplies by the exponent instead of raising to the power.

16. **B** -19

1 Find each power first. $(-3)^3 = -27$ (an odd number of negative factors gives a negative result).

2 $2^3 = 8$.

3 Combine: $-27 + 8 = -19$.

WHY NOT (A) subtracts 2^3 instead. (C) multiplies each base by its exponent. (D) gets the sign of $(-3)^3$ wrong.

17. **C** 52,900,000

1 Multiplying by 10^7 moves the decimal point 7 places to the right.

2 5.29 becomes 52,900,000 (fill the empty places with zeros).

WHY NOT (A) multiplies by the exponent instead of by a power of 10. (B) moves the decimal point one place too few. (D) moves the decimal point one place too many.

18. **D** 4^{11}

1 Power of a power: multiply the exponents. $(4^3)^3 = 4^9$.

2 Product of powers: add the exponents. $4^9 \times 4^2 = 4^{11}$.

WHY NOT (A) multiplies the bases as well. (B) multiplies all the exponents. (C) adds all the exponents; the power of a power should be multiplied.

19. **D** 1.8×10^4

1 Divide the first factors: $9 \div 5 = 1.8$.

2 Divide the powers of 10 by subtracting exponents: $10^{11} \div 10^7 = 10^4$.

3 1.8 is between 1 and 10, so the answer is 1.8×10^4.

WHY NOT (A) subtracts the exponents in the wrong order.

20. **A** 1.1×10^{-3}

1 Compare the powers of 10 first. Because every first factor is between 1 and 10, the number with the larger power of 10 is always the larger number (and remember that −5 is larger than −6).

2 1.1×10^{-3} has the same power of 10 as 2×10^{-3}, so compare the first factors: 1.1 is smaller than 2.

3 Each of the other choices is greater than 2×10^{-3}.

WHY NOT (B) has a positive exponent, so it is greater than 1. (C) has a smaller first factor, but its power of 10 is larger, so it is greater than 2×10^{-3}. (D) has the same power of 10 but a larger first factor.

21. **A** 5

1 Write 9 as a power of 3: $9 = 3^2$.

2 Now $3^2 \times 3^3 = 3^5$, so $n = 2 + 3 = 5$.

WHY NOT (B) multiplies the exponents instead of adding them. (C) adds 9 to the exponent instead of writing 9 as a power of 3.

22. **C** 2.4×10^{-5}

1 Compare the powers of 10 first. Because every first factor is between 1 and 10, the number with the larger power of 10 is always the larger number (and remember that −5 is larger than −6).

2 2.4×10^{-5} has 10^{-5}, a larger power of 10 than 10^{-6}, so it is greater even though its first factor is smaller.

3 Each of the other choices is less than 7.7×10^{-6}.

WHY NOT (A) has a larger first factor, but its power of 10 is smaller. (B) has a larger first factor, but its power of 10 is smaller, so it is less than 7.7×10^{-6}. (D) has the same power of 10 but a smaller first factor.

23. **A** 8

1 Write 4 as a power of 2: $4 = 2^2$.

2 Now $2^2 \times 2^6 = 2^8$, so $n = 2 + 6 = 8$.

WHY NOT (B) adds 4 to the exponent instead of writing 4 as a power of 2. (C) multiplies the exponents instead of adding them.

24. **D** 4×10^7

1 Multiply the first factors: $5 \times 8 = 40$.

2 Add the exponents: $-4 + 10 = 6$, giving 40×10^6.

3 40 is not less than 10, so write it as 4×10^1 and add 1 to the exponent: the answer is 4×10^7.

WHY NOT (B) rewrites 40 as 4 but forgets to add 1 to the exponent.

25. **B** 3

1 Same base, so add the exponents: $-6 + 7 = 1$, which gives 3^1.

2 $3^1 = 3$.

WHY NOT (A) gets the sign of the exponent wrong: $-6 + 7$ is 1, not −1. (D) multiplies the bases as well as adding the exponents.

CHAPTER 9

Square Roots & Radicals

Tested on: Mathematics Knowledge

Key Concepts

The *square root* of a number is the value that, multiplied by itself, gives the number: $\sqrt{49} = 7$ because $7 \times 7 = 49$. The symbol $\sqrt{\ }$ always means the positive root. Knowing the perfect squares by heart makes every problem in this chapter faster.

Perfect squares and cubes to memorize

n	1	2	3	4	5	6	7	8
n^2	1	4	9	16	25	36	49	64
n^3	1	8	27	64	125	216	343	512
n	9	10	11	12	13	14	15	20
n^2	81	100	121	144	169	196	225	400

The *cube root* undoes a cube: $\sqrt[3]{125} = 5$ because $5^3 = 125$.

Working with radicals

- Simplify: pull out the *largest* perfect-square factor. $\sqrt{72} = \sqrt{36 \cdot 2} = 6\sqrt{2}$.
- Add or subtract only *like* radicals (same number under the root): $3\sqrt{5} + 4\sqrt{5} = 7\sqrt{5}$.
- Multiply the numbers under the roots: $\sqrt{6} \cdot \sqrt{15} = \sqrt{90} = 3\sqrt{10}$.
- Rationalize a denominator: multiply top and bottom by the root. $\frac{6}{\sqrt{3}} = \frac{6\sqrt{3}}{3} = 2\sqrt{3}$.

Fractions, decimals, and equations

$\sqrt{\frac{49}{64}} = \frac{7}{8}$ (root of top over root of bottom), and $\sqrt{0.09} = \sqrt{\frac{9}{100}} = \frac{3}{10} = 0.3$. The equation $x^2 = 81$ has *two* solutions, $x = 9$ and $x = -9$; read the question to see whether it wants both or only the positive one.

WORKED EXAMPLE

Simplify $\sqrt{12} + \sqrt{27}$.
Solution. The radicals are not alike yet, so simplify each: $\sqrt{12} = \sqrt{4 \cdot 3} = 2\sqrt{3}$ and $\sqrt{27} = \sqrt{9 \cdot 3} = 3\sqrt{3}$. Now they are like radicals: $2\sqrt{3} + 3\sqrt{3} = 5\sqrt{3}$.

No-Calculator Tip Estimate a root by trapping it between perfect squares: $49 < 53 < 64$, so $\sqrt{53}$ is between 7 and 8, a little above 7.

Common Traps

- Splitting a root over addition: $\sqrt{9+16} = \sqrt{25} = 5$, not $3 + 4 = 7$.
- Adding unlike radicals: $\sqrt{2} + \sqrt{3}$ is *not* $\sqrt{5}$.
- Decimal places: $\sqrt{0.09} = 0.3$, not 0.03 (check: $0.3^2 = 0.09$).
- Forgetting the negative solution of $x^2 = 81$ when the question asks for all solutions.

Practice Set 9

Score: _____ / 25 Target time: 25 minutes

Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

1. Which of the following is equal to $\sqrt{4} \times \sqrt{100}$?

A 16 **B** 19 **C** 20 **D** 400

2. $\sqrt{222}$ is between which two consecutive whole numbers?

A 12 and 13 **B** 13 and 14
C 14 and 15 **D** 17 and 18

3. Evaluate $\sqrt{64}$.

A 7 **B** 8 **C** 9 **D** 32

4. What is the value of $\sqrt{\frac{81}{100}}$?

A $\frac{9}{100}$ **B** $\frac{9}{10}$ **C** $\frac{10}{9}$ **D** $\frac{81}{10}$

5. A square garden plot has an area of 49 square yards. How long is each side, in yards?

A 5 **B** 6 **C** 7 **D** 8

6. Which of the following is equal to $\sqrt{\frac{121}{81}}$?

A $\frac{11}{81}$ **B** $\frac{11}{9}$ **C** $\frac{14}{9}$ **D** $\frac{121}{9}$

7. Between which two consecutive whole numbers does $\sqrt{176}$ lie?

A 13 and 14 **B** 14 and 15
C 15 and 16 **D** 16 and 17

8. If $x^2 - 400 = 0$ and $x > 0$, what is x?

A -20 **B** 12 **C** 14 **D** 20

9. Simplify $\sqrt{147}$.

A $3\sqrt{7}$ **B** $7\sqrt{3}$ **C** $10\sqrt{3}$ **D** $49\sqrt{3}$

10. What is $4\sqrt{5} + 9\sqrt{5}$?

A $5\sqrt{5}$ **B** $13\sqrt{5}$ **C** $13\sqrt{10}$ **D** $36\sqrt{5}$

11. Write $\sqrt{350}$ in simplest form.

A 5 **B** $3\sqrt{14}$ **C** $4\sqrt{14}$ **D** $5\sqrt{14}$

12. Which of the following is equal to $\sqrt{15} \cdot \sqrt{12}$?

A $5\sqrt{5}$ **B** $6\sqrt{5}$ **C** $36\sqrt{5}$ **D** 180

13. Simplify $\sqrt{252}$.

A $6\sqrt{7}$ **B** $7\sqrt{6}$ **C** $9\sqrt{7}$ **D** $36\sqrt{7}$

14. Which of the following is equal to $\sqrt{8} \cdot \sqrt{3}$?

A $2\sqrt{6}$ **B** $3\sqrt{6}$ **C** $4\sqrt{6}$ **D** 24

15. Which of the following is equal to $3\sqrt{3} - \sqrt{3}$?

A $\sqrt{3}$ **B** 2 **C** $2\sqrt{3}$ **D** $3\sqrt{3}$

16. What is the square root of 0.0001?

A 0.01 **B** 0.02 **C** 0.04 **D** 0.1

17. A square storage pad has an area of 56 square feet. To the nearest foot, how long is each side?

A 6 **B** 7 **C** 8 **D** 28

18. If $3x^2 = 147$ and $x < 0$, what is the value of x?

A -49 **B** -21 **C** -7 **D** 7

19. Simplify $\sqrt{72} + \sqrt{32}$.

A $10\sqrt{2}$ **B** $12\sqrt{2}$ **C** 20 **D** $24\sqrt{2}$

20. Rationalize the denominator: $\frac{18}{\sqrt{6}}$.

A 3 **B** $3\sqrt{6}$ **C** 18 **D** $18\sqrt{6}$

21. What is $\sqrt{180} - \sqrt{45}$ in simplest form?

A $3\sqrt{5}$ **B** $5\sqrt{5}$ **C** $9\sqrt{5}$ **D** $18\sqrt{5}$

22. Rationalize the denominator: $\frac{14}{\sqrt{2}}$.

A $\frac{\sqrt{2}}{14}$ **B** $\frac{7\sqrt{2}}{2}$ **C** 7 **D** $7\sqrt{2}$

23. What is $(5\sqrt{12})(2\sqrt{5})$ in simplest form?

A $10\sqrt{15}$ **B** $10\sqrt{17}$ **C** $14\sqrt{15}$ **D** $20\sqrt{15}$

24. What are all the solutions of $5x^2 = 20$?

A -2 only **B** 2 only

C 2 and -2 **D** 4 and -4

25. Evaluate $\sqrt{25 \times 81}$.

A 45 **B** 50 **C** 60 **D** $2{,}025$

Answers & Explanations 9

1 C	**6** B	**11** D	**16** A	**21** A
2 C	**7** A	**12** B	**17** B	**22** D
3 B	**8** D	**13** A	**18** C	**23** D
4 B	**9** B	**14** A	**19** A	**24** C
5 C	**10** B	**15** C	**20** B	**25** A

1. **C** 20

1 Find each square root: $\sqrt{4} = 2$ and $\sqrt{100} = 10$.

2 Then $2 \times 10 = 20$.

WHY NOT (D) forgets to take the square roots.

2. **C** 14 and 15

1 Find the perfect squares on either side of 222: $14^2 = 196$ and $15^2 = 225$.

2 Since $196 < 222 < 225$, the square root is between 14 and 15.

3. **B** 8

1 Look for the number that, multiplied by itself, gives 64.

2 $8 \times 8 = 64$, so the square root is 8.

WHY NOT (D) divides by 2 instead of taking the square root.

4. **B** $\frac{9}{10}$

1 Take the square root of the top and of the bottom: $\sqrt{\frac{81}{100}} = \frac{\sqrt{81}}{\sqrt{100}}$.

2 $\sqrt{81} = 9$ and $\sqrt{100} = 10$, so the answer is $\frac{9}{10}$.

WHY NOT (A) takes the square root of the numerator only. (C) flips the fraction. (D) takes the square root of the denominator only.

5. **C** 7

1 Look for the number that, multiplied by itself, gives 49.

2 $7 \times 7 = 49$, so the square root is 7.

6. **B** $\frac{11}{9}$

1 Take the square root of the top and of the bottom: $\sqrt{\frac{121}{81}} = \frac{\sqrt{121}}{\sqrt{81}}$.

2 $\sqrt{121} = 11$ and $\sqrt{81} = 9$, so the answer is $\frac{11}{9}$.

WHY NOT (A) takes the square root of the numerator only. (D) takes the square root of the denominator only.

7. **A** 13 and 14

1 Find the perfect squares on either side of 176: $13^2 = 169$ and $14^2 = 196$.

2 Since $169 < 176 < 196$, the square root is between 13 and 14.

8. **D** 20

1 Add 400 to both sides: $x^2 = 400$.

2 Undo the square by taking the square root: $x = \sqrt{400}$ or $x = -\sqrt{400}$.

3 $\sqrt{400} = 20$, and the positive solution is $x = 20$.

WHY NOT (A) is the negative solution, but the question asks for the positive one.

9. **B** $7\sqrt{3}$

1 Find the largest perfect square that divides 147: it is 49, because $147 = 49 \times 3$.

2 Split the root: $\sqrt{147} = \sqrt{49} \times \sqrt{3} = 7\sqrt{3}$.

TIP Check: $(7\sqrt{3})^2 = 49 \times 3 = 147$. ✓

WHY NOT (A) swaps the numbers: writes $3\sqrt{7}$ instead of $7\sqrt{3}$. (D) pulls out 49 instead of its square root, 7.

10. **B** $13\sqrt{5}$

1 Both terms have $\sqrt{5}$, so they are like radicals: combine the numbers in front, just like $4x + 9x$.

2 $4 + 9 = 13$, so the answer is $13\sqrt{5}$. The $\sqrt{5}$ stays the same.

WHY NOT (A) subtracts instead of adding. (C) adds the numbers under the radicals too. (D) multiplies the numbers in front instead of combining them.

11. **D** $5\sqrt{14}$

1 Find the largest perfect square that divides 350: it is 25, because $350 = 25 \times 14$.

2 Split the root: $\sqrt{350} = \sqrt{25} \times \sqrt{14} = 5\sqrt{14}$.

TIP Check: $(5\sqrt{14})^2 = 25 \times 14 = 350$. ✓

WHY NOT (A) drops the leftover $\sqrt{14}$.

12. **B** $6\sqrt{5}$

1 Multiply the numbers under the radicals: $\sqrt{15} \cdot \sqrt{12} = \sqrt{15 \times 12} = \sqrt{180}$.

2 Simplify: $\sqrt{180} = \sqrt{36 \cdot 5} = 6\sqrt{5}$.

WHY NOT (C) pulls out 36 instead of its square root, 6. (D) drops the radical sign.

13. **A** $6\sqrt{7}$

1 Find the largest perfect square that divides 252: it is 36, because $252 = 36 \times 7$.

2 Split the root: $\sqrt{252} = \sqrt{36} \times \sqrt{7} = 6\sqrt{7}$.

TIP Check: $(6\sqrt{7})^2 = 36 \times 7 = 252$. ✓

WHY NOT (B) swaps the numbers: writes $7\sqrt{6}$ instead of $6\sqrt{7}$. (D) pulls out 36 instead of its square root, 6.

14. **A** $2\sqrt{6}$

1 Multiply the numbers under the radicals: $\sqrt{8} \cdot \sqrt{3} = \sqrt{8 \times 3} = \sqrt{24}$.

2 Simplify: $\sqrt{24} = \sqrt{4 \cdot 6} = 2\sqrt{6}$.

WHY NOT (C) pulls out 4 instead of its square root, 2. (D) drops the radical sign.

15. **C** $2\sqrt{3}$

1 Both terms have $\sqrt{3}$, so they are like radicals: combine the numbers in front, just like $3x - x$.

2 $3 - 1 = 2$, so the answer is $2\sqrt{3}$. The $\sqrt{3}$ stays the same.

WHY NOT (B) drops the $\sqrt{3}$ after subtracting. (D) multiplies the numbers in front instead of combining them.

16. **A** 0.01

1 Write the decimal as a fraction: $0.0001 = \frac{1}{10{,}000}$.

2 Take the root of the top and the bottom: $\sqrt{\frac{1}{10{,}000}} = \frac{1}{100} = 0.01$.

3 Check: $0.01 \times 0.01 = 0.0001$. ✓

WHY NOT (D) puts the decimal point too far to the right.

17. **B** 7

1 Trap the root between perfect squares: $7^2 = 49$ and $8^2 = 64$, so $\sqrt{56}$ is between 7 and 8.

2 Compare distances: $56 - 49 = 7$ and $64 - 56 = 8$. 56 is closer to 49.

3 So $\sqrt{56}$ is closest to 7.

WHY NOT (C) is the other neighbor, but 56 is closer to 49. (D) divides 56 by 2 instead of taking the square root.

18. **C** -7

1 Divide both sides by 3: $x^2 = 49$.

2 Take the square root: $x = 7$ or $x = -7$. Since $x < 0$, $x = -7$.

WHY NOT (A) forgets to take the square root. (B) multiplies by 3 instead of dividing. (D) is the positive solution, but the question says $x < 0$.

19. **A** $10\sqrt{2}$

1 The numbers under the radicals are different, so simplify each radical first.

2 $\sqrt{72} = \sqrt{36 \cdot 2} = 6\sqrt{2}$ and $\sqrt{32} = \sqrt{16 \cdot 2} = 4\sqrt{2}$.

3 Now they are like radicals: $6\sqrt{2} + 4\sqrt{2} = 10\sqrt{2}$.

WHY NOT (C) adds the numbers under the radicals after simplifying. (D) multiplies the numbers in front instead of adding.

20. **B** $3\sqrt{6}$

1 Multiply the top and bottom by $\sqrt{6}$: $\frac{18}{\sqrt{6}} \times \frac{\sqrt{6}}{\sqrt{6}} = \frac{18\sqrt{6}}{6}$, because $\sqrt{6} \cdot \sqrt{6} = 6$.

2 Simplify $\frac{18}{6}$: divide top and bottom by 6, giving $3\sqrt{6}$.

WHY NOT (A) multiplies only the bottom by $\sqrt{6}$. (C) cancels the 6 on the bottom against the $\sqrt{6}$ on top. (D) multiplies the top by $\sqrt{6}$ but forgets that the bottom becomes 6.

21. **A** $3\sqrt{5}$

1 The numbers under the radicals are different, so simplify each radical first.

2 $\sqrt{180} = \sqrt{36 \cdot 5} = 6\sqrt{5}$ and $\sqrt{45} = \sqrt{9 \cdot 5} = 3\sqrt{5}$.

3 Now they are like radicals: $6\sqrt{5} - 3\sqrt{5} = 3\sqrt{5}$.

WHY NOT (C) adds instead of subtracting. (D) multiplies the numbers in front instead of subtracting.

22. **D** $7\sqrt{2}$

1 Multiply the top and bottom by $\sqrt{2}$: $\frac{14}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{14\sqrt{2}}{2}$, because $\sqrt{2} \cdot \sqrt{2} = 2$.

2 Simplify $\frac{14}{2}$: divide top and bottom by 2, giving $7\sqrt{2}$.

WHY NOT (A) turns the fraction upside down. (B) thinks $\sqrt{2} \cdot \sqrt{2} = 4$. (C) multiplies only the bottom by $\sqrt{2}$.

23. **D** $20\sqrt{15}$

1 Multiply the numbers in front and the numbers under the radicals separately: $5 \times 2 = 10$ and $\sqrt{12} \cdot \sqrt{5} = \sqrt{60}$.

2 Simplify the radical: $\sqrt{60} = \sqrt{4 \cdot 15} = 2\sqrt{15}$.

3 Multiply: $10 \times 2\sqrt{15} = 20\sqrt{15}$.

WHY NOT (A) multiplies by only one of the numbers in front. (B) adds the numbers under the radicals. (C) adds the numbers in front instead of multiplying them.

24. **C** 2 and −2

1 Divide both sides by 5: $x^2 = 4$.

2 Take the square root of both sides. A positive number has two square roots: $x = 2$ or $x = -2$.

3 Check: $5(2)^2 = 20$, and $(-2)^2$ is also 4. ✓

WHY NOT (A) keeps only the negative solution. (B) forgets the negative solution. (D) forgets to take the square root.

25. **A** 45

1 The square root of a product is the product of the square roots: $\sqrt{25 \times 81} = \sqrt{25} \times \sqrt{81}$.

2 $\sqrt{25} = 5$ and $\sqrt{81} = 9$, so the answer is $5 \times 9 = 45$.

WHY NOT (D) forgets to take the square root.

PART II

Algebra

Expressions, equations, inequalities, systems, polynomials, and quadratics: the algebra on the Mathematics Knowledge subtest.

10 Algebraic Expressions
11 Linear Equations
12 Inequalities
13 Systems of Equations
14 Polynomials & Exponent Rules
15 Factoring & Quadratic Equations

CHAPTER 10

Algebraic Expressions

Tested on: Mathematics Knowledge • Arithmetic Reasoning

Key Concepts

An *algebraic expression* combines numbers and letters (variables) with operations, such as $3x + 5$ or $2a^2 - 3b$. This chapter covers the four skills the test checks most: plugging in values, combining like terms, distributing, and turning words into expressions.

Vocabulary

In $4x^2 - 7x + 9$ there are three *terms*. The number in front of a variable is its *coefficient* (4 and -7); 9 is the *constant*. *Like terms* have exactly the same variable part: $5x$ and $-2x$ are like terms, but $5x$ and $5x^2$ are not, and neither are $3x$ and $3y$.

Evaluating: substitute, then follow the order of operations

Replace each variable with its value *in parentheses*, then do powers, then multiplication, then addition and subtraction. For $a = -2$ and $b = 4$:

$$2a^2 - 3b = 2(-2)^2 - 3(4) = 2(4) - 12 = 8 - 12 = -4.$$

Note that $(-2)^2 = 4$, but $-x^2$ with $x = 2$ is $-(2^2) = -4$.

Simplifying

- Combine like terms by adding their coefficients: $5x + 3y - 2x + 7y = 3x + 10y$.
- Distribute: multiply *every* term inside the parentheses, signs included. $-2(x + 5) = -2x - 10$ and $-(x - 4) = -x + 4$.

Words into symbols

sum, more than, increased by	$+$
difference, less than, decreased by	$-$
product, times, twice	$\times$
quotient, divided by, half of	$\div$

"5 less than twice a number" is $2n - 5$: *less than* reverses the order, so it is *not* $5 - 2n$.

WORKED EXAMPLE

Simplify $3(2x - 4) - 2(x + 5)$.
Solution. Distribute each number: $3(2x - 4) = 6x - 12$ and $-2(x + 5) = -2x - 10$. Combine like terms: $6x - 2x = 4x$ and $-12 - 10 = -22$. The result is $4x - 22$.

No-Calculator Tip Check a simplification by plugging in an easy number. With $x = 1$: $3(2 - 4) - 2(1 + 5) = -6 - 12 = -18$, and $4(1) - 22 = -18$ too.

Common Traps

- Squaring a negative without parentheses: $(-3)^2 = 9$, not -9.
- Distributing a minus sign to only the first term: $-2(x + 5)$ is $-2x - 10$, not $-2x + 10$.
- Combining unlike terms: $3x + 4y$ cannot become $7xy$.

Practice Set 10

Score: _____ / 25 Target time: 26 minutes

Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

1. Evaluate $9x + 8y$ for $x = 9$ and $y = 3$.

A 29 **B** 99 **C** 105 **D** 204

2. Combine like terms: $5a + 4b - 6a - 6b$

A $-a + 10b$ **B** $-a - 2b$
C $11a - 2b$ **D** $-3ab$

3. What is the value of $4x - 6$ when $x = 6$?

A 0 **B** 4 **C** 18 **D** 40

4. Simplify: $7x + 9y + 3x - 4y$

A $10x + 5y$ **B** $11x + 5y$
C $10x + 13y$ **D** $15xy$

5. Which expression means "10 more than a number"?

A $n - 10$ **B** $n + 10$ **C** $10 - n$ **D** $10n$

6. Which expression means "8 times a number, increased by 13"?

A $8n + 13$ **B** $21n$
C $8(n + 13)$ **D** $13n + 8$

7. The area of a triangle is $A = \frac{1}{2}bh$. What is A when $b = 8$ and $h = 14$?

A 11 **B** 46 **C** 51 **D** 56

8. Distance traveled is given by $d = rt$, where r is the rate in miles per hour and t is the time in hours. What is d when $r = 50$ and $t = 4$?

A 12.5 **B** 54 **C** 200 **D** 250

9. If $x = 4$ and $y = -4$, what is the value of $5x - 3y$?

A 8 **B** 32 **C** 52 **D** 56

10. Which of the following is equal to $6(4x - 9) - 3(2x - 2)$?

A $18x - 7$ **B** $18x - 48$
C $18x - 60$ **D** $30x - 60$

11. Evaluate $4a^2 - 2b$ for $a = -4$ and $b = 4$.

A 56 **B** 72 **C** 112 **D** 248

12. Simplify: $3(4x + 8) - 4(4x - 3)$

A $-4x + 36$ **B** $28x + 12$
C $-4x + 11$ **D** $-4x + 12$

13. Simplify: $3x^2 + 2x + 2x^2 - 7x + 3$

A $5x^2 + 9x + 3$ **B** $5x^2 - 2x$
C $5x^2 - 5x + 3$ **D** $5x^3 - 5x + 3$

14. If n stands for a number, which expression represents "10 subtracted from twice a number"?

A $n^2 - 10$ **B** $2n - 10$
C $10 - 2n$ **D** $2(n - 10)$

15. Which expression means "the sum of 10 times a number and 7"?

A $17n$ **B** $10 + 7n$
C $10(n + 7)$ **D** $10n + 7$

16. Shirts cost \$15 each and hats cost \$14 each. Which expression gives the total cost, in dollars, of x shirts and y hats?

A $29xy$ **B** $29(x + y)$
C $15x + 14y$ **D** $14x + 15y$

17. A gym charges a \$37 sign-up fee plus \$22 per month. Which expression gives the total cost, in dollars, of a membership for t months?

A $59t$ **B** $37 + 22 + t$
C $37 + 22t$ **D** $37t + 22$

18. Simple interest is given by $I = Prt$, where P is the amount invested, r is the yearly rate as a decimal, and t is the time in years. What is I when $P = 1{,}000$, $r = 0.02$, and $t = 3$?

A \$60 **B** \$600 **C** \$1,060 **D** \$6,000

19. Which of the following is equal to $x(x + 8) - 3(x + 9)$?

A $x^2 + 11x - 27$ **B** $7x - 27$
C $x^2 + 5x + 27$ **D** $x^2 + 5x - 27$

20. If $a = -3$ and $b = -5$, what is the value of $4ab - b^2$?

A -35 **B** 35 **C** 55 **D** 85

21. Which of the following is equal to $5(4x + 5) - (5x + 1)$?

A $15x + 24$ **B** $25x + 26$
C $15x + 4$ **D** $15x + 26$

22. Evaluate $2x^2 - 2xy$ for $x = -4$ and $y = 4$.

A -64 **B** 0 **C** 34 **D** 64

23. The volume of a box is $V = lwh$. If $V = 400$, $l = 10$, and $w = 8$, what is h?

A 3 **B** 3.5 **C** 5 **D** 320

24. The sides of a triangle have lengths $2x+8$, $4x-4$, and $4x - 7$. Which expression represents the perimeter of the triangle?

A $20x - 6$ **B** $10x + 19$
C $7x$ **D** $10x - 3$

25. A rental truck costs \$59 per day plus \$0.50 per mile. Which expression gives the cost, in dollars, of renting the truck for d days and driving it m miles?

A $(59 + 0.50)dm$ **B** $59m + 0.50d$
C $59 + 0.50m$ **D** $59d + 0.50m$

Answers & Explanations 10

1 C	**6** A	**11** A	**16** C	**21** A
2 B	**7** D	**12** A	**17** C	**22** D
3 C	**8** C	**13** C	**18** A	**23** C
4 A	**9** B	**14** B	**19** D	**24** D
5 B	**10** B	**15** D	**20** B	**25** D

1. **C** 105

1 Substitute $x = 9$ and $y = 3$: $9(9) + 8(3)$.

2 Multiply first: $81 + 24$.

3 Then add or subtract: $81 + 24 = 105$.

WHY NOT (A) adds the numbers instead of multiplying. (B) swaps the values of x and y.

2. **B** $-a - 2b$

1 Group the like terms: a-terms $5a - 6a$ and b-terms $4b - 6b$.

2 Add the coefficients: $5 - 6 = -1$ and $4 - 6 = -2$.

3 The simplified expression is $-a - 2b$.

WHY NOT (A) ignores the minus sign in front of $6b$. (C) ignores the minus sign in front of $6a$. (D) combines unlike terms; a-terms and b-terms cannot be added together.

3. **C** 18

1 Substitute $x = 6$: $4(6) - 6$.

2 Multiply first: $24 - 6$.

3 Then add or subtract: $24 - 6 = 18$.

WHY NOT (A) subtracts before multiplying. (B) adds 4 and 6 instead of multiplying. (D) reads $4x$ as the two-digit number 46.

4. **A** $10x + 5y$

1 Group the like terms: x-terms $7x + 3x$ and y-terms $9y - 4y$.

2 Add the coefficients: $7 + 3 = 10$ and $9 - 4 = 5$.

3 The simplified expression is $10x + 5y$.

WHY NOT (C) ignores the minus sign in front of $4y$. (D) combines unlike terms; x-terms and y-terms cannot be added together.

5. **B** $n + 10$

1 "More than" means add: start with the number and add 10.

2 With n for the number, the expression is $n + 10$.

WHY NOT (A) subtracts instead of adding. (D) multiplies instead of adding.

6. **A** $8n + 13$

1 "8 times a number" is $8n$; "increased by 13" adds 13 to that.

2 With n for the number, the expression is $8n + 13$.

WHY NOT (C) adds before multiplying. (D) swaps the two numbers.

7. **D** 56

1 Substitute: $A = \frac{1}{2}(8)(14)$.

2 Multiply: $8 \times 14 = 112$, and half of 112 is 56.

WHY NOT (A) adds the base and height instead of multiplying.

8. **C** 200

1 Substitute: $d = 50 \times 4$.

2 Multiply: $50 \times 4 = 200$ miles.

WHY NOT (A) divides the rate by the time. (B) adds the rate and time instead of multiplying.

9. **B** 32

1 Substitute, putting each value in parentheses: $5(4) - 3(-4)$.

2 Multiply each term (watch the signs): $20 + 12$.

3 Add and subtract from left to right: $20 + 12 = 32$.

WHY NOT (A) forgets that a negative times a negative is positive.

10. **B** $18x - 48$

1 Distribute 6: $6(4x - 9) = 24x - 54$.

2 Distribute -3 (the minus sign comes with it): $-3(2x - 2) = -6x + 6$.

3 Combine like terms: $24x - 6x = 18x$ and $-54 + 6 = -48$. The result is $18x - 48$.

TIP Check with $x = 1$: the original and $18x - 48$ both equal -30.

WHY NOT (A) multiplies only the first term in each set of parentheses. (C) multiplies -2 by 3 instead of -3; the minus sign goes to every term. (D) adds the second product instead of subtracting it.

11. **A** 56

1 Substitute, putting each value in parentheses: $4(-4)^2 - 2(4)$.

2 Powers first: $(-4)^2 = 16$.

3 Multiply each term (watch the signs): $64 - 8$.

4 Add and subtract from left to right: $64 - 8 = 56$.

WHY NOT (D) squares $4a$ instead of only a.

12. **A** $-4x + 36$

1 Distribute 3: $3(4x + 8) = 12x + 24$.

2 Distribute -4 (the minus sign comes with it): $-4(4x - 3) = -16x + 12$.

3 Combine like terms: $12x - 16x = -4x$ and $24 + 12 = 36$. The result is $-4x + 36$.

TIP Check with $x = 1$: the original and $-4x + 36$ both equal 32.

WHY NOT (B) adds the second product instead of subtracting it. (C) multiplies only the first term in each set of parentheses. (D) multiplies -3 by 4 instead of -4; the minus sign goes to every term.

13. **C** $5x^2 - 5x + 3$

1 Like terms have the same variable part. The x^2-terms are $3x^2 + 2x^2$; the x-terms are $2x - 7x$; the constant is 3.

2 Add the coefficients: $3 + 2 = 5$ and $2 - 7 = -5$. The exponent stays the same.

3 The simplified expression is $5x^2 - 5x + 3$.

WHY NOT (A) ignores the minus sign in front of $7x$. (B) adds the constant 3 to the x-terms. (D) adds the exponents when combining like terms.

14. **B** $2n - 10$

1 "Twice a number" is $2n$. "10 subtracted from" it means take 10 away from $2n$, so $2n$ comes first.

2 With n for the number, the expression is $2n - 10$.

WHY NOT (A) reads *twice* as squaring. (C) reverses the order; *subtracted from* means take it away from the other amount. (D) subtracts before doubling.

15. **D** $10n + 7$

1 The two amounts being added are "10 times a number" ($10n$) and 7.

2 With n for the number, the expression is $10n + 7$.

WHY NOT (B) attaches the variable to the wrong number. (C) multiplies the whole sum $n + 7$ by 10; only the number should be multiplied by 10.

16. **C** $15x + 14y$

1 Each shirt costs \$15, so x of them cost $15x$ dollars.

2 Each hat costs \$14, so y of them cost $14y$ dollars.

3 Add the two costs: $15x + 14y$.

WHY NOT (A) combines unlike terms. (B) charges both prices for every item. (D) matches each price with the wrong item.

17. **C** $37 + 22t$

1 The \$37 is charged once, no matter what.

2 The \$22 is charged for each month, so t months cost $22t$ dollars.

3 Total: $37 + 22t$.

WHY NOT (A) charges the one-time \$37 for every month too. (B) adds t instead of multiplying it by \$22. (D) swaps the one-time charge and the rate.

18. **A** \$60

1 Substitute: $I = 1{,}000 \times 0.02 \times 3$.

2 $1{,}000 \times 0.02 = 20$, and $20 \times 3 = 60$. The interest is \$60.

WHY NOT (C) adds the amount invested to the interest. (D) uses 2 instead of 0.02.

19. **D** $x^2 + 5x - 27$

1 Distribute x: $x(x + 8) = x^2 + 8x$.

2 Distribute -3: $-3(x + 9) = -3x - 27$.

3 Combine the x-terms: $8x - 3x = 5x$. The result is $x^2 + 5x - 27$.

TIP Check with $x = 1$: the original and $x^2 + 5x - 27$ both equal -21.

WHY NOT (A) adds $3x$ instead of subtracting it. (B) writes $x \cdot x$ as $2x$ instead of x^2. (C) multiplies 9 by 3 instead of -3.

20. **B** 35

1 Substitute, putting each value in parentheses: $4(-3)(-5) - (-5)^2$.

2 Powers first: $(-5)^2 = 25$.

3 Multiply each term (watch the signs): $60 - 25$.

4 Add and subtract from left to right: $60 - 25 = 35$.

WHY NOT (D) treats $(-5)^2$ as -25.

21. **A** $15x + 24$

1 Distribute 5: $5(4x + 5) = 20x + 25$.

2 A minus sign in front of parentheses means multiply by -1: $-(5x + 1) = -5x - 1$. *Both* signs change.

3 Combine like terms: $15x + 24$.

TIP Check with $x = 1$: the original and $15x + 24$ both equal 39.

WHY NOT (B) drops the parentheses without changing any signs. (C) multiplies only $4x$ by 5. (D) changes the sign of only the first term inside the second parentheses.

22. **D** 64

1 Substitute, putting each value in parentheses: $2(-4)^2 - 2(-4)(4)$.

2 Powers first: $(-4)^2 = 16$.

3 Multiply each term (watch the signs): $32 + 32$.

4 Add and subtract from left to right: $32 + 32 = 64$.

WHY NOT (B) treats $(-4)^2$ as -16.

23. **C** 5

1 Substitute: $400 = 10 \times 8 \times h$, so $400 = 80h$.

2 Divide by 80: $h = 400 \div 80 = 5$.

WHY NOT (D) subtracts instead of dividing.

24. **D** $10x - 3$

1 The perimeter is the sum of the three sides.

2 Add the x-terms: $2+4+4 = 10$. Add the constants: $8 - 4 - 7 = -3$.

3 The perimeter is $10x - 3$.

WHY NOT (A) doubles the sum, as if the figure were a rectangle. (B) adds every constant, ignoring the minus signs. (C) combines unlike terms.

25. **D** $59d + 0.50m$

1 Daily charges: $59d$ dollars.

2 Mileage charges: $0.50m$ dollars.

3 Total: $59d + 0.50m$.

WHY NOT (A) multiplies everything together. (B) matches each rate with the wrong quantity. (C) charges the daily rate only once instead of d times.

CHAPTER 11

Linear Equations

Tested on: Mathematics Knowledge • Arithmetic Reasoning

Key Concepts

An *equation* says that two expressions are equal. To *solve* it, find the value of the variable that makes both sides equal. The golden rule: **whatever you do to one side, do to the other side**, so the two sides stay balanced.

Undo operations in reverse order

Get the variable alone by undoing what was done to it, last operation first: addition ↔ subtraction, multiplication ↔ division.

$$3x - 7 = 11 \implies 3x = 18 \implies x = 6$$

(add 7 to both sides, then divide both sides by 3). **Check** by substituting: $3(6) - 7 = 18 - 7 = 11$. ✓

Clean up first

- **Parentheses:** distribute to *every* term inside: $4(x - 2) = 4x - 8$ and $-2(x + 5) = -2x - 10$.
- **Variables on both sides:** add or subtract to move all the x-terms to one side and all the plain numbers to the other.
- **Fractions:** multiply *every* term by the least common denominator (LCD). **Decimals:** multiply every term by 10 or 100.
- **Formulas:** to solve $d = rt$ for t, treat the other letters like numbers and divide: $t = \frac{d}{r}$.

Translating words into equations

is, equals, gives, the result is	$=$
more than, increased by, sum, plus	$+$
less than, decreased by, difference	$-$
times, twice, tripled, product	$\times$
consecutive integers	$n,\ n+1,\ n+2,\ \ldots$
consecutive even (or odd) integers	$n,\ n+2,\ n+4,\ \ldots$

Careful: "5 less than x" is $x - 5$, *not* $5 - x$.

WORKED EXAMPLE

Solve $2(x + 3) = 5x - 9$.

Solution. Distribute: $2x + 6 = 5x - 9$. Subtract $2x$ from both sides: $6 = 3x - 9$. Add 9 to both sides: $15 = 3x$. Divide by 3: $x = 5$. Check: $2(5 + 3) = 16$ and $5(5) - 9 = 16$. ✓

No-Calculator Tip Multiple choice lets you *backsolve*: plug each choice into the equation; the one that makes both sides equal is the answer. When x-terms are on both sides, move them toward the side with the larger x-coefficient so that coefficient stays positive.

Common Traps

- Distributing to only the first term: $3(x + 4)$ is $3x + 12$, not $3x + 4$.
- Moving a term across the equal sign without changing its sign.
- Answering x when the question asks for something else, such as $2x + 1$.
- Reversing "less than": "7 less than $3n$" is $3n - 7$.

Practice Set 11

Score: _____ / 25 Target time: 26 minutes

Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

1. What value of x makes $\frac{x}{2} = 10$ true?

A 5 **B** 8 **C** 12 **D** 20

2. "A number quadrupled and then decreased by 15 is 1." If n stands for the number, which equation represents this sentence?

A $4(n - 15) = 1$ **B** $15 - 4n = 1$
C $4n - 15 = 1$ **D** $4n + 15 = 1$

3. Solve for x: $x + 17 = 26$.

A -9 **B** 7 **C** 8 **D** 9

4. When 20 is added to twice a number, the result is 24. What is the number?

A 2 **B** 3 **C** 4 **D** 22

5. Solve for x: $2x + 12 = 32$.

A 4 **B** 8 **C** 9 **D** 10

6. This month a recruiting station signed up 16 more recruits than four times the number it signed up last month. It signed up 84 recruits this month. How many did it sign up last month?

A 5 **B** 16 **C** 17 **D** 25

7. "A number tripled and then decreased by 12 is 9." If n stands for the number, which equation represents this sentence?

A $12 - 3n = 9$ **B** $3(n - 12) = 9$
C $3n - 12 = 9$ **D** $3n + 12 = 9$

8. If $\frac{x}{8} + 4 = 15$, what is the value of x?

A $\frac{11}{8}$ **B** 88 **C** 116 **D** 152

9. Solve for x: $8x - 15 = 3x - 45$.

A -6 **B** -5 **C** $-\frac{30}{11}$ **D** 6

10. A gym charges a one-time sign-up fee of \$50 plus \$25 per month. Ana has paid \$200 in all. For how many months has she been a member?

A 6 months **B** 7 months
C 8 months **D** 10 months

11. If $6x + 3 = 2x + 7$, what is the value of x?

A -1 **B** $\frac{1}{2}$ **C** 1 **D** $\frac{5}{2}$

12. A plumber charges a \$60 service fee plus \$75 per hour. Carlos's bill was \$210. How many hours did the plumber work?

A 2 hours **B** 2.8 hours
C 3.6 hours **D** 4 hours

13. What value of x makes $23 - 2x = 17$ true?

A -20 **B** -6 **C** -3 **D** 3

14. Solve for x: $5(x + 9) - 7 = 78$.

A 7 **B** 8 **C** $\frac{76}{5}$ **D** 40

15. Solve for x: $2.5x - 1.4 = 26.1$.

A 1.1 **B** 9.88 **C** 11 **D** 110

16. If $-16x + 4y = 12$, which of the following expresses y in terms of x?

A $y = 16x + 3$ **B** $y = 4x + 3$
C $y = -4x + 3$ **D** $y = 16x + 12$

17. If $6x - 15 = 45$, then $3x - 2$ equals which of the following?

A 28 **B** 29 **C** 30 **D** 31

18. The sum of three consecutive integers is 39. What is the largest of these integers?

A 10 **B** 12 **C** 13 **D** 14

19. What value of x makes $6(x-6) - 5(x+9) = -82$ true?

A -91 **B** -67 **C** -1 **D** $-\frac{1}{11}$

20. If $\frac{x}{5} - \frac{x}{10} = 6$, what is the value of x?

A 30 **B** 60 **C** 61 **D** 300

21. The volume of a box is $V = lwh$. Which equation gives h in terms of the other variables?

A $h = Vlw$ **B** $h = V - lw$
C $h = \frac{lw}{V}$ **D** $h = \frac{V}{lw}$

22. If $6x - 6 = 12$, what is the value of $2x - 2$?

A 3 **B** 4 **C** 12 **D** 36

23. What value of x makes $14 + x = 7x - 52$ true?

A $-\frac{33}{4}$ **B** 11 **C** 12 **D** 66

24. The sum of two consecutive even integers is 18. What is the larger of these integers?

A 2 **B** 8 **C** 9 **D** 10

25. "Seven times the sum of a number and 5 is 56." If n stands for the number, which equation represents this sentence?

A $7(n + 5) = 56$ **B** $7 + n + 5 = 56$
C $\frac{n+5}{7} = 56$ **D** $7n + 5 = 56$

Answers & Explanations 11

1 D	**6** C	**11** C	**16** B	**21** D
2 C	**7** C	**12** A	**17** A	**22** B
3 D	**8** B	**13** D	**18** D	**23** B
4 A	**9** A	**14** B	**19** C	**24** D
5 D	**10** A	**15** C	**20** B	**25** A

1. **D** 20

1 $\frac{x}{2}$ means x divided by 2. Undo the division by multiplying both sides by 2.

2 $x = 10 \times 2 = 20$.

WHY NOT (A) divides by 2 instead of multiplying. (B) subtracts 2 instead of multiplying by it. (C) adds 2 instead of multiplying by it.

2. **C** $4n - 15 = 1$

1 Translate piece by piece. Let n be the number; "is," "gives," and "equals" all become "=."

2 "Quadrupled" means multiplied by 4: $4n$. Then "decreased by 15" means subtract 15: $4n - 15$.

3 The equation is $4n - 15 = 1$.

WHY NOT (A) subtracts 15 before multiplying by 4. (B) subtracts in the wrong order. (D) adds instead of subtracting.

3. **D** 9

1 The equation adds 17 to x. Undo that by subtracting 17 from both sides: $x = 26 - 17 = 9$.

2 Check: $9 + 17 = 26$. ✓

WHY NOT (A) subtracts in the wrong order.

4. **A** 2

1 Let n be the number. Write the words as an equation: $2n + 20 = 24$.

2 Undo the addition: subtract 20 from both sides. $2n = 24 - 20 = 4$.

3 Divide both sides by 2: $n = \frac{4}{2} = 2$.

WHY NOT (C) forgets to divide by 2. (D) adds 20 instead of subtracting it.

5. **D** 10

1 Undo the addition: subtract 12 from both sides. $2x = 32 - 12 = 20$.

2 Undo the multiplication: divide both sides by 2. $x = \frac{20}{2} = 10$.

3 Check: $2(10) + 12 = 20 + 12 = 32$. ✓

WHY NOT (A) divides only 32 by 2, not the 12: $x + 12 = \frac{32}{2}$.

6. **C** 17

1 Let n be the number of recruits last month. Write the words as an equation: $4n + 16 = 84$.

2 Undo the addition: subtract 16 from both sides. $4n = 84 - 16 = 68$.

3 Divide both sides by 4: $n = \frac{68}{4} = 17$.

WHY NOT (A) divides only 84 by 4, not the 16: $n + 16 = \frac{84}{4}$. (D) adds 16 instead of subtracting it.

7. **C** $3n - 12 = 9$

1 Translate piece by piece. Let n be the number; "is," "gives," and "equals" all become "=."

2 "Tripled" means multiplied by 3: $3n$. Then "decreased by 12" means subtract 12: $3n - 12$.

3 The equation is $3n - 12 = 9$.

WHY NOT (A) subtracts in the wrong order. (B) subtracts 12 before multiplying by 3. (D) adds instead of subtracting.

8. **B** 88

1 Undo the addition: subtract 4 from both sides. $\frac{x}{8} = 15 - 4 = 11$.

2 Undo the division: multiply both sides by 8. $x = 11 \times 8 = 88$.

3 Check: $\frac{88}{8} + 4 = 11 + 4 = 15$. ✓

WHY NOT (A) divides by 8 instead of multiplying. (C) multiplies only 15 by 8, not the 4: $x + 4 = 120$. (D) moves 4 across without changing its sign.

9. **A** -6

1 Get the x-terms on the left: subtract $3x$ from both sides. $5x - 15 = -45$.

2 Undo the subtraction: add 15 to both sides. $5x = -45 + 15 = -30$.

3 Undo the multiplication: divide both sides by 5. $x = \frac{-30}{5} = -6$.

4 Check with $x = -6$: the left side is $8(-6) - 15 = -63$ and the right side is $3(-6) - 45 = -63$. ✓

WHY NOT (C) adds the x-terms instead of subtracting one from the other. (D) has the right size but the wrong sign.

10. **A** 6 months

1 Let m be the number of months. The total is the fixed fee plus \$25 for each month: $50 + 25m = 200$.

2 Subtract the fee from both sides: $25m = 200 - 50 = 150$.

3 Divide by the rate: $m = 150 \div 25 = 6$ months.

WHY NOT (C) ignores the fixed fee. (D) adds the fee instead of subtracting it.

11. **C** 1

1 Get the x-terms on the left: subtract $2x$ from both sides. $4x + 3 = 7$.

2 Undo the addition: subtract 3 from both sides. $4x = 7 - 3 = 4$.

3 Undo the multiplication: divide both sides by 4. $x = \frac{4}{4} = 1$.

4 Check with $x = 1$: the left side is $6(1) + 3 = 9$ and the right side is $2(1) + 7 = 9$. ✓

WHY NOT (A) has the right size but the wrong sign. (B) adds the x-terms instead of subtracting one from the other. (D) moves 3 across without changing its sign.

12. **A** 2 hours

1 Let h be the number of hours. The total is the fixed fee plus \$75 for each hour: $60 + 75h = 210$.

2 Subtract the fee from both sides: $75h = 210 - 60 = 150$.

3 Divide by the rate: $h = 150 \div 75 = 2$ hours.

WHY NOT (B) ignores the fixed fee. (C) adds the fee instead of subtracting it.

13. **D** 3

1 Move the 23 away from the x-term: subtract 23 from both sides. $-2x = 17 - 23 = -6$.

2 The x-term is $-2x$, so divide both sides by -2: $x = \frac{-6}{-2} = 3$.

3 Check: $23 - 2(3) = 23 - 6 = 17$. ✓

TIP Dividing by a negative number changes the sign of the result.

WHY NOT (A) adds 23 to both sides instead of subtracting it. (B) forgets to divide by -2. (C) divides by 2 instead of -2.

14. **B** 8

1 Distribute 5 to *both* terms in the parentheses: $5 \cdot x = 5x$ and $5 \cdot 9 = 45$. The equation becomes $5x + 45 - 7 = 78$.

2 Combine the plain numbers: $45 - 7 = 38$, so $5x + 38 = 78$.

3 Undo the addition: subtract 38 from both sides. $5x = 78 - 38 = 40$.

4 Undo the multiplication: divide both sides by 5. $x = \frac{40}{5} = 8$.

WHY NOT (C) multiplies only the x by 5, not the 9. (D) forgets to divide by 5.

15. **C** 11

1 Clear the decimals: multiply *every* term by 10. $25x - 14 = 261$.

2 Undo the subtraction: add 14 to both sides. $25x = 261 + 14 = 275$.

3 Divide both sides by 25: $x = \frac{275}{25} = 11$.

TIP You can also work with the decimals directly: $2.5x = 27.5$, and $27.5 \div 2.5 = 11$.

WHY NOT (A) slips the decimal point. (B) moves -1.4 across without changing its sign. (D) slips the decimal point.

16. **B** $y = 4x + 3$

1 Move the x-term to the right side: add $16x$ to both sides. $4y = 16x + 12$.

2 Divide both sides by 4: $y = \frac{16x+12}{4}$.

3 Divide *each* term on top by 4: $y = 4x + 3$.

WHY NOT (A) divides only 12 by 4, not the x-term. (C) does not change the sign of $-16x$ when moving it to the other side. (D) forgets to divide by 4.

17. **A** 28

1 First solve for x. Undo the subtraction: add 15 to both sides. $6x = 45 + 15 = 60$.

2 Undo the multiplication: divide both sides by 6. $x = \frac{60}{6} = 10$.

3 The question asks for $3x - 2$, not x. Substitute: $3(10) - 2 = 30 - 2 = 28$.

TIP Always reread the question before choosing: finding x is only part of the job.

WHY NOT (C) forgets the -2 in $3x - 2$.

18. **D** 14

1 Let n be the smallest. Consecutive integers go up by 1, so the numbers are $n, n + 1, n + 2$.

2 Add them: $3n + 3 = 39$.

3 Subtract 3: $3n = 36$. Divide by 3: $n = 12$.

4 The numbers are 12, 13, 14. The largest is 14.

WHY NOT (B) is the smallest number, not the largest. (C) is the middle number, not the largest.

19. **C** -1

1 Distribute each number to both terms in its parentheses. The -5 multiplies *both* x and 9: $-5(x+9) = -5x - 45$.

2 So $6x - 36 - 5x - 45 = -82$.

3 Combine like terms: $x - 81 = -82$.

4 Undo the subtraction: add 81 to both sides. $x = -82 + 81 = -1$.

WHY NOT (A) writes +45 instead of -45 (does not distribute the minus sign). (B) multiplies only the x-terms, not the numbers in parentheses. (D) adds $6x$ and $5x$ instead of subtracting.

20. **B** 60

1 Clear the fractions: multiply *every* term by the LCD of 5 and 10, which is 10.

2 $10 \cdot \frac{x}{5} = 2x$, $10 \cdot \frac{x}{10} = x$, and $10 \cdot 6 = 60$. So $2x - x = 60$.

3 Combine like terms: $x = 60$.

4 Check: $\frac{60}{5} - \frac{60}{10} = 12 - 6 = 6$. ✓

WHY NOT (A) subtracts the denominators, as if the result were $\frac{x}{5}$.

21. **D** $h = \frac{V}{lw}$

1 Treat every letter except h as if it were a number. Undo what is being done to h, in reverse order, doing the same thing to both sides.

2 h is multiplied by l and by w, so divide both sides by lw: $\frac{V}{lw} = h$.

3 So $h = \frac{V}{lw}$.

TIP Check with easy numbers: pick values for the other letters, find h, and see that the original formula holds.

WHY NOT (A) multiplies by lw instead of dividing. (B) subtracts lw instead of dividing by it. (C) divides in the wrong order.

22. **B** 4

1 Notice that $2x - 2$ is exactly $\frac{1}{3}$ of $6x - 6$: $6x - 6 = 3(2x - 2)$.

2 So $2x - 2 = 12 \div 3 = 4$. No need to find x.

3 Check by solving: $x = 3$, and $2(3) - 2 = 4$. ✓

WHY NOT (A) gives the value of x. (C) gives the value of $6x - 6$. (D) multiplies by 3 instead of dividing.

23. **B** 11

1 Get the x-terms on the right, where the coefficient is larger: subtract x from both sides. $14 = 6x - 52$.

2 Add 52 to both sides: $14 + 52 = 6x$, so $66 = 6x$.

3 Undo the multiplication: divide both sides by 6. $x = \frac{66}{6} = 11$.

4 Check with $x = 11$: the left side is $14 + 11 = 25$ and the right side is $7(11) - 52 = 25$. ✓

WHY NOT (A) adds the x-terms instead of subtracting one from the other. (D) forgets to divide by 6.

24. **D** 10

1 Let n be the smallest. Consecutive even integers go up by 2, so the numbers are $n, n + 2$.

2 Add them: $2n + 2 = 18$.

3 Subtract 2: $2n = 16$. Divide by 2: $n = 8$.

4 The numbers are 8, 10. The larger is 10.

WHY NOT (B) is the smaller number, not the larger. (C) divides the sum by the number of integers and stops.

25. **A** $7(n + 5) = 56$

1 Translate piece by piece. Let n be the number; "is," "gives," and "equals" all become "=."

2 "The sum of a number and 5" is $n + 5$. "Seven times the sum" multiplies the *whole* sum, so it needs parentheses: $7(n + 5)$.

3 The equation is $7(n + 5) = 56$.

WHY NOT (B) adds 7 instead of multiplying by it. (C) divides by 7 instead of multiplying. (D) multiplies only the number by 7, not the whole sum.

CHAPTER 12

Inequalities

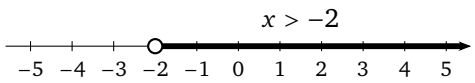
Tested on: Mathematics Knowledge • Arithmetic Reasoning

Key Concepts

An *inequality* compares two quantities that need not be equal. Its solution is usually a whole range of numbers, not a single value. You solve it almost exactly like an equation, with **one** extra rule.

Symbols and the number line

$x < 3$	x is less than 3	open circle at 3, shade left
$x > 3$	x is greater than 3	open circle at 3, shade right
$x \le 3$	x is at most 3 (no more than 3)	closed dot at 3, shade left
$x \ge 3$	x is at least 3 (no less than 3)	closed dot at 3, shade right



Solving: one extra rule

Add or subtract any number on both sides, and multiply or divide both sides by a *positive* number, exactly as with equations. But when you **multiply or divide both sides by a negative number, reverse the inequality symbol**:

$$-2x < 6 \implies x > -3$$

Compound inequalities have three parts; do the same thing to all three: $-3 < 2x + 1 \le 9 \Rightarrow -4 < 2x \le 8 \Rightarrow -2 < x \le 4$.

Absolute value

$|x - 3|$ is the distance between x and 3 on the number line.

- $|x - 3| < 5$: within 5 of 3, so $-2 < x < 8$ (one "between" piece).
- $|x - 3| > 5$: more than 5 away, so $x < -2$ or $x > 8$ (two pieces).

WORKED EXAMPLE

Solve $5 - 3x \ge 17$.
Solution. Subtract 5 from both sides: $-3x \ge 12$. Divide both sides by -3 and *reverse* the inequality symbol: $x \le -4$. Check with $x = -5$: $5 - 3(-5) = 20$, and $20 \ge 17$. ✓

No-Calculator Tip Test a point. Pick an easy number in your answer (such as 0, if it is included) and substitute it into the *original* inequality. If it makes a true statement, your inequality symbol points the right way.

Common Traps

- Forgetting to reverse the inequality symbol after dividing by a negative number.
- Reversing it when you only *subtracted* a number: $x - 5 > 2$ is simply $x > 7$.
- Mixing up $<$ and $\leq$: "at least 10" includes 10.
- Rounding the wrong way in word problems: to earn *at least* a goal, round the number of hours *up*.

Practice Set 12

Score: _____ / 25 Target time: 26 minutes

Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

1. Which of the following is the solution of $9x - 9 < 0$?

A $x < 9$ **B** $x < 1$ **C** $x > 1$ **D** $x < -1$

2. "A number n decreased by 2 is at least 36." Which inequality represents this statement?

A $n - 2 \leq 36$ **B** $n - 2 < 36$
C $n - 2 > 36$ **D** $n - 2 \geq 36$

3. Which inequality describes all values of x that make $8x < 56$ true?

A $x > 7$ **B** $x < 7$ **C** $x < 448$ **D** $x < 48$

4. What is the largest integer x that satisfies $4x - 4 < -16$?

A -5 **B** -4 **C** -2 **D** -1

5. Which of the following is the solution of $x + 11 > 23$?

A $x > 12$ **B** $x \geq 12$ **C** $x < 12$ **D** $x > 34$

6. Which number makes $3x + 1 > -11$ true?

A -6 **B** -5 **C** -4 **D** -1

7. Which inequality represents the statement "Seven times a number n is greater than 11"?

A $7n \geq 11$ **B** $7n < 11$ **C** $7n \leq 11$ **D** $7n > 11$

8. How many integers x satisfy $-3 < x < 6$?

A 6 **B** 7 **C** 8 **D** 10

9. Which inequality describes all values of x that make $-5x - 15 > -35$ true?

A $x > 4$ **B** $x < 4$ **C** $x < -4$ **D** $x < 10$

10. Which of the following is the solution of $7x + 7 \geq 6x - 2$?

A $x \leq -9$ **B** $x \geq -\frac{9}{13}$ **C** $x \geq 5$ **D** $x \geq -9$

11. Solve the inequality $9 \leq 3x + 3 \leq 18$.

A $2 \leq x \leq 5$ **B** $2 \leq x \leq 15$
C $3 \leq x \leq 6$ **D** $4 \leq x \leq 7$

12. A trailer can carry at most 3,000 pounds. It already holds 350 pounds of tools. What is the greatest number of 65-pound ammunition crates that can be added?

A 40 crates **B** 41 crates
C 46 crates **D** 51 crates

13. Which of the following is the solution of $-22 \leq 2x - 6 < -12$?

A $-8 < x \leq -3$ **B** $-16 \leq x < -3$
C $-8 \leq x < -3$ **D** $-14 \leq x < -9$

14. Before leaving for basic training, Diego has \$325 in the bank and deposits \$75 every week. What is the fewest number of weeks until he has at least \$2,900?

A 33 weeks **B** 34 weeks
C 35 weeks **D** 39 weeks

15. Solve the inequality $8x + 10 \geq 4x - 2$.

A $x \geq 2$ **B** $x \geq -1$ **C** $x \geq -3$ **D** $x \leq -3$

16. Solve the inequality $-2 - 9x \geq -29$.

A $x \leq \frac{31}{9}$ **B** $x \leq 3$ **C** $x \geq 3$ **D** $x \leq -3$

17. Which value of x satisfies $-1 < 5x + 9 \leq 24$?

A -4 **B** -3 **C** -2 **D** 3

18. Corporal Haddad needs a total score of at least 250 points on a fitness test. Haddad has already earned 160 points on the first events. Which inequality shows the points p that Haddad still needs from the last event?

A $160 + p < 250$ **B** $160p \geq 250$
C $160 + p \geq 250$ **D** $160 + p > 250$

19. Which of the following is the solution of $|x+5| > 7$?

A $x < -2$ or $x > 12$ **B** $x < -12$ or $x > 2$
C $-12 \leq x \leq 2$ **D** $x > 2$

20. Phone plan A costs \$10 per month plus \$0.10 per text message. Plan B costs \$45 per month with unlimited texts. What is the greatest number of texts in a month for which plan A costs less than plan B?

A 349 texts **B** 350 texts
C 449 texts **D** 549 texts

21. Which of the following is the solution of $|2x + 2| \leq 2$?

A $-2 \leq x \leq 0$ **B** $x \leq 0$
C $x < -2$ or $x > 0$ **D** $0 \leq x \leq 2$

22. Car rental A costs \$20 per day plus \$0.25 per mile. Car rental B costs \$60 per day with unlimited miles. For a one-day rental, what is the greatest whole number of miles for which rental A costs less than rental B?

A 159 miles **B** 160 miles
C 239 miles **D** 319 miles

23. Solve the inequality $-1 \leq 8 - 3x < 17$.

A $-3 < x \leq 3$ **B** $-\frac{25}{3} < x \leq -\frac{7}{3}$
C $-3 \leq x < 3$ **D** $x \leq -3$ or $x > 3$

24. Which of the following is the solution of $-2x - 14 < 2x - 10$?

A $x > 6$ **B** $x < -1$ **C** $x > 1$ **D** $x > -1$

25. How many integer values of x make $-19 \leq 4x + 1 < -7$ true?

A 1 **B** 2 **C** 3 **D** 4

Answers & Explanations 12

1 B	**6** D	**11** A	**16** B	**21** A
2 D	**7** D	**12** A	**17** D	**22** A
3 B	**8** C	**13** C	**18** C	**23** A
4 B	**9** B	**14** C	**19** B	**24** D
5 A	**10** D	**15** C	**20** A	**25** C

1. **B** $x < 1$

1 Add 9 to both sides (adding or subtracting never changes the direction): $9x < 0 + 9$, so $9x < 9$.

2 Divide both sides by 9. Dividing by a positive number keeps the inequality symbol the same: $x < \frac{9}{9}$, so $x < 1$.

3 Check with a number in the answer, $x = 0$: $9(0) - 9 = -9$, and $-9 < 0$ is true. ✓

WHY NOT (A) forgets to divide by 9. (C) reverses the inequality symbol, but that happens only when you divide by a negative number. (D) moves −9 across without changing its sign.

2. **D** $n - 2 \geq 36$

1 "At least 36" means the quantity is greater than or equal to 36: it includes 36 itself, so the inequality symbol is ≥.

2 The quantity is $n-2$, so the inequality is $n-2 \geq 36$.

WHY NOT (A) points the inequality symbol the wrong way: "at least 36" means 36 or more. (B) points the inequality symbol the wrong way and also gets the endpoint 36 wrong: "at least 36" means 36 or more. (C) uses >, which leaves out exactly 36, but 36 meets "at least 36".

3. **B** $x < 7$

1 Divide both sides by 8. Dividing by a positive number keeps the inequality symbol the same: $x < \frac{56}{8}$, so $x < 7$.

2 Check with a number in the answer, $x = 6$: $8(6) = 48$, and $48 < 56$ is true. ✓

WHY NOT (A) reverses the inequality symbol, but 8 is positive. (C) multiplies by 8 instead of dividing. (D) subtracts 8 instead of dividing by it.

4. **B** -4

1 Solve first. Add 4 to both sides (adding or subtracting never changes the direction): $4x < -16 + 4$, so $4x < -12$.

2 Divide both sides by 4 (positive, so the inequality symbol stays the same): $x < -3$.

3 −3 itself is *not* allowed (<), so the largest integer is −4.

WHY NOT (A) is a solution, but not the largest one. (C) gives $4(-2) - 4 = -12$, which is not less than −16. (D) gives $4(-1) - 4 = -8$, which is not less than −16.

5. **A** $x > 12$

1 Subtract 11 from both sides (adding or subtracting never changes the direction): $x > 23 - 11$, so $x > 12$.

2 Check with a number in the answer, $x = 13$: $13 + 11 = 24$, and $24 > 23$ is true. ✓

WHY NOT (B) uses ≥ instead of >. (C) reverses the inequality symbol, but that happens only when you multiply or divide by a negative number. (D) adds 11 instead of subtracting it.

6. **D** -1

1 Solve first. Subtract 1 from both sides (adding or subtracting never changes the direction): $3x > -11 - 1$, so $3x > -12$.

2 Divide both sides by 3 (positive, so the inequality symbol stays the same): $x > -4$.

3 Only −1 is greater than −4. Check: $3(-1)+1 = -2$, which is greater than −11. ✓

WHY NOT (A) gives $3(-6) + 1 = -17$, which is not greater than −11. (B) gives $3(-5) + 1 = -14$, which is not greater than −11. (C) makes the left side equal to −11, but the inequality is strict.

7. **D** $7n > 11$

1 "Greater than 11" means the quantity is greater than 11: it does not include 11 itself, so the inequality symbol is >.

2 The quantity is $7n$, so the inequality is $7n > 11$.

WHY NOT (A) uses ≥, which includes exactly 11, but 11 is not "greater than 11". (B) points the inequality symbol the wrong way: "greater than 11" means bigger than 11. (C) points the inequality symbol the wrong way and also gets the endpoint 11 wrong: "greater than 11" means bigger than 11.

8. **C** 8

1 −3 is not included (<), and 6 is not included (<). So the integers run from −2 to 5.

2 List them: −2, −1, 0, 1, 2, 3, 4, 5. That is 8 integers.

TIP Count from −2 to 5 with last − first +1: $5 - (-2) + 1 = 8$.

WHY NOT (B) forgets to count 0, which is an integer. (D) counts both endpoints, but one or both are not included.

9. **B** $x < 4$

1 Add 15 to both sides (adding or subtracting never changes the direction): $-5x > -35 + 15$, so $-5x > -20$.

2 Divide both sides by -5. Dividing by a *negative* number reverses the inequality symbol: $>$ becomes $<$: $x < \frac{-20}{-5}$, so $x < 4$.

3 Check with a number in the answer, $x = 3$: $-5(3) - 15 = -30$, and $-30 > -35$ is true. ✓

WHY NOT (A) forgets to reverse the inequality symbol when dividing by -5. (C) gets the number's sign wrong (positive vs. negative) when dividing by -5. (D) moves -15 across without changing its sign.

10. **D** $x \geq -9$

1 Collect the x-terms on the left: subtract $6x$ from both sides. $x + 7 \geq -2$.

2 Subtract 7 from both sides (adding or subtracting never changes the direction): $x \geq -2 - 7$, so $x \geq -9$.

WHY NOT (A) reverses the inequality symbol, but no division by a negative number happened. (B) adds the x-terms instead of subtracting, getting $13x$. (C) moves 7 across without changing its sign.

11. **A** $2 \leq x \leq 5$

1 Do the same thing to all three parts. Subtract 3 from each part: $6 \leq 3x \leq 15$.

2 Divide each part by 3 (positive, so the inequality symbols stay the same): $2 \leq x \leq 5$.

3 Check with $x = 3$: the middle is $3(3) + 3 = 12$, and $9 \leq 12 \leq 18$ is true. ✓

WHY NOT (B) forgets to divide the right part by 3. (C) divides by 3 without first removing the 3. (D) adds 3 instead of subtracting it.

12. **A** 40 crates

1 Let c be the number of crates. The total must stay at or under the limit: $350 + 65c \leq 3{,}000$.

2 Subtract 350: $65c \leq 2{,}650$. Divide by 65: $c \leq \frac{2{,}650}{65} = 40\frac{10}{13}$.

3 c must be a whole number no bigger than that, so round *down*: 40. Check: 41 would make 3,015, over the limit.

WHY NOT (B) rounds up, which goes over the limit. (C) forgets the 350 pounds already counted. (D) adds the starting amount instead of subtracting it.

13. **C** $-8 \leq x < -3$

1 Do the same thing to all three parts. Add 6 to each part: $-16 \leq 2x < -6$.

2 Divide each part by 2 (positive, so the inequality symbols stay the same): $-8 \leq x < -3$.

3 Check with $x = -7$: the middle is $2(-7) - 6 = -20$, and $-22 \leq -20 < -12$ is true. ✓

WHY NOT (A) swaps which endpoint is included. (B) forgets to divide the left part by 2. (D) subtracts 6 instead of adding it.

14. **C** 35 weeks

1 Let w be the number of weeks. The total must be at least the goal: $325 + 75w \geq 2900$.

2 Subtract 325: $75w \geq 2575$. Divide by 75: $w \geq \frac{2575}{75} = 34\frac{1}{3}$.

3 w must be a whole number at least that big, so round *up*: 35. Check: $325 + 75(34) = 2875$ is too little, but $325 + 75(35) = 2950$ is enough.

WHY NOT (B) rounds down, which falls just short of the goal. (D) ignores the $325 already saved.

15. **C** $x \geq -3$

1 Collect the x-terms on the left: subtract $4x$ from both sides. $4x + 10 \geq -2$.

2 Subtract 10 from both sides (adding or subtracting never changes the direction): $4x \geq -2 - 10$, so $4x \geq -12$.

3 Divide both sides by 4. Dividing by a positive number keeps the inequality symbol the same: $x \geq \frac{-12}{4}$, so $x \geq -3$.

WHY NOT (A) moves 10 across without changing its sign. (B) adds the x-terms instead of subtracting, getting $12x$. (D) reverses the inequality symbol, but no division by a negative number happened.

16. **B** $x \leq 3$

1 Add 2 to both sides (adding or subtracting never changes the direction): $-9x \geq -29+2$, so $-9x \geq -27$.

2 Divide both sides by -9. Dividing by a *negative* number reverses the inequality symbol: $\geq$ becomes $\leq$: $x \leq \frac{-27}{-9}$, so $x \leq 3$.

3 Check with a number in the answer, $x = 2$: $-2 - 9(2) = -20$, and $-20 \geq -29$ is true. ✓

WHY NOT (A) moves -2 across without changing its sign. (C) forgets to reverse the inequality symbol when dividing by -9. (D) gets the number's sign wrong (positive vs. negative) when dividing by -9.

17. **D** 3

1 Solve first. Subtract 9 from all three parts: $-10 < 5x \leq 15$, then divide all three by 5: $-2 < x \leq 3$.

2 Only 3 is in this range. Check: $5(3) + 9 = 24$, and $-1 < 24 \leq 24$. ✓

WHY NOT (A) gives $5(-4) + 9 = -11$, which is outside the range. (B) gives $5(-3) + 9 = -6$, which is outside the range. (C) makes the middle equal to -1, but that end is not included.

18. **C** $160 + p \geq 250$

1 "At least 250" means the quantity is greater than or equal to 250: it includes 250 itself, so the inequality symbol is $\geq$.

2 The quantity is $160 + p$, so the inequality is $160 + p \geq 250$.

WHY NOT (A) points the inequality symbol the wrong way and also gets the endpoint 250 wrong: "at least 250" means 250 or more. (B) multiplies instead of adding the points. (D) uses $>$, which leaves out exactly 250, but 250 meets "at least 250".

19. **B** $x < -12$ or $x > 2$

1 $|x + 5| > 7$ means $x + 5$ is *more than* 7 away from zero, so there are two pieces: $x + 5 < -7$ or $x + 5 > 7$.

2 Solve each piece: $x < -12$ or $x > 2$.

3 Check with $x = 3$: $|3 + 5| = 8$, and $8 > 7$ is true. ✓

WHY NOT (A) uses the wrong sign: $x + 5$ is zero when $x = -5$, not $x = 5$. (C) gives the "between" set, which would solve the inequality with $\leq$. (D) keeps only the positive case; the inside can also be very negative.

20. **A** 349 texts

1 Let t be the number of texts. Plan A costs less when $10 + 0.10t < 45$.

2 Subtract 10: $0.10t < 35$.

3 Divide by 0.10: $t < 35 \div 0.10 = 350$ (dividing by 0.10 is the same as multiplying by 10).

4 At exactly 350 texts the two plans cost the same, so the greatest whole number that is *less* is 349.

WHY NOT (B) is where the plans cost the same or more, not less. (C) ignores the $10 fee. (D) adds the fee instead of subtracting it.

21. **A** $-2 \leq x \leq 0$

1 $|2x + 2| \leq 2$ means $2x + 2$ is within 2 of zero, so it is *between* -2 and 2: $-2 \leq 2x + 2 \leq 2$.

2 Subtract 2 from all three parts: $-4 \leq 2x \leq 0$. Divide by 2: $-2 \leq x \leq 0$.

3 Check with $x = -1$: $|2(-1) + 2| = 0$, and $0 \leq 2$ is true. ✓

WHY NOT (B) keeps only the positive case; the inside must also be greater than -2. (C) gives the "outside" set, which would solve the inequality with $>$. (D) uses the wrong sign: $2x + 2$ is zero when $x = -1$, not $x = 1$.

22. **A** 159 miles

1 Let m be the number of miles. Rental A costs less when $20 + 0.25m < 60$.

2 Subtract 20: $0.25m < 40$.

3 Divide by 0.25: $m < 40 \div 0.25 = 160$ (dividing by 0.25 is the same as multiplying by 4).

4 At exactly 160 miles the two plans cost the same, so the greatest whole number that is *less* is 159.

WHY NOT (B) is where the plans cost the same or more, not less. (C) ignores the $20 fee. (D) adds the fee instead of subtracting it.

23. **A** $-3 < x \leq 3$

1 Do the same thing to all three parts. Subtract 8 from each part: $-9 \leq -3x < 9$.

2 Divide each part by -3. Dividing by a negative number reverses *both* inequality symbols: $3 \geq x > -3$.

3 Rewrite it with the smaller number on the left: $-3 < x \leq 3$.

4 Check with $x = -2$: the middle is $8 - 3(-2) = 14$, and $-1 \leq 14 < 17$ is true. ✓

WHY NOT (B) adds 8 instead of subtracting it. (C) moves the numbers but leaves each symbol in its old place. (D) gives the numbers *outside* the answer.

24. **D** $x > -1$

1 Collect the x-terms on the left: subtract $2x$ from both sides. $-4x - 14 < -10$.

2 Add 14 to both sides (adding or subtracting never changes the direction): $-4x < -10 + 14$, so $-4x < 4$.

3 Divide both sides by -4. Dividing by a *negative* number reverses the inequality symbol: $<$ becomes $>$: $x > \frac{4}{-4}$, so $x > -1$.

TIP To avoid dividing by a negative, collect the x-terms on the right instead: $-4 < 4x$, so $-1 < x$, which means the same as $x > -1$.

WHY NOT (A) moves -14 across without changing its sign. (B) forgets to reverse the inequality symbol when dividing by -4. (C) gets the number's sign wrong (positive vs. negative) when dividing by -4.

25. **C** 3

1 Solve first. Subtract 1 from all three parts: $-20 \leq 4x < -8$. Divide all three parts by 4: $-5 \leq x < -2$.

2 -5 is included ($\leq$), and -2 is not included ($<$). So the integers run from -5 to -3.

3 List them: $-5, -4, -3$. That is 3 integers.

TIP Count from -5 to -3 with last − first +1: $-3 - (-5) + 1 = 3$.

WHY NOT (B) leaves out both endpoints. (D) counts both endpoints, but one or both are not included.

CHAPTER 13

Systems of Equations

Tested on: Mathematics Knowledge • Arithmetic Reasoning

Key Concepts

A *system of equations* is two equations that share the same variables. Its solution is the pair (x, y) that makes *both* equations true at the same time. On the ASVAB you solve systems by elimination or by substitution, and you set them up yourself in word problems with two unknowns.

Elimination: add or subtract the equations

Line the equations up and add (or subtract) them so one variable cancels.

$$\begin{array}{r} x + y = 10 \\ x - y = 4 \\ \hline 2x = 14 \end{array} \qquad \Rightarrow \qquad x = 7, \;\; y = 10 - 7 = 3$$

If nothing cancels right away, first multiply one equation (*every* term, including the number on the right) so that a pair of coefficients match or are opposites.

Substitution

When one equation is already solved for a variable, such as $y = 2x - 1$, replace that variable in the other equation, *in parentheses*: $3x + 2y = 19$ becomes $3x + 2(2x - 1) = 19$, so $7x - 2 = 19$ and $x = 3$. Then $y = 2(3) - 1 = 5$.

Setting up word problems

Two unknowns need two equations: usually one counts the *items* and one adds up their *value*. For 200 tickets, adults \$12 and children \$5, total \$1,560:

$$a + c = 200 \qquad 12a + 5c = 1{,}560$$

WORKED EXAMPLE

Solve the ticket system above for the number of adult tickets.
Solution. From the first equation, $c = 200 - a$. Substitute: $12a + 5(200 - a) = 1{,}560$, so $12a + 1{,}000 - 5a = 1{,}560$, which gives $7a = 560$ and $a = 80$. (Then $c = 120$. Check: $960 + 600 = 1{,}560$.)

No-Calculator Tip Read what is asked. If the question wants $x + y$, try adding or subtracting the equations first: you may get $x + y$ directly. You can also backsolve: a choice is correct only if it works in *both* equations.

Common Traps

- Answering with the wrong variable (y when the question asks for x).
- Subtracting equations but changing the sign of only the first term.
- Multiplying one side of an equation but not the other side.
- Dropping the parentheses in substitution: $3x - (2x - 1)$ is $3x - 2x + 1$.

Practice Set 13

Score: _____ / 25 Target time: 27 minutes

Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

1. If $3x + y = 50$ and $x - y = 10$, what is the value of x?

A -15 **B** 5 **C** 10 **D** 15

2. Which ordered pair (x, y) is the solution of this system?

$$x + y = 2$$
$$x - 3y = 10$$

A $(7, -1)$ **B** $(-2, 4)$ **C** $(5, -3)$ **D** $(4, -2)$

3. Solve the system below. What is the value of x?

$$x + y = 12$$
$$x - y = 6$$

A 3 **B** 6 **C** 9 **D** 18

4. Which ordered pair (x, y) satisfies both $x + y = -1$ and $x - 2y = 2$?

A $(1, -2)$ **B** $(0, -1)$ **C** $(2, 0)$ **D** $(-1, 0)$

5. The sum of two numbers is 91, and their difference is 17. What is the larger number?

A 37 **B** 51 **C** 52 **D** 54

6. Which ordered pair (x, y) satisfies both $x + y = 17$ and $2x - y = 7$?

A $(8, 9)$ **B** $(9, 11)$ **C** $(8, -9)$ **D** $(9, 8)$

7. The sum of two numbers is 19, and their difference is 7. What is the larger number?

A 6 **B** 13 **C** 16 **D** 26

8. If $x + y = 6$ and $x - y = 4$, what is the value of y?

A -1 **B** 0 **C** 1 **D** 5

9. What is the value of x in the solution of this system?

$$x + 3y = -4$$
$$5x - 3y = 34$$

A 5 **B** 6 **C** 8 **D** 30

10. Solve the system for y:

$$y = 3x + 8$$
$$6x + y = -1$$

A -1 **B** 3 **C** 4 **D** 5

11. If $4x + y = -9$ and $2x - y = 3$, what is the value of x?

A -6 **B** -5 **C** -1 **D** 4

12. A water park sold 220 day passes on Saturday. Adult passes cost \$38 and child passes cost \$16. Pass sales were \$6,226. How many adult passes were sold?

A 97 **B** 110 **C** 113 **D** 123

13. Two recruiters signed up 55 recruits last quarter. One recruiter signed up 9 more than the other. How many recruits did the more successful recruiter sign up?

A 32 **B** 35 **C** 46 **D** 64

14. The base movie theater sold 92 tickets on Sunday. Evening tickets cost \$14 and matinee tickets cost \$6. The theater took in \$960. How many evening tickets were sold?

A 41 **B** 46 **C** 48 **D** 51

15. Ryan is 22 years older than Marcus. The sum of their ages is 100. How old is Marcus?

A 36 **B** 37 **C** 39 **D** 61

16. If $5x - 3y = 21$ and $2x + 3y = 42$, what is the value of $x + y$?

A 1 **B** 8 **C** 9 **D** 17

17. If $y = -3x + 6$ and $4x + 3y = 13$, what is the value of y?

A 1 **B** 3 **C** $\frac{51}{5}$ **D** $\frac{123}{5}$

18. A jar holds 35 coins, all quarters and dimes. The coins are worth \$4.70 in all. How many quarters are in the jar?

A 4 **B** 8 **C** 12 **D** 27

19. If $5x - 4y = 12$ and $2x + 2y = 30$, what is the value of y?

A 1 **B** 7 **C** 8 **D** 15

20. If $3x + 4y = 61$ and $4x + 3y = 65$, what is the value of $x - y$?

A 2 **B** 3 **C** 4 **D** 18

21. A tea shop mixes black tea that costs \$6 per pound with green tea that costs \$16 per pound to make 30 pounds of a tea blend that is worth \$15 per pound. How many pounds of black tea are in the mix?

A 3 pounds **B** 5 pounds
C 15 pounds **D** 27 pounds

22. If $x + 5y = 65$ and $5x + y = 37$, what is the value of $x - y$?

A -7 **B** 5 **C** 17 **D** 28

23. A feed store mixes cracked corn that costs \$1 per pound with sunflower seeds that cost \$6 per pound to make 15 pounds of a bird-feed mix that is worth \$4 per pound. How many pounds of sunflower seeds are in the mix?

A 6 pounds **B** 7 pounds
C 9 pounds **D** 12 pounds

24. If $3x + 6y = -33$ and $5x - 2y = 17$, what is the value of y?

A -7 **B** -6 **C** -5 **D** 1

25. A unit's booth at a 5K race sold 106 items, all T-shirts and caps. T-shirts cost \$18 and caps cost \$10. Sales came to \$1,412. How many more caps than T-shirts were sold?

A 14 **B** 18 **C** 44 **D** 62

Answers & Explanations 13

1 D	**6** A	**11** C	**16** D	**21** A
2 D	**7** B	**12** D	**17** B	**22** A
3 C	**8** C	**13** A	**18** B	**23** C
4 B	**9** A	**14** D	**19** B	**24** B
5 D	**10** D	**15** C	**20** C	**25** B

1. **D** 15

1 Add the equations. The y-terms cancel: $4x = 60$.

2 Divide both sides by 4: $x = \frac{60}{4} = 15$.

3 Substitute $x = 15$ into $x - y = 10$: $15 - y = 10$, so $-y = -5$ and $y = 5$.

WHY NOT (A) has the right size but the wrong sign. (B) gives the value of y, not x. (C) gives the value of $x - y$, not x.

2. **D** $(4, -2)$

1 A solution must make *both* equations true. Test $(4, -2)$, that is, $x = 4$ and $y = -2$.

2 First equation: $4 + (-2) = 2$. ✓

3 Second equation: $4 - 3(-2) = 10$. ✓

WHY NOT (A) works in the second equation but not the first. (B) swaps the values of x and y. (C) works in the first equation but not the second.

3. **C** 9

1 Add the equations. The y-terms cancel: $2x = 18$.

2 Divide both sides by 2: $x = \frac{18}{2} = 9$.

3 Substitute $x = 9$ into $x + y = 12$: $9 + y = 12$, so $y = 3$.

WHY NOT (A) gives the value of y, not x. (B) gives the value of $x - y$, not x. (D) forgets to divide by 2.

4. **B** $(0, -1)$

1 A solution must make *both* equations true. Test $(0, -1)$, that is, $x = 0$ and $y = -1$.

2 First equation: $0 + (-1) = -1$. ✓

3 Second equation: $0 - 2(-1) = 2$. ✓

WHY NOT (A) works in the first equation but not the second. (C) works in the second equation but not the first. (D) swaps the values of x and y.

5. **D** 54

1 Let L be the larger number and s be the smaller number. Then $L + s = 91$ and $L - s = 17$.

2 Add the two equations; s cancels: $2L = 108$, so $L = 54$.

3 Check: the other is $91 - 54 = 37$, and $54 - 37 = 17$. ✓

WHY NOT (A) gives the smaller number instead.

6. **A** $(8, 9)$

1 A solution must make *both* equations true. Test $(8, 9)$, that is, $x = 8$ and $y = 9$.

2 First equation: $8 + 9 = 17$. ✓

3 Second equation: $2(8) - 9 = 7$. ✓

TIP You can also solve it directly: adding the equations cancels y.

WHY NOT (B) works in the second equation but not the first. (C) gives y the wrong sign. (D) swaps the values of x and y.

7. **B** 13

1 Let L be the larger number and s be the smaller number. Then $L + s = 19$ and $L - s = 7$.

2 Add the two equations; s cancels: $2L = 26$, so $L = 13$.

3 Check: the other is $19 - 13 = 6$, and $13 - 6 = 7$. ✓

WHY NOT (A) gives the smaller number instead. (D) finds $2L = 26$ but forgets to divide by 2.

8. **C** 1

1 Add the equations. The y-terms cancel: $2x = 10$.

2 Divide both sides by 2: $x = \frac{10}{2} = 5$.

3 Substitute $x = 5$ into $x + y = 6$: $5 + y = 6$, so $y = 1$.

WHY NOT (A) has the right size but the wrong sign. (D) gives the value of x, not y.

9. **A** 5

1 Add the equations. The y-terms cancel: $6x = 30$.

2 Divide both sides by 6: $x = \frac{30}{6} = 5$.

3 Substitute $x = 5$ into $x + 3y = -4$: $5 + 3y = -4$, so $3y = -9$ and $y = -3$.

WHY NOT (C) gives the value of $x - y$, not x. (D) forgets to divide by 6.

10. **D** 5

1 The first equation tells you what y equals. Replace y in the second equation with $(3x + 8)$: $6x + (3x + 8) = -1$.

2 Distribute: $6x + 3x + 8 = -1$, so $9x + 8 = -1$.

3 Solve: $9x = -9$, so $x = -1$.

4 Now use the first equation: $y = 3(-1) + 8 = 5$.

WHY NOT (A) gives the value of x, not y. (C) gives $x + y$.

11. **C** -1

1 Add the equations. The y-terms cancel: $6x = -6$.

2 Divide both sides by 6: $x = \frac{-6}{6} = -1$.

3 Substitute $x = -1$ into $2x - y = 3$: $2(-1) - y = 3$, so $-y = 5$ and $y = -5$.

WHY NOT (A) forgets to divide by 6. (B) gives the value of y, not x. (D) gives the value of $x - y$, not x.

12. **D** 123

1 Let a be the number of adult passes. The rest are child passes: $220 - a$.

2 Add up the money: $38a + 16(220 - a) = 6{,}226$.

3 Distribute and combine: $38a + 3{,}520 - 16a = 6{,}226$, so $22a + 3{,}520 = 6{,}226$.

4 Subtract 3,520: $22a = 2{,}706$. Divide by 22: $a = 123$.

5 Check: $38(123) + 16(97) = 4{,}674 + 1{,}552 = 6{,}226$. ✓

WHY NOT (A) gives the number of child passes. (B) assumes the two kinds were sold in equal numbers.

13. **A** 32

1 Let L be the larger count and s be the smaller count. Then $L + s = 55$ and $L - s = 9$.

2 Add the two equations; s cancels: $2L = 64$, so $L = 32$.

3 Check: the other is $55 - 32 = 23$, and $32 - 23 = 9$. ✓

WHY NOT (D) finds $2L = 64$ but forgets to divide by 2.

14. **D** 51

1 Let e be the number of evening tickets. The rest are matinee tickets: $92 - e$.

2 Add up the money: $14e + 6(92 - e) = 960$.

3 Distribute and combine: $14e + 552 - 6e = 960$, so $8e + 552 = 960$.

4 Subtract 552: $8e = 408$. Divide by 8: $e = 51$.

5 Check: $14(51) + 6(41) = 714 + 246 = 960$. ✓

WHY NOT (A) gives the number of matinee tickets. (B) assumes the two kinds were sold in equal numbers. (C) divides the total by the sum of the two prices.

15. **C** 39

1 Let L be Ryan's age and s be Marcus's age. Then $L + s = 100$ and $L - s = 22$.

2 Add the two equations; s cancels: $2L = 122$, so $L = 61$.

3 Then $s = 100 - 61 = 39$.

WHY NOT (D) gives Ryan's age instead.

16. **D** 17

1 Add the equations. The y-terms cancel: $7x = 63$.

2 Divide both sides by 7: $x = \frac{63}{7} = 9$.

3 Substitute $x = 9$ into $2x + 3y = 42$: $2(9) + 3y = 42$, so $3y = 24$ and $y = 8$.

4 The question asks for $x + y$: $9 + 8 = 17$.

WHY NOT (A) gives the value of $x - y$, not $x + y$. (B) gives the value of y, not $x + y$. (C) gives the value of x, not $x + y$.

17. **B** 3

1 The first equation tells you what y equals. Replace y in the second equation with $(-3x + 6)$: $4x + 3(-3x + 6) = 13$.

2 Distribute: $4x - 9x + 18 = 13$, so $-5x + 18 = 13$.

3 Solve: $-5x = -5$, so $x = 1$.

4 Now use the first equation: $y = -3(1) + 6 = 3$.

WHY NOT (A) gives the value of x, not y. (C) multiplies only $-3x$ by 3, not 6. (D) makes a sign error with the 18.

18. **B** 8

1 Work in cents so there are no decimals: a quarter is 25 cents, a dime is 10 cents, and \$4.70 is 470 cents.

2 Let q be the number of quarters; then there are $35 - q$ dimes. Total value: $25q + 10(35 - q) = 470$.

3 Distribute: $25q + 350 - 10q = 470$, so $15q = 470 - 350 = 120$.

4 Divide by 15: $q = 8$.

WHY NOT (D) gives the number of dimes.

19. **B** 7

1 Multiply *every* term of the second equation by 2 so the y-terms will cancel: $4x + 4y = 60$.

2 Add the equations. The y-terms cancel: $9x = 72$.

3 Divide both sides by 9: $x = \frac{72}{9} = 8$.

4 Substitute $x = 8$ into $2x + 2y = 30$: $2(8) + 2y = 30$, so $2y = 14$ and $y = 7$.

WHY NOT (A) gives the value of $x - y$, not y. (C) gives the value of x, not y. (D) gives the value of $x + y$, not y.

20. **C** 4

1 Look at the coefficients: they are swapped. Subtract the second equation from the first: $-x + y = -4$.

2 Divide by -1: $x - y = \frac{-4}{-1} = 4$.

TIP No need to find x and y separately (they are 11 and 7).

WHY NOT (D) adds the equations, which gives $x + y$.

21. **A** 3 pounds

1 Let b be the pounds of black tea; then $30-b$ pounds are green tea.

2 The value of the mix is $30 \times 15 = 450$ dollars, so $6b + 16(30 - b) = 450$.

3 Distribute and combine: $6b + 480 - 16b = 450$, so $-10b + 480 = 450$.

4 Subtract 480: $-10b = -30$. Divide by -10: $b = 3$ pounds of black tea.

TIP Sense check: \$15 per pound is closer to the price of green tea, so the mix must contain more green tea.

WHY NOT (B) divides by \$6 instead of by the price difference \$10. (C) assumes equal amounts of each. (D) gives the pounds of green tea.

22. **A** -7

1 Look at the coefficients: they are swapped. Subtract the second equation from the first: $-4x + 4y = 28$.

2 Divide by -4: $x - y = \frac{28}{-4} = -7$.

TIP No need to find x and y separately (they are 5 and 12).

WHY NOT (B) gives the value of x. (C) adds the equations, which gives $x + y$. (D) forgets to divide by -4.

23. **C** 9 pounds

1 Let c be the pounds of cracked corn; then $15 - c$ pounds are sunflower seeds.

2 The value of the mix is $15 \times 4 = 60$ dollars, so $1c + 6(15 - c) = 60$.

3 Distribute and combine: $1c + 90 - 6c = 60$, so $-5c + 90 = 60$.

4 Subtract 90: $-5c = -30$. Divide by -5: $c = 6$ pounds of cracked corn.

5 The question asks for sunflower seeds: $15 - 6 = 9$ pounds.

TIP Sense check: \$4 per pound is closer to the price of sunflower seeds, so the mix must contain more sunflower seeds.

WHY NOT (A) gives the pounds of cracked corn.

24. **B** -6

1 Multiply *every* term of the second equation by 3 so the y-terms will cancel: $15x - 6y = 51$.

2 Add the equations. The y-terms cancel: $18x = 18$.

3 Divide both sides by 18: $x = \frac{18}{18} = 1$.

4 Substitute $x = 1$ into $5x - 2y = 17$: $5(1) - 2y = 17$, so $-2y = 12$ and $y = -6$.

WHY NOT (C) gives the value of $x + y$, not y. (D) gives the value of x, not y.

25. **B** 18

1 Let t be the number of T-shirts. The rest are caps: $106 - t$.

2 Add up the money: $18t + 10(106 - t) = 1{,}412$.

3 Distribute and combine: $18t + 1{,}060 - 10t = 1{,}412$, so $8t + 1{,}060 = 1{,}412$.

4 Subtract 1,060: $8t = 352$. Divide by 8: $t = 44$.

5 So there were $106 - 44 = 62$ caps. The question asks how many more: $62 - 44 = 18$.

6 Check: $18(44) + 10(62) = 792 + 620 = 1{,}412$. ✓

WHY NOT (C) gives the number of T-shirts, not the difference. (D) gives the number of caps, not the difference.

CHAPTER 14

Polynomials & Exponent Rules

Tested on: Mathematics Knowledge

Key Concepts

A *polynomial* is a sum of terms such as $3x^2$, $-5x$, and 7. Each term has a *coefficient* (the number in front) and a variable part. *Like terms* have exactly the same variable part: x^2 and $4x^2$ are like terms; x^2 and x^3 are not.

Exponent rules

$x^a \cdot x^b = x^{a+b}$	add exponents	$x^0 = 1$	$(x \neq 0)$
$\dfrac{x^a}{x^b} = x^{a-b}$	subtract exponents	$x^{-a} = \dfrac{1}{x^a}$	negative exponent
$(x^a)^b = x^{ab}$	multiply exponents	$(xy)^a = x^a y^a$	every factor

A variable with no exponent has exponent 1: $x = x^1$.

Adding, subtracting, and vocabulary

Add or subtract polynomials by combining like terms: add the *coefficients* and keep the exponent ($2x^2 + 5x^2 = 7x^2$). To subtract, distribute the minus sign to *every* term: $-(x^2 - 4x + 1) = -x^2 + 4x - 1$. The *degree* is the highest exponent; the *leading coefficient* is the coefficient of that term. In $5x^3 - 2x^7 + x$ the degree is 7 and the leading coefficient is -2.

Multiplying

Multiply every term by every term. For two binomials use FOIL (First, Outer, Inner, Last):

$$(x + 3)(x - 5) = x^2 - 5x + 3x - 15 = x^2 - 2x - 15$$

Special products worth memorizing:

$$(a + b)^2 = a^2 + 2ab + b^2 \quad (a - b)^2 = a^2 - 2ab + b^2 \quad (a + b)(a - b) = a^2 - b^2$$

WORKED EXAMPLE

Simplify $(2x^3y)^2 \cdot 3xy^4$.

Solution. Square every factor: $(2x^3y)^2 = 2^2x^{3\cdot 2}y^2 = 4x^6y^2$. Then multiply coefficients and add exponents: $4x^6y^2 \cdot 3xy^4 = 12x^{6+1}y^{2+4} = 12x^7y^6$.

No-Calculator Tip Check with a number. Put $x = 2$ (and $y = 1$) into the original expression and into your answer: the two values must match. It takes seconds and catches most sign and exponent slips.

Common Traps

- $(x+4)^2$ is *not* x^2+16: the middle term $8x$ is missing.
- $(2x^3)^2$ is $4x^6$, not $2x^6$: the coefficient is squared too.
- Adding exponents when you add like terms: $x^2+x^2=2x^2$, not x^4.
- Forgetting to change *every* sign when subtracting a polynomial.

Practice Set 14

Score: _____ / 25 Target time: 25 minutes

Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

1. Simplify: $(7x^2-3x-12)+(2x^2-7x-11)$

A $9x^2+4x-23$ **B** $9x^4-10x^2-23$
C $-x^2-23$ **D** $9x^2-10x-23$

2. Simplify: $(3x^3)(7x^6)$

A $10x^{18}$ **B** $21x^{18}$ **C** $21x^9$ **D** $10x^9$

3. Simplify $\frac{25x^5}{5x^4}$. Assume $x \neq 0$.

A $5x$ **B** $5x^9$ **C** $20x$ **D** $5x^2$

4. What is the degree of the polynomial $-4-2x^5-8x^6+7x^8$?

A 0 **B** 4 **C** 8 **D** 19

5. Simplify $\frac{24x^{10}}{6x^4}$. Assume $x \neq 0$.

A $4x^7$ **B** $18x^6$ **C** $4x^6$ **D** $4x^{14}$

6. What is the leading coefficient of $7x+8x^2+3x^4$?

A -3 **B** 3 **C** 4 **D** 7

7. Simplify: $(4x^4)(4x^6)$

A $8x^{24}$ **B** $16x^{10}$ **C** $8x^{10}$ **D** $16x^{24}$

8. What is the sum of $6x^2-x+5$ and $x^2+5x+12$?

A $7x^4+4x^2+17$ **B** $7x^2+4x+17$
C $7x^2-6x+17$ **D** $11x^2+17$

9. Subtract $7x^2+5x-4$ from x^2+2x-3.

A $6x^2+3x-1$ **B** $-6x^2+7x-7$
C $-6x^2-3x+1$ **D** $8x^2+7x-7$

10. What is $(-6x^3y)^2$?

A $36x^6y^2$ **B** $-6x^6y^2$
C $36x^5y^3$ **D** $-36x^6y^2$

11. Multiply: $(x+8)(x-7)$

A x^2-x-56 **B** x^2+x+56
C x^2-56 **D** x^2+x-56

12. Multiply: $(x-2)(x+2)$

A x^2+4 **B** x^2-4x-4
C x^2-4 **D** x^2-4x+4

13. Expand and simplify $(x+9)(x-8)$.

A x^2-x-72 **B** x^2-72
C x^2+x+72 **D** x^2+x-72

14. Multiply: $(x-12)^2$

A $x^2-24x+144$ **B** $x^2-24x-144$
C x^2+144 **D** x^2-144

15. Simplify: $(5x^3)^2$

A $25x^6$ **B** $5x^6$ **C** $10x^6$ **D** $25x^5$

16. Simplify: $(2x^2+2x-4)-(7x^2-9x+2)$

A $-5x^2+11x-2$ **B** $-5x^2-7x-2$
C $9x^2-7x-2$ **D** $-5x^2+11x-6$

17. Divide $9x^3+15x^2-15x$ by $3x$. (Assume $x \neq 0$.)

A $3x^2+5x-5$ **B** $3x^2+5x$
C $3x^2$ **D** $3x^3+5x^2-5x$

18. Simplify: $(-2x^4y)(8x^6y^3)$

A $-16x^{24}y^3$ **B** $-16x^{10}y^4$
C $6x^{10}y^4$ **D** $-16x^{10}y^3$

19. Multiply: $(x-5)(x^2+3x-6)$

A $x^3-2x^2+9x+30$ **B** $x^3+3x^2-6x+30$
C $x^3-2x^2-21x+30$ **D** x^3-2x^2+30

20. Which expression is equal to $(2x^{-6})(4x)$?

A $\frac{6}{x^5}$ **B** $\frac{8}{x^6}$ **C** $8x^7$ **D** $\frac{8}{x^5}$

21. When $(x-4)(x^2+x+2)$ is multiplied out and simplified, what is the coefficient of x^2?

A -4 **B** -3 **C** 1 **D** 5

22. Which expression is equal to $(5x^{-3})(4x^6)$?

A $20x^9$ **B** $20x^3$ **C** $\frac{20}{x^{18}}$ **D** $9x^3$

23. What is $(-3x^3y^2)^3$?

A $27x^9y^6$ **B** $-27x^6y^5$
C $-27x^9y^6$ **D** $-3x^9y^6$

24. Expand and simplify $(5x+1)(4x-8)$.

A $20x^2-36x-8$ **B** $20x^2-36x+8$
C $20x^2+36x-8$ **D** $20x^2-8$

25. Which expression is equal to $(6x+3)^2$?

A $6x^2+36x+9$ **B** $36x^2+36x+9$
C $36x^2+18x+9$ **D** $36x^2+9$

Answers & Explanations 14

1 D	**6** B	**11** D	**16** D	**21** B
2 C	**7** B	**12** C	**17** A	**22** B
3 A	**8** B	**13** D	**18** B	**23** C
4 C	**9** C	**14** A	**19** C	**24** A
5 C	**10** A	**15** A	**20** D	**25** B

1. **D** $9x^2 - 10x - 23$

1 Group like terms: $(7+2)x^2+(-3+(-7))x+(-12+(-11))$.

2 Add the coefficients and keep each exponent: $9x^2 - 10x - 23$.

WHY NOT (A) subtracts the x-terms instead of adding them. (B) adds the exponents as well as the coefficients. (C) combines unlike terms (x^2-terms with x-terms).

2. **C** $21x^9$

1 Multiply the coefficients: $3 \cdot 7 = 21$.

2 Multiply like bases by *adding* exponents: $x^3 \cdot x^6 = x^{3+6} = x^9$.

3 So the product is $21x^9$.

WHY NOT (A) adds the coefficients and multiplies the exponents. (B) multiplies the exponents instead of adding them. (D) adds the coefficients instead of multiplying them.

3. **A** $5x$

1 Divide the coefficients: $25 \div 5 = 5$.

2 Divide like bases by *subtracting* exponents: $\frac{x^5}{x^4} = x^{5-4} = x$.

3 So the result is $5x$.

WHY NOT (B) adds the exponents instead of subtracting. (C) subtracts the coefficients instead of dividing.

4. **C** 8

1 The degree is the *highest* exponent, no matter where the term is written.

2 The exponents are 0, 5, 6, 8. The highest is 8.

WHY NOT (A) gives the exponent of the first term written, not the highest exponent. (B) counts the terms. (D) adds all the exponents.

5. **C** $4x^6$

1 Divide the coefficients: $24 \div 6 = 4$.

2 Divide like bases by *subtracting* exponents: $\frac{x^{10}}{x^4} = x^{10-4} = x^6$.

3 So the result is $4x^6$.

WHY NOT (B) subtracts the coefficients instead of dividing. (D) adds the exponents instead of subtracting.

6. **B** 3

1 Rewrite in standard form (highest exponent first): $3x^4 + 8x^2 + 7x$.

2 The leading coefficient is the number in front of the highest-power term, $3x^4$: it is 3, sign included.

WHY NOT (A) has the wrong sign; the sign is part of the coefficient. (C) gives the degree instead of the coefficient. (D) takes the coefficient of the first term written.

7. **B** $16x^{10}$

1 Multiply the coefficients: $4 \cdot 4 = 16$.

2 Multiply like bases by *adding* exponents: $x^4 \cdot x^6 = x^{4+6} = x^{10}$.

3 So the product is $16x^{10}$.

WHY NOT (A) adds the coefficients and multiplies the exponents. (C) adds the coefficients instead of multiplying them. (D) multiplies the exponents instead of adding them.

8. **B** $7x^2 + 4x + 17$

1 Group like terms: $(6+1)x^2 + (-1+5)x + (5+12)$.

2 Add the coefficients and keep each exponent: $7x^2 + 4x + 17$.

WHY NOT (A) adds the exponents as well as the coefficients. (C) subtracts the x-terms instead of adding them. (D) combines unlike terms (x^2-terms with x-terms).

9. **C** $-6x^2 - 3x + 1$

1 "Subtract $7x^2 + 5x - 4$ from $x^2 + 2x - 3$" means $(x^2 + 2x - 3) - (7x^2 + 5x - 4)$.

2 Distribute the minus sign to *every* term of the second polynomial: $-(7x^2 + 5x - 4) = -7x^2 - 5x + 4$.

3 Combine like terms: $(1 - 7)x^2 + (2 - 5)x + (-3 - (-4)) = -6x^2 - 3x + 1$.

WHY NOT (A) subtracts in the wrong order: "subtract A from B" means $B - A$. (B) changes the sign of only the first term of the second polynomial. (D) adds the polynomials instead of subtracting.

10. **A** $36x^6y^2$

1 Raise *every* factor inside the parentheses to the power 2.

2 Coefficient: $(-6)^2 = 36$. Variables: multiply exponents, $(x^3)^2 = x^6$ and $(y)^2 = y^2$.

3 So the result is $36x^6y^2$.

WHY NOT (B) forgets to raise the coefficient −6 to the power. (C) adds the exponents instead of multiplying. (D) keeps the negative sign, but a negative number squared is positive.

11. **D** $x^2 + x - 56$

1 Use FOIL. First: $x \cdot x = x^2$. Outer: $x \cdot (-7) = -7x$. Inner: $8 \cdot x = 8x$. Last: $8 \cdot (-7) = -56$.

2 Add them and combine the two middle terms: $x^2 - 7x + 8x - 56 = x^2 + x - 56$.

WHY NOT (A) gets the sign of the middle term wrong. (B) gets the sign of the last term wrong. (C) multiplies only the First and Last terms and leaves out the middle.

12. **C** $x^2 - 4$

1 This is a sum times a difference of the same two terms, so use $(A + B)(A - B) = A^2 - B^2$ with $A = x$ and $B = 2$.

2 The middle terms cancel ($+2x$ and $-2x$), leaving $(x)^2 - 2^2 = x^2 - 4$.

TIP Check with $x = 1$: the original and your answer must give the same number.

WHY NOT (A) makes the last term positive, but a positive times a negative is negative. (B) gets the sign of one middle product wrong ($-2x$ instead of $2x$), so the middle terms add to $-4x$ instead of canceling. (D) treats it like a perfect square.

13. **D** $x^2 + x - 72$

1 Use FOIL. First: $x \cdot x = x^2$. Outer: $x \cdot (-8) = -8x$. Inner: $9 \cdot x = 9x$. Last: $9 \cdot (-8) = -72$.

2 Add them and combine the two middle terms: $x^2 - 8x + 9x - 72 = x^2 + x - 72$.

WHY NOT (A) gets the sign of the middle term wrong. (B) multiplies only the First and Last terms and leaves out the middle. (C) gets the sign of the last term wrong.

14. **A** $x^2 - 24x + 144$

1 Use the pattern $(A - B)^2 = A^2 - 2AB + B^2$ with $A = x$ and $B = 12$.

2 $A^2 = (x)^2 = x^2$, $-2AB = -2(x)(12) = -24x$, and $B^2 = 12^2 = 144$.

3 So $(x - 12)^2 = x^2 - 24x + 144$.

TIP Check with $x = 1$: the original and your answer must give the same number.

WHY NOT (B) makes the last term negative, but $(-12)^2 = 144$. (C) squares each term but leaves out the middle term $-24x$. (D) squares each term, leaves out the middle term, and makes the last term negative.

15. **A** $25x^6$

1 Raise *every* factor inside the parentheses to the power 2.

2 Coefficient: $5^2 = 25$. Variables: multiply exponents, $(x^3)^2 = x^6$.

3 So the result is $25x^6$.

WHY NOT (B) forgets to raise the coefficient 5 to the power. (C) multiplies the coefficient by 2 instead of raising it to the power 2. (D) adds the exponents instead of multiplying.

16. **D** $-5x^2 + 11x - 6$

1 Distribute the minus sign to *every* term of the second polynomial: $-(7x^2 - 9x + 2) = -7x^2 + 9x - 2$.

2 Combine like terms: $(2-7)x^2 + (2-(-9))x + (-4-2) = -5x^2 + 11x - 6$.

WHY NOT (A) forgets to change the sign of the last term. (B) changes the sign of only the first term of the second polynomial. (C) adds the polynomials instead of subtracting.

17. **A** $3x^2 + 5x - 5$

1 Divide *each* term of the top by $3x$.

2 $\frac{9x^3}{3x} = 3x^2$, $\frac{15x^2}{3x} = 5x$, and $\frac{-15x}{3x} = -5$.

3 So the result is $3x^2 + 5x - 5$.

WHY NOT (B) drops the last term, but $\frac{-15x}{3x}$ is -5, not 0. (C) leaves an x on the last term. (D) divides the coefficients but forgets to subtract the exponents.

18. **B** $-16x^{10}y^4$

1 Multiply the coefficients: $(-2) \cdot 8 = -16$.

2 Multiply like bases by *adding* exponents: $x^4 \cdot x^6 = x^{4+6} = x^{10}$ and $y^1 \cdot y^3 = y^{1+3} = y^4$ (a letter with no exponent has exponent 1).

3 So the product is $-16x^{10}y^4$.

WHY NOT (A) multiplies the exponents instead of adding them. (C) adds the coefficients instead of multiplying them. (D) treats a y with no exponent as y^0 instead of y^1.

19. **C** $x^3 - 2x^2 - 21x + 30$

1 Multiply each term of $(x - 5)$ by every term of $(x^2 + 3x - 6)$.

2 $x(x^2 + 3x - 6) = x^3 + 3x^2 - 6x$.

3 $(-5)(x^2 + 3x - 6) = -5x^2 - 15x + 30$.

4 Add and combine like terms: $x^3 + 3x^2 - 6x - 5x^2 - 15x + 30 = x^3 - 2x^2 - 21x + 30$.

WHY NOT (A) makes a sign error combining the x-terms. (B) multiplies -5 by only -6, the last term of the trinomial. (D) drops the x-terms.

20. **D** $\frac{8}{x^5}$

1 Multiply the coefficients: $2 \cdot 4 = 8$.

2 Add the exponents: $x^{-6} \cdot x^1 = x^{-6+1} = x^{-5}$, and $x^{-5} = \frac{1}{x^5}$.

3 So the product is $\frac{8}{x^5}$.

WHY NOT (A) adds the coefficients instead of multiplying them. (B) multiplies the exponents instead of adding them. (C) ignores the negative sign on the exponent.

21. **B** -3

1 Multiply each term of $(x - 4)$ by every term of $(x^2 + x + 2)$.

2 $x(x^2 + x + 2) = x^3 + x^2 + 2x$.

3 $(-4)(x^2 + x + 2) = -4x^2 - 4x - 8$.

4 Add and combine like terms: $x^3 + x^2 + 2x - 4x^2 - 4x - 8 = x^3 - 3x^2 - 2x - 8$.

WHY NOT (A) uses only the x^2-term from $(-4)(x^2 + x + 2)$. (C) uses only the x^2-term from $x(x^2 + x + 2)$. (D) subtracts instead of adding the like terms.

22. **B** $20x^3$

1 Multiply the coefficients: $5 \cdot 4 = 20$.

2 Add the exponents: $x^{-3} \cdot x^6 = x^{-3+6} = x^3$.

3 So the product is $20x^3$.

WHY NOT (A) ignores the negative sign on the exponent. (C) multiplies the exponents instead of adding them. (D) adds the coefficients instead of multiplying them.

23. **C** $-27x^9y^6$

1 Raise *every* factor inside the parentheses to the power 3.

2 Coefficient: $(-3)^3 = -27$. Variables: multiply exponents, $(x^3)^3 = x^9$ and $(y^2)^3 = y^6$.

3 So the result is $-27x^9y^6$.

WHY NOT (A) loses the negative sign: a negative number to an odd power stays negative. (B) adds the exponents instead of multiplying. (D) forgets to raise the coefficient -3 to the power.

24. **A** $20x^2 - 36x - 8$

1 Use FOIL. First: $5x \cdot 4x = 20x^2$. Outer: $5x \cdot (-8) = -40x$. Inner: $1 \cdot 4x = 4x$. Last: $1 \cdot (-8) = -8$.

2 Add them and combine the two middle terms: $20x^2 - 40x + 4x - 8 = 20x^2 - 36x - 8$.

WHY NOT (B) gets the sign of the last term wrong. (C) gets the sign of the middle term wrong. (D) multiplies only the First and Last terms and leaves out the middle.

25. **B** $36x^2 + 36x + 9$

1 Use the pattern $(A + B)^2 = A^2 + 2AB + B^2$ with $A = 6x$ and $B = 3$.

2 $A^2 = (6x)^2 = 36x^2$, $2AB = 2(6x)(3) = 36x$, and $B^2 = 3^2 = 9$.

3 So $(6x + 3)^2 = 36x^2 + 36x + 9$.

TIP Check with $x = 1$: the original and your answer must give the same number.

WHY NOT (A) does not square the coefficient 6. (C) forgets to double the middle term. (D) squares each term but leaves out the middle term $36x$.

CHAPTER 15

Factoring & Quadratic Equations

Tested on: Mathematics Knowledge

Key Concepts

Factoring is multiplication in reverse: it rewrites a sum as a product. It is also the fastest way to solve a *quadratic equation* (an equation with an x^2 term) without a calculator.

Factoring patterns

- **Greatest common factor (GCF) first:** $6x^2 + 9x = 3x(2x + 3)$.
- **Trinomials** $x^2 + bx + c$: find two numbers that *multiply* to c and *add* to b. For $x^2 + 7x + 12$: $3 \cdot 4 = 12$ and $3 + 4 = 7$, so $(x + 3)(x + 4)$.
- **Difference of squares:** $a^2 - b^2 = (a + b)(a - b)$, so $x^2 - 49 = (x + 7)(x - 7)$.
- **Perfect squares:** $x^2 + 10x + 25 = (x + 5)^2$.
- **Leading coefficient not 1:** try factor pairs and check the middle term with FOIL: $2x^2 + 7x + 3 = (2x + 1)(x + 3)$.

Solving quadratic equations

1. Move everything to one side so the other side is 0.
2. Factor, then set *each* factor equal to 0 (a product is 0 only when one of its factors is 0).

If there is no x-term, take square roots and keep both signs: $x^2 = 49$ gives $x = 7$ or $x = -7$. When factoring is hard, use the quadratic formula for $ax^2 + bx + c = 0$:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Function notation

$f(x) = x^2 - 3x + 2$ is a rule; $f(-2)$ means "replace every x with (-2)": $f(-2) = (-2)^2 - 3(-2) + 2 = 4 + 6 + 2 = 12$.

WORKED EXAMPLE

Solve $x^2 - 5x = 14$.
Solution. Subtract 14: $x^2 - 5x - 14 = 0$. Two numbers that multiply to -14 and add to -5 are -7 and 2, so $(x - 7)(x + 2) = 0$. Then $x - 7 = 0$ or $x + 2 = 0$: $x = 7$ or $x = -2$. Check: $49 - 35 = 14$ and $4 + 10 = 14$. ✓

No-Calculator Tip To find the pair for $x^2 + bx + c$, list the factor pairs of c. If c is positive, both numbers have the sign of b; if c is negative, the signs differ and the larger number takes the sign of b.

Common Traps

- $(x + 3)(x - 5) = 0$ gives $x = -3$ or $x = 5$: use the *opposites* of the numbers in the factors.
- Forgetting the negative root of $x^2 = 49$.
- $x(x - 3) = 10$ does *not* mean $x = 10$ or $x - 3 = 10$; you need 0 on one side first.
- $(-3)^2 = 9$, but $-3^2 = -9$: keep the parentheses when you substitute.

Practice Set 15

Score: _____ / 25 Target time: 25 minutes

Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

1. What is the greatest common factor of the terms of $14x^2 + 12x$?

A $2x$ **B** $2x^2$ **C** $12x$ **D** $84x^2$

2. Factor: $x^2 - 8x + 15$

A $(x - 9)(x + 1)$ **B** $(x + 3)(x + 5)$
C $(x - 15)(x - 1)$ **D** $(x - 5)(x - 3)$

3. Solve $x(x + 2) = 0$.

A $x = 0$ or $x = 2$ **B** $x = -2$
C $x = -2$ or $x = 1$ **D** $x = -2$ or $x = 0$

4. Factor: $x^2 - 17x + 72$

A $(x + 3)(x + 24)$ **B** $(x - 9)(x - 8)$
C $(x + 2)(x + 36)$ **D** $(x + 8)(x + 9)$

5. What are the solutions of $(x - 8)(x - 7) = 0$?

A $x = 8$ **B** $x = -8$ or $x = 7$
C $x = 7$ or $x = 8$ **D** $x = -8$ or $x = -7$

6. What is the greatest common factor of the terms of $16x^3 - 36x^2$?

A $144x^3$ **B** $4x^3$ **C** $4x^2$ **D** $16x^2$

7. What are the solutions of $n^2 - 49 = 0$?

A $n = -7$ or $n = 7$ **B** $n = -49$ or $n = 49$
C $n = 7$ **D** $n = -8$ or $n = 8$

8. Given $f(x) = x^2 + x - 3$, find $f(-5)$.

A -33 **B** 17 **C** 27 **D** 34

9. Which is the factored form of $x^2 - 4x - 77$?

A $(x - 77)(x + 1)$ **B** $(x - 7)(x + 11)$
C $(x - 5)(x + 1)$ **D** $(x - 11)(x + 7)$

10. Factor completely: $t^2 + 16t + 64$

A $(t - 8)(t + 8)$ **B** $(t + 1)(t + 64)$
C $(t + 8)^2$ **D** $(t - 8)^2$

11. Which values of x satisfy $x^2 - 2x - 35 = 0$?

A $x = -5$ **B** $x = -7$ or $x = 5$
C $x = -5$ or $x = 7$ **D** $x = -1$ or $x = 35$

12. Which of the following is a solution of $x^2 - 6x - 16 = 0$?

A -16 **B** -8 **C** 2 **D** 8

13. Solve for x: $x^2 - 3x - 54 = 0$.

A $x = -6$ or $x = 9$ **B** $x = -9$ or $x = 6$
C $x = -1$ or $x = 54$ **D** $x = -2$ or $x = 27$

14. Which value of x makes $x^2 + 15x + 54 = 0$ true?

A -6 **B** 6 **C** 9 **D** 54

15. Which expression is equal to $m^2 - 6m + 9$?

A $(m - 9)(m - 1)$ **B** $(m + 3)^2$
C $(m - 3)(m + 3)$ **D** $(m - 3)^2$

16. Factor: $x^2 + 4x - 45$

A $(x - 9)(x + 5)$ **B** $(x - 5)(x + 9)$
C $(x - 15)(x + 3)$ **D** $(x - 3)(x + 15)$

17. Solve for x: $4x^2 = 484$.

A $x = -11$ or $x = 11$
B $x = -22$ or $x = 22$
C $x = -121$ or $x = 121$
D $x = 11$

18. Given $f(x) = -x^2 - 2x + 3$, find $f(-3)$.

A -12 **B** -4 **C** -2 **D** 0

19. Which is the factored form of $5x^2 - 11x + 6$?

A $(x - 6)(x - 1)$ **B** $(5x + 6)(x + 1)$
C $(5x - 6)(x - 1)$ **D** $(5x - 1)(x - 6)$

20. What are the solutions of $6x^2 + x - 5 = 0$?

A $x = -1$ or $x = \frac{5}{6}$ **B** $x = -\frac{5}{6}$ or $x = 1$
C $x = -2$ or $x = \frac{5}{3}$ **D** $x = \frac{5}{6}$

21. Factor: $3x^2 + 11x - 42$

A $(3x - 7)(x + 6)$ **B** $(3x + 7)(x - 6)$
C $(3x + 6)(x - 7)$ **D** $(x - 7)(x + 6)$

22. Use the quadratic formula to solve $2x^2 + 13x + 15 = 0$.

A $x = -10$ or $x = -3$ **B** $x = -5$ or $x = -\frac{3}{2}$
C $x = \frac{3}{2}$ or $x = 5$ **D** $x = -\frac{15}{2}$ or $x = 1$

23. Which is the factored form of $25x^2 - 36$?

A $(5x + 6)^2$ **B** $(5x - 6)(5x + 6)$
C $(25x - 6)(25x + 6)$ **D** $(5x - 6)^2$

24. Which is the completely factored form of $5x^2 + 5x - 100$?

A $5(x - 4)(x + 5)$ **B** $5(x - 5)(x + 4)$
C $(x - 4)(x + 5)$ **D** $5(x - 20)(x + 1)$

25. Which values of x satisfy $x^2 = 10x - 21$?

A $x = 7$ **B** $x = 3$ or $x = 7$
C $x = -7$ or $x = -3$ **D** $x = 1$ or $x = 21$

Answers & Explanations 15

1 A	**6** C	**11** C	**16** B	**21** A
2 D	**7** A	**12** D	**17** A	**22** B
3 D	**8** B	**13** A	**18** D	**23** B
4 B	**9** D	**14** A	**19** C	**24** A
5 C	**10** C	**15** D	**20** A	**25** B

1. **A** $2x$

1 Coefficients: the largest number that divides both 14 and 12 is 2.

2 Variables: both terms contain x; use the *lower* power, x.

3 So the GCF is $2x$. (Check: $14x^2+12x = 2x(7x+6)$.)

WHY NOT (B) uses the higher power of x; the GCF takes the lower power. (C) uses the smaller coefficient, which does not divide both terms. (D) gives the least common multiple, not the greatest common factor.

2. **D** $(x-5)(x-3)$

1 Two numbers that multiply to 15 and add to -8 are -3 and -5: $-3\cdot(-5)=15$ and $-3+(-5)=-8$.

2 So $x^2-8x+15=(x-5)(x-3)$.

3 Check with FOIL: $x^2-5x-3x+15=x^2-8x+15$. ✓

WHY NOT (A) uses 1 and -9, which add to -8 but multiply to -9, not 15. (B) has both signs backwards. (C) uses -15 and -1, which multiply to 15 but add to -16, not -8.

3. **D** $x=-2$ or $x=0$

1 A product is 0 only if one of its factors is 0, so set each factor equal to 0.

2 $x=0$ or $x+2=0$, which gives $x=-2$.

3 The solutions are $x=-2$ or $x=0$.

WHY NOT (A) uses the number in the factor instead of its opposite. (B) forgets that the factor x gives the solution $x=0$. (C) sets the factor x equal to 1 instead of 0.

4. **B** $(x-9)(x-8)$

1 Two numbers that multiply to 72 and add to -17 are -8 and -9: $-8\cdot(-9)=72$ and $-8+(-9)=-17$.

2 So $x^2-17x+72=(x-9)(x-8)$.

3 Check with FOIL: $x^2-9x-8x+72=x^2-17x+72$. ✓

WHY NOT (A) uses 3 and 24, which multiply to 72 but add to 27, not -17. (C) uses 2 and 36, which multiply to 72 but add to 38, not -17. (D) has both signs backwards.

5. **C** $x=7$ or $x=8$

1 A product is 0 only if one of its factors is 0, so set each factor equal to 0.

2 $x-8=0$ gives $x=8$; $x-7=0$ gives $x=7$.

3 The solutions are $x=7$ or $x=8$. (Notice they are the *opposites* of the numbers in the factors.)

WHY NOT (A) forgets the second factor. (B) gets the sign of one solution wrong. (D) uses the numbers in the factors instead of their opposites.

6. **C** $4x^2$

1 Coefficients: the largest number that divides both 16 and 36 is 4.

2 Variables: both terms contain x^2; use the *lower* power, x^2.

3 So the GCF is $4x^2$. (Check: $16x^3-36x^2=4x^2(4x-9)$.)

WHY NOT (A) gives the least common multiple, not the greatest common factor. (B) uses the higher power of x; the GCF takes the lower power. (D) uses the smaller coefficient, which does not divide both terms.

7. **A** $n=-7$ or $n=7$

1 Add 49 to both sides: $n^2=49$.

2 Take the square root of both sides and keep *both* signs: $n=7$ or $n=-7$, because $7^2=49$ and $(-7)^2=49$.

WHY NOT (B) forgets to take the square root. (C) forgets the negative solution.

8. **B** 17

1 Replace every x with (-5): $f(-5)=(-5)^2+(-5)-3$.

2 Square first (order of operations): $(-5)^2=25$. So $f(-5)=25-5-3$.

3 Combine: $f(-5)=17$.

WHY NOT (A) treats $(-5)^2$ as -25. (C) makes a sign error with x.

9. **D** $(x - 11)(x + 7)$

1 Two numbers that multiply to -77 and add to -4 are 7 and -11: $7 \cdot (-11) = -77$ and $7 + (-11) = -4$.

2 So $x^2 - 4x - 77 = (x - 11)(x + 7)$.

3 Check with FOIL: $x^2 - 11x + 7x - 77 = x^2 - 4x - 77$. ✓

WHY NOT (A) uses -77 and 1, which multiply to -77 but add to -76, not -4. (B) has the signs switched, which makes the middle term $4x$. (C) uses 1 and -5, which add to -4 but multiply to -5, not -77.

10. **C** $(t + 8)^2$

1 The first and last terms are perfect squares: $t^2 = (t)^2$ and $64 = 8^2$.

2 The middle term is $2(t)(8) = 16t$, so it is a perfect square: $t^2 + 16t + 64 = (t + 8)^2$.

WHY NOT (A) is a difference of squares, which has no middle term. (B) uses 64 and 1, which multiply correctly but give the wrong middle term. (D) gets the sign wrong; the middle term tells you the sign.

11. **C** $x = -5$ or $x = 7$

1 Factor: two numbers that multiply to -35 and add to -2 are 5 and -7, so $(x + 5)(x - 7) = 0$.

2 Set each factor equal to 0: $x = -5$ or $x = 7$.

WHY NOT (A) finds only one of the two solutions. (B) uses the numbers in the factors instead of their opposites. (D) factors with -35 and 1, which multiply to -35 but do not add to -2.

12. **D** 8

1 Factor: two numbers that multiply to -16 and add to -6 are -8 and 2, so $(x - 8)(x + 2) = 0$.

2 The solutions are $x = 8$ and $x = -2$. Only 8 is among the choices.

3 Check: $(8)^2 - 6(8) - 16 = 64 - 48 - 16 = 0$. ✓

WHY NOT (A) is the constant term, not a solution. (B) has the wrong sign: it is the number in a factor, not a solution. (C) is the opposite of the other solution.

13. **A** $x = -6$ or $x = 9$

1 Factor: two numbers that multiply to -54 and add to -3 are 6 and -9, so $(x + 6)(x - 9) = 0$.

2 Set each factor equal to 0: $x = -6$ or $x = 9$.

WHY NOT (B) uses the numbers in the factors instead of their opposites. (C) factors with -54 and 1, which multiply to -54 but do not add to -3. (D) factors with -27 and 2, which multiply to -54 but do not add to -3.

14. **A** -6

1 Factor: two numbers that multiply to 54 and add to 15 are 6 and 9, so $(x + 6)(x + 9) = 0$.

2 The solutions are $x = -6$ and $x = -9$. Only -6 is among the choices.

3 Check: $(-6)^2 + 15(-6) + 54 = 36 - 90 + 54 = 0$. ✓

WHY NOT (B) has the wrong sign: it is the number in a factor, not a solution. (C) is the opposite of the other solution. (D) is the constant term, not a solution.

15. **D** $(m - 3)^2$

1 The first and last terms are perfect squares: $m^2 = (m)^2$ and $9 = 3^2$.

2 The middle term is $-2(m)(3) = -6m$, so it is a perfect square: $m^2 - 6m + 9 = (m - 3)^2$.

WHY NOT (A) uses 9 and 1, which multiply correctly but give the wrong middle term. (B) gets the sign wrong; the middle term tells you the sign. (C) is a difference of squares, which has no middle term.

16. **B** $(x - 5)(x + 9)$

1 Two numbers that multiply to -45 and add to 4 are 9 and -5: $9 \cdot (-5) = -45$ and $9 + (-5) = 4$.

2 So $x^2 + 4x - 45 = (x - 5)(x + 9)$.

3 Check with FOIL: $x^2 - 5x + 9x - 45 = x^2 + 4x - 45$. ✓

WHY NOT (A) has the signs switched, which makes the middle term $-4x$. (C) uses -15 and 3, which multiply to -45 but add to -12, not 4. (D) uses -3 and 15, which multiply to -45 but add to 12, not 4.

17. **A** $x = -11$ or $x = 11$

1 Divide both sides by 4: $x^2 = 121$.

2 Take the square root and keep both signs: $x = 11$ or $x = -11$.

WHY NOT (B) forgets to divide by 4. (C) forgets to take the square root. (D) forgets the negative solution.

18. **D** 0

1 Replace every x with (-3): $f(-3) = -(-3)^2 - 2(-3) + 3$.

2 Square first (order of operations): $(-3)^2 = 9$. So $f(-3) = -9 + 6 + 3$.

3 Combine: $f(-3) = 0$.

WHY NOT (A) makes a sign error with $-2x$.

19. **C** $(5x - 6)(x - 1)$

1 The first terms must multiply to $5x^2$, so try $(5x + _)(x + _)$. The last numbers must multiply to 6.

2 Test pairs until the Outer + Inner products give $-11x$: with -6 and -1, $5x \cdot (-1) + (-6) \cdot x = -5x - 6x = -11x$.

3 So $5x^2 - 11x + 6 = (5x - 6)(x - 1)$.

WHY NOT (A) ignores the leading coefficient 5. (B) has the signs backwards. (D) puts -6 and -1 in the wrong factors, so the middle term comes out $-31x$.

20. **A** $x = -1$ or $x = \frac{5}{6}$

1 Here $a = 6$, $b = 1$, $c = -5$.

2 Compute $b^2 - 4ac$: $1^2 - 4(6)(-5) = 1 + 120 = 121$, and $\sqrt{121} = 11$.

3 $x = \frac{-1 \pm 11}{12}$: either $x = \frac{-1+11}{12} = \frac{5}{6}$ or $x = \frac{-1-11}{12} = -1$.

TIP You can check by factoring: $6x^2 + x - 5 = (6x - 5)(x + 1)$.

WHY NOT (B) uses 1 instead of -1 in the formula. (C) divides by 6 instead of by $2a = 12$. (D) uses only the + sign of $\pm$.

21. **A** $(3x - 7)(x + 6)$

1 The first terms must multiply to $3x^2$, so try $(3x + _)(x + _)$. The last numbers must multiply to -42.

2 Test pairs until the Outer + Inner products give $11x$: with -7 and 6, $3x \cdot 6 + (-7) \cdot x = 18x - 7x = 11x$.

3 So $3x^2 + 11x - 42 = (3x - 7)(x + 6)$.

WHY NOT (B) has the signs backwards. (C) puts -7 and 6 in the wrong factors, so the middle term comes out $-15x$. (D) ignores the leading coefficient 3.

22. **B** $x = -5$ or $x = -\frac{3}{2}$

1 Here $a = 2$, $b = 13$, $c = 15$.

2 Compute $b^2 - 4ac$: $13^2 - 4(2)(15) = 169 - 120 = 49$, and $\sqrt{49} = 7$.

3 $x = \frac{-13 \pm 7}{4}$: either $x = \frac{-13+7}{4} = -\frac{3}{2}$ or $x = \frac{-13-7}{4} = -5$.

TIP You can check by factoring: $2x^2 + 13x + 15 = (2x + 3)(x + 5)$.

WHY NOT (A) divides by 2 instead of by $2a = 4$. (C) uses 13 instead of -13 in the formula. (D) adds $4ac$ instead of subtracting it.

23. **B** $(5x - 6)(5x + 6)$

1 $25x^2 - 36$ is a difference of squares: $(5x)^2 - 6^2$.

2 Use $A^2 - B^2 = (A + B)(A - B)$: $25x^2 - 36 = (5x - 6)(5x + 6)$.

WHY NOT (A) is a perfect square, not a difference of squares. (C) does not take the square root of 25. (D) is a perfect square, not a difference of squares.

24. **A** $5(x - 4)(x + 5)$

1 First take out the GCF, 5: $5x^2 + 5x - 100 = 5(x^2 + x - 20)$.

2 Now factor $x^2 + x - 20$: find two numbers that multiply to -20 and add to 1. They are 5 and -4.

3 So $5x^2 + 5x - 100 = 5(x - 4)(x + 5)$.

WHY NOT (B) has the signs of both numbers backwards. (C) leaves out the common factor. (D) uses numbers that multiply to -20 but add to -19, not 1.

25. **B** $x = 3$ or $x = 7$

1 Get 0 on one side: move every term to the left. $x^2 - 10x + 21 = 0$.

2 Factor: two numbers that multiply to 21 and add to -10 are -7 and -3, so $(x - 7)(x - 3) = 0$.

3 Set each factor equal to 0: $x = 7$ or $x = 3$.

WHY NOT (A) finds only one of the two solutions. (C) uses the numbers in the factors instead of their opposites. (D) factors with -21 and -1, which multiply to 21 but do not add to -10.

PART III

Geometry

Angles, triangles, area, circles, volume, and the coordinate plane.

16 Lines & Angles
17 Triangles & the Pythagorean Theorem
18 Perimeter & Area
19 Circles
20 Volume & Surface Area
21 Coordinate Geometry

CHAPTER 16

Lines & Angles

Tested on: Mathematics Knowledge

Key Concepts

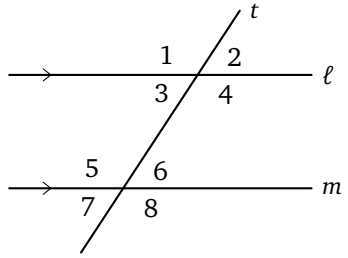
Angles are measured in degrees (°). Almost every angle question on the ASVAB comes down to a few facts: a straight line is 180°, a full turn is 360°, and parallel lines create pairs of equal angles.

Angle vocabulary and angle pairs

acute	less than 90°	**straight**	exactly 180°
right	exactly 90° (marked with a small square)	**reflex**	between 180° and 360°
obtuse	between 90° and 180°		

- **Complementary** angles add up to 90°; **supplementary** angles add up to 180°.
- Angles side by side on a straight line add up to 180°; angles all the way around a point add up to 360°.
- When two lines cross, the angles directly across from each other are **vertical angles**, and they are *equal*.

Parallel lines cut by a transversal



When $\ell \parallel m$:

- **Corresponding** angles (same position at each crossing) are equal: $\angle 1 = \angle 5$, $\angle 4 = \angle 8$.
- **Alternate interior** angles are equal: $\angle 3 = \angle 6$, $\angle 4 = \angle 5$.
- **Alternate exterior** angles are equal: $\angle 1 = \angle 8$, $\angle 2 = \angle 7$.
- **Same-side interior** angles add up to 180°: $\angle 3 + \angle 5 = 180°$.

Angles of a polygon (n = number of sides)

- Sum of the interior angles: $(n - 2) \times 180°$ (triangle 180°, quadrilateral 360°, pentagon 540°, hexagon 720°, octagon 1,080°).
- In a *regular* polygon all angles are equal: each interior angle is $\dfrac{(n-2)\times 180°}{n}$.
- The exterior angles of any polygon add up to 360°, so each exterior angle of a regular polygon is $360° \div n$.

WORKED EXAMPLE

Two angles form a straight line. One measures $(2x + 10)°$ and the other $(3x - 5)°$. What is x?
Solution. Angles on a straight line add up to 180°: $(2x+10)+(3x-5) = 180$, so $5x+5 = 180$, $5x = 175$, and $x = 35$. Check: the angles are 80° and 100°, and $80 + 100 = 180$.

No-Calculator Tip In a parallel-lines figure there are only *two* angle sizes: every small (acute) angle is equal to every other small angle, every large (obtuse) angle is equal to every other large angle, and small + large = 180°.

Common Traps

- Mixing up the totals: **c**omplementary comes before **s**upplementary, just as 90 comes before 180.
- Treating same-side interior angles as equal. They add up to 180°.
- Finding x and stopping when the question asks for the *angle*: plug x back in.
- Using $n \times 180°$ instead of $(n - 2) \times 180°$ for a polygon.

Practice Set 16

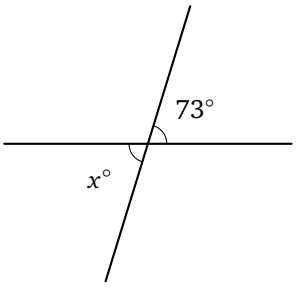
Score: _____ / 25 Target time: 25 minutes

Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

1. Two angles are supplementary. One of them measures 46°. What is the measure of the other angle?

A 134° **B** 147° **C** 226° **D** 314°

2. In the figure, two straight lines intersect, forming the 73° angle shown. What is the value of x?

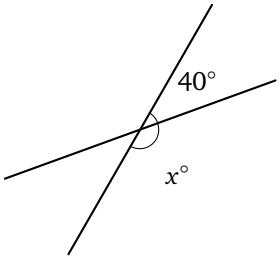


A 73 **B** 83 **C** 107 **D** 287

3. What is the measure of the complement of a 40° angle?

A 37° **B** 40° **C** 50° **D** 140°

4. Two lines cross, and one of the four angles they form measures 40°, as shown. Find x.

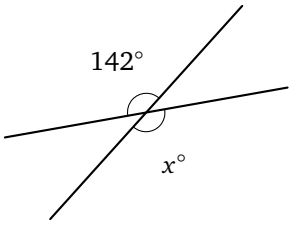


A 140 **B** 155 **C** 160 **D** 320

5. What type of angle has a measure of 317°?

A right **B** reflex **C** acute **D** straight

6. The figure shows two intersecting straight lines and a 142° angle. What is the value of x?



A 38 **B** 127 **C** 142 **D** 218

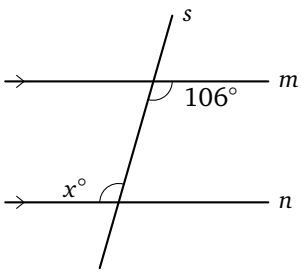
7. Which term describes an angle that measures 54°?

A reflex **B** acute **C** right **D** obtuse

8. $\angle 1$ and $\angle 2$ are supplementary. If $\angle 1$ measures 76°, what is the measure of $\angle 2$?

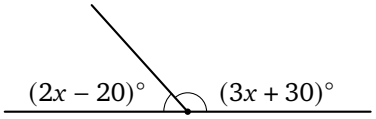
A 14° **B** 76° **C** 101° **D** 104°

9. Parallel lines m and n are cut by transversal s, forming the 106° angle shown. What is the value of x?



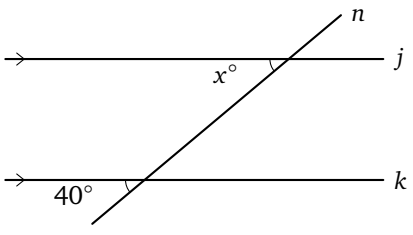
A 106 **B** 111 **C** 116 **D** 254

10. In the figure, a ray meets a straight line, forming angles of $(3x + 30)°$ and $(2x - 20)°$. What is the value of x?



A 16 **B** 32 **C** 34 **D** 38

11. In the figure, $j \parallel k$, and one of the angles measures 40°. What is the value of x?

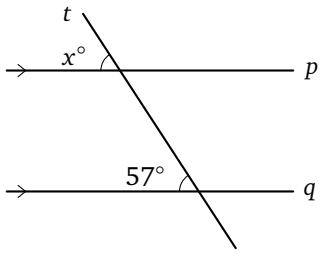


A 40 **B** 50 **C** 140 **D** 320

12. Find the sum of the interior angles of a polygon with 8 sides.

A 135° **B** 360° **C** 1,080° **D** 1,440°

13. In the figure, lines p and q are parallel, and one angle formed by transversal t measures 57°. What is the value of x?



A 33 **B** 47 **C** 57 **D** 123

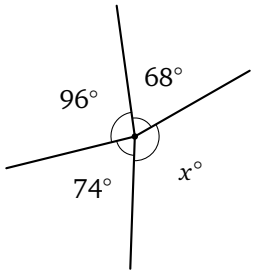
14. What is the sum of the measures of the interior angles of a 13-sided polygon?

A 1,980° **B** 2,160° **C** 2,340° **D** 2,580°

15. $\angle J$ and $\angle K$ are complementary. $\angle J$ measures $(2x + 17)°$, and $\angle K$ measures $(6x + 25)°$. What is the value of x?

A 3 **B** 6 **C** 9 **D** 29

16. Four angles meet at a point, as shown. Three of them measure 68°, 96°, and 74°. What is the value of x?

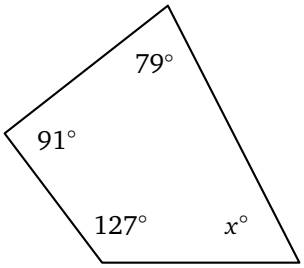


A 16 **B** 122 **C** 196 **D** 238

17. What is the measure of each interior angle of a regular nonagon (9 sides)?

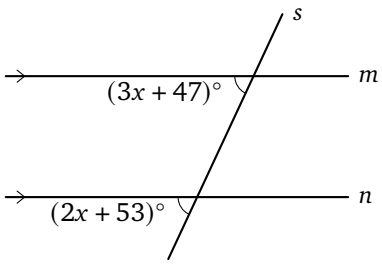
A 40° **B** 140° **C** 160° **D** 1,260°

18. In the quadrilateral shown, three of the interior angles measure 127°, 79°, and 91°. What is the value of x?



A 48 **B** 63 **C** 90 **D** 243

19. In the figure, lines m and n are parallel. The marked angles measure $(3x + 47)°$ and $(2x + 53)°$. What is the value of x?

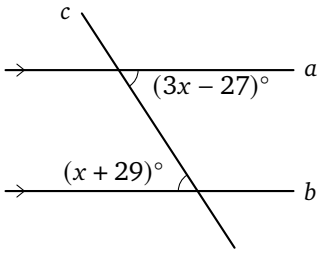


A 6 **B** 16 **C** 65 **D** 100

20. What is the measure of the smaller angle between the hour hand and the minute hand of a clock at 2:55?

A 62.5° **B** 90° **C** 102.5° **D** 117.5°

21. In the figure, lines a and b are parallel. The marked angles measure $(x + 29)°$ and $(3x - 27)°$. What is the measure of the $(x + 29)°$ angle?



A 28° **B** 42° **C** 52° **D** 57°

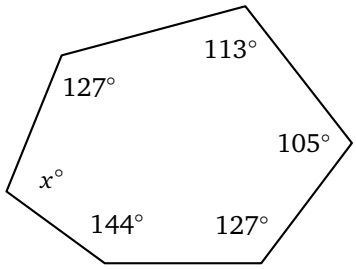
22. A clock shows 6:25. What is the smaller angle formed by its hour hand and minute hand?

A 17.5° **B** 30° **C** 42.5° **D** 317.5°

23. $\angle P$ and $\angle Q$ are supplementary. $\angle P$ measures $(5x - 40)°$, and $\angle Q$ measures $(4x + 13)°$. What is the measure of $\angle Q$?

A 105° **B** 110° **C** 120° **D** 225°

24. In the hexagon shown, five of the interior angles measure 144°, 127°, 105°, 113°, and 127°. What is the value of x?



A 104 **B** 119 **C** 120 **D** 284

25. Petty Officer Novak sketches a regular polygon whose interior angles each measure 144°. How many sides does the polygon have?

A 5 **B** 7 **C** 8 **D** 10

Answers & Explanations 16

1 A	**6** C	**11** A	**16** B	**21** D
2 A	**7** B	**12** C	**17** B	**22** C
3 C	**8** D	**13** C	**18** B	**23** A
4 A	**9** A	**14** A	**19** A	**24** A
5 B	**10** C	**15** B	**20** D	**25** D

1. **A** 134°

1 Supplementary angles add up to 180°.

2 Subtract the angle you know: 180° − 46° = 134°.

3 Check: 46° + 134° = 180°. ✓

TIP **C**omplementary comes before **s**upplementary in the alphabet, just as 90 comes before 180.

WHY NOT (C) adds the angle to 180° instead of subtracting it. (D) subtracts from 360°, the total all the way around a point.

2. **A** 73

1 The $x°$ angle is directly across from the 73° angle where the lines cross, so the two are *vertical angles*.

2 Vertical angles are equal, so $x = 73$.

WHY NOT (C) assumes the two angles add up to 180°, but vertical angles are equal. (D) assumes the two angles add up to 360°, but vertical angles are equal.

3. **C** 50°

1 Complementary angles add up to 90°.

2 Subtract the angle you know: 90° − 40° = 50°.

3 Check: 40° + 50° = 90°. ✓

TIP **C**omplementary comes before **s**upplementary in the alphabet, just as 90 comes before 180.

WHY NOT (B) repeats the given angle, but complementary angles are not equal; they add up to 90°. (D) finds the supplement (total 180°) instead of the complement (total 90°).

4. **A** 140

1 The $x°$ angle and the 40° angle sit side by side along one straight line, so together they make 180°.

2 $x + 40 = 180$, so $x = 180 - 40 = 140$.

WHY NOT (D) assumes the two angles add up to 360° instead of 180°.

5. **B** reflex

1 Compare with the benchmarks: acute is less than 90°, right is exactly 90°, obtuse is between 90° and 180°, straight is exactly 180°, and reflex is between 180° and 360°.

2 317° is between 180° and 360°, so the angle is *reflex*.

WHY NOT (A) describes an angle of exactly 90°. (C) describes an angle smaller than 90°. (D) describes an angle of exactly 180°.

6. **C** 142

1 The $x°$ angle is directly across from the 142° angle where the lines cross, so the two are *vertical angles*.

2 Vertical angles are equal, so $x = 142$.

WHY NOT (A) assumes the two angles add up to 180°, but vertical angles are equal. (D) assumes the two angles add up to 360°, but vertical angles are equal.

7. **B** acute

1 Compare with the benchmarks: acute is less than 90°, right is exactly 90°, obtuse is between 90° and 180°, straight is exactly 180°, and reflex is between 180° and 360°.

2 54° is less than 90°, so the angle is *acute*.

WHY NOT (A) describes an angle between 180° and 360°. (C) describes an angle of exactly 90°. (D) describes an angle between 90° and 180°.

8. **D** 104°

1 Supplementary angles add up to 180°.

2 Subtract the angle you know: 180° − 76° = 104°.

3 Check: 76° + 104° = 180°. ✓

TIP **C**omplementary comes before **s**upplementary in the alphabet, just as 90 comes before 180.

WHY NOT (A) finds the complement (total 90°) instead of the supplement (total 180°). (B) repeats the given angle, but supplementary angles are not equal; they add up to 180°.

9. **A** 106

1 The 106° angle and the $x°$ angle are *alternate interior* angles: they are between the parallel lines, on opposite sides of the transversal.

2 When the lines are parallel, alternate interior angles are equal, so $x = 106$.

TIP Every acute angle in this figure is 74° and every obtuse angle is 106°. The $x°$ angle is obtuse, so $x = 106$.

WHY NOT (D) assumes the two angles add up to 360°; they are equal.

10. **C** 34

1 Angles that form a straight line add up to 180°: $(3x + 30) + (2x - 20) = 180$.

2 Combine like terms: $5x + 10 = 180$.

3 Subtract 10 from both sides: $5x = 180 - 10 = 170$.

4 Divide both sides by 5: $x = 170 \div 5 = 34$.

TIP Check: with $x = 34$ the angles are 132° and 48°, and $132 + 48 = 180$.

WHY NOT (A) uses 90° instead of 180°. (D) makes a sign error when moving the constant term.

11. **A** 40

1 The 40° angle and the $x°$ angle are *corresponding* angles: they sit in the same position at the two crossings.

2 When the lines are parallel, corresponding angles are equal, so $x = 40$.

TIP Every acute angle in this figure is 40° and every obtuse angle is 140°. The $x°$ angle is acute, so $x = 40$.

WHY NOT (B) assumes the two angles add up to 90°; they are equal. (C) assumes the two angles add up to 180°, but corresponding angles are equal. (D) assumes the two angles add up to 360°; they are equal.

12. **C** 1,080°

1 The polygon has $n = 8$ sides. Use: sum of interior angles = $(n - 2) \times 180°$.

2 $(8 - 2) \times 180° = 6 \times 180° = 1{,}080°$.

TIP Why it works: diagonals from one corner cut the polygon into $8 - 2 = 6$ triangles, and each triangle holds 180°.

WHY NOT (A) gives one angle of a regular polygon, not the sum of all the angles. (B) gives the sum of the exterior angles, which is always 360°. (D) multiplies by the number of sides instead of $n - 2$.

13. **C** 57

1 The 57° angle and the $x°$ angle are *corresponding* angles: they sit in the same position at the two crossings.

2 When the lines are parallel, corresponding angles are equal, so $x = 57$.

TIP Every acute angle in this figure is 57° and every obtuse angle is 123°. The $x°$ angle is acute, so $x = 57$.

WHY NOT (A) assumes the two angles add up to 90°; they are equal. (D) assumes the two angles add up to 180°, but corresponding angles are equal.

14. **A** 1,980°

1 The polygon has $n = 13$ sides. Use: sum of interior angles = $(n - 2) \times 180°$.

2 $(13 - 2) \times 180° = 11 \times 180° = 1{,}980°$.

TIP Why it works: diagonals from one corner cut the polygon into $13 - 2 = 11$ triangles, and each triangle holds 180°.

WHY NOT (B) uses $n - 1$ instead of $n - 2$. (C) multiplies by the number of sides instead of $n - 2$.

15. **B** 6

1 Complementary angles add up to 90°: $(2x + 17) + (6x + 25) = 90$.

2 Combine like terms: $8x + 42 = 90$.

3 Subtract 42 from both sides: $8x = 90 - 42 = 48$.

4 Divide both sides by 8: $x = 48 \div 8 = 6$.

TIP Check: with $x = 6$ the angles are 29° and 61°, and $29 + 61 = 90$.

WHY NOT (D) is the measure of an angle, not the value of x.

16. **B** 122

1 Angles all the way around a point add up to 360°: $68 + 96 + 74 + x = 360$.

2 Add the known angles: $68 + 96 + 74 = 238$.

3 Subtract: $x = 360 - 238 = 122$.

WHY NOT (A) uses 180° and leaves out one angle. (C) leaves out one of the given angles. (D) adds the three given angles and stops.

17. **B** 140°

1 Sum of the interior angles: $(9 - 2) \times 180° = 1{,}260°$.

2 A regular polygon has 9 equal angles: $1{,}260° \div 9 = 140°$.

TIP Faster: each exterior angle is $360° \div 9 = 40°$, so each interior angle is $180° - 40° = 140°$.

WHY NOT (A) is the exterior angle, not the interior angle. (C) uses $n - 1$ instead of $n - 2$. (D) is the sum of all the interior angles, not one angle.

18. **B** 63

1 A quadrilateral has 4 sides, so its interior angles add up to $(4 - 2) \times 180° = 360°$.

2 Add the known angles: $127 + 79 + 91 = 297$.

3 Subtract: $x = 360 - 297 = 63$.

WHY NOT (C) divides the angle sum by the number of angles, as if all the angles were equal. (D) uses $(n - 1) \times 180°$ for the angle sum instead of $(n - 2) \times 180°$.

19. **A** 6

1 These are corresponding angles (they sit in the same position at the two crossings). Corresponding angles are equal, so set the expressions equal: $3x + 47 = 2x + 53$.

2 Subtract $2x$ from both sides: $x + 47 = 53$.

3 Subtract 47 from both sides: $x = 53 - 47 = 6$.

WHY NOT (B) assumes the two angles add up to 180°, but these angles are equal. (C) is the measure of an angle, not the value of x. (D) makes a sign error when moving the constant term.

20. **D** 117.5°

1 The minute hand moves $360° \div 60 = 6°$ per minute. At 55 minutes past the hour it is $55 \times 6° = 330°$ past the 12.

2 The hour hand moves 30° per hour plus 0.5° per minute. At 2:55 it is $2{\times}30°{+}55{\times}0.5° = 60°{+}27.5° = 87.5°$ past the 12.

3 The hands are $330° - 87.5° = 242.5°$ apart.

4 That is more than 180°, so the smaller angle is $360° - 242.5° = 117.5°$.

TIP Each hour mark on a clock face is $360° \div 12 = 30°$ from the next.

WHY NOT (B) forgets that the hour hand moves past the hour mark (0.5° per minute).

21. **D** 57°

1 These are alternate interior angles (they are between the parallel lines, on opposite sides of the transversal). Alternate interior angles are equal, so set the expressions equal: $3x - 27 = x + 29$.

2 Subtract x from both sides: $2x - 27 = 29$.

3 Add 27 to both sides: $2x = 29 + 27 = 56$.

4 Divide both sides by 2: $x = 56 \div 2 = 28$.

5 Plug $x = 28$ into $x + 29$: $28 + 29 = 57$, so the angle measures 57°.

WHY NOT (A) stops at the value of x instead of finding the angle.

22. **C** 42.5°

1 The minute hand moves $360° \div 60 = 6°$ per minute. At 25 minutes past the hour it is $25 \times 6° = 150°$ past the 12.

2 The hour hand moves 30° per hour plus 0.5° per minute. At 6:25 it is $6{\times}30°{+}25{\times}0.5° = 180°{+}12.5° = 192.5°$ past the 12.

3 The hands are $192.5° - 150° = 42.5°$ apart.

TIP Each hour mark on a clock face is $360° \div 12 = 30°$ from the next.

WHY NOT (B) forgets that the hour hand moves past the hour mark (0.5° per minute). (D) gives the larger angle, not the smaller one.

23. **A** 105°

1 Supplementary angles add up to 180°: $(5x - 40) + (4x + 13) = 180$.

2 Combine like terms: $9x - 27 = 180$.

3 Add 27 to both sides: $9x = 180 + 27 = 207$.

4 Divide both sides by 9: $x = 207 \div 9 = 23$.

5 Plug $x = 23$ into $4x + 13$: $4(23) + 13 = 105$, so the angle measures 105°.

WHY NOT (D) sets the two angles equal instead of adding them.

24. **A** 104

1 A hexagon has 6 sides, so its interior angles add up to $(6 - 2) \times 180° = 720°$.

2 Add the known angles: $144{+}127{+}105{+}113{+}127 = 616$.

3 Subtract: $x = 720 - 616 = 104$.

WHY NOT (C) divides the angle sum by the number of angles, as if all the angles were equal. (D) uses $(n - 1) \times 180°$ for the angle sum instead of $(n - 2) \times 180°$.

25. **D** 10

1 Each exterior angle is $180° - 144° = 36°$.

2 The exterior angles add up to 360°, so the number of sides is $360 \div 36 = 10$.

3 Check: $(10 - 2) \times 180° \div 10 = 144°$. ✓

WHY NOT (A) divides 180° instead of 360° by the exterior angle.

CHAPTER 17

Triangles & the Pythagorean Theorem

Tested on: Mathematics Knowledge • Arithmetic Reasoning

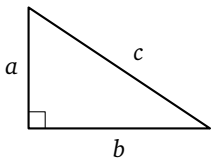
Key Concepts

A triangle has three sides and three angles. Two facts solve most triangle questions on the ASVAB: the angles always add up to 180°, and in a right triangle $a^2 + b^2 = c^2$.

Angles in a triangle

- The three angles add up to 180°. Missing angle = 180° − (the other two).
- **Isosceles** triangle: two equal sides, and the two angles opposite them (the *base angles*) are equal. **Equilateral**: three equal sides and three 60° angles. **Scalene**: no equal sides.
- By angles: **acute** (all angles less than 90°), **right** (one 90° angle), **obtuse** (one angle more than 90°).
- An **exterior angle** (made by extending one side) equals the sum of the two *remote* interior angles.
- **Triangle inequality:** any two sides must add up to *more* than the third side.

The Pythagorean theorem (right triangles only)



$$a^2 + b^2 = c^2$$

c is the **hypotenuse**: the longest side, across from the right angle. To find a leg, subtract: $a^2 = c^2 - b^2$. Memorize the common **Pythagorean triples** and their multiples: 3-4-5 (6-8-10, 9-12-15, . . .), 5-12-13, 8-15-17, 7-24-25.

Special right triangles and similar triangles

- 45°-45°-90°: sides in the ratio $1 : 1 : \sqrt{2}$ (legs s, hypotenuse $s\sqrt{2}$).
- 30°-60°-90°: sides in the ratio $1 : \sqrt{3} : 2$ (short leg s, long leg $s\sqrt{3}$, hypotenuse $2s$).
- **Similar** triangles have the same shape: equal angles and proportional sides. Shadows cast at the same time of day make similar triangles.

WORKED EXAMPLE

A 13-foot ladder leans against a wall. Its foot is 5 feet from the wall. How high up the wall does the ladder reach?

Solution. The ladder is the hypotenuse: $5^2 + h^2 = 13^2$, so $h^2 = 169 - 25 = 144$ and $h = \sqrt{144} = 12$ feet.

No-Calculator Tip Spot the triple before you square anything. For legs 15 and 20, divide by 5 to get 3 and 4; the hypotenuse is $5 \times 5 = 25$.

Common Traps

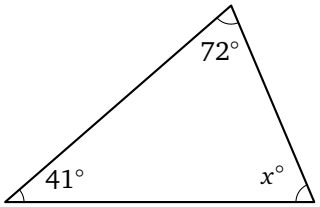
- Forgetting the square root: $a^2 + b^2$ is c^2, not c.
- Adding the squares when you need a leg; subtract from the hypotenuse instead.
- Mixing up $\sqrt{2}$ (45°-45°-90°) and $\sqrt{3}$ (30°-60°-90°).
- In similar triangles, *multiplying* by the scale factor, not *adding* the difference.

Practice Set 17

Score: _____ / 25 Target time: 26 minutes

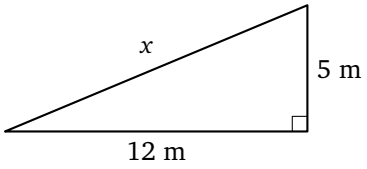
Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

1. In the triangle shown, two angles measure 41° and 72°. What is the value of x?



A 57 **B** 62 **C** 67 **D** 247

2. The legs of a right triangle measure 12 meters and 5 meters. How long is the hypotenuse?

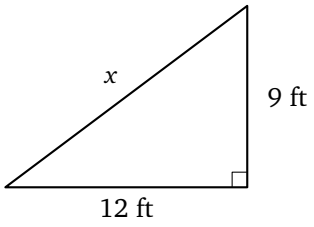


A 13 m **B** 17 m **C** 18 m **D** 169 m

3. Two angles of a triangle measure 45° and 79°. What is the measure of the third angle?

A 51° **B** 56° **C** 124° **D** 236°

4. A right triangle has legs of 12 feet and 9 feet. What is the length of the hypotenuse?

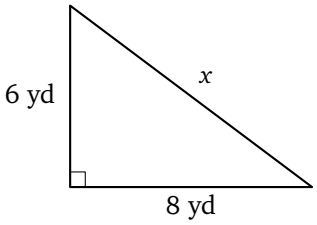


A 10 ft **B** 13 ft **C** 15 ft **D** 21 ft

5. A triangle has angles of 54° and 78°. Which of the following describes the triangle?

A an equilateral triangle
B an acute triangle
C an obtuse triangle
D a right triangle

6. In right triangle RST, the legs are 8 yards and 6 yards long. What is the length x of the hypotenuse?



A 9 yd **B** 10 yd **C** 14 yd **D** 100 yd

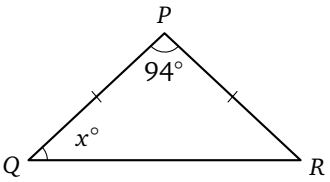
7. A triangle has angles of 24° and 66°. Which of the following describes the triangle?

A a right triangle
B an acute triangle
C an equilateral triangle
D an obtuse triangle

8. In triangle RST, $\angle R$ measures 70° and $\angle S$ measures 58°. What is the measure of $\angle T$?

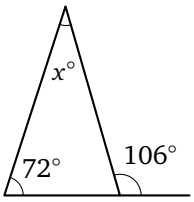
A 52° **B** 110° **C** 128° **D** 232°

9. The triangle shown is isosceles, with the two marked sides equal. The angle between the equal sides measures 94°. What is the value of x?



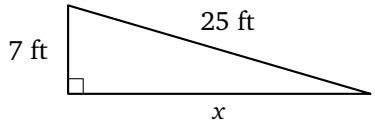
A 43 **B** 86 **C** 94 **D** 133

10. In the figure, one side of the triangle is extended to form a 106° exterior angle. One remote interior angle measures 72°. What is the value of x?



A 34 **B** 74 **C** 108 **D** 178

11. A right triangle has a hypotenuse of 25 feet and a leg of 7 feet. How long is its other leg?



A 18 ft **B** 19 ft **C** 23 ft **D** 24 ft

12. A rectangular TV screen measures 25 inches on the diagonal and is 20 inches wide. How tall is the screen?

A 5 in **B** 14 in **C** 15 in **D** 225 in

13. Two sides of a triangular garden measure 14 yards and 21 yards. Which of the following could be the length, in yards, of the third side?

A 1 **B** 4 **C** 7 **D** 11

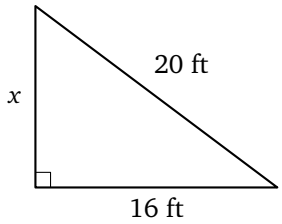
14. Petty Officer Brooks's patrol moves 8 kilometers east, then 6 kilometers north. What is the straight-line distance from the patrol back to its starting point?

A 10 km **B** 13 km **C** 14 km **D** 100 km

15. Carlos has two boards, 20 feet and 21 feet long, and wants to make a triangular frame with a third board. Which of these board lengths, in feet, would work?

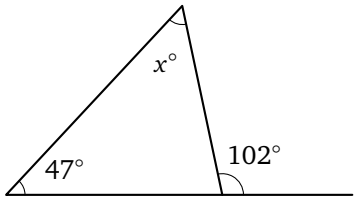
A 1 **B** 30 **C** 41 **D** 48

16. In the right triangle shown, the hypotenuse is 20 feet and one leg is 16 feet. What is the value of x?



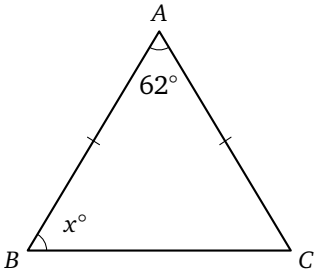
A 4 ft **B** 12 ft **C** 15 ft **D** 144 ft

17. The exterior angle shown measures 102°, and one of the remote interior angles measures 47°. What is the value of x?



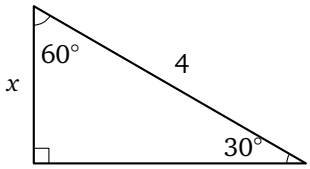
A 35 **B** 40 **C** 55 **D** 149

18. In isosceles triangle ABC, the vertex angle $\angle A$ is 62°. What is the value of x, the measure of a base angle?



A 59 **B** 62 **C** 118 **D** 149

19. In a 30°-60°-90° triangle, the hypotenuse is 4 units long. How many units long is the shorter leg?

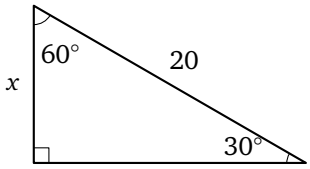


A 2 **B** $2\sqrt{2}$ **C** $2\sqrt{3}$ **D** 8

20. A 4-foot fence post casts a shadow 7 feet long. At the same time, a building casts a shadow 203 feet long. How tall is the building?

A 112 ft **B** 116 ft
C 200 ft **D** 355.25 ft

21. The figure shows a 30°-60°-90° triangle in which the hypotenuse is 20 units long. What is the value of x?

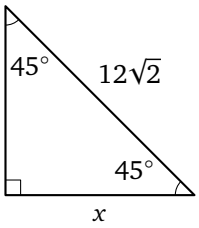


A 10 **B** $10\sqrt{2}$ **C** $10\sqrt{3}$ **D** 40

22. Andre, who is 5 feet tall, casts a shadow 12 feet long. At the same time, a flagpole casts a shadow 108 feet long. How tall is the flagpole?

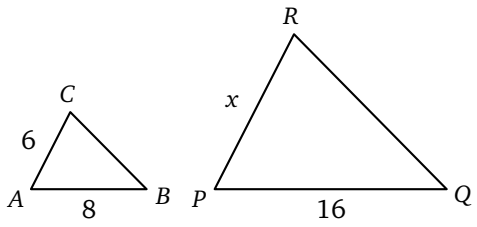
A 45 ft **B** 101 ft **C** 259.2 ft **D** 540 ft

23. In the 45°-45°-90° triangle shown, the hypotenuse measures $12\sqrt{2}$ units. What is the length x of each leg?



A $6\sqrt{2}$ **B** 12 **C** $12\sqrt{2}$ **D** 24

24. $\triangle ABC \sim \triangle PQR$. If $AB = 8$, $AC = 6$, and $PQ = 16$, what is the length x of PR?



A 3 **B** 8 **C** 10 **D** 12

25. A rectangular parking lot is 100 feet long and 75 feet wide. Tyrone walks diagonally across it instead of along two sides. How many feet shorter is the diagonal route?

A 25 ft **B** 30 ft **C** 50 ft **D** 125 ft

Answers & Explanations 17

1 C	**6** B	**11** D	**16** B	**21** A
2 A	**7** A	**12** C	**17** C	**22** A
3 B	**8** A	**13** D	**18** A	**23** B
4 C	**9** A	**14** A	**19** A	**24** D
5 B	**10** A	**15** B	**20** B	**25** C

1. **C** 67

1 The three angles of any triangle add up to 180°.

2 Add the two known angles: 41 + 72 = 113.

3 Subtract from 180: $x = 180 - 113 = 67$.

WHY NOT (D) uses 360° instead of 180° for the angle sum.

2. **A** 13 m

1 Use the Pythagorean theorem: $a^2 + b^2 = c^2$, where c is the hypotenuse.

2 Square the legs and add: $5^2+12^2 = 25+144 = 169$.

3 Take the square root: $c = \sqrt{169} = 13$ m.

WHY NOT (B) adds the legs instead of using $a^2+b^2 = c^2$. (D) forgets to take the square root.

3. **B** 56°

1 The three angles of any triangle add up to 180°.

2 Add the two known angles: 45 + 79 = 124.

3 Subtract from 180: 180 – 124 = 56, so the third angle is 56°.

WHY NOT (C) adds the two given angles and stops. (D) uses 360° instead of 180° for the angle sum.

4. **C** 15 ft

1 Use the Pythagorean theorem: $a^2 + b^2 = c^2$, where c is the hypotenuse.

2 Square the legs and add: $9^2+12^2 = 81+144 = 225$.

3 Take the square root: $c = \sqrt{225} = 15$ ft.

TIP Spot the triple: 9-12-? is 3-4-5 times 3, so the hypotenuse is $5 \times 3 = 15$.

WHY NOT (D) adds the legs instead of using $a^2 + b^2 = c^2$.

5. **B** an acute triangle

1 Find the third angle: 180° – 54° – 78° = 48°.

2 The largest angle is 78°, which is less than 90°, so all three angles are acute. It is an acute triangle.

WHY NOT (A) would need all three angles to be 60°. (C) would need an angle greater than 90°, but the largest angle is 78°. (D) would need a 90° angle, but the angles are 54°, 78°, and 48°.

6. **B** 10 yd

1 Use the Pythagorean theorem: $a^2 + b^2 = c^2$, where c is the hypotenuse.

2 Square the legs and add: $6^2 + 8^2 = 36 + 64 = 100$.

3 Take the square root: $c = \sqrt{100} = 10$ yd.

TIP Spot the triple: 6-8-? is 3-4-5 times 2, so the hypotenuse is $5 \times 2 = 10$.

WHY NOT (C) adds the legs instead of using $a^2+b^2 = c^2$. (D) forgets to take the square root.

7. **A** a right triangle

1 Find the third angle: 180° – 24° – 66° = 90°.

2 The largest angle is 90°, which is exactly 90°. It is a right triangle.

WHY NOT (B) would need all three angles to be less than 90°, but one angle is 90°. (C) would need all three angles to be 60°. (D) would need an angle greater than 90°, but the largest angle is 90°.

8. **A** 52°

1 The three angles of any triangle add up to 180°.

2 Add the two known angles: 70 + 58 = 128.

3 Subtract from 180: 180 – 128 = 52, so the third angle is 52°.

WHY NOT (B) subtracts only one of the given angles. (C) adds the two given angles and stops. (D) uses 360° instead of 180° for the angle sum.

9. **A** 43

1 In an isosceles triangle the two base angles (opposite the equal sides) are equal. Call each one x.

2 The angles add up to 180: $x + x + 94 = 180$, so $2x = 180 - 94 = 86$.

3 Divide by 2: $x = 86 \div 2 = 43$.

WHY NOT (B) forgets to split what is left between the two equal base angles. (C) assumes the base angle equals the vertex angle. (D) uses 360° instead of 180°.

10. **A** 34

1 An exterior angle of a triangle equals the sum of the two remote interior angles.

2 So $72 + x = 106$, and $x = 106 - 72 = 34$.

WHY NOT (B) gives the interior angle next to the exterior angle. (C) subtracts the known angle from 180 and ignores the exterior angle. (D) adds the angles instead of subtracting.

11. **D** 24 ft

1 The hypotenuse is the longest side, so use $a^2 + b^2 = c^2$ and solve for the missing leg: $\text{leg}^2 = c^2 - (\text{other leg})^2$.

2 $25^2 - 7^2 = 625 - 49 = 576$.

3 Take the square root: $\sqrt{576} = 24$ ft.

WHY NOT (A) subtracts the lengths instead of their squares.

12. **C** 15 in

1 The width, the height, and the diagonal of the screen form a right triangle with the diagonal as the hypotenuse.

2 Missing leg: $25^2 - 20^2 = 625 - 400 = 225$.

3 $\sqrt{225} = 15$, so the answer is 15 inches.

TIP Shortcut: divide by 5 and you get the 3-4-5 triple, so the missing side is $3 \times 5 = 15$.

WHY NOT (A) subtracts the lengths instead of their squares. (D) forgets to take the square root.

13. **D** 11

1 Triangle inequality: any two sides must add up to *more* than the third side.

2 So the third side must be longer than $21 - 14 = 7$ and shorter than $14 + 21 = 35$: between 7 and 35, not equal to either.

3 Only 11 is in that range.

WHY NOT (A) is too short: $1 + 14 < 21$. (C) equals $21 - 14$; the two shorter sides would lie flat along the longest side.

14. **A** 10 km

1 Traveling east and then north makes a right angle, so the two legs of the trip and the straight-line distance form a right triangle.

2 $8^2 + 6^2 = 64 + 36 = 100$.

3 $\sqrt{100} = 10$, so the answer is 10 kilometers.

TIP Shortcut: divide by 2 and you get the 3-4-5 triple, so the hypotenuse is $5 \times 2 = 10$.

WHY NOT (C) adds the two distances instead of using the Pythagorean theorem. (D) forgets to take the square root.

15. **B** 30

1 Triangle inequality: any two sides must add up to *more* than the third side.

2 So the third side must be longer than $21 - 20 = 1$ and shorter than $20 + 21 = 41$: between 1 and 41, not equal to either.

3 Only 30 is in that range.

WHY NOT (A) equals $21 - 20$; the two shorter sides would lie flat along the longest side. (C) equals $20 + 21$; the two given sides would lie flat along the third side. (D) is too long: $20 + 21 < 48$.

16. **B** 12 ft

1 The hypotenuse is the longest side, so use $a^2 + b^2 = c^2$ and solve for the missing leg: $\text{leg}^2 = c^2 - (\text{other leg})^2$.

2 $20^2 - 16^2 = 400 - 256 = 144$.

3 Take the square root: $\sqrt{144} = 12$ ft.

TIP Shortcut: divide by 4 and you get the 3-4-5 triple, so the missing side is $3 \times 4 = 12$.

WHY NOT (A) subtracts the lengths instead of their squares. (D) forgets to take the square root.

17. **C** 55

1 An exterior angle of a triangle equals the sum of the two remote interior angles.

2 So $47 + x = 102$, and $x = 102 - 47 = 55$.

WHY NOT (D) adds the angles instead of subtracting.

18. **A** 59

1 In an isosceles triangle the two base angles (opposite the equal sides) are equal. Call each one x.

2 The angles add up to 180: $x + x + 62 = 180$, so $2x = 180 - 62 = 118$.

3 Divide by 2: $x = 118 \div 2 = 59$.

WHY NOT (B) assumes the base angle equals the vertex angle. (C) forgets to split what is left between the two equal base angles. (D) uses 360° instead of 180°.

19. **A** 2

1 In a 30°-60°-90° triangle the sides are in the ratio $1 : \sqrt{3} : 2$ (short leg : long leg : hypotenuse). The short leg is across from the 30° angle.

2 Short leg = hypotenuse $\div 2 = 4 \div 2 = 2$.

WHY NOT (B) divides the hypotenuse by $\sqrt{2}$, the 45°-45°-90° rule for a leg. (C) gives the long leg, not the short leg. (D) doubles the hypotenuse instead of halving it.

20. **B** 116 ft

1 Objects and their shadows at the same time of day form similar triangles, so $\frac{\text{height}}{\text{shadow}}$ is the same for both: $\frac{h}{203} = \frac{4}{7}$.

2 Cross-multiply: $7h = 4 \times 203 = 812$.

3 Divide by 7: $h = 812 \div 7 = 116$ feet.

WHY NOT (C) assumes the building is 3 feet shorter than its shadow, just as the post is, instead of using a ratio. (D) sets up the proportion upside down.

21. **A** 10

1 In a 30°-60°-90° triangle the sides are in the ratio $1 : \sqrt{3} : 2$ (short leg : long leg : hypotenuse). The short leg is across from the 30° angle.

2 Short leg = hypotenuse $\div 2 = 20 \div 2 = 10$.

WHY NOT (B) divides the hypotenuse by $\sqrt{2}$, the 45°-45°-90° rule for a leg. (C) gives the long leg, not the short leg. (D) doubles the hypotenuse instead of halving it.

22. **A** 45 ft

1 Objects and their shadows at the same time of day form similar triangles, so $\frac{\text{height}}{\text{shadow}}$ is the same for both: $\frac{h}{108} = \frac{5}{12}$.

2 Cross-multiply: $12h = 5 \times 108 = 540$.

3 Divide by 12: $h = 540 \div 12 = 45$ feet.

WHY NOT (B) assumes the flagpole is 7 feet shorter than its shadow, just as Andre is, instead of using a ratio. (C) sets up the proportion upside down. (D) multiplies by the height but forgets to divide by the shadow.

23. **B** 12

1 In a 45°-45°-90° triangle the sides are in the ratio $1 : 1 : \sqrt{2}$ (leg : leg : hypotenuse).

2 Leg = hypotenuse $\div \sqrt{2} = 12\sqrt{2} \div \sqrt{2} = 12$.

WHY NOT (A) halves the hypotenuse, which is the 30°-60°-90° rule for the short leg. (C) solves $x^2 = c^2$ instead of $x^2 + x^2 = c^2$, forgetting the second leg. (D) multiplies the hypotenuse by $\sqrt{2}$ instead of dividing.

24. **D** 12

1 The order of the letters gives the matching vertices: $A \leftrightarrow P$, $B \leftrightarrow Q$, $C \leftrightarrow R$. So x is side PR, which matches AC (6), and PQ matches AB.

2 Corresponding sides are proportional: $\frac{PR}{AC} = \frac{PQ}{AB}$, so $\frac{x}{6} = \frac{16}{8}$.

3 The scale factor is $16 \div 8 = 2$.

4 Multiply: $x = 6 \times 2 = 12$.

WHY NOT (A) divides by the scale factor instead of multiplying.

25. **C** 50 ft

1 Going along two sides: $75 + 100 = 175$ feet.

2 The diagonal is the hypotenuse: $\sqrt{75^2 + 100^2} = \sqrt{15{,}625} = 125$ feet.

3 The savings: $175 - 125 = 50$ feet.

TIP Shortcut: divide by 25 and you get the 3-4-5 triple, so the hypotenuse is $5 \times 25 = 125$.

WHY NOT (A) subtracts the width from the length. (D) gives the length of the diagonal, not the savings.

CHAPTER 18

Perimeter & Area

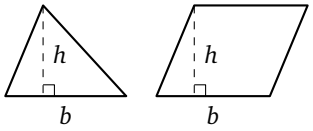
Tested on: Mathematics Knowledge • Arithmetic Reasoning

Key Concepts

Perimeter is the distance around a figure: add up the sides, and the answer is in plain units (feet, meters). **Area** is the amount of surface inside: it is measured in *square* units (ft^2, or square feet).

Formulas

Shape	Perimeter	Area
Rectangle	$2l + 2w$	$l \times w$
Square	$4s$	s^2
Triangle	add the sides	$\frac{1}{2} \times b \times h$
Parallelogram	add the sides	$b \times h$
Trapezoid	add the sides	$\frac{1}{2}(b_1 + b_2) \times h$



The height h is always measured at a *right angle* to the base (the dashed lines). It is not a slanted side.

Composite shapes, units, and scaling

- Split an odd shape into rectangles and add the areas, or take a big rectangle and subtract the missing piece.
- 1 yard = 3 feet, so 1 square yard = $3 \times 3 = 9$ square feet. Likewise 1 square foot = $12 \times 12 = 144$ square inches.
- If every side is multiplied by k, the perimeter is multiplied by k and the area by k^2: doubling the sides makes the area 4 times as large.

WORKED EXAMPLE

Carpet costs \$25 per square yard. How much does it cost to carpet a room that is 15 feet by 12 feet?
Solution. Area = $15 \times 12 = 180$ square feet. Change to square yards: $180 \div 9 = 20$ square yards. Cost = $20 \times \$25 = \$500$.

No-Calculator Tip Convert lengths *before* you multiply when the numbers divide evenly: 15 ft × 12 ft is 5 yd × 4 yd = 20 square yards, with no large division at all.

Common Traps

- Mixing up perimeter (add the sides) and area (multiply).
- Forgetting the $\frac{1}{2}$ in the triangle and trapezoid formulas.
- Using a slanted side instead of the height.

- Dividing by 3 instead of 9 to change square feet to square yards.

Practice Set 18

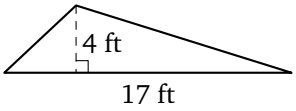
Score: _____ / 25 Target time: 26 minutes

Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

1. A square has sides of 10 centimeters. What is its area?

A 20 cm^2 **B** 40 cm^2 **C** 50 cm^2 **D** 100 cm^2

2. What is the area of the triangle shown, with base 17 feet and height 4 feet?

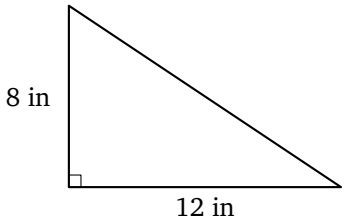


A 21 ft^2 **B** 34 ft^2 **C** 44 ft^2 **D** 68 ft^2

3. A square has sides of 12 meters. What is its area?

A 24 m^2 **B** 48 m^2 **C** 144 m^2 **D** 148 m^2

4. A right triangle has legs of 12 inches and 8 inches. What is its area?

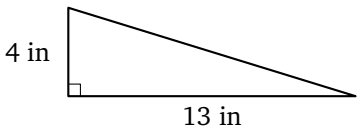


A 20 in^2 **B** 24 in^2 **C** 48 in^2 **D** 96 in^2

5. A square has an area of 64 square inches. What is the length of one side?

A 6 in **B** 7 in **C** 8 in **D** 16 in

6. What is the area of the right triangle shown, whose legs measure 13 inches and 4 inches?



A 13 in^2 **B** 17 in^2 **C** 20 in^2 **D** 26 in^2

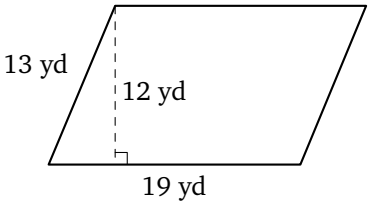
7. The area of a square is 169 square feet. How long is each side?

A 9 ft **B** 12 ft **C** 13 ft **D** 42.25 ft

8. A barracks room floor is a rectangle 18 feet long and 16 feet wide. What is its area?

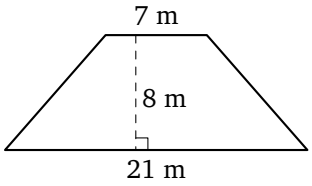
A 34 ft^2 **B** 68 ft^2 **C** 144 ft^2 **D** 288 ft^2

9. The parallelogram shown has a base of 19 yards, a height of 12 yards, and slanted sides of 13 yards. What is its perimeter?



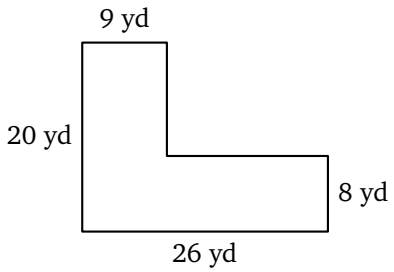
A 62 yd **B** 64 yd **C** 70 yd **D** 228 yd

10. The trapezoid shown has parallel bases of 21 meters and 7 meters and a height of 8 meters. What is its area?



A 112 m^2 **B** 168 m^2 **C** 224 m^2 **D** 588 m^2

11. The figure shows a lawn. All corners are right angles. The sides shown measure 26 yards, 20 yards, 9 yards, and 8 yards. What is the area?

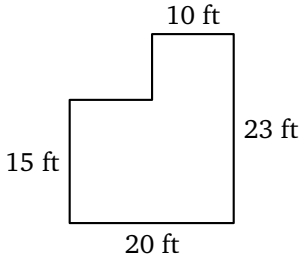


A 316 yd^2 **B** 320 yd^2 **C** 448 yd^2 **D** 520 yd^2

12. A barracks hallway floor is a rectangle 16 feet by 10 feet. It will be covered with square tiles that measure 2 feet on each side. How many tiles are needed?

A 26 **B** 30 **C** 38 **D** 40

13. The figure shows the floor of a storage building. All corners are right angles. The sides shown measure 20 feet, 23 feet, 10 feet, and 15 feet. What is the area?

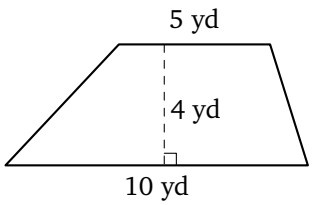


A 380 ft^2 **B** 388 ft^2 **C** 390 ft^2 **D** 460 ft^2

14. Airman Okafor must fence a rectangular equipment yard on base that measures 80 feet by 55 feet. The fencing costs $11 per foot. What is the total cost of the fence?

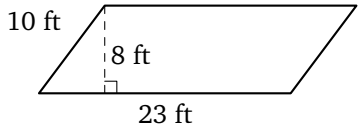
A $1,485 **B** $2,365 **C** $2,970 **D** $48,400

15. What is the area of a trapezoid with bases of 5 yards and 10 yards and a height of 4 yards?



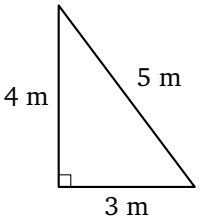
A 20 yd^2 **B** 30 yd^2 **C** 40 yd^2 **D** 60 yd^2

16. What is the area of a parallelogram with a base of 23 feet, a height of 8 feet, and slanted sides of 10 feet, as shown?



A 66 ft^2 **B** 92 ft^2 **C** 184 ft^2 **D** 230 ft^2

17. A right triangle has sides of 4 meters, 3 meters, and 5 meters. What is its area?



A 4 m^2 **B** 6 m^2 **C** 10 m^2 **D** 12 m^2

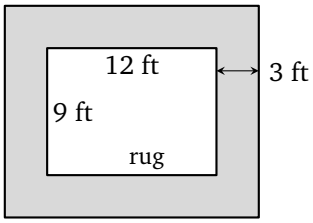
18. The area of a rectangle is 108 square feet, and the rectangle is 12 feet long. What is its perimeter?

A 9 ft **B** 21 ft **C** 38 ft **D** 42 ft

19. Grace is buying carpet for a recreation room that measures 12 feet by 18 feet. The carpet costs $18 per square yard. What is the cost of the carpet?

A $24 **B** $432 **C** $1,296 **D** $3,888

20. A 12-foot by 9-foot rug is centered in a rectangular room, leaving a strip of bare floor 3 feet wide along every wall. What is the area of the bare floor alone?

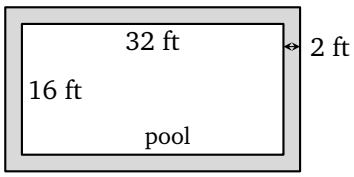


A 72 ft^2 **B** 162 ft^2 **C** 178 ft^2 **D** 270 ft^2

21. Each side of a square is multiplied by the same number, and the area of the square becomes 36 times as large. By what number was each side multiplied?

A 6 **B** 9 **C** 18 **D** 36

22. A rectangular swimming pool is 32 feet long and 16 feet wide. A concrete walkway 2 feet wide surrounds the pool. What is the area of the walkway alone?



A 100 ft^2 **B** 192 ft^2 **C** 200 ft^2 **D** 208 ft^2

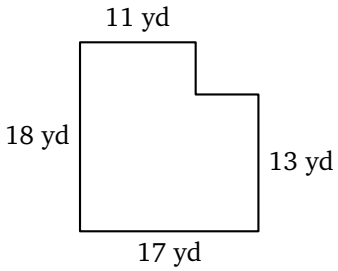
23. A rectangle that is 9 centimeters by 3 centimeters has a perimeter of 24 centimeters. If its length and width are both multiplied by 4, what is the perimeter of the new rectangle?

A 24 cm **B** 40 cm **C** 78 cm **D** 96 cm

24. Brandon is buying carpet for a bedroom that measures 24 feet by 15 feet. The carpet costs $32 per square yard. What is the cost of the carpet?

A $1,280 **B** $1,320 **C** $3,840 **D** $11,520

25. The figure shows a garden. All corners are right angles. The sides shown measure 17 yards, 18 yards, 11 yards, and 13 yards. What is the perimeter?



A 70 yd **B** 78 yd **C** 81 yd **D** 276 yd

Answers & Explanations 18

1 D	**6** D	**11** A	**16** C	**21** A
2 B	**7** C	**12** D	**17** B	**22** D
3 C	**8** D	**13** A	**18** D	**23** D
4 C	**9** B	**14** C	**19** B	**24** A
5 C	**10** A	**15** B	**20** B	**25** A

1. **D** 100 cm^2

1 Area of a square = side × side = s^2.

2 $10 \times 10 = 100$, so the area is 100 square centimeters.

WHY NOT (A) doubles the side instead of squaring it. (B) gives the perimeter, not the area.

2. **B** 34 ft^2

1 The base and the height (the dashed line, at a right angle to the base) are given.

2 Area of a triangle = $\frac{1}{2} \times$ base × height.

3 $\frac{1}{2} \times 17 \times 4 = \frac{1}{2} \times 68 = 34$, so the area is 34 square feet.

WHY NOT (A) adds the base and height. (D) forgets to multiply by $\frac{1}{2}$.

3. **C** 144 m^2

1 Area of a square = side × side = s^2.

2 $12 \times 12 = 144$, so the area is 144 square meters.

WHY NOT (A) doubles the side instead of squaring it. (B) gives the perimeter, not the area.

4. **C** 48 in^2

1 In a right triangle the two legs are a base and a height.

2 Area of a triangle = $\frac{1}{2} \times$ base × height.

3 $\frac{1}{2} \times 12 \times 8 = \frac{1}{2} \times 96 = 48$, so the area is 48 square inches.

WHY NOT (A) adds the base and height. (B) takes half of the base and half of the height. (D) forgets to multiply by $\frac{1}{2}$.

5. **C** 8 in

1 Area of a square = $s \times s$, so the side is the number that times itself gives the area.

2 $8 \times 8 = 64$, so each side is 8 inches.

WHY NOT (D) divides the area by 4, as if it were the perimeter.

6. **D** 26 in^2

1 In a right triangle the two legs are a base and a height.

2 Area of a triangle = $\frac{1}{2} \times$ base × height.

3 $\frac{1}{2} \times 13 \times 4 = \frac{1}{2} \times 52 = 26$, so the area is 26 square inches.

WHY NOT (A) takes half of the base and half of the height. (B) adds the base and height.

7. **C** 13 ft

1 Area of a square = $s \times s$, so the side is the number that times itself gives the area.

2 $13 \times 13 = 169$, so each side is 13 feet.

WHY NOT (D) divides the area by 4, as if it were the perimeter.

8. **D** 288 ft^2

1 Area of a rectangle = length × width.

2 $18 \times 16 = 288$, so the area is 288 square feet.

WHY NOT (A) adds the length and width instead of multiplying. (B) gives the perimeter, not the area. (C) takes half, as if the shape were a triangle.

9. **B** 64 yd

1 The perimeter is the distance around: two bases and two slanted sides. The height is inside the figure, so it is not part of the perimeter.

2 $19+13+19+13 = 64$, so the perimeter is 64 yards.

WHY NOT (A) uses the height instead of the slanted side. (D) gives the area, not the perimeter.

10. **A** 112 m^2

1 Area of a trapezoid = $\frac{1}{2} \times (b_1 + b_2) \times h$: average the two parallel bases, then multiply by the height.

2 Add the bases: 21 + 7 = 28.

3 $\frac{1}{2} \times 28 \times 8 = 14 \times 8 = 112$, so the area is 112 square meters.

WHY NOT (B) uses only the longer base. (C) forgets to multiply by $\frac{1}{2}$. (D) multiplies the two bases instead of adding them.

11. **A** 316 yd^2

1 Find the missing corner: it is 26 – 9 = 17 yd wide and 20 – 8 = 12 yd tall.

2 Area of the whole 26×20 rectangle: 26×20 = 520.

3 Subtract the missing corner: 520 – 17 × 12 = 520 – 204 = 316 square yards.

TIP Or split the shape into two rectangles: 9 × 20 = 180 and 17 × 8 = 136; 180 + 136 = 316.

WHY NOT (C) subtracts the wrong rectangle. (D) forgets to subtract the missing corner.

12. **D** 40

1 Area of the floor: 16 × 10 = 160 square feet.

2 Each tile covers 2 × 2 = 4 square feet.

3 Number of tiles: 160 ÷ 4 = 40.

TIP Or count tiles along each side: 16 ÷ 2 = 8 by 10 ÷ 2 = 5, and 8 × 5 = 40.

WHY NOT (A) works with the perimeter instead of the area.

13. **A** 380 ft^2

1 Find the missing corner: it is 20 – 10 = 10 ft wide and 23 – 15 = 8 ft tall.

2 Area of the whole 20×23 rectangle: 20×23 = 460.

3 Subtract the missing corner: 460 – 10 × 8 = 460 – 80 = 380 square feet.

TIP Or split the shape into two rectangles: 10 × 23 = 230 and 10 × 15 = 150; 230 + 150 = 380.

WHY NOT (D) forgets to subtract the missing corner.

14. **C** $2,970

1 A fence goes around the edge, so find the perimeter: 2 × 80 + 2 × 55 = 160 + 110 = 270 feet.

2 Multiply by the price per foot: 270 × 11 = 2,970, so the fence costs $2,970.

WHY NOT (A) fences only one length and one width (half the perimeter). (B) leaves out one side. (D) uses the area instead of the perimeter.

15. **B** 30 yd^2

1 Area of a trapezoid = $\frac{1}{2} \times (b_1 + b_2) \times h$: average the two parallel bases, then multiply by the height.

2 Add the bases: 10 + 5 = 15.

3 $\frac{1}{2} \times 15 \times 4 = 15 \times 2 = 30$, so the area is 30 square yards.

WHY NOT (A) uses only the shorter base. (C) uses only the longer base. (D) forgets to multiply by $\frac{1}{2}$.

16. **C** 184 ft^2

1 Area of a parallelogram = base × height, where the height is the dashed line at a right angle to the base (not the slanted side).

2 23 × 8 = 184, so the area is 184 square feet.

TIP Why it works: cut the triangle off one end and slide it to the other end. You get a 23 × 8 rectangle.

WHY NOT (A) gives the perimeter, not the area. (B) takes $\frac{1}{2}$, as for a triangle. (D) multiplies by the slanted side instead of the height.

17. **B** 6 m^2

1 In a right triangle the two legs (the sides that form the right angle) are the base and the height. The longest side, 5 meters, is the hypotenuse and is not used.

2 Area of a triangle = $\frac{1}{2}$ × base × height.

3 $\frac{1}{2} \times 3 \times 4 = \frac{1}{2} \times 12 = 6$, so the area is 6 square meters.

WHY NOT (C) uses the hypotenuse as the height. (D) forgets to multiply by $\frac{1}{2}$.

18. **D** 42 ft

1 Find the width: 108 ÷ 12 = 9 ft.

2 Perimeter = 2 × 12 + 2 × 9 = 24 + 18 = 42, so the perimeter is 42 feet.

WHY NOT (A) finds the width and stops. (B) adds only one length and one width.

19. **B** $432

1 Change each length to yards: 12 ÷ 3 = 4 yd and 18 ÷ 3 = 6 yd.

2 Area in square yards: 4 × 6 = 24 square yards.

3 Cost: 24 × 18 = 432, so the carpet costs $432.

TIP Or find square feet first (12 × 18 = 216) and divide by 9: 216 ÷ 9 = 24 square yards.

WHY NOT (A) finds the number of square yards but forgets to multiply by the price. (C) divides by 3 instead of 9 (a square yard is 3 ft by 3 ft). (D) forgets to change square feet to square yards.

20. **B** 162 ft^2

1 The bare floor adds 3 feet on *both* ends of each side, so the outer rectangle is 12 + 2 × 3 = 18 by 9 + 2 × 3 = 15 feet.

2 Outer area: 18 × 15 = 270 square feet.

3 Rug area: 12 × 9 = 108 square feet.

4 Bare floor alone: 270 – 108 = 162 square feet.

WHY NOT (A) adds the bare floor width to only one end of each side. (D) gives the total area, including the rug.

21. **A** 6

1 If each side is multiplied by k, the area is multiplied by k^2.

2 So $k^2 = 36$, and $k = \sqrt{36} = 6$.

TIP Check: $6 \times 6 = 36$.

WHY NOT (B) divides the area factor by 4 (the number of sides) instead of taking its square root. (C) halves the area factor instead of taking its square root. (D) gives the area factor itself.

22. **D** 208 ft^2

1 The walkway adds 2 feet on *both* ends of each side, so the outer rectangle is $32 + 2 \times 2 = 36$ by $16 + 2 \times 2 = 20$ feet.

2 Outer area: $36 \times 20 = 720$ square feet.

3 Pool area: $32 \times 16 = 512$ square feet.

4 Walkway alone: $720 - 512 = 208$ square feet.

WHY NOT (A) adds the walkway width to only one end of each side. (B) multiplies the perimeter by the width and misses the four corner squares.

23. **D** 96 cm

1 Perimeter is a length (a sum of sides), so when every side is multiplied by 4, the perimeter is multiplied by 4 too.

2 $24 \times 4 = 96$, so the new perimeter is 96 centimeters.

WHY NOT (A) gives the original perimeter. (B) adds 4 to the length and to the width instead of multiplying. (C) multiplies only the length by 4.

24. **A** $1,280

1 Change each length to yards: $24 \div 3 = 8$ yd and $15 \div 3 = 5$ yd.

2 Area in square yards: $8 \times 5 = 40$ square yards.

3 Cost: $40 \times 32 = 1{,}280$, so the carpet costs $1,280.

TIP Or find square feet first ($24 \times 15 = 360$) and divide by 9: $360 \div 9 = 40$ square yards.

WHY NOT (C) divides by 3 instead of 9 (a square yard is 3 ft by 3 ft). (D) forgets to change square feet to square yards.

25. **A** 70 yd

1 Find the missing corner: it is $17 - 11 = 6$ yd wide and $18 - 13 = 5$ yd tall.

2 Go around the shape and add all six sides: $17 + 13 + 6 + 5 + 11 + 18 = 70$ yards.

TIP Shortcut: the two inside edges together make up for the missing pieces of the top and side, so the perimeter equals that of a 17×18 rectangle: $2 \times (17 + 18) = 70$.

WHY NOT (C) counts the two inside edges twice. (D) gives the area, not the perimeter.

CHAPTER 19

Circles

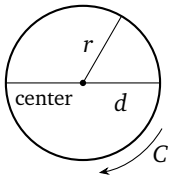
Tested on: Mathematics Knowledge • Arithmetic Reasoning

Key Concepts

Circle questions use just two formulas, and the ASVAB does *not* give them to you, so learn them by heart. Almost every mistake comes from mixing up the radius and the diameter, or the circumference and the area.

Parts of a circle

- The **radius** r runs from the center to the circle.
- The **diameter** d goes all the way across through the center: $d = 2r$ and $r = d \div 2$.
- The **circumference** C is the distance around the circle (its perimeter).
- The **area** A is the space inside, in square units.



The two formulas

$$C = 2\pi r = \pi d \qquad A = \pi r^2$$

$\pi \approx 3.14 \approx \frac{22}{7}$. Most answers are left "in terms of π": 36π simply means $36 \times \pi$. Use 3.14 or $\frac{22}{7}$ only when the question tells you to. In πr^2 only the radius is squared: if $r = 6$, then $A = \pi \times 36 = 36\pi$.

Parts of a circle's area and perimeter

- A **semicircle** is half a circle; a **quarter circle** is one fourth. Their perimeters also include the straight edges.
- A **sector** (a pie slice) with central angle $a°$ is the fraction $\frac{a}{360}$ of the circle:

$$\text{arc length} = \frac{a}{360} \times 2\pi r \qquad \text{sector area} = \frac{a}{360} \times \pi r^2$$

- **Shaded regions:** area of the whole shape minus the area of the part that is cut out. A ring between two circles is $\pi R^2 - \pi r^2$.

WORKED EXAMPLE

A circular garden has a diameter of 10 feet. Using 3.14 for π, find its area and the length of fence needed to go around it.

Solution. The radius is $10 \div 2 = 5$ feet. Area: $A = \pi r^2 \approx 3.14 \times 5^2 = 3.14 \times 25 = 78.5$ square feet. Fence: $C = \pi d \approx 3.14 \times 10 = 31.4$ feet.

No-Calculator Tip To multiply by 3.14, multiply by 3 and by 0.14 and add: $3.14 \times 25 = 75 + 3.5 = 78.5$. Scaling: if the radius is multiplied by k, the circumference is multiplied by k but the area by k^2, so doubling the radius makes the area 4 times as large.

Common Traps

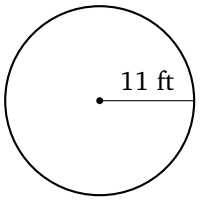
- Putting the diameter into πr^2. Halve the diameter *first*.
- Mixing up the formulas: $2\pi r$ is the distance around, πr^2 is the space inside.
- Halving after squaring: with $d = 10$, the area is $\pi \times 5^2 = 25\pi$, not $\pi \times 100 \div 2 = 50\pi$.
- Leaving out the straight edges in the perimeter of a semicircle or quarter circle.

Practice Set 19

Score: _____ / 25 Target time: 26 minutes

Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

1. In the figure, the radius of the circle is 11 feet. What is the area of the circle?



A 11π ft^2 **B** 22π ft^2 **C** 121π ft^2 **D** 484π ft^2

2. A circle has a radius of 11 feet. What is its circumference?

A 11π ft **B** 20π ft **C** 22π ft **D** 121π ft

3. What is the area of a circle with a radius of 3 centimeters?

A 6π cm^2 **B** 9π cm^2
C 16π cm^2 **D** 36π cm^2

4. A round wall clock has a radius of 5 inches. What is the distance around the edge of the clock?

A 10π in **B** 12π in **C** 20π in **D** 25π in

5. A goat is tied to a stake in an open field with a rope 14 feet long. How many square feet of grass can the goat reach? Use $\frac{22}{7}$ for π.

A 616 ft^2 **B** 638 ft^2
C 660 ft^2 **D** 2,464 ft^2

6. What is the circumference of a circle with a diameter of 38 centimeters?

A 19π cm **B** 34π cm **C** 36π cm **D** 38π cm

7. Airman Patel marks the edge of a circular landing zone with a diameter of 70 yards using a rope. How many yards of rope are needed to go all the way around? Use $\frac{22}{7}$ for π.

A 110 yd **B** 220 yd
C 440 yd **D** 3,850 yd

8. What is the area of a circle with a radius of 18 yards?

A 36π yd^2 **B** 324π yd^2
C 361π yd^2 **D** $1{,}296\pi$ yd^2

9. The circumference of a circle is 18π centimeters. What is the radius of the circle?

A 4.5 cm **B** 7 cm **C** 9 cm **D** 18 cm

10. A flower bed in the corner of a yard is shaped like a quarter circle with a radius of 8 feet. What is the area of the flower bed?

A 4π ft^2 **B** 16π ft^2 **C** 32π ft^2 **D** 64π ft^2

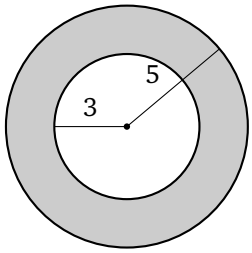
11. A circular pond has a radius of 12 feet. A gravel path 2 feet wide is built all the way around it. What is the area of the path?

A 52π ft^2 **B** 112π ft^2 **C** 196π ft^2 **D** 340π ft^2

12. Circle A has a diameter of 6 inches, and circle B has a diameter of 24 inches. The area of circle B is how many times the area of circle A?

A 4 **B** 8 **C** 16 **D** 64

13. The figure shows two circles with the same center. The large circle has a radius of 5 meters, and the small circle has a radius of 3 meters. What is the area of the shaded ring?



A 4π m^2 **B** 12π m^2 **C** 14π m^2 **D** 16π m^2

14. If the diameter of a circle is multiplied by 4, the circumference of the circle is multiplied by what number?

A 2 **B** 3 **C** 4 **D** 16

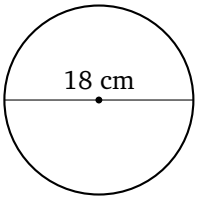
15. A window is shaped like a semicircle with a diameter of 8 feet. What is the area of the window?

A 4π ft^2 **B** 6π ft^2 **C** 7π ft^2 **D** 8π ft^2

16. A circle has an area of 36π square inches. What is the circumference of the circle?

A 6π in **B** 11π in **C** 12π in **D** 24π in

17. In the figure, the diameter of the circle is 18 centimeters. What is the area of the circle?

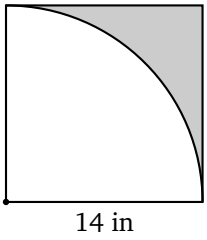


A 81π cm^2 **B** 100π cm^2
C 162π cm^2 **D** 324π cm^2

18. Ana's lawn sprinkler waters a circle with a diameter of 16 feet. How many square feet of lawn does it water? Use 3.14 for π.

A 200.96 ft^2 **B** 213.52 ft^2
C 401.92 ft^2 **D** 803.84 ft^2

19. The figure shows a square with sides of 14 inches. A quarter circle is drawn inside it with its center at the lower-left corner of the square and a radius of 14 inches. What is the area of the shaded region?



A $(196 - 59\pi)$ in^2 **B** $(196 - 49\pi)$ in^2
C 49π in^2 **D** $(196 - 7\pi)$ in^2

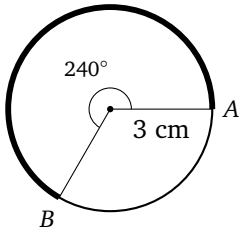
20. A lawn sprinkler sprays water 18 feet and turns back and forth through an angle of 60°. What is the area of the lawn it waters?

A 6π ft^2 **B** 54π ft^2 **C** 108π ft^2 **D** 324π ft^2

21. The diameter of a tire on Specialist Carter's cargo truck is 42 inches. About how many complete turns does it make in 1 mile? (1 mile = 5,280 feet; use $\frac{22}{7}$ for π.)

A 40 **B** 240 **C** 360 **D** 480

22. In the circle shown, the radius is 3 centimeters and the central angle is 240°. What is the length of the arc from A to B shown in bold?

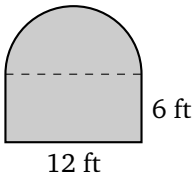


A 2π cm **B** 4π cm **C** 5π cm **D** 6π cm

23. A wheel on Hannah's lawn mower has a diameter of 7 inches. How many feet does it roll in 12 complete turns? Use $\frac{22}{7}$ for π.

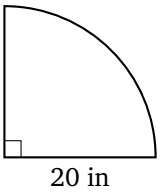
A 22 ft **B** 38.5 ft **C** 44 ft **D** 264 ft

24. The figure shows a rectangle 12 feet wide and 6 feet tall with a semicircle on top. What is the area of the whole figure?



A 72 ft^2
B $(6\pi + 72)$ ft^2
C $(14\pi + 72)$ ft^2
D $(18\pi + 72)$ ft^2

25. The figure shows one fourth of a circle. Each straight side is 20 inches long. What is the perimeter of the region?



A 10π in
B $(10\pi + 20)$ in
C $(10\pi + 40)$ in
D $(20\pi + 40)$ in

Answers & Explanations 19

1 C	**6** D	**11** A	**16** C	**21** D
2 C	**7** B	**12** C	**17** A	**22** B
3 B	**8** B	**13** D	**18** A	**23** A
4 A	**9** C	**14** C	**19** B	**24** D
5 A	**10** B	**15** D	**20** B	**25** C

1. **C** 121π ft^2

1 The area of a circle is $A = \pi r^2$. Here the radius is 11 feet.

2 Square the radius: $11^2 = 121$.

3 So $A = \pi \times 121 = 121\pi$ square feet.

WHY NOT (A) forgets to square the radius. (B) uses the circumference formula $2\pi r$ instead of πr^2. (D) uses the diameter in place of the radius.

2. **C** 22π ft

1 The circumference is the distance around: $C = 2\pi r$.

2 Substitute $r = 11$: $C = 2 \times \pi \times 11 = 22\pi$ feet.

WHY NOT (A) uses the radius in $C = \pi d$; the diameter is twice the radius. (D) finds the area, not the circumference.

3. **B** 9π cm^2

1 The area of a circle is $A = \pi r^2$. Here the radius is 3 centimeters.

2 Square the radius: $3^2 = 9$.

3 So $A = \pi \times 9 = 9\pi$ square centimeters.

WHY NOT (A) uses the circumference formula $2\pi r$ instead of πr^2. (D) uses the diameter in place of the radius.

4. **A** 10π in

1 The circumference is the distance around: $C = 2\pi r$.

2 Substitute $r = 5$: $C = 2 \times \pi \times 5 = 10\pi$ inches.

WHY NOT (C) doubles the radius and then uses that diameter in $2\pi r$. (D) finds the area, not the circumference.

5. **A** 616 ft^2

1 The goat can reach every point within 14 feet of the stake, so the region is a circle with a radius of 14 feet.

2 The area of a circle is $A = \pi r^2$. Square the radius: $14^2 = 196$.

3 $A \approx \frac{22}{7} \times 196 = 22 \times 28 = 616$ square feet.

WHY NOT (D) uses the diameter in place of the radius.

6. **D** 38π cm

1 When you know the diameter, use $C = \pi d$.

2 Substitute $d = 38$: $C = \pi \times 38 = 38\pi$ centimeters.

WHY NOT (A) uses the radius in $C = \pi d$.

7. **B** 220 yd

1 The distance around a circle is its circumference: $C = \pi d$.

2 $C \approx \frac{22}{7} \times 70 = 22 \times 10 = 220$ yards.

WHY NOT (A) uses the radius instead of the diameter. (C) uses the diameter in $2\pi r$, which doubles the answer. (D) finds the area, not the distance around.

8. **B** 324π yd^2

1 The area of a circle is $A = \pi r^2$. Here the radius is 18 yards.

2 Square the radius: $18^2 = 324$.

3 So $A = \pi \times 324 = 324\pi$ square yards.

WHY NOT (A) uses the circumference formula $2\pi r$ instead of πr^2. (D) uses the diameter in place of the radius.

9. **C** 9 cm

1 Use $C = 2\pi r$: $2\pi r = 18\pi$.

2 Divide both sides by π: $2r = 18$. (This is the diameter.)

3 Divide by 2: $r = 18 \div 2 = 9$ centimeters.

WHY NOT (A) divides by 4 instead of by 2. (D) gives the diameter; divide it by 2 to get the radius.

10. **B** 16π ft^2

1 A whole circle with radius 8 has area $\pi \times 8^2 = 64\pi$.

2 A quarter circle is one fourth of it: $64\pi \div 4 = 16\pi$ square feet.

WHY NOT (A) finds the length of the curved edge, not the area. (C) takes half of the circle instead of one fourth. (D) finds the area of the whole circle.

11. **A** 52π ft^2

1 The path and the pond together form a large circle with radius $12 + 2 = 14$ feet.

2 Large circle: $\pi \times 14^2 = 196\pi$. The pond: $\pi \times 12^2 = 144\pi$.

3 The path is the difference: $196\pi - 144\pi = 52\pi$ square feet.

WHY NOT (B) adds the width twice when finding the outer radius. (C) includes the pond itself. (D) adds the two circle areas instead of subtracting.

12. **C** 16

1 Radii: $6 \div 2 = 3$ and $24 \div 2 = 12$ inches.

2 Areas: circle A has $\pi \times 3^2 = 9\pi$, and circle B has $\pi \times 12^2 = 144\pi$.

3 Divide: $144\pi \div 9\pi = 16$.

TIP Shortcut: the diameter is 4 times as large, so the area is $4^2 = 16$ times as large.

WHY NOT (A) compares the diameters instead of the areas. (B) doubles the ratio instead of squaring it. (D) cubes the ratio.

13. **D** 16π m^2

1 Shaded ring = large circle − small circle.

2 Large circle: $\pi \times 5^2 = 25\pi$. Small circle: $\pi \times 3^2 = 9\pi$.

3 Subtract: $25\pi - 9\pi = 16\pi$ square meters.

WHY NOT (A) subtracts the radii before squaring.

14. **C** 4

1 Circumference depends on the radius itself (not squared): $C = 2\pi r$.

2 Replace r with $4r$: $2\pi(4r) = 4 \times 2\pi r$.

3 So the circumference is multiplied by 4.

WHY NOT (D) squares the factor; only the area grows by the square.

15. **D** 8π ft^2

1 The radius is half the diameter: $r = 8 \div 2 = 4$ feet.

2 A whole circle with this radius has area $\pi \times 4^2 = 16\pi$.

3 A semicircle is half of a circle: $16\pi \div 2 = 8\pi$ square feet.

WHY NOT (A) finds the length of the curved edge, not the area.

16. **C** 12π in

1 First find the radius: $\pi r^2 = 36\pi$, so $r^2 = 36$ and $r = 6$.

2 Then $C = 2\pi r = 2 \times \pi \times 6 = 12\pi$ inches.

WHY NOT (A) finds the radius correctly but then uses πr instead of $2\pi r$. (D) uses the diameter in $2\pi r$.

17. **A** 81π cm^2

1 The formula $A = \pi r^2$ needs the radius, which is half the diameter: $r = 18 \div 2 = 9$ centimeters.

2 Square the radius: $9^2 = 81$.

3 So $A = \pi \times 81 = 81\pi$ square centimeters.

WHY NOT (C) squares the diameter and then halves it; halve first, then square. (D) uses the diameter in place of the radius.

18. **A** 200.96 ft^2

1 The area formula $A = \pi r^2$ needs the radius: $r = 16 \div 2 = 8$ feet.

2 Square the radius: $8^2 = 64$.

3 $A \approx 3.14 \times 64 = 200.96$ square feet.

TIP Multiply by 3 and by 0.14, then add: $3 \times 64 = 192$ and $0.14 \times 64 = 8.96$, so the total is 200.96.

WHY NOT (C) squares the diameter and then halves it; halve first, then square. (D) uses the diameter in place of the radius.

19. **B** $(196 - 49\pi)$ in^2

1 Square: $14^2 = 196$.

2 Quarter circle with radius 14: $\frac{1}{4} \times \pi \times 14^2 = 49\pi$.

3 Shaded region = square − quarter circle = $196 - 49\pi$ square inches.

WHY NOT (C) finds the area of the quarter circle, not the shaded region. (D) subtracts the length of the arc instead of the area.

20. **B** 54π ft^2

1 The watered region is a sector of a circle with radius 18 feet and angle 60°.

2 That is $\frac{60}{360} = \frac{1}{6}$ of a whole circle. Whole circle: $\pi \times 18^2 = 324\pi$.

3 Sector: $\frac{1}{6} \times 324\pi = 54\pi$ square feet.

WHY NOT (A) finds the length of the arc, not the area. (C) divides the angle by 180° instead of 360°. (D) finds the area of the whole circle.

21. **D** 480

1 One turn covers one circumference: $C = \pi d \approx \frac{22}{7} \times 42 = 22 \times 6 = 132$ inches.

2 Change to feet: $132 \div 12 = 11$ feet.

3 Turns = distance ÷ circumference: $5{,}280 \div 11 = 480$.

WHY NOT (A) forgets to convert the circumference from inches to feet. (B) uses the diameter in $2\pi r$.

22. **B** 4π cm

1 The arc is $\frac{240}{360} = \frac{2}{3}$ of the whole circle.

2 Circumference: $2\pi \times 3 = 6\pi$.

3 Arc length: $\frac{2}{3} \times 6\pi = 4\pi$ centimeters.

WHY NOT (A) uses πr instead of $2\pi r$ for the circumference. (D) finds the circumference of the whole circle.

23. **A** 22 ft

1 In one turn the wheel rolls one circumference: $C = \pi d \approx \frac{22}{7} \times 7 = 22$ inches.

2 In 12 turns: $22 \times 12 = 264$ inches.

3 Convert to feet (12 inches = 1 foot): $264 \div 12 = 22$ feet.

WHY NOT (B) uses the area of the wheel instead of its circumference. (C) uses the diameter in $2\pi r$. (D) forgets to convert inches to feet.

24. **D** $(18\pi + 72)$ ft^2

1 Rectangle: $12 \times 6 = 72$ square feet.

2 The semicircle's diameter is the width, 12, so its radius is 6. Its area is half a circle: $\frac{1}{2} \times \pi \times 6^2 = 18\pi$.

3 Total area: $18\pi + 72$ square feet.

WHY NOT (A) leaves out the semicircle. (B) adds the length of the curved edge instead of the area.

25. **C** $(10\pi + 40)$ in

1 The distance around is the curved edge plus two straight edges, each a radius (20 inches).

2 The curved edge is one fourth of the circumference: $\frac{1}{4} \times 2\pi \times 20 = 10\pi$ inches.

3 Add the straight edges: $20 + 20 = 40$, so the total is $10\pi + 40$ inches.

WHY NOT (A) forgets the two straight edges (the radii). (B) adds only one of the two straight edges. (D) uses half of the circumference instead of one fourth.

CHAPTER 20

Volume & Surface Area

Tested on: Mathematics Knowledge • Arithmetic Reasoning

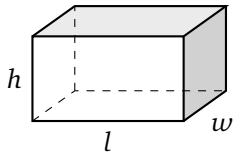
Key Concepts

Volume is the amount of space inside a solid, measured in cubic units (in^3, ft^3). *Surface area* is the total area of all the faces you could paint or wrap, measured in square units (in^2, ft^2).

Boxes and cubes

A rectangular box (prism) with length l, width w, and height h:

$$V = l \times w \times h \qquad SA = 2lw + 2lh + 2wh$$



The surface area adds three *pairs* of matching faces: top and bottom, front and back, left and right.
A cube with edge s has six identical square faces:

$$V = s^3 \qquad SA = 6s^2$$

Cylinders, cones, and spheres

Any solid with the same shape all the way up has V = (area of the base) × height. For a cylinder the base is a circle, so

$$V = \pi r^2 h$$

A cone holds exactly $\frac{1}{3}$ as much as a cylinder with the same base and height: $V = \frac{1}{3}\pi r^2 h$. A sphere of radius r has $V = \frac{4}{3}\pi r^3$. (In this book, the cone and sphere formulas are given in the problem when you need them.)

Units

Use one unit for every length *before* you multiply: 6 inches = $\frac{1}{2}$ foot, 4 inches = $\frac{1}{3}$ foot.

- 1 ft^3 = 12 × 12 × 12 = 1,728 in^3
- 1 yd^3 = 3 × 3 × 3 = 27 ft^3 (concrete, gravel, and soil are sold by the cubic yard)
- 1 ft^3 of water is about 7.5 gallons

WORKED EXAMPLE

A fish tank is 3 feet long, 2 feet wide, and 2 feet tall. If 1 cubic foot holds 7.5 gallons, how many gallons does the tank hold?
Solution. $V = 3 \times 2 \times 2 = 12$ cubic feet, and $12 \times 7.5 = 90$ gallons.

No-Calculator Tip Multiply in the easiest order: $25 \times 7 \times 4 = (25 \times 4) \times 7 = 700$. Scaling: if every edge is doubled, the surface area is multiplied by $2^2 = 4$ and the volume by $2^3 = 8$.

Common Traps

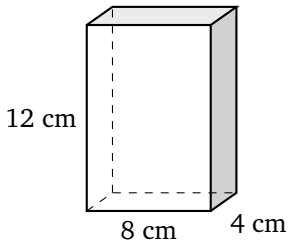
- Adding the dimensions instead of multiplying them.
- Mixing inches and feet in one calculation.
- Counting only three faces for surface area (each face has a twin).
- Using the diameter instead of the radius in $\pi r^2 h$.
- Dividing cubic feet by 9 (or 3) instead of 27 to get cubic yards.

Practice Set 20

Score: _____ / 25 Target time: 26 minutes

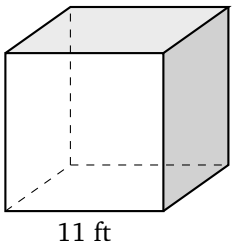
Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

1. The rectangular box shown is 8 centimeters long, 4 centimeters wide, and 12 centimeters high. What is its volume?



A 24 cm^3 **B** 32 cm^3 **C** 352 cm^3 **D** 384 cm^3

2. A cube has edges that are each 11 feet long. What is its volume?



A 33 ft^3 **B** 121 ft^3
C 726 ft^3 **D** 1,331 ft^3

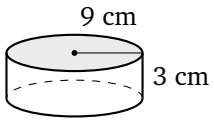
3. A rectangular box is 12 centimeters long, 10 centimeters wide, and 11 centimeters high. What is its volume?

A 33 cm^3 **B** 1,320 cm^3
C 1,430 cm^3 **D** 1,440 cm^3

4. A cube-shaped box has edges that are each 7 centimeters long. What is its total surface area?

A 196 cm^2 **B** 294 cm^2 **C** 343 cm^2 **D** 392 cm^2

5. What is the volume of the cylinder shown? The radius is 9 centimeters and the height is 3 centimeters. (Figure not drawn to scale.)

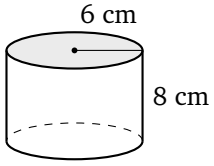


A 27π cm^3 **B** 54π cm^3
C 81π cm^3 **D** 243π cm^3

6. A cube has edges that are each 11 inches long. What is its total surface area?

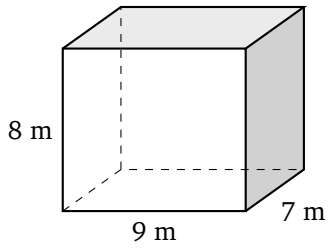
A 66 in^2 **B** 484 in^2
C 726 in^2 **D** 1,331 in^2

7. What is the volume of the cylinder shown? The radius is 6 centimeters and the height is 8 centimeters.



A 36π cm^3 **B** 48π cm^3
C 96π cm^3 **D** 288π cm^3

8. The rectangular box shown is 9 meters long, 7 meters wide, and 8 meters high. What is its volume?



A 24 m^3 **B** 504 m^3 **C** 560 m^3 **D** 567 m^3

9. Yuki wants to cover a gift box with wrapping paper with no overlap. The box is 13 inches long, 8 inches wide, and 5 inches tall. How many square inches of paper will cover the whole box?

A 209 in^2 **B** 210 in^2 **C** 418 in^2 **D** 520 in^2

10. The volume of a rectangular prism is 900 cubic feet. Its length is 15 feet and its width is 6 feet. What is its height?

A 9 ft **B** 10 ft **C** 60 ft **D** 90 ft

11. A water storage tank at a forward operating base is 9 feet long, 6 feet wide, and 4 feet deep. How many gallons of water does it hold when it is full? (1 cubic foot holds 7.5 gallons.)

A 216 gal **B** 1,512 gal
C 1,620 gal **D** 1,710 gal

12. Private Nguyen is loading cube-shaped crates that measure 3 feet on each side into a cargo container that is 15 feet long, 6 feet wide, and 6 feet high. What is the greatest number of crates that will fit inside?

A 10 **B** 20 **C** 180 **D** 540

13. Yuki's rectangular fish tank is 4 feet long, 2 feet wide, and 2 feet deep. How many gallons of water does it hold when it is full? (1 cubic foot holds 7.5 gallons.)

A 16 gal **B** 60 gal **C** 112 gal **D** 120 gal

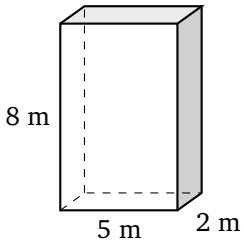
14. Ana is packing wooden blocks shaped like cubes, 3 inches on each side, into a toy chest that is 33 inches long, 15 inches wide, and 21 inches deep. How many blocks will fill the chest?

A 385 **B** 420 **C** 3,465 **D** 10,395

15. An aquarium is 36 inches long and 10 inches wide. It holds 5,760 cubic inches of water. How deep is the water?

A 14 in **B** 15 in **C** 16 in **D** 160 in

16. The closed rectangular box shown is 5 meters long, 2 meters wide, and 8 meters high. What is its total surface area? (Figure not drawn to scale.)

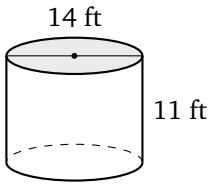


A 66 m^2 **B** 132 m^2 **C** 152 m^2 **D** 164 m^2

17. The volume of a cube-shaped storage crate is 64 cubic feet. What is the total area of its six faces?

A 16 ft^2 **B** 24 ft^2 **C** 64 ft^2 **D** 96 ft^2

18. The cylinder shown has a diameter of 14 feet and a height of 11 feet. What is its volume?



A 539π ft^3 **B** 704π ft^3
C $1{,}078\pi$ ft^3 **D** $2{,}156\pi$ ft^3

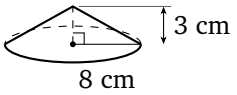
19. Ana is filling a sandbox that is 6 feet long and 5 feet wide with sand 4 inches deep. Sand is sold in bags that each hold half a cubic foot. How many bags does she need to buy?

A 5 **B** 20 **C** 60 **D** 240

20. If every edge of a cube is tripled, its surface area is multiplied by what number?

A 3 **B** 6 **C** 9 **D** 27

21. The volume of a cone is $V = \frac{1}{3}\pi r^2 h$. What is the volume of a cone with radius 8 centimeters and height 3 centimeters? (Figure not drawn to scale.)



A 8π cm^3 **B** 16π cm^3
C 64π cm^3 **D** 192π cm^3

22. The height of a cylinder is tripled, and its radius stays the same. The volume of the new cylinder is how many times the volume of the original?

A 3 **B** 6 **C** 9 **D** 27

23. A cone and a cylinder have the same radius and the same height. The cone holds 33π cubic centimeters. How much does the cylinder hold? (Cone: $V = \frac{1}{3}\pi r^2 h$.)

A 11π cm^3 **B** 33π cm^3
C 66π cm^3 **D** 99π cm^3

24. Kevin is filling a sandbox that is 10 feet long and 6 feet wide with sand 4 inches deep. Sand is sold in bags that each hold half a cubic foot. How many bags does he need to buy?

A 40 **B** 42 **C** 120 **D** 480

25. Ana's backyard pond is 6 feet long, 4 feet wide, and 2 feet deep. It is being filled with a garden hose that delivers 10 gallons per minute. How many minutes will it take to fill it to a depth of 1 foot? (1 cubic foot holds 7.5 gallons.)

A 2.4 min **B** 18 min **C** 36 min **D** 180 min

Answers & Explanations 20

1 D	**6** C	**11** C	**16** B	**21** C
2 D	**7** D	**12** B	**17** D	**22** A
3 B	**8** B	**13** D	**18** A	**23** D
4 B	**9** C	**14** A	**19** B	**24** A
5 D	**10** B	**15** C	**20** C	**25** B

1. **D** 384 cm^3

1 The volume of a rectangular box is $V = l \times w \times h$.

2 Multiply in a convenient order: $8 \times 4 = 32$, then $32 \times 12 = 384$.

3 The volume is 384 cubic centimeters.

WHY NOT (A) adds the three dimensions instead of multiplying them. (B) finds the area of the base only and forgets the height. (C) finds the surface area, not the volume.

2. **D** 1,331 ft^3

1 All three dimensions of a cube equal the edge: $V = s^3 = s \times s \times s$.

2 $11 \times 11 = 121$, and $121 \times 11 = 1{,}331$.

3 The volume is 1,331 cubic feet.

WHY NOT (A) multiplies the edge by 3 instead of cubing it. (B) squares the edge instead of cubing it. (C) finds the surface area, not the volume.

3. **B** 1,320 cm^3

1 The volume of a rectangular box is $V = l \times w \times h$.

2 Multiply in a convenient order: $10 \times 11 = 110$, then $110 \times 12 = 1{,}320$.

3 The volume is 1,320 cubic centimeters.

WHY NOT (A) adds the three dimensions instead of multiplying them.

4. **B** 294 cm^2

1 A cube has 6 identical square faces. One face has area $7^2 = 49$ square centimeters.

2 All six faces: $6 \times 49 = 294$ square centimeters.

WHY NOT (A) counts only four of the six faces. (C) finds the volume, not the surface area.

5. **D** 243π cm^3

1 The volume of a cylinder is (area of the base) × height: $V = \pi r^2 h$.

2 Base: $\pi \times 9^2 = 81\pi$.

3 Volume: $81\pi \times 3 = 243\pi$ cubic centimeters.

WHY NOT (A) forgets to square the radius. (B) uses $2\pi r$ instead of πr^2 for the base. (C) finds the area of the base only.

6. **C** 726 in^2

1 A cube has 6 identical square faces. One face has area $11^2 = 121$ square inches.

2 All six faces: $6 \times 121 = 726$ square inches.

WHY NOT (A) forgets to square the edge. (B) counts only four of the six faces. (D) finds the volume, not the surface area.

7. **D** 288π cm^3

1 The volume of a cylinder is (area of the base) × height: $V = \pi r^2 h$.

2 Base: $\pi \times 6^2 = 36\pi$.

3 Volume: $36\pi \times 8 = 288\pi$ cubic centimeters.

WHY NOT (A) finds the area of the base only. (B) forgets to square the radius. (C) uses $2\pi r$ instead of πr^2 for the base.

8. **B** 504 m^3

1 The volume of a rectangular box is $V = l \times w \times h$.

2 Multiply in a convenient order: $7 \times 8 = 56$, then $56 \times 9 = 504$.

3 The volume is 504 cubic meters.

WHY NOT (A) adds the three dimensions instead of multiplying them.

9. **C** 418 in^2

1 The box has three pairs of matching faces: top and bottom ($13 \times 8 = 104$), front and back ($13 \times 5 = 65$), and the two ends ($8 \times 5 = 40$).

2 Add one face of each pair: $104 + 65 + 40 = 209$.

3 Double it for the matching faces: $2 \times 209 = 418$ square inches.

WHY NOT (A) adds three faces but forgets that each face has a matching opposite face. (B) leaves out the top and bottom. (D) finds the volume, not the surface area.

10. **B** 10 ft

1 Use $V = l \times w \times h$: $900 = 15 \times 6 \times h$.

2 Area of the base: $15 \times 6 = 90$.

3 Divide: $h = 900 \div 90 = 10$ feet.

WHY NOT (C) divides by only one of the two given dimensions. (D) gives the area of the base.

11. **C** 1,620 gal

1 Volume: $9 \times 6 \times 4 = 216$ cubic feet.

2 Each cubic foot holds 7.5 gallons: $216 \times 7.5 =$ 1,620 gallons.

TIP Multiplying by 7.5 is the same as multiplying by 15 and halving: $216 \times 15 = 3{,}240$, and $3{,}240 \div 2 =$ 1,620.

WHY NOT (A) finds the volume in cubic feet but forgets to change it to gallons. (B) uses 7 gallons per cubic foot instead of 7.5. (D) uses the surface area instead of the volume.

12. **B** 20

1 Count how many fit along each edge: $15 \div 3 = 5$, $6 \div 3 = 2$, and $6 \div 3 = 2$.

2 Multiply: $5 \times 2 \times 2 = 20$.

TIP Check with volumes: the container holds $15 \times 6 \times 6 = 540$ cubic feet, and one cube is $3^3 = 27$, so $540 \div 27 = 20$.

WHY NOT (A) counts only the bottom layer. (C) divides the volume by the edge length instead of by the volume of one cube. (D) finds the volume of the container but forgets to divide.

13. **D** 120 gal

1 Volume: $4 \times 2 \times 2 = 16$ cubic feet.

2 Each cubic foot holds 7.5 gallons: $16 \times 7.5 = 120$ gallons.

TIP Multiplying by 7.5 is the same as multiplying by 15 and halving: $16 \times 15 = 240$, and $240 \div 2 = 120$.

WHY NOT (A) finds the volume in cubic feet but forgets to change it to gallons. (C) uses 7 gallons per cubic foot instead of 7.5.

14. **A** 385

1 Count how many fit along each edge: $33 \div 3 = 11$, $15 \div 3 = 5$, and $21 \div 3 = 7$.

2 Multiply: $11 \times 5 \times 7 = 385$.

TIP Check with volumes: the container holds $33 \times 15 \times 21 = 10{,}395$ cubic inches, and one cube is $3^3 = 27$, so $10{,}395 \div 27 = 385$.

WHY NOT (C) divides the volume by the edge length instead of by the volume of one cube. (D) finds the volume of the container but forgets to divide.

15. **C** 16 in

1 Use $V = l \times w \times h$: $5{,}760 = 36 \times 10 \times h$.

2 Area of the base: $36 \times 10 = 360$.

3 Divide: $h = 5{,}760 \div 360 = 16$ inches.

WHY NOT (D) divides by only one of the two given dimensions.

16. **B** 132 m^2

1 The box has three pairs of matching faces: top and bottom ($5 \times 2 = 10$), front and back ($5 \times 8 = 40$), and the two ends ($2 \times 8 = 16$).

2 Add one face of each pair: $10 + 40 + 16 = 66$.

3 Double it for the matching faces: $2 \times 66 = 132$ square meters.

WHY NOT (A) adds three faces but forgets that each face has a matching opposite face.

17. **D** 96 ft^2

1 Find the edge: which number cubed is 64? $4 \times 4 \times 4 = 64$, so $s = 4$ feet.

2 One face: $4^2 = 16$. Six faces: $6 \times 16 = 96$ square feet.

WHY NOT (A) finds the area of only one face. (B) forgets to square the edge. (C) counts only four of the six faces.

18. **A** 539π ft^3

1 The radius is half the diameter: $r = 14 \div 2 = 7$ feet.

2 Base: $\pi r^2 = \pi \times 7^2 = 49\pi$ square feet.

3 Volume: $V = \pi r^2 h = 49\pi \times 11 = 539\pi$ cubic feet.

WHY NOT (C) squares the diameter and then halves it; halve first. (D) uses the diameter in place of the radius.

19. **B** 20

1 Change the depth to feet: 4 in $= \frac{4}{12}$ ft $= \frac{1}{3}$ ft.

2 Volume: $6 \times 5 \times \frac{1}{3} = 10$ cubic feet.

3 Each bag holds $\frac{1}{2}$ cubic foot, so it takes 2 bags for every cubic foot: $10 \times 2 = 20$ bags.

WHY NOT (A) multiplies by the bag size instead of dividing by it. (C) leaves out the depth. (D) uses the depth in inches as if it were feet.

20. **C** 9

1 Each face's area multiplies two lengths, and each one is multiplied by 3.

2 So every face, and the total surface area, is multiplied by $3 \times 3 = 9$.

WHY NOT (A) assumes the surface area grows by the same factor as the edges. (D) cubes the factor; that is how volume grows.

21. **C** 64π cm^3

1 Square the radius: $8^2 = 64$.

2 Substitute: $V = \frac{1}{3} \times \pi \times 64 \times 3 = \frac{1}{3} \times 192\pi = 64\pi$ cubic centimeters.

WHY NOT (A) forgets to square the radius. (B) uses $2r$ instead of r^2. (D) forgets the $\frac{1}{3}$ (that is the volume of a cylinder).

22. **A** 3

1 In $V = \pi r^2 h$, the height is not squared.

2 Replace h with $3h$: $\pi r^2(3h) = 3\pi r^2 h$, so the volume is multiplied by 3.

WHY NOT (C) squares the factor; only the radius is squared in the formula. (D) cubes the factor.

23. **D** 99π cm^3

1 A cone holds $\frac{1}{3}$ as much as a cylinder with the same base and height, so the cylinder holds 3 times as much as the cone.

2 $3 \times 33\pi = 99\pi$ cubic centimeters.

WHY NOT (A) divides by 3 instead of multiplying by 3. (B) assumes the cylinder holds the same amount. (C) doubles instead of tripling.

24. **A** 40

1 Change the depth to feet: 4 in = $\frac{4}{12}$ ft = $\frac{1}{3}$ ft.

2 Volume: $10 \times 6 \times \frac{1}{3} = 20$ cubic feet.

3 Each bag holds $\frac{1}{2}$ cubic foot, so it takes 2 bags for every cubic foot: $20 \times 2 = 40$ bags.

WHY NOT (C) leaves out the depth. (D) uses the depth in inches as if it were feet.

25. **B** 18 min

1 Volume of water: $6 \times 4 \times 1 = 24$ cubic feet (use the water depth, not the tank's full depth).

2 Gallons: $24 \times 7.5 = 180$.

3 Time: $180 \div 10 = 18$ minutes.

WHY NOT (A) forgets to change cubic feet to gallons. (C) uses the tank's full depth of 2 ft instead of the water depth of 1 ft. (D) finds the number of gallons, not the number of minutes.

CHAPTER 21

Coordinate Geometry

Tested on: Mathematics Knowledge

Key Concepts

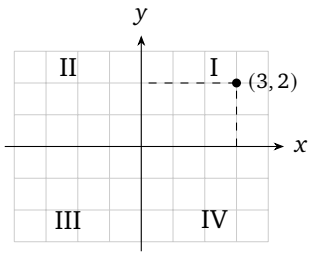
The coordinate plane is a map: every point has an address (x, y). The x-coordinate tells you how far to go right (+) or left (−); the y-coordinate tells you how far to go up (+) or down (−). Always x first, then y.

Points and quadrants

The axes split the plane into four **quadrants**, numbered counterclockwise starting at the upper right:

- Quadrant I: $(+, +)$ Quadrant II: $(-, +)$
- Quadrant III: $(-, -)$ Quadrant IV: $(+, -)$

A point on an axis is in no quadrant. The point $(3, 2)$ is 3 units right and 2 units up from the origin $(0, 0)$.



Slope

Slope measures steepness: **rise over run**.

$$m = \frac{\text{rise}}{\text{run}} = \frac{y_2 - y_1}{x_2 - x_1}$$

A positive slope goes up to the right, a negative slope goes down to the right, a horizontal line has slope 0, and a vertical line has no slope (undefined). **Parallel** lines have equal slopes. **Perpendicular** lines have slopes that are negative reciprocals: flip the fraction *and* change the sign ($\frac{2}{3}$ and $-\frac{3}{2}$).

Equations of lines

In **slope-intercept form** $y = mx + b$, m is the slope and b is the y-intercept (where the line crosses the y-axis). To read the slope of an equation like $2x + 3y = 12$, solve for y first: $3y = -2x + 12$, so $y = -\frac{2}{3}x + 4$. To find the x-intercept, set $y = 0$; for the y-intercept, set $x = 0$. A point is on a line exactly when its coordinates make the equation true.

Distance and midpoint

For points (x_1, y_1) and (x_2, y_2):

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \qquad M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

The distance formula is the Pythagorean theorem: the horizontal and vertical changes are the legs of a right triangle.

WORKED EXAMPLE

Find the equation of the line through $(1, 2)$ and $(3, 8)$.
Solution. Slope: $m = \frac{8-2}{3-1} = \frac{6}{2} = 3$. Put $(1, 2)$ into $y = 3x + b$: $2 = 3 + b$, so $b = -1$. The line is $y = 3x - 1$. Check with $(3, 8)$: $3 \times 3 - 1 = 8$. ✓

No-Calculator Tip Check a line by plugging in a point. For distances, look for Pythagorean triples: 3-4-5, 6-8-10, 5-12-13, 8-15-17. If the legs are 6 and 8, the distance is 10, with no square roots needed.

Common Traps

- Putting the run on top: slope is $\frac{\text{change in } y}{\text{change in } x}$.
- Subtracting in different orders on top and bottom (sign error).
- Perpendicular slope: changing the sign *or* flipping, instead of both.
- Reading $y = 4 - 2x$ as slope 4; the slope is the number with x, so it is -2.

Practice Set 21

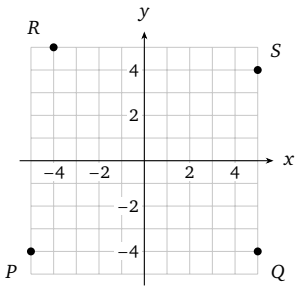
Score: _____ / 25 Target time: 25 minutes

Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

1. The point $(10, 2)$ is plotted on a coordinate plane. In which quadrant is it?

A Quadrant I **B** Quadrant II
C Quadrant III **D** Quadrant IV

2. Which point in the figure has coordinates $(5, -4)$? (Each grid square is 1 unit.)



A point S **B** point R **C** point Q **D** point P

3. What is the slope of the line that passes through $(5, 1)$ and $(9, 10)$?

A $-\frac{9}{4}$ **B** $\frac{4}{9}$ **C** $\frac{11}{14}$ **D** $\frac{9}{4}$

4. What is the y-intercept of the line $y = -\frac{3}{4}x - 7$?

A $-\frac{28}{3}$ **B** -7 **C** $-\frac{3}{4}$ **D** 7

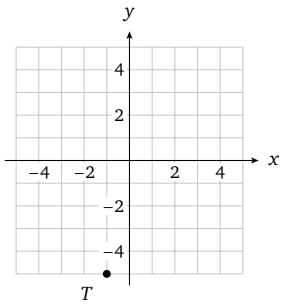
5. A line passes through the points $(3, 1)$ and $(8, 10)$. What is its slope?

A $-\frac{9}{5}$ **B** $\frac{5}{9}$ **C** $\frac{9}{5}$ **D** 9

6. What is the y-intercept of the line $y = 8x - 9$?

A -9 **B** $\frac{9}{8}$ **C** 8 **D** 9

7. A map is drawn on a coordinate grid, and point T shows the location of a lookout post. What are the coordinates of point T? (Each grid square is 1 unit.)

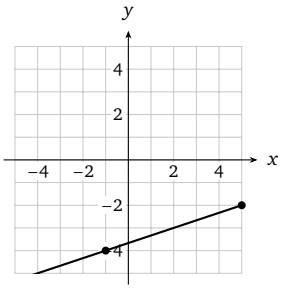


A $(-5, -1)$ **B** $(-1, -5)$ **C** $(1, -5)$ **D** $(-1, 5)$

8. In which quadrant does the point $(3, -9)$ lie?

A Quadrant I **B** Quadrant II
C Quadrant III **D** Quadrant IV

9. The line in the figure passes through the points $(-1, -4)$ and $(5, -2)$. What is the slope of the line?



A $-\frac{1}{3}$ **B** $\frac{1}{3}$ **C** 2 **D** 3

10. The midpoint of segment AB is $M(6, 2)$. If point A is $(2, -4)$, what are the coordinates of point B?

A $(8, -2)$ **B** $(-2, -10)$
C $(10, 8)$ **D** $(4, 6)$

11. What is the distance between the points $(-1, 0)$ and $(11, -5)$?

A 13 **B** 17 **C** 60 **D** 169

12. What is the x-intercept of the line $y = 3x + 21$?

A -7 **B** $\frac{1}{7}$ **C** 7 **D** 21

13. The graph of $y = 2x + 8$ passes through which of these points?

A $(3, 13)$ **B** $(3, -2)$ **C** $(14, 3)$ **D** $(3, 14)$

14. What is the x-intercept of the line $y = 4x - 8$?

A -8 **B** -2 **C** $-\frac{1}{2}$ **D** 2

15. Which ordered pair is a solution of the equation $y = -4x - 3$?

A $(-19, 4)$ **B** $(4, -3)$ **C** $(4, -13)$ **D** $(4, -19)$

16. What is the distance between the points $(4, 1)$ and $(-2, -7)$?

A 10 **B** 14 **C** 48 **D** 100

17. What is the midpoint of the segment joining $(6, 4)$ and $(-8, 8)$?

A $(6, -1)$ **B** $(-1, 6)$ **C** $(-7, 2)$ **D** $(-2, 12)$

18. A line passes through the points $(-4, -8)$ and $(-3, 1)$. What is its slope?

A -9 **B** $\frac{1}{9}$ **C** 9 **D** 10

19. A line has a slope of 5 and passes through the point $(-1, -4)$. What is its equation?

A $y = 5x - 1$ **B** $y = 5x - 4$
C $y = 5x + 1$ **D** $y = -5x - 9$

20. What is the slope of a line that is perpendicular to the line $y = -2x + 4$?

A -2 **B** $-\frac{1}{2}$ **C** $\frac{1}{2}$ **D** 2

21. Which equation describes the line that passes through $(-2, 10)$ and $(2, 4)$?

A $y = -\frac{2}{3}x + \frac{26}{3}$ **B** $y = -\frac{3}{2}x - 7$
C $y = -\frac{3}{2}x + 10$ **D** $y = -\frac{3}{2}x + 7$

22. What is the y-intercept of the line $6x - y = -19$?

A -19 **B** $-\frac{19}{6}$ **C** 6 **D** 19

23. A line has a slope of 4 and passes through the point $(2, 7)$. What is its equation?

A $y = 4x - 1$ **B** $y = -4x + 15$
C $y = 4x + 1$ **D** $y = 4x + 7$

24. What is the slope of the line $6x - 4y = 24$?

A $-\frac{3}{2}$ **B** $\frac{2}{3}$ **C** $\frac{3}{2}$ **D** 6

25. What is the slope of a line that is parallel to the line $x + 3y = 3$?

A -3 **B** $-\frac{1}{3}$ **C** $\frac{1}{3}$ **D** 3

Answers & Explanations 21

1 A	**6** A	**11** A	**16** A	**21** D
2 C	**7** B	**12** A	**17** B	**22** D
3 D	**8** D	**13** D	**18** C	**23** A
4 B	**9** B	**14** D	**19** C	**24** C
5 C	**10** C	**15** D	**20** C	**25** B

1. **A** Quadrant I

1 The x-coordinate 10 is positive, so the point is right of the y-axis. The y-coordinate 2 is positive, so it is above the x-axis.

2 That is the upper right region. The quadrants are numbered counterclockwise from the upper right (I, II, III, IV), so this is Quadrant I.

WHY NOT (B) reads the sign of the x-coordinate backwards. (C) reverses both signs. (D) reads the sign of the y-coordinate backwards.

2. **C** point Q

1 The x-coordinate comes first: go 5 units right from the origin.

2 Then go 4 units down. That is point Q.

WHY NOT (A) reads the sign of the y-coordinate backwards. (B) swaps the x- and y-coordinates. (D) reads the sign of the x-coordinate backwards.

3. **D** $\frac{9}{4}$

1 Use $m = \frac{y_2 - y_1}{x_2 - x_1}$ with $(5, 1)$ and $(9, 10)$.

2 Rise: $10 - 1 = 9$. Run: $9 - 5 = 4$.

3 Slope: $\frac{9}{4}$.

TIP Quick check: as x increases, y increases, so the slope must be positive.

WHY NOT (A) subtracts the coordinates in opposite orders on the top and bottom. (B) divides the change in x by the change in y (run over rise). (C) adds the coordinates instead of subtracting them.

4. **B** -7

1 In $y = mx + b$, the y-intercept is b, the number added on its own.

2 Check by setting $x = 0$: $y = -\frac{3}{4}(0) - 7 = -7$. So the y-intercept is -7.

WHY NOT (A) finds the x-intercept instead. (C) gives the slope instead of the y-intercept. (D) changes the sign of the y-intercept.

5. **C** $\frac{9}{5}$

1 Use $m = \frac{y_2 - y_1}{x_2 - x_1}$ with $(3, 1)$ and $(8, 10)$.

2 Rise: $10 - 1 = 9$. Run: $8 - 3 = 5$.

3 Slope: $\frac{9}{5}$.

TIP Quick check: as x increases, y increases, so the slope must be positive.

WHY NOT (A) subtracts the coordinates in opposite orders on the top and bottom. (B) divides the change in x by the change in y (run over rise). (D) finds the rise but forgets to divide by the run.

6. **A** -9

1 In $y = mx + b$, the y-intercept is b, the number added on its own.

2 Check by setting $x = 0$: $y = 8(0) - 9 = -9$. So the y-intercept is -9.

WHY NOT (B) finds the x-intercept instead. (C) gives the slope instead of the y-intercept. (D) changes the sign of the y-intercept.

7. **B** $(-1, -5)$

1 Start at the origin. Point T is 1 unit left, so $x = -1$.

2 It is 5 units down, so $y = -5$. Write x first: $(-1, -5)$.

WHY NOT (A) swaps the x- and y-coordinates. (C) reads the sign of the x-coordinate backwards. (D) reads the sign of the y-coordinate backwards.

8. **D** Quadrant IV

1 The x-coordinate 3 is positive, so the point is right of the y-axis. The y-coordinate -9 is negative, so it is below the x-axis.

2 That is the lower right region. The quadrants are numbered counterclockwise from the upper right (I, II, III, IV), so this is Quadrant IV.

WHY NOT (A) reads the sign of the y-coordinate backwards. (B) swaps the x- and y-coordinates. (C) reads the sign of the x-coordinate backwards.

9. **B** $\frac{1}{3}$

1 Use $m = \frac{y_2 - y_1}{x_2 - x_1}$ with $(-1, -4)$ and $(5, -2)$.

2 Rise: $-2 - (-4) = 2$. Run: $5 - (-1) = 6$.

3 Slope: $\frac{2}{6} = \frac{1}{3}$.

TIP Quick check: as x increases, y increases, so the slope must be positive.

WHY NOT (A) subtracts the coordinates in opposite orders on the top and bottom. (C) finds the rise but forgets to divide by the run. (D) divides the change in x by the change in y (run over rise).

10. **C** $(10, 8)$

1 From A to M, x changes by $6 - 2 = 4$ and y changes by $2 - (-4) = 6$.

2 B is the same distance past M: $x = 6 + 4 = 10$ and $y = 2 + 6 = 8$.

3 Check: the midpoint of $(2, -4)$ and $(10, 8)$ is $\left(\frac{12}{2}, \frac{4}{2}\right) = (6, 2)$. ✓

WHY NOT (A) adds A and M without doubling M. (B) goes from M through A instead of from A through M. (D) finds only the change from A to M, not the endpoint.

11. **A** 13

1 Horizontal change: $|11 - (-1)| = 12$. Vertical change: $|-5 - 0| = 5$.

2 These are the legs of a right triangle, and the distance is the hypotenuse: $d = \sqrt{12^2 + 5^2} = \sqrt{144 + 25} = \sqrt{169} = 13$.

TIP The legs 5 and 12 belong to the Pythagorean triple 5-12-13.

WHY NOT (B) adds the horizontal and vertical distances instead of using the Pythagorean theorem. (C) multiplies the horizontal and vertical distances. (D) forgets to take the square root.

12. **A** -7

1 The x-intercept is where the line crosses the x-axis, so set $y = 0$: $0 = 3x + 21$.

2 $3x = -21$, so $x = -21 \div 3 = -7$.

WHY NOT (B) divides the slope by the y-intercept instead of dividing $-b$ by the slope. (C) makes a sign error when solving for x. (D) gives the y-intercept instead of the x-intercept.

13. **D** $(3, 14)$

1 A point is on the line if its coordinates make the equation true. Substitute each x and see whether you get the matching y.

2 For $(3, 14)$: $y = 2(3) + 8 = 6 + 8 = 14$. It matches, so $(3, 14)$ is on the line.

3 Each of the other points gives a different y when you substitute its x.

WHY NOT (A) adds the slope to x instead of multiplying. (B) uses the wrong sign for the y-intercept. (C) swaps the x- and y-coordinates of a point on the line.

14. **D** 2

1 The x-intercept is where the line crosses the x-axis, so set $y = 0$: $0 = 4x - 8$.

2 $4x = 8$, so $x = 8 \div 4 = 2$.

WHY NOT (A) gives the y-intercept instead of the x-intercept. (B) makes a sign error when solving for x. (C) divides the slope by the y-intercept instead of dividing $-b$ by the slope.

15. **D** $(4, -19)$

1 A point is on the line if its coordinates make the equation true. Substitute each x and see whether you get the matching y.

2 For $(4, -19)$: $y = -4(4) - 3 = -16 - 3 = -19$. It matches, so $(4, -19)$ is on the line.

3 Each of the other points gives a different y when you substitute its x.

WHY NOT (A) swaps the x- and y-coordinates of a point on the line. (B) adds the slope to x instead of multiplying. (C) uses the wrong sign for the y-intercept.

16. **A** 10

1 Horizontal change: $|-2 - 4| = 6$. Vertical change: $|-7 - 1| = 8$.

2 These are the legs of a right triangle, and the distance is the hypotenuse: $d = \sqrt{6^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100} = 10$.

TIP The legs 6 and 8 belong to the Pythagorean triple 6-8-10.

WHY NOT (B) adds the horizontal and vertical distances instead of using the Pythagorean theorem. (C) multiplies the horizontal and vertical distances. (D) forgets to take the square root.

17. **B** $(-1, 6)$

1 Average the x-coordinates: $\frac{6 + (-8)}{2} = \frac{-2}{2} = -1$.

2 Average the y-coordinates: $\frac{4 + 8}{2} = \frac{12}{2} = 6$.

3 The midpoint is $(-1, 6)$.

WHY NOT (A) swaps the x- and y-coordinates. (C) subtracts the coordinates instead of adding them. (D) adds the coordinates but forgets to divide by 2.

18. **C** 9

1 Use $m = \frac{y_2 - y_1}{x_2 - x_1}$ with $(-4, -8)$ and $(-3, 1)$.

2 Rise: $1 - (-8) = 9$. Run: $-3 - (-4) = 1$.

3 Slope: $\frac{9}{1} = 9$.

TIP Quick check: as x increases, y increases, so the slope must be positive.

WHY NOT (A) subtracts the coordinates in opposite orders on the top and bottom. (B) divides the change in x by the change in y (run over rise).

19. **C** $y = 5x + 1$

1 Start with $y = 5x + b$, since the slope is 5.

2 Substitute the point $(-1, -4)$ to find b: $-4 = 5(-1) + b = -5 + b$, so $b = -4 - (-5) = 1$.

3 The equation is $y = 5x + 1$.

WHY NOT (A) gets the sign of the y-intercept wrong. (B) uses -4, the y-coordinate of a point, as the y-intercept. (D) makes a sign error in the slope.

20. **C** $\frac{1}{2}$

1 The line $y = -2x + 4$ has slope -2 (the coefficient of x).

2 A perpendicular slope is the negative reciprocal: flip -2 and change its sign to get $\frac{1}{2}$.

3 Check: $(-2) \times \frac{1}{2} = -1$. ✓

WHY NOT (A) gives the slope of a parallel line. (B) flips the fraction but forgets to change the sign. (D) changes the sign but forgets to flip the fraction.

21. **D** $y = -\frac{3}{2}x + 7$

1 Slope: $m = \frac{4 - 10}{2 - (-2)} = \frac{-6}{4} = -\frac{3}{2}$.

2 Substitute the point $(-2, 10)$ to find b: $10 = -\frac{3}{2}(-2) + b = 3 + b$, so $b = 10 - 3 = 7$.

3 The equation is $y = -\frac{3}{2}x + 7$.

4 Check with $(2, 4)$: $-\frac{3}{2}(2) + 7 = 4$. ✓

WHY NOT (A) uses run over rise for the slope. (B) gets the sign of the y-intercept wrong. (C) uses 10, the y-coordinate of a point, as the y-intercept.

22. **D** 19

1 Solve for y. Move the x-term to the other side: $-y = -6x - 19$.

2 Divide every term by -1: $y = 6x + 19$.

3 The y-intercept is the constant term: 19.

WHY NOT (A) forgets to divide by the coefficient of y. (B) finds the x-intercept instead. (C) gives the slope instead of the y-intercept.

23. **A** $y = 4x - 1$

1 Start with $y = 4x + b$, since the slope is 4.

2 Substitute the point $(2, 7)$ to find b: $7 = 4(2) + b = 8 + b$, so $b = 7 - 8 = -1$.

3 The equation is $y = 4x - 1$.

WHY NOT (B) makes a sign error in the slope. (C) gets the sign of the y-intercept wrong. (D) uses 7, the y-coordinate of a point, as the y-intercept.

24. **C** $\frac{3}{2}$

1 Solve for y. Move the x-term to the other side: $-4y = -6x + 24$.

2 Divide every term by -4: $y = \frac{3}{2}x - 6$.

3 The slope is the coefficient of x: $\frac{3}{2}$.

WHY NOT (A) forgets to change the sign when moving the x-term. (B) flips the fraction (puts the x-coefficient on the bottom). (D) uses the coefficient of x without solving for y.

25. **B** $-\frac{1}{3}$

1 Solve for y to find the slope: $3y = -x + 3$, so $y = -\frac{1}{3}x + 1$ and the slope is $-\frac{1}{3}$.

2 Parallel lines have the same slope, so the answer is $-\frac{1}{3}$.

WHY NOT (A) flips the slope upside down. (C) changes the sign of the slope. (D) gives the perpendicular slope.

PART IV

Word Problems

The heart of the Arithmetic Reasoning subtest: money, percents, rates, work, mixtures, statistics, probability, and multi-step problems.

22 Money: Shopping, Discounts, Tax & Wages
23 Interest, Commission & Percent Change
24 Rate, Distance & Time
25 Work Rates & Mixtures
26 Averages, Statistics & Probability
27 Measurement, Units & Multi-Step Problems

CHAPTER 22

Money: Shopping, Discounts, Tax & Wages

Tested on: Arithmetic Reasoning

Key Concepts

Money problems are the most common word problems on the Arithmetic Reasoning test. They use only a handful of ideas (add up a bill, take a percent off, add a percent on, figure out pay), but they chain two or three of them together. Label every number you write with what it means, and always re-read what the question actually asks for.

Shopping

- **Total cost** = quantity × price for each item, added up. **Change** = amount paid − total cost.
- **Sale price:** subtract the discount, or multiply by the percent you still pay. 25% off means you pay 75%: $0.75 \times \$80 = \$60$.
- **Sales tax:** add the tax on. A 6% tax means you pay 106%: $1.06 \times \$50 = \$53$.
- **Discount and tax together:** take the discount first, then figure the tax on the *sale* price.
- **Unit price** = price ÷ number of units. The *better buy* is the one with the lower unit price.

Wages and paychecks

Hourly pay = hours × rate. **Overtime** at "time-and-a-half" pays 1.5× the regular rate, but *only* for the hours over 40 in a week. A salary is split evenly over the paychecks in a year:

Paid . . .	Paychecks per year
weekly	52
every two weeks (biweekly)	26
twice a month (semimonthly)	24
monthly	12

A full-time year is $40 \times 52 = 2{,}080$ hours.

Rounding to fit the situation

"How many can you buy?" rounds *down*: you cannot buy part of an item. "How many weeks until you can afford it?" rounds *up*: after a partial week you are still short.

WORKED EXAMPLE

Kevin earns \$18 per hour and gets time-and-a-half for hours over 40. Last week he worked 46 hours. How much did he earn?
Solution. Regular pay: $40 \times \$18 = \$720$. Overtime rate: $1.5 \times \$18 = \$27$ per hour, and $46 - 40 = 6$ overtime hours earn $6 \times \$27 = \$162$. Total: $\$720 + \$162 = \$882$.

No-Calculator Tip Build percents from 10% (move the decimal point one place left). For \$64: 10% = \$6.40, so 5% = \$3.20, 15% = \$9.60, and 20% = \$12.80. To multiply by 1.5, add half: $1.5 \times \$18 = \$18 + \$9 = \$27$.

Common Traps

- Stopping a step early: giving the discount instead of the sale price, or the tax instead of the total.
- Paying time-and-a-half for *all* the hours instead of only the hours over 40.
- Reading "every two weeks" as twice a month: that is 26 paychecks a year, not 24.
- Figuring the tax on the original price when the item is on sale.

Practice Set 22

Score: _____ / 40 Target time: 48 minutes

Problems 1–12 are warm-ups, 13–28 are test level, and 29–40 are challenge problems.

1. At a hardware store, Tyrone buys a box of screws for \$6.75 and a paintbrush for \$6.99. He pays with a \$50 bill. How much change should he receive?

A \$36.26 **B** \$37.26 **C** \$43.25 **D** \$51.26

2. Shopping for school supplies, Victor buys a pack of pens for \$5.99 and a binder for \$8.49. He pays with a \$50 bill. How much change does he get back?

A \$14.48 **B** \$20.52 **C** \$35.00 **D** \$35.52

3. At a grocery store, Jamal buys a loaf of bread for \$3 and a box of cereal for \$5.99. He pays with a \$20 bill. How much change should he receive?

A \$8.99 **B** \$10.01 **C** \$11.00 **D** \$11.01

4. The base exchange is having a 25%-off sale. Private Brooks buys a smartwatch that regularly costs \$390. How much does she pay? (There is no sales tax at the exchange.)

A \$292.50 **B** \$365.00 **C** \$380.25 **D** \$487.50

5. A winter coat regularly sells for \$195. This week it is on sale for 20% off. What is the sale price?

A \$39 **B** \$96 **C** \$136 **D** \$156

6. A pair of running shoes regularly sells for \$110. This week it is on sale for 10% off. What is the sale price?

A \$11 **B** \$79 **C** \$89 **D** \$99

7. Ryan buys a printer online for \$130. The sales tax rate is 4%. What is the total cost, including tax?

A \$5.20 **B** \$135.20 **C** \$155.20 **D** \$182.00

8. Ethan buys a laptop at a department store for \$390. The sales tax rate is 6%. What is the total cost, including tax?

A \$23.40 **B** \$413.40 **C** \$563.40 **D** \$624.00

9. A recruiter has \$330 to spend on T-shirts to give away at a high school career fair. Each T-shirt costs \$6.50. How many T-shirts can the recruiter buy?

A 47 **B** 48 **C** 50 **D** 51

10. A softball coach has \$129 to buy caps for the team. Each cap costs \$6.50. How many caps can the coach buy?

A 17 **B** 18 **C** 19 **D** 20

11. Lance Corporal Nguyen's take-home pay is \$3,540 a month. He sends $\frac{1}{5}$ of it home to his family. How much does he send home to his family each month?

A \$508 **B** \$708 **C** \$908 **D** \$2,832

12. Sofia takes home $4,260 a month. She spends $\frac{1}{6}$ of it on groceries. How much of the month's pay is left?

A $710 **B** $2,550 **C** $3,050 **D** $3,550

13. Shopping for school supplies, Victor buys 2 sets of highlighters at $5.95 each and 3 packs of pens at $7 each. He pays with a $100 bill. How much change does he get back?

A $67.10 **B** $68.10 **C** $81.10 **D** $87.05

14. At a hardware store, Ethan buys 3 cans of spray paint at $5.75 each and a tape measure for $10.99. He pays with a $50 bill. How much change does he get back?

A $17.76 **B** $19.76 **C** $21.00 **D** $21.76

15. At a warehouse store, a 12-ounce jar of peanut butter costs $1.68, and a 28-ounce jar of peanut butter costs $4.20. Priya wants the better buy, so she compares the prices per ounce. What is the lower price per ounce?

A $0.14 **B** $0.15 **C** $0.29 **D** $2.52

16. At the base commissary, a pack of 4 AA batteries costs $3.04, and a pack of 12 AA batteries costs $7.44. What is the price per battery of the better buy?

A $0.14 **B** $0.48 **C** $0.62 **D** $0.76

17. Lance Corporal Murphy buys 4 houseplants at $15 each and a bag of potting soil for $10 in town on a weekend pass. The sales tax rate is 6%. What is the total cost, including tax?

A $4.20 **B** $74.20 **C** $76.00 **D** $112.00

18. Brandon works as a security guard and earns $15 per hour, with time-and-a-half for every hour over 40 in a week. Last week he worked 42 hours. How much did he earn last week?

A $645 **B** $660 **C** $745 **D** $945

19. Nora works as a line cook and earns $15 per hour, with time-and-a-half for every hour over 40 in a week. Last week she worked 43 hours. How much did she earn last week?

A $645.00 **B** $667.50 **C** $690.00 **D** $967.50

20. Mei works as a dental assistant for a salary of $46,800 a year and is paid twice a month, on the 15th and on the last day of the month. How much is each paycheck before taxes and other deductions?

A $900 **B** $1,350 **C** $1,800 **D** $1,950

21. Staff Sergeant Carter's yearly base pay is $46,800. Service members are paid twice a month, on the 1st and the 15th. How much is each paycheck before taxes and other deductions?

A $900 **B** $1,800 **C** $1,950 **D** $3,900

22. Four roommates order takeout. The food costs $57, and they add a 20% tip for the delivery driver. They split the total equally. What is each roommate's share?

A $2.85 **B** $11.10 **C** $14.25 **D** $17.10

23. A squad of 8 soldiers orders pizza for a movie night. The order costs $117, and they give the driver a 20% tip. If they split the cost evenly, how much does each soldier pay?

A $11.55 **B** $15.55 **C** $17.55 **D** $140.40

24. Private Carter is saving for the security deposit and first month's rent on an apartment off base, which costs $2,400. She has already saved $1,050 and can save $200 a week. How many weeks will it take her to have enough money?

A 3 weeks **B** 6 weeks
C 7 weeks **D** 12 weeks

25. Tyrone is saving for a new phone, which costs $700. He has already saved $200 and can save $60 a week. How many weeks will it take him to have enough money?

A 5 weeks **B** 7 weeks **C** 8 weeks **D** 9 weeks

26. Private Haddad's take-home pay is $2,040 a month. She puts $\frac{1}{10}$ of it into savings and pays $\frac{1}{6}$ of it toward the car loan. How much is left each month for everything else?

A $1,496 **B** $1,700 **C** $1,785 **D** $1,836

27. The platoon has $210 for a barbecue. After spending $45 on charcoal and paper plates, it will spend the rest on packs of hamburger patties at $16 each. How many packs can it buy?

A 10 **B** 11 **C** 13 **D** 15

28. At a sporting-goods store, energy bars cost $2.25 per bar, or $18.48 for a box of 12. How much is saved by buying a box of 12 instead of 12 single bars?

A $0.71 **B** $1.54 **C** $5.52 **D** $8.52

29. Lance Corporal Lopez needs new tires before driving to a new duty station. A set of four tires that regularly costs $780 is on sale for 15% off. A sales tax of 8% is charged on the sale price. How much does she pay in all?

A $663.00 **B** $716.04 **C** $725.40 **D** $842.40

30. A dishwasher is priced at $560. It is on sale for 10% off, and an 8% sales tax is charged on the sale price. What is the total cost?

A $463.68 **B** $504.00 **C** $544.32 **D** $548.80

31. James buys a recliner during a 10%-off sale. The regular price is $740, and 5% sales tax is added to the sale price. How much does he pay, including tax?

A $632.70 **B** $666.00 **C** $699.30 **D** $703.00

32. Marcus works as a warehouse worker and earns $19 per hour, with time-and-a-half for every hour over 40 in a week. Last week he worked 9 hours a day Monday through Friday and 6 hours on Saturday. How much did he earn for the week?

A $1,073.50 **B** $1,178.00
C $1,373.50 **D** $1,453.50

33. Andre works as a line cook and earns $15 per hour, with time-and-a-half for every hour over 40 in a week. Last week his pay was $780. How many hours did he work last week?

A 8 hours **B** 48 hours
C 51 hours **D** 52 hours

34. At a grocery store, bottled water comes in three sizes: a case of 12 bottles of water for $3.48, a case of 24 bottles of water for $5.28, and a case of 32 bottles of water for $6.72. Which size is the best buy?

A The case of 32
B The case of 24
C The case of 12
D All three cost the same per bottle

35. Online, coffee comes in three sizes: a 12-ounce bag of coffee for $4.44, a 24-ounce bag of coffee for $6.96, and a 40-ounce bag of coffee for $12.40. Which size is the best buy?

A The 12-ounce bag
B The 40-ounce bag
C The 24-ounce bag
D All three cost the same per ounce

36. Lance Corporal Murphy's take-home pay is $2,940 a month. He pays $\frac{1}{10}$ of the pay toward the car loan, and then puts $\frac{1}{6}$ of what is left into savings. How much money does he put into savings?

A $294 **B** $441 **C** $490 **D** $2,205

37. Omar works as a dental assistant for a salary of $46,800 a year and is paid every two weeks. Of each paycheck, 25% is withheld for taxes. How much does he take home from each paycheck?

A $450 **B** $1,350 **C** $1,800 **D** $2,925

38. Emily has two job offers. Job A pays $17 per hour, and Job B pays a salary of $36,920 a year. At either job she would work 40 hours a week, 52 weeks a year. How much more per year does the better-paying job pay?

A $1,560 **B** $2,920 **C** $4,280 **D** $35,360

39. Specialist Walker is leaving the Army after four years and has two civilian job offers. Job A pays $20 per hour, and Job B pays a salary of $40,040 a year. At either job he would work 40 hours a week, 52 weeks a year. How much more per year does the better-paying job pay?

A $30 **B** $40 **C** $1,560 **D** $41,600

40. At a warehouse store, paper towels cost $2.29 per roll, or $17.40 for a pack of 12. A camp director needs 31 rolls. The plan is to buy as many full packs as possible and the rest as single rolls. What is the total cost?

A $34.80 **B** $50.83 **C** $52.20 **D** $70.99

Answers & Explanations 22

1 A	**9** C	**17** B	**25** D	**33** B
2 D	**10** C	**18** A	**26** A	**34** A
3 D	**11** B	**19** B	**27** A	**35** C
4 A	**12** D	**20** D	**28** D	**36** B
5 D	**13** A	**21** C	**29** B	**37** B
6 D	**14** D	**22** D	**30** C	**38** A
7 B	**15** A	**23** C	**31** C	**39** C
8 B	**16** C	**24** C	**32** A	**40** B

1. **A** $36.26

1 Add the prices to find the total cost: $6.75 + $6.99 = $13.74.

2 Subtract the total from the amount paid: $50.00 – $13.74 = $36.26.

TIP Check by counting up: $13.74+$36.26 = $50.00.

WHY NOT (B) borrows for the cents but forgets to take that dollar away from the dollars. (C) leaves out the price of the paintbrush.

2. **D** $35.52

1 Add the prices to find the total cost: $5.99 + $8.49 = $14.48.

2 Subtract the total from the amount paid: $50.00 – $14.48 = $35.52.

TIP Check by counting up: $14.48+$35.52 = $50.00.

WHY NOT (A) is the total cost, not the change. (C) rounds each price up to a whole dollar instead of using the exact prices.

3. **D** $11.01

1 Add the prices to find the total cost: $3 + $5.99 = $8.99.

2 Subtract the total from the amount paid: $20.00 – $8.99 = $11.01.

TIP Check by counting up: $8.99 + $11.01 = $20.00.

WHY NOT (A) is the total cost, not the change. (C) rounds each price up to a whole dollar instead of using the exact prices.

4. **A** $292.50

1 Find the discount: 25% of $390 = 0.25 × 390 = $97.50.

2 Subtract it from the regular price: $390 – $97.50 = $292.50.

TIP Shortcut: 25% off means you pay 100% – 25% = 75%, and 0.75 × $390 = $292.50.

WHY NOT (B) subtracts the percent as if it were dollars. (C) moves the decimal point one place too far when finding the discount. (D) adds the discount instead of subtracting it.

5. **D** $156

1 Find the discount: 20% of $195 = 0.2×195 = $39.

2 Subtract it from the regular price: $195 – $39 = $156.

TIP Shortcut: 20% off means you pay 100% – 20% = 80%, and 0.8 × $195 = $156.

WHY NOT (A) is the amount of the discount, not the sale price.

6. **D** $99

1 Find the discount: 10% of $110 = 0.1×110 = $11.

2 Subtract it from the regular price: $110 – $11 = $99.

TIP Shortcut: 10% off means you pay 100% – 10% = 90%, and 0.9 × $110 = $99.

WHY NOT (A) is the amount of the discount, not the sale price.

7. **B** $135.20

1 Find the tax: 4% of $130 = 0.04 × 130 = $5.20.

2 Add the tax to the price: $130 + $5.20 = $135.20.

TIP In one step: with tax you pay 104%, so multiply by 1.04.

WHY NOT (A) is the tax alone, not the total cost. (D) moves the decimal point only one place when finding the tax.

8. **B** $413.40

1 Find the tax: 6% of $390 = 0.06 × 390 = $23.40.

2 Add the tax to the price: $390+$23.40 = $413.40.

TIP In one step: with tax you pay 106%, so multiply by 1.06.

WHY NOT (A) is the tax alone, not the total cost. (D) moves the decimal point only one place when finding the tax.

9. **C** 50

1 Divide by the price of one: $330 ÷ $6.50 is 50 with some money left over, because 50 × $6.50 = $325.

2 One more would cost $331.50, which is more than $330. Round *down*: 50 T-shirts.

TIP Left over: $330 – $325 = $5, not enough for one more.

WHY NOT (A) rounds the price up to $7 before dividing. (D) rounds up, but 51 T-shirts would cost $331.50.

10. **C** 19

1 Divide by the price of one: $129 ÷ $6.50 is 19 with some money left over, because 19 × $6.50 = $123.50.

2 One more would cost $130, which is more than $129. Round *down*: 19 caps.

TIP Left over: $129 – $123.50 = $5.50, not enough for one more.

WHY NOT (B) rounds the price up to $7 before dividing. (D) rounds up, but 20 caps would cost $130.

11. **B** $708

1 *Of* means multiply: $\frac{1}{5} \times \$3{,}540$. To find $\frac{1}{5}$ of an amount, divide by 5.

2 $3,540 ÷ 5 = $708.

TIP Check: 5 × $708 = $3,540.

WHY NOT (D) is the amount left over, not the amount sent home.

12. **D** $3,550

1 Find $\frac{1}{6}$ of the pay: $4,260 ÷ 6 = $710. That is the amount spent on groceries.

2 Subtract it from the pay: $4,260 – $710 = $3,550.

TIP Or: $1 - \frac{1}{6} = \frac{5}{6}$ of the pay is left, and $\frac{5}{6} \times \$4{,}260 = \$3{,}550$.

WHY NOT (A) is the amount spent on groceries, not the amount left.

13. **A** $67.10

1 Find the cost of each kind of item: 2 × $5.95 = $11.90, and 3 × $7 = $21.

2 Add to get the total cost: $11.90 + $21 = $32.90.

3 Subtract the total from the amount paid: $100.00 – $32.90 = $67.10.

WHY NOT (B) borrows for the cents but forgets to take that dollar away from the dollars. (C) counts only one of the packs of pens. (D) prices only one of each item and ignores the quantities.

14. **D** $21.76

1 Find the cost of each kind of item: 3 × $5.75 = $17.25, and the tape measure costs $10.99.

2 Add to get the total cost: $17.25 + $10.99 = $28.24.

3 Subtract the total from the amount paid: $50.00 – $28.24 = $21.76.

WHY NOT (C) rounds each price up to a whole dollar instead of using the exact prices.

15. **A** $0.14

1 Divide each price by its number of ounces. Smaller size: $1.68 ÷ 12 = $0.14 per ounce.

2 Larger size: $4.20 ÷ 28 = $0.15 per ounce.

3 The better buy is the one with the lower price per ounce: the 12-ounce jar, at $0.14 per ounce.

WHY NOT (B) is the price per ounce of the worse buy. (D) subtracts the two total prices instead of finding each price per ounce.

16. **C** $0.62

1 Divide each price by its number of batteries. Smaller size: $3.04 ÷ 4 = $0.76 per battery.

2 Larger size: $7.44 ÷ 12 = $0.62 per battery.

3 The better buy is the one with the lower price per battery: the pack of 12, at $0.62 per battery.

WHY NOT (A) is the difference between the two prices per battery. (D) is the price per battery of the worse buy.

17. **B** $74.20

1 Find the cost before tax: 4×$15+$10 = $60+$10 = $70.

2 Find the tax: 6% of $70 = 0.06 × 70 = $4.20.

3 Add the tax to the price: $70 + $4.20 = $74.20.

WHY NOT (A) is the tax alone, not the total cost. (C) adds the tax rate as if it were dollars. (D) moves the decimal point only one place when finding the tax.

18. **A** $645

1 Only hours over 40 are overtime: 42 – 40 = 2 overtime hours.

2 Regular pay: 40 × $15 = $600. Overtime rate: 1.5 × $15 = $22.50 per hour.

3 Overtime pay: 2 × $22.50 = $45.

4 Total: $600 + $45 = $645.

WHY NOT (B) pays double time instead of time-and-a-half. (D) pays time-and-a-half for all the hours, not just the hours over 40.

19. **B** $667.50

1 Only hours over 40 are overtime: 43 – 40 = 3 overtime hours.

2 Regular pay: 40 × $15 = $600. Overtime rate: 1.5 × $15 = $22.50 per hour.

3 Overtime pay: 3 × $22.50 = $67.50.

4 Total: $600 + $67.50 = $667.50.

WHY NOT (A) pays every hour at the regular rate. (C) pays double time instead of time-and-a-half. (D) pays time-and-a-half for all the hours, not just the hours over 40.

20. **D** $1,950

1 Twice a month means 12 × 2 = 24 paychecks a year.

2 Divide the yearly pay by the number of paychecks: $46,800 ÷ 24 = $1,950.

TIP Check: 24 × $1,950 = $46,800.

WHY NOT (A) divides by 52, the number of weeks, not the number of paychecks. (C) uses 26 paychecks, but twice a month gives 24 paychecks a year.

21. **C** $1,950

1 Twice a month means 12 × 2 = 24 paychecks a year.

2 Divide the yearly pay by the number of paychecks: $46,800 ÷ 24 = $1,950.

TIP Check: 24 × $1,950 = $46,800.

WHY NOT (A) divides by 52, the number of weeks, not the number of paychecks. (B) uses 26 paychecks, but twice a month gives 24 paychecks a year. (D) divides by 12, which gives one month's pay.

22. **D** $17.10

1 Find the tip: 0.2 × $57 = $11.40.

2 Total with tip: $57 + $11.40 = $68.40.

3 Split it 4 ways: $68.40 ÷ 4 = $17.10.

WHY NOT (A) is one roommate's share of the tip only. (C) forgets to add the tip.

23. **C** $17.55

1 Find the tip: 0.2 × $117 = $23.40.

2 Total with tip: $117 + $23.40 = $140.40.

3 Split it 8 ways: $140.40 ÷ 8 = $17.55.

WHY NOT (D) is the total for the whole group, not one soldier's share.

24. **C** 7 weeks

1 Money still needed: $2,400 – $1,050 = $1,350.

2 Divide by the weekly savings: 1,350 ÷ 200 is 6 with 150 left over. After 6 weeks she has saved 6 × $200 = $1,200, still $150 short.

3 So she needs one more week: 6 + 1 = 7 weeks.

WHY NOT (B) rounds down, but after 6 weeks she is still $150 short. (D) ignores the $1,050 already saved.

25. **D** 9 weeks

1 Money still needed: $700 – $200 = $500.

2 Divide by the weekly savings: 500 ÷ 60 is 8 with 20 left over. After 8 weeks he has saved 8 × $60 = $480, still $20 short.

3 So he needs one more week: 8 + 1 = 9 weeks.

WHY NOT (C) rounds down, but after 8 weeks he is still $20 short.

26. **A** $1,496

1 Amount saved: $\frac{1}{10} \times \$2{,}040 = \$204$.

2 Amount paid on the car loan: $\frac{1}{6} \times \$2{,}040 = \$340$.

3 What is left: $2,040 – $204 – $340 = $1,496.

WHY NOT (B) subtracts only the amount paid on the car loan. (C) adds the fractions by adding the tops and the bottoms. (D) subtracts only the amount saved.

27. **A** 10

1 First take out the fixed cost: $210 – $45 = $165 is left to spend.

2 Divide by the price of one: $165 ÷ $16 is 10 with some money left over, because 10 × $16 = $160.

3 One more would cost $176, which is more than $165. Round *down*: 10 packs.

TIP Left over: $165 – $160 = $5, not enough for one more.

WHY NOT (B) rounds up, but 11 packs would cost $176. (C) forgets to subtract the $45 first. (D) adds the $45 instead of subtracting it.

28. **D** $8.52

1 Cost of 12 single bars: 12 × $2.25 = $27.

2 Savings: $27 – $18.48 = $8.52.

WHY NOT (A) is the savings on one bar, not on the whole box. (B) is the price per bar in the box.

29. **B** $716.04

1 Find the discount: $0.15 \times \$780 = \$117$.

2 Sale price: $\$780 - \$117 = \$663$.

3 Tax on the *sale* price: $0.08 \times \$663 = \$53.04$.

4 Total: $\$663 + \$53.04 = \$716.04$.

TIP Or chain the multipliers: $\$780 \times 0.85 \times 1.08 = \$716.04$.

WHY NOT (A) forgets to add the sales tax. (C) figures the tax on the regular price instead of the sale price. (D) adds the tax but forgets the discount.

30. **C** $544.32

1 Find the discount: $0.1 \times \$560 = \$56$.

2 Sale price: $\$560 - \$56 = \$504$.

3 Tax on the *sale* price: $0.08 \times \$504 = \$40.32$.

4 Total: $\$504 + \$40.32 = \$544.32$.

TIP Or chain the multipliers: $\$560 \times 0.9 \times 1.08 = \$544.32$.

WHY NOT (A) subtracts the tax instead of adding it. (B) forgets to add the sales tax. (D) figures the tax on the regular price instead of the sale price.

31. **C** $699.30

1 Find the discount: $0.1 \times \$740 = \$74$.

2 Sale price: $\$740 - \$74 = \$666$.

3 Tax on the *sale* price: $0.05 \times \$666 = \$33.30$.

4 Total: $\$666 + \$33.30 = \$699.30$.

TIP Or chain the multipliers: $\$740 \times 0.9 \times 1.05 = \$699.30$.

WHY NOT (A) subtracts the tax instead of adding it. (B) forgets to add the sales tax. (D) figures the tax on the regular price instead of the sale price.

32. **A** $1,073.50

1 Total hours: $9 \times 5 + 6 = 51$ hours.

2 Only hours over 40 are overtime: $51 - 40 = 11$ overtime hours.

3 Regular pay: $40 \times \$19 = \$760$. Overtime rate: $1.5 \times \$19 = \$28.50$ per hour.

4 Overtime pay: $11 \times \$28.50 = \$313.50$.

5 Total: $\$760 + \$313.50 = \$1,073.50$.

WHY NOT (B) pays double time instead of time-and-a-half. (D) pays time-and-a-half for all the hours, not just the hours over 40.

33. **B** 48 hours

1 Pay for the first 40 hours: $40 \times \$15 = \$600$.

2 The rest of the pay is overtime: $\$780 - \$600 = \$180$.

3 Overtime rate: $1.5 \times \$15 = \$22.50$ per hour. Overtime hours: $\$180 \div \$22.50 = 8$.

4 Total hours: $40 + 8 = 48$ hours.

WHY NOT (A) gives only the overtime hours, not the total. (D) divides the overtime pay by the regular rate instead of the overtime rate.

34. **A** The case of 32

1 Find each price per bottle (price ÷ number of bottles): $\$3.48 \div 12 = \$0.29$; $\$5.28 \div 24 = \$0.22$; $\$6.72 \div 32 = \$0.21$.

2 The lowest price per bottle is $0.21. The case of 32 is the best buy.

WHY NOT (B) costs $0.22 per bottle, more than $0.21. (C) has the lowest price tag, but it costs $0.29 per bottle, more than $0.21. (D) is not true: the prices per bottle are $0.29, $0.22, and $0.21.

35. **C** The 24-ounce bag

1 Find each price per ounce (price ÷ number of ounces): $\$4.44 \div 12 = \$0.37$; $\$6.96 \div 24 = \$0.29$; $\$12.40 \div 40 = \$0.31$.

2 The lowest price per ounce is $0.29. The 24-ounce bag is the best buy.

TIP The biggest package is not always the best deal, so always compare unit prices.

WHY NOT (A) has the lowest price tag, but it costs $0.37 per ounce, more than $0.29. (B) is the largest size, but it costs $0.31 per ounce, more than $0.29. (D) is not true: the prices per ounce are $0.37, $0.29, and $0.31.

36. **B** $441

1 First amount (paid on the car loan): $\frac{1}{10} \times \$2{,}940 = \$294$.

2 What is left: $\$2{,}940 - \$294 = \$2{,}646$.

3 Second amount is $\frac{1}{6}$ of *that*: $\frac{1}{6} \times \$2{,}646 = \$441$.

WHY NOT (A) is the amount paid on the car loan. (C) takes $\frac{1}{6}$ of the whole pay instead of $\frac{1}{6}$ of what is left. (D) is the amount remaining, not the amount asked for.

37. **B** $1,350

1 Every two weeks means 52 ÷ 2 = 26 paychecks a year (*not* 24).

2 Divide the yearly pay by the number of paychecks: $46,800 ÷ 26 = $1,800.

3 Taxes withheld: 0.25 × $1,800 = $450.

4 Take-home pay: $1,800 – $450 = $1,350.

WHY NOT (A) is the amount withheld, not the take-home pay. (C) forgets to subtract the taxes withheld. (D) divides by 12 months instead of the number of paychecks.

38. **A** $1,560

1 Hours in a year: 40 × 52 = 2,080 hours.

2 Job A per year: 2,080 × $17 = $35,360.

3 Compare with Job B's $36,920: $36,920 – $35,360 = $1,560. Job B pays $1,560 more.

WHY NOT (B) uses 2,000 hours a year instead of 2,080. (C) assumes 4 weeks in every month (48 weeks), but a year has 52 weeks. (D) is Job A's yearly pay, not the difference.

39. **C** $1,560

1 Hours in a year: 40 × 52 = 2,080 hours.

2 Job A per year: 2,080 × $20 = $41,600.

3 Compare with Job B's $40,040: $41,600 – $40,040 = $1,560. Job A pays $1,560 more.

WHY NOT (A) is the difference for one week, not for a year. (B) uses 2,000 hours a year instead of 2,080. (D) is Job A's yearly pay, not the difference.

40. **B** $50.83

1 Full packs: 31 ÷ 12 is 2 with 7 left over, so buy 2 packs and 7 single rolls.

2 Packs: 2 × $17.40 = $34.80. Singles: 7 × $2.29 = $16.03.

3 Total: $34.80 + $16.03 = $50.83.

WHY NOT (A) forgets the 7 single rolls. (C) buys one more full pack instead of 7 single rolls. (D) buys every roll singly.

CHAPTER 23

Interest, Commission & Percent Change

Tested on: Arithmetic Reasoning

Key Concepts

Chapter 6 covered the basic percent questions. This chapter puts percents to work with money: interest on savings and loans, commission and markup, raises, and the two kinds of problems that trip up the most test takers, *chains* of percents and *working backward* from a percent.

Simple interest

$$I = P \times r \times t$$

P = principal (the amount saved or borrowed), r = yearly rate as a decimal, t = time *in years*. The balance (or the amount owed) is $P + I$. Change months to years by dividing by 12: 6 months = $\frac{6}{12} = \frac{1}{2}$ year, 8 months = $\frac{8}{12} = \frac{2}{3}$ year, 18 months = $1\frac{1}{2}$ years.

Commission, markup, and raises

- **Commission** = rate × sales. Total pay = base pay + commission.
- **Markup** = rate × cost; selling price = cost + markup; profit = selling price − cost.
- **Raise:** new pay = old pay + rate × old pay = (1 + rate)× old pay.

Chains of percents and working backward

- Percent changes in a row **multiply**: 20% off and then 10% off leaves $0.80 \times 0.90 = 0.72$ of the price, a 28% discount, not 30%.
- **Compound interest** works the same way: each year's interest is added before the next year's interest is figured.
- **Working backward:** if the price *after* 20% off is \$64, then \$64 is 80% of the original. Divide: $64 \div 0.80 = \$80$.

WORKED EXAMPLE

After a 25% raise, a store manager earns \$20 per hour. What did she earn before the raise?
Solution. The new wage is 125% of the old wage: $1.25 \times \text{old} = 20$, so old = $20 \div 1.25 = \$16$. Check: 25% of \$16 is \$4, and \$16 + \$4 = \$20. (Taking 25% off \$20 gives \$15, which is wrong: the raise was 25% of the *old* wage.)

No-Calculator Tip Turn friendly decimals into fractions: dividing by 0.80 is multiplying by $\frac{5}{4}$, and dividing by 1.25 is multiplying by $\frac{4}{5}$. For interest, find one year's interest first, then multiply by the number of years.

Common Traps

- Using months as if they were years in $I = Prt$.
- Working backward by adding the percent to the *new* amount ($64 after 20% off is not 64 + 12.80).
- Adding two discounts in a row, or adding a raise and a cut that follow each other.
- Paying commission on *all* sales when it is earned only above a set amount.

Practice Set 23

Score: _____ / 35 Target time: 42 minutes

Problems 1–10 are warm-ups, 11–24 are test level, and 25–35 are challenge problems.

1. Carlos borrows $5,000 from a credit union to buy a used car. The loan charges 4% simple interest per year. How much interest will he pay in 4 years?

A $200 **B** $600 **C** $800 **D** $5,800

2. Specialist Walker puts a $5,000 enlistment bonus into a savings account at the base credit union. The account pays 2% simple interest per year. How much interest will he earn in 4 years?

A $400 **B** $4,000 **C** $5,400 **D** $40,000

3. Mei buys a certificate of deposit (CD) for $5,500. The CD pays 3% simple interest per year. How much interest will she earn in 2 years?

A $165 **B** $280 **C** $330 **D** $5,830

4. Grace, an Army veteran, sells insurance and earns a 5% commission on the yearly premiums she sells. This month she sold $29,000 in premiums. What is her commission?

A $1,450 **B** $14,500
C $27,550 **D** $145,000

5. Emily sells cars and earns a 4% commission on the price of each car she sells. What is her commission on a car that sells for $22,500?

A $90 **B** $800 **C** $900 **D** $9,000

6. When Sergeant Carter is promoted, her monthly base pay goes up 6%. Before the promotion it was $2,950. What is it after the promotion?

A $177 **B** $2,773 **C** $2,956 **D** $3,127

7. Tyrone earns a salary of $70,000 a year. He gets a 6% raise. What is his new yearly salary?

A $4,200 **B** $74,200
C $104,200 **D** $112,000

8. Specialist Kim's monthly base pay is $2,950. This year's military pay raise is 3%. What will his new monthly base pay be?

A $88.50 **B** $3,038.50
C $3,835.00 **D** $4,538.50

9. A battalion's goal for its charity drive is $2,500. So far it has raised $1,000. What percent of the goal has it reached?

A 40% **B** 50% **C** 60% **D** 250%

10. A recruiting station's goal is 40 enlistments this quarter. So far it has signed 22 recruits. What percent of its goal has it met?

A 0.55% **B** 40% **C** 45% **D** 55%

11. Mei deposits $3,100 in a savings account that pays 5% simple interest per year. How much interest will she earn in 18 months?

A $232.50 **B** $279.00
C $2,790.00 **D** $3,332.50

12. Hannah borrows $11,000 from a credit union to buy a used car. The loan charges 6% simple interest per year. How much interest will she pay in 10 months?

A $55 **B** $550 **C** $660 **D** $6,600

13. Jamal deposits $500 in a savings account that pays 5% simple interest per year. How much money will be in the account after 6 months?

A $12.50 **B** $512.50 **C** $525.00 **D** $625.00

14. Seaman Novak borrows $5,500 from the base credit union to furnish his first apartment off base. The loan charges 6% simple interest per year. What is the total amount he must pay back at the end of 3 years?

A $990 **B** $5,830 **C** $6,490 **D** $15,400

15. Brandon works at a home-security company and earns a base salary of $500 per week plus a 5% commission on his sales. His sales this week total $6,100. What is his total pay for the week?

A $305 **B** $805 **C** $830 **D** $3,550

16. Derek works at a furniture store and earns a base salary of $1,500 per month plus a 2% commission on his sales. His sales this month total $75,000. What is his total pay for the month?

A $3,000 **B** $3,030 **C** $16,500 **D** $76,500

17. After a promotion, Sergeant Patel's monthly base pay went up 12% to $3,248. What was his monthly base pay before the promotion?

A $389.76 **B** $2,858.24
C $2,900.00 **D** $3,236.00

18. A unit's blood drive collected 96 pints this year, 50% more than last year. How many pints did it collect last year?

A 48 **B** 64 **C** 65 **D** 69

19. A food truck spends $2 to make each burrito and marks up the cost by 150%. What price is put on each burrito?

A $2.30 **B** $3.00 **C** $4.00 **D** $5.00

20. A vendor at the base's Fourth of July fair buys T-shirts for $8 each and marks up the cost by 50%. What price is put on each T-shirt?

A $12 **B** $13 **C** $14 **D** $58

21. A car dealer lowers the price of a $19,000 used car by 5%. When the car does not sell, the dealer cuts the new price by another 5%. What is the final price of the used car?

A $17,100.00 **B** $17,147.50
C $18,050.00 **D** $18,952.50

22. A store takes 50% off a $150 winter coat. Service members get an extra 10% military discount on the sale price. What is the price after both discounts?

A $37.50 **B** $57.50 **C** $60.00 **D** $67.50

23. On a weekend pass, a group of airmen order a meal that costs $116. They pay a 5% sales tax and leave a 20% tip, both figured on the $116. What is the total cost?

A $121.80 **B** $139.20 **C** $145.00 **D** $146.16

24. After raising $1,800, a battalion has reached 30% of the goal for its charity drive. What is the goal?

A $3,060 **B** $6,000 **C** $7,000 **D** $7,500

25. Tyrone deposited $6,000 in an account that pays simple interest. After 4 years the account had earned $1,200 in interest. What is the yearly interest rate?

A 0.05% **B** 1.25% **C** 5% **D** 20%

26. Seaman Walker borrows $4,000 from the base credit union to furnish her first apartment off base. The loan charges 6.5% simple interest per year. After how many years will the interest on the loan add up to $1,300?

A 0.05 years **B** 3 years
C 4 years **D** 5 years

27. Sofia works at a home-security company. She earns $450 per week plus a 10% commission on all sales *over* $2,000 for the week. Her sales this week total $5,800. What is her total pay for the week?

A $380 **B** $830 **C** $930 **D** $1,030

28. Maria earns $300 per week plus a 5% commission on her sales at a car dealership. How much must she sell in a week to earn a total of $1,000?

A $700 **B** $14,000 **C** $18,000 **D** $20,000

29. Hannah bought a pair of headphones on sale for 40% off. With a 6% sales tax added to the sale price, she paid $57.24. What was the original price of the pair of headphones?

A $54 **B** $60 **C** $70 **D** $90

30. Marcus bought a jacket on sale for 10% off. With a 6% sales tax added to the sale price, he paid $124.02. What was the original price of the jacket?

A $110.00 **B** $117.00 **C** $130.00 **D** $137.80

31. During a clearance sale, a sporting-goods store marks a tent down 40%. At the register, customers get an additional 10% off the sale price. The two discounts together are the same as a single discount of what percent?

A 4% **B** 31% **C** 46% **D** 50%

32. A hardware store buys hammers from a supplier for $8 each and marks up the cost by 75%. Later, the hammer goes on sale for 30% off the marked price. How much profit does the store make on each hammer sold at the sale price?

A $1.80 **B** $3.60 **C** $6.00 **D** $9.80

33. Lance Corporal Kim deposits a $10,400 reenlistment bonus in an account that pays 10% interest per year, compounded once a year. How much interest will the account earn in 2 years?

A $1,040 **B** $2,080 **C** $2,184 **D** $12,584

34. Petty Officer Chen deposits a $14,800 reenlistment bonus in an account that pays 2% yearly interest, compounded every six months (so 1% is added every six months). How much interest will the account earn in 1 year?

A $148.00 **B** $197.48 **C** $296.00 **D** $297.48

35. Hannah borrows $9,600 to pay for a welding certification course. The loan charges 4% simple interest per year, figured on the full $9,600 for all 4 years. She will repay the principal plus interest in equal monthly payments over 4 years. How much is each monthly payment?

A $32 **B** $200 **C** $208 **D** $232

Answers & Explanations 23

1 C	**8** B	**15** B	**22** D	**29** D
2 A	**9** A	**16** A	**23** C	**30** C
3 C	**10** D	**17** C	**24** B	**31** C
4 A	**11** A	**18** B	**25** C	**32** A
5 C	**12** B	**19** D	**26** D	**33** C
6 D	**13** B	**20** A	**27** B	**34** D
7 B	**14** C	**21** B	**28** B	**35** D

1. **C** $800

1 Use $I = P \times r \times t$ with $r = 4\% = 0.04$ and $t = 4$ years.

2 One year's interest: $\$5{,}000 \times 0.04 = \$200$.

3 For 4 years: $\$200 \times 4 = \$800$.

WHY NOT (A) finds the interest for only one year. (D) is the total with interest, not the interest alone.

2. **A** $400

1 Use $I = P \times r \times t$ with $r = 2\% = 0.02$ and $t = 4$ years.

2 One year's interest: $\$5{,}000 \times 0.02 = \$100$.

3 For 4 years: $\$100 \times 4 = \$400$.

WHY NOT (B) writes 2% as 0.2 instead of 0.02. (C) is the total with interest, not the interest alone. (D) forgets to change the percent to a decimal.

3. **C** $330

1 Use $I = P \times r \times t$ with $r = 3\% = 0.03$ and $t = 2$ years.

2 One year's interest: $\$5{,}500 \times 0.03 = \$165$.

3 For 2 years: $\$165 \times 2 = \$330$.

WHY NOT (A) finds the interest for only one year. (D) is the total with interest, not the interest alone.

4. **A** $1,450

1 Commission = rate × sales. Change the rate to a decimal: 5% = 0.05.

2 Multiply: $0.05 \times \$29{,}000 = \$1{,}450$.

WHY NOT (B) writes 5% as 0.5 instead of 0.05. (C) subtracts the commission from the premiums sold. (D) forgets to change the percent to a decimal.

5. **C** $900

1 Commission = rate × sales. Change the rate to a decimal: 4% = 0.04.

2 Multiply: $0.04 \times \$22{,}500 = \$900$.

WHY NOT (A) moves the decimal point three places instead of two. (D) writes 4% as 0.4 instead of 0.04.

6. **D** $3,127

1 Find the raise: $0.06 \times \$2{,}950 = \$177$.

2 Add it to the old pay: $\$2{,}950 + \$177 = \$3{,}127$.

TIP In one step: $1.06 \times \$2{,}950 = \$3{,}127$.

WHY NOT (A) is the amount of the raise, not the new monthly base pay. (B) subtracts the change instead of adding it. (C) adds the percent as if it were dollars.

7. **B** $74,200

1 Find the raise: $0.06 \times \$70{,}000 = \$4{,}200$.

2 Add it to the old pay: $\$70{,}000 + \$4{,}200 = \$74{,}200$.

TIP In one step: $1.06 \times \$70{,}000 = \$74{,}200$.

WHY NOT (A) is the amount of the raise, not the new salary. (D) moves the decimal point only one place.

8. **B** $3,038.50

1 Find the raise: $0.03 \times \$2{,}950 = \$88.50$.

2 Add it to the old pay: $\$2{,}950 + \$88.50 = \$3{,}038.50$.

TIP In one step: $1.03 \times \$2{,}950 = \$3{,}038.50$.

WHY NOT (A) is the amount of the raise, not the new monthly base pay. (C) moves the decimal point only one place.

9. **A** 40%

1 Write the part done over the goal: $\frac{1{,}000}{2{,}500}$.

2 Divide and change to a percent: $1{,}000 \div 2{,}500 = 0.4 = 40\%$.

WHY NOT (C) is the percent still to go, not the percent done. (D) divides the goal by the amount done instead of the amount done by the goal.

10. **D** 55%

1 Write the part done over the goal: $\frac{22}{40}$.

2 Divide and change to a percent: $22 \div 40 = 0.55 = 55\%$.

WHY NOT (A) forgets to multiply by 100. (C) is the percent still to go, not the percent done.

11. **A** $232.50

1 Time must be in years: 18 months = $\frac{18}{12} = 1\frac{1}{2} =$ 1.5 years.

2 One year's interest: $\$3{,}100 \times 0.05 = \$155$.

3 For $1\frac{1}{2}$ years: $\$155 \times 1\frac{1}{2} = \$232.50$.

WHY NOT (B) writes 18 months as 1.8 years. (C) uses 18 years instead of 18 months. (D) is the total with interest, not the interest alone.

12. **B** $550

1 Time must be in years: 10 months = $\frac{10}{12} = \frac{5}{6}$ year.

2 One year's interest: $\$11{,}000 \times 0.06 = \$660$.

3 For $\frac{5}{6}$ of a year: $\$660 \times \frac{5}{6} = \$550$.

WHY NOT (A) finds the interest for just one month. (C) finds the interest for a full year. (D) uses 10 years instead of 10 months.

13. **B** $512.50

1 Change the time to years: 6 months = $\frac{6}{12} = \frac{1}{2} =$ 0.5 year.

2 One year's interest: $\$500 \times 0.05 = \$25$.

3 Interest for the whole time: $\$25 \times \frac{1}{2} = \$12.50$.

4 Add it to the principal: $\$500 + \$12.50 = \$512.50$.

WHY NOT (A) is only the interest, not the total. (C) adds only one year of interest. (D) writes 5% as 0.5 instead of 0.05.

14. **C** $6,490

1 One year's interest: $\$5{,}500 \times 0.06 = \$330$.

2 Interest for the whole time: $\$330 \times 3 = \$990$.

3 Add it to the principal: $\$5{,}500 + \$990 = \$6{,}490$.

WHY NOT (A) is only the interest, not the total. (B) adds only one year of interest. (D) writes 6% as 0.6 instead of 0.06.

15. **B** $805

1 Commission: $0.05 \times \$6{,}100 = \$305$.

2 Add the base salary: $\$500 + \$305 = \$805$.

WHY NOT (A) is the commission only; it leaves out the base salary. (C) figures the commission on the base salary plus the sales. (D) writes 5% as 0.5.

16. **A** $3,000

1 Commission: $0.02 \times \$75{,}000 = \$1{,}500$.

2 Add the base salary: $\$1{,}500 + \$1{,}500 = \$3{,}000$.

WHY NOT (B) figures the commission on the base salary plus the sales. (C) writes 2% as 0.2. (D) adds the sales instead of the commission.

17. **C** $2,900.00

1 The new pay is 112% of the old pay: $1.12 \times$ old pay = $3,248.

2 Divide: $\$3{,}248 \div 1.12 = \$2{,}900$.

TIP Check: $1.12 \times \$2{,}900 = \$3{,}248$.

WHY NOT (A) finds 12% of the new pay instead of removing the raise. (B) takes 12% off the new pay, but the raise was 12% of the old pay. (D) subtracts the percent as if it were dollars.

18. **B** 64

1 The new number is 150% of the old number: $1.5 \times$ old = 96.

2 Divide: $96 \div 1.5 = 64$.

TIP Check: 50% of 64 is 32, and $64 + 32 = 96$.

WHY NOT (A) takes 50% off this year's number, but the increase was 50% of the earlier number.

19. **D** $5.00

1 Markup: $1.5 \times \$2 = \$3$.

2 Selling price: $\$2 + \$3 = \$5$.

TIP In one step: $2.5 \times \$2 = \$5$.

WHY NOT (A) moves the decimal point one place too far when finding the markup. (B) is the markup, not the selling price.

20. **A** $12

1 Markup: $0.5 \times \$8 = \$4$.

2 Selling price: $\$8 + \$4 = \$12$.

TIP In one step: $1.5 \times \$8 = \$12$.

WHY NOT (D) adds the percent as if it were dollars.

21. **B** $17,147.50

1 First discount: $\$19{,}000 - 0.05 \times \$19{,}000 = \$19{,}000 - \$950 = \$18{,}050$.

2 The second discount is taken from the *new* price: $\$18{,}050 - 0.05 \times \$18{,}050 = \$18{,}050 - \$902.50 = \$17{,}147.50$.

TIP In one line: $\$19{,}000 \times 0.95 \times 0.95 = \$17{,}147.50$.

WHY NOT (A) adds the discounts to get 10% off. (C) stops after the first discount. (D) multiplies the two rates ($5\% \times 5\% = 0.25\%$) and takes only that much off.

22. **D** $67.50

1 First discount: $\$150 - 0.5 \times \$150 = \$150 - \$75 = \$75$.

2 The second discount is taken from the *new* price: $\$75 - 0.1 \times \$75 = \$75 - \$7.50 = \$67.50$.

TIP In one line: $\$150 \times 0.5 \times 0.9 = \$67.50$.

WHY NOT (C) adds the discounts to get 60% off.

23. **C** $145.00

1 Tax: $0.05 \times \$116 = \$5.80$.

2 Tip: $0.2 \times \$116 = \$23.20$.

3 Total: $\$116 + \$5.80 + \$23.20 = \$145$.

TIP Both percents are of the same amount, so add them first: 5%+20% = 25%, and $1.25 \times \$116 = \$145$.

WHY NOT (A) forgets the tip. (B) forgets the tax. (D) figures the tip on the total with tax, but the tip is on the food cost only.

24. **B** $6,000

1 $1,800 is 30% of the goal: $0.3 \times \text{goal} = \$1{,}800$.

2 Divide: $\$1{,}800 \div 0.3 = \$6{,}000$.

TIP Check: 30% of $6,000 is $1,800.

WHY NOT (A) adds 70% of $1,800 instead of dividing.

25. **C** 5%

1 Interest for one year: $\$1{,}200 \div 4 = \$300$.

2 Divide by the principal: $\$300 \div \$6{,}000 = 0.05$.

3 Change to a percent: $0.05 = 5\%$.

TIP Check: $\$6{,}000 \times 0.05 \times 4 = \$1{,}200$.

WHY NOT (A) forgets to change the decimal to a percent. (B) divides by 4 twice. (D) forgets to divide by the 4 years.

26. **D** 5 years

1 One year's interest: $\$4{,}000 \times 0.065 = \$260$.

2 Number of years: $\$1{,}300 \div \$260 = 5$.

TIP Check: $\$260 \times 5 = \$1{,}300$.

WHY NOT (A) forgets to change the percent to a decimal.

27. **B** $830

1 Sales that earn commission: $\$5{,}800 - \$2{,}000 = \$3{,}800$.

2 Commission: $0.1 \times \$3{,}800 = \$380$.

3 Total pay: $\$450 + \$380 = \$830$.

WHY NOT (A) is the commission only; it leaves out the base pay. (D) pays commission on all the sales, not just the sales over $2,000.

28. **B** $14,000

1 Commission needed: $\$1{,}000 - \$300 = \$700$.

2 That commission is 5% of sales: $0.05 \times \text{sales} = \$700$.

3 Divide: $\$700 \div 0.05 = \$14{,}000$.

TIP Check: $0.05 \times \$14{,}000 = \$700$, and $\$300 + \$700 = \$1{,}000$.

WHY NOT (A) is the commission needed, not the sales needed. (D) forgets to subtract the base pay first.

29. **D** $90

1 Remove the tax first. The amount paid is 106% of the sale price: $\$57.24 \div 1.06 = \$54$.

2 The sale price is 60% of the original: $\$54 \div 0.6 = \$90$.

TIP Check: $\$90 \times 0.6 = \$54$, and $\$54 \times 1.06 = \$57.24$.

WHY NOT (A) is the sale price, not the original price.

30. **C** $130.00

1 Remove the tax first. The amount paid is 106% of the sale price: $\$124.02 \div 1.06 = \$117$.

2 The sale price is 90% of the original: $\$117 \div 0.9 = \$130$.

TIP Check: $\$130 \times 0.9 = \$117$, and $\$117 \times 1.06 = \$124.02$.

WHY NOT (B) is the sale price, not the original price. (D) forgets to remove the sales tax.

31. **C** 46%

1 After 40% off you pay 60% of the price, and after another 10% off you pay 90% of *that*.

2 Multiply: $0.6 \times 0.9 = 0.54$, so you pay 54% of the original price.

3 The single discount is $100\% - 54\% = 46\%$.

TIP Test it on $100: 40% off gives $60, and 10% off that gives $54, which is $46 off.

WHY NOT (A) multiplies the two percents. (D) adds the two discounts, but the second is taken from a smaller price.

32. **A** $1.80

1 Marked price: $\$8 + 0.75 \times \$8 = \$8 + \$6 = \$14$.

2 Sale price: $\$14 - 0.3 \times \$14 = \$14 - \$4.20 = \$9.80$.

3 Profit: $\$9.80 - \$8 = \$1.80$.

WHY NOT (B) combines 75% up and 30% off into one 45% markup. (C) is the original markup; it forgets the sale. (D) is the sale price, not the profit.

33. **C** $2,184

1 Rate for each period: 10%.

2 After the first year: $\$10{,}400 + 0.1 \times \$10{,}400 = \$10{,}400 + \$1{,}040 = \$11{,}440$.

3 After the second: $\$11{,}440 + 0.1 \times \$11{,}440 = \$11{,}440 + \$1{,}144 = \$12{,}584$.

4 Interest earned: $\$12{,}584 - \$10{,}400 = \$2{,}184$.

TIP The second year earns more ($1,144 vs. $1,040) because it earns interest on the interest.

WHY NOT (A) stops after the first period. (B) uses simple interest; it leaves out the interest earned on the first interest. (D) is the balance, not the interest.

34. **D** $297.48

1 Rate for each period: 1% (half of the yearly rate).

2 After the first six months: $14,800 + 0.01 × $14,800 = $14,800 + $148 = $14,948.

3 After the second: $14,948 + 0.01 × $14,948 = $14,948 + $149.48 = $15,097.48.

4 Interest earned: $15,097.48 – $14,800 = $297.48.

TIP The second period earns more ($149.48 vs. $148) because it earns interest on the interest.

WHY NOT (A) stops after the first period. (C) uses simple interest; it leaves out the interest earned on the first interest.

35. **D** $232

1 Interest: $9,600 × 0.04 × 4 = $1,536.

2 Total to repay: $9,600 + $1,536 = $11,136.

3 Number of payments: 4 × 12 = 48 months.

4 Each payment: $11,136 ÷ 48 = $232.

WHY NOT (A) gives only the interest part of each payment. (B) forgets to add the interest. (C) adds only one year of interest.

CHAPTER 24

Rate, Distance & Time

Tested on: Arithmetic Reasoning

Key Concepts

Anything that moves at a steady rate (a car, a convoy, a runner, a printer turning out pages) follows one formula. The skill is in setting it up with matching units and in knowing which rates to add, subtract, or leave alone.

The distance formula

$$d = r \times t \qquad r = \frac{d}{t} \qquad t = \frac{d}{r}$$

d = distance, r = rate (speed), t = time. The units must match: with a speed in miles per hour, the time must be in *hours*. The same formula works for any rate: amount = rate × time (words typed, pages printed, gallons used).

Minutes and hours

Divide minutes by 60 to get hours: 15 min = $\frac{1}{4}$ h = 0.25 h, 20 min = $\frac{1}{3}$ h, 30 min = 0.5 h, 45 min = 0.75 h. Going the other way, multiply by 60: 0.4 h = 24 min. Beware: 2.5 hours is 2 hours *30* minutes, not 2 hours 50 minutes.

Two movers

- Moving **toward** each other or in **opposite** directions: the gap changes at the *sum* of the speeds.
- Moving in the **same** direction (one catching up): the gap changes at the *difference* of the speeds.
- **Average speed** = $\dfrac{\text{total distance}}{\text{total time}}$. It is *not* the average of the speeds unless the times are equal.

WORKED EXAMPLE

A convoy drives 120 miles to a range at 40 mph and returns at 60 mph. What is its average speed for the round trip?

Solution. Out: 120 ÷ 40 = 3 hours. Back: 120 ÷ 60 = 2 hours. Total: 240 miles in 5 hours, so the average speed is 240 ÷ 5 = 48 mph, not 50 mph: the convoy spends more time at the slower speed.

No-Calculator Tip For pace and speed, use 60: a pace of 8 minutes per mile is 60 ÷ 8 = 7.5 mph, and 10 mph is 60 ÷ 10 = 6 minutes per mile.

Common Traps

- Multiplying a speed in miles per hour by a time in *minutes*.
- Writing 45 minutes as 0.45 hour (it is 0.75 hour).

- Averaging two speeds instead of dividing total distance by total time.
- Adding speeds when one vehicle chases another (subtract them).

Practice Set 24

Score: _____ / 40 Target time: 48 minutes

Problems 1–12 are warm-ups, 13–28 are test level, and 29–40 are challenge problems.

1. Sofia drives at an average speed of 61 miles per hour for 6 hours. How far does she drive?

A 67 miles **B** 366 miles
C 427 miles **D** 466 miles

2. A supply convoy travels 144 miles in 4 hours. What is the convoy's average speed?

A 31 mph **B** 36 mph
C 140 mph **D** 576 mph

3. A freight train travels at an average speed of 55 miles per hour for 8 hours. How far does the train travel?

A 63 miles **B** 290 miles
C 440 miles **D** 495 miles

4. Leila rides a bicycle at an average speed of 15 miles per hour for 4 hours. How far does she ride?

A 19 miles **B** 45 miles **C** 60 miles **D** 75 miles

5. Victor rides a bicycle at 18 miles per hour for 20 minutes. How many miles does he ride?

A 3.6 miles **B** 6 miles
C 18 miles **D** 360 miles

6. The supply convoy Staff Sergeant Okafor is driving in travels at 42 miles per hour for 50 minutes. How many miles does the convoy travel?

A 21 miles **B** 25 miles
C 35 miles **D** 2,100 miles

7. Diego's delivery van used 14 gallons of gas to travel 252 miles. How many miles per gallon did it get?

A 18 **B** 21 **C** 22 **D** 238

8. Omar's pickup truck used 15 gallons of gas to travel 255 miles. How many miles per gallon did it get?

A 17 **B** 18 **C** 19 **D** 240

9. On a field trip, a school bus used 21 gallons of diesel to travel 168 miles. How many miles per gallon did it get?

A 5 **B** 6 **C** 8 **D** 147

10. Working steadily, a soldier can fill 24 sandbags per hour. How many sandbags can the soldier fill in 3 hours?

A 27 **B** 72 **C** 82 **D** 102

11. A field kitchen crew can make 50 sandwiches per hour. How many hours will it take to make 200 sandwiches?

A 1 hour **B** 3 hours
C 4 hours **D** 150 hours

12. A conveyor belt at a warehouse moves 15 packages per minute. How many packages does it move in 20 minutes?

A 35 **B** 300 **C** 350 **D** 375

13. Ryan rides a bicycle at 16 miles per hour for 1 hour 45 minutes. How many miles does he ride?

A 13 miles **B** 16 miles
C 23.2 miles **D** 28 miles

14. A train covers 7 miles in 10 minutes. What is the train's average speed in miles per hour?

A 0.7 mph **B** 42 mph **C** 47 mph **D** 70 mph

15. Tyrone lives 30 miles from work and drives at an average of 60 miles per hour. How many minutes does the drive take?

A 30 minutes **B** 35 minutes
C 50 minutes **D** 120 minutes

16. Two buses start 115 miles apart and travel toward each other on the same road. One goes 55 miles per hour and the other goes 60 miles per hour. In how many hours will they meet?

A 1 hour **B** 2 hours
C 3 hours **D** 23 hours

17. Two cyclists start 116 miles apart and travel toward each other on the same trail. One goes 11 miles per hour and the other goes 18 miles per hour. In how many hours will they meet?

A 1 hour **B** 3 hours **C** 4 hours **D** 8 hours

18. Two cars leave a town at the same time and travel in opposite directions. One goes 50 miles per hour and the other goes 65 miles per hour. How far apart are they after 5 hours?

A 75 miles **B** 575 miles
C 625 miles **D** 675 miles

19. James's delivery van gets 14 miles per gallon. Gas costs $3.35 per gallon. How much will gas cost for a 182-mile trip to visit family?

A $43.55 **B** $46.90 **C** $48.55 **D** $609.70

20. Victor's pickup truck gets 15 miles per gallon. Gas costs $3.05 per gallon. How much will gas cost for a 75-mile drive to the beach?

A $5.00 **B** $9.25 **C** $15.25 **D** $45.75

21. Tyrone jogs 4 miles in 30 minutes. What is his average speed in miles per hour?

A 7.5 mph **B** 8 mph
C 11 mph **D** 120 mph

22. Emily runs at a steady 7.5 miles per hour. How many minutes does it take her to run 5 miles?

A 8 minutes **B** 40 minutes
C 45 minutes **D** 66 minutes

23. Tasha walks at a steady 3 miles per hour. How many minutes does it take her to walk 3 miles?

A 1 minute **B** 20 minutes
C 60 minutes **D** 65 minutes

24. On a training exercise, a squad marches 10 miles at 4 miles per hour, then 2 miles through rough terrain at 2 miles per hour. How many hours does the whole trip take?

A 3.5 hours **B** 4 hours
C 4.5 hours **D** 6 hours

25. Ana runs 25 miles at 10 miles per hour, then walks 6 miles at 3 miles per hour. How many hours does the whole trip take?

A 2.5 hours **B** 3.1 hours
C 4.5 hours **D** 6 hours

26. Andre leaves home at 10:20 a.m. and drives 195 miles at an average speed of 60 miles per hour. At what time does he arrive?

A 12:35 p.m. **B** 1:05 p.m.
C 1:20 p.m. **D** 1:35 p.m.

27. Luis leaves for the airport at 8:15 a.m. and drives 180 miles at an average speed of 48 miles per hour. At what time does he arrive?

A 11:15 a.m. **B** 11:45 a.m.
C 12:00 p.m. **D** 1:00 p.m.

28. A water purification unit at a field camp produces 3,000 gallons of drinking water per hour. How many gallons does it produce in 10 minutes?

A 300 **B** 350 **C** 400 **D** 500

29. Jasmine rides a bicycle 18 miles to a lake at 18 miles per hour and rides back at 9 miles per hour. What is the average speed for the round trip?

A 6 mph **B** 9 mph
C 12 mph **D** 13.5 mph

30. Corporal Reyes leads a convoy 48 miles to a training range at 24 miles per hour. On the way back, the convoy travels the same road at 48 miles per hour. What is the average speed for the round trip?

A 16 mph **B** 32 mph **C** 36 mph **D** 72 mph

31. Warrant Officer Kim flies a supply helicopter 300 miles to an outpost at 150 miles per hour and flies back at 100 miles per hour. What is the average speed for the whole round trip?

A 120 mph **B** 125 mph
C 130 mph **D** 250 mph

32. Victor leaves home on a bicycle at 12 miles per hour. An hour and a half later, his sister follows on a scooter at 21 miles per hour. How many hours after the sister leaves will she catch up?

A 1 hour **B** 2 hours
C 3 hours **D** 3.5 hours

33. A cargo ship leaves port at 16 miles per hour. Two hours later, a supply boat leaves the same port on the same course at 24 miles per hour. How many hours after the supply boat leaves will it catch up?

A 0.8 hours **B** 4 hours
C 6 hours **D** 7 hours

34. A platoon starts a road march at 4 miles per hour. One hour later, Corporal Carter follows the same route in a Humvee carrying water at 20 miles per hour. How far from the start will the Humvee catch up?

A 1 mile **B** 3 miles **C** 4 miles **D** 5 miles

35. A pitcher throws a fastball at 75 miles per hour. What is this speed in feet per second? (45 miles per hour is the same as 66 feet per second.)

A 96 **B** 102 **C** 110 **D** 121

36. A runner reaches a top speed of 30 miles per hour. What is this speed in feet per second? (90 miles per hour is the same as 132 feet per second.)

A 36 **B** 44 **C** 72 **D** 396

37. James bikes 30 miles on a flat trail at 15 miles per hour, then 3 miles uphill at 6 miles per hour. What is the average speed for the whole trip?

A 7.2 mph **B** 10.5 mph
C 13.2 mph **D** 21 mph

38. A delivery truck travels 210 miles on the interstate at 70 miles per hour, then 50 miles in city traffic at 25 miles per hour. What is the average speed for the whole trip?

A 37 mph **B** 42 mph
C 47.5 mph **D** 52 mph

39. For a trip of 504 miles, Victor can take either a car that gets 28 miles per gallon or a truck that gets 12 miles per gallon. Gas costs $4.25 per gallon. How much will he save on gas by taking the car?

A $102.00 **B** $112.00 **C** $132.00 **D** $178.50

40. A charter bus carrying new recruits leaves the processing station at 7:40 a.m. and travels 110 miles at 55 miles per hour. It makes a 45-minute rest stop, then travels another 65 miles at 65 miles per hour to the training base. At what time does it arrive at the base?

A 10:40 a.m. **B** 11:25 a.m.
C 11:55 a.m. **D** 12:10 p.m.

Answers & Explanations 24

1 B	**9** C	**17** C	**25** C	**33** B
2 B	**10** B	**18** B	**26** D	**34** D
3 C	**11** C	**19** A	**27** C	**35** C
4 C	**12** B	**20** C	**28** D	**36** B
5 B	**13** D	**21** B	**29** C	**37** C
6 C	**14** B	**22** B	**30** B	**38** D
7 A	**15** A	**23** C	**31** A	**39** A
8 A	**16** A	**24** A	**32** B	**40** B

1. **B** 366 miles

1 Use $d = r \times t$: distance = speed × time.

2 $d = 61 \times 6 = 366$ miles.

WHY NOT (A) adds the speed and the time instead of multiplying.

2. **B** 36 mph

1 Use $r = d \div t$: speed = distance ÷ time.

2 $r = 144 \div 4 = 36$ miles per hour.

WHY NOT (C) subtracts the time from the distance. (D) multiplies the distance by the time instead of dividing.

3. **C** 440 miles

1 Use $d = r \times t$: distance = speed × time.

2 $d = 55 \times 8 = 440$ miles.

WHY NOT (A) adds the speed and the time instead of multiplying.

4. **C** 60 miles

1 Use $d = r \times t$: distance = speed × time.

2 $d = 15 \times 4 = 60$ miles.

WHY NOT (A) adds the speed and the time instead of multiplying.

5. **B** 6 miles

1 Change the time to hours: 20 min = $\frac{20}{60} = \frac{1}{3}$ hour.

2 Distance: $18 \times \frac{1}{3} = 6$ miles.

TIP Or think per minute: 18 miles per hour is $18 \div 60 = 0.3$ mile per minute, and $0.3 \times 20 = 6$.

WHY NOT (A) treats 20 minutes as 0.2 hour. (C) gives the distance for a full hour. (D) uses the minutes as if they were hours.

6. **C** 35 miles

1 Change the time to hours: 50 min = $\frac{50}{60} = \frac{5}{6}$ hour.

2 Distance: $42 \times \frac{5}{6} = 35$ miles.

TIP Or think per minute: 42 miles per hour is $42 \div 60 = 0.7$ mile per minute, and $0.7 \times 50 = 35$.

WHY NOT (A) treats 50 minutes as 0.5 hour. (D) uses the minutes as if they were hours.

7. **A** 18

1 Miles per gallon = miles ÷ gallons.

2 $252 \div 14 = 18$ miles per gallon.

TIP Check: $18 \times 14 = 252$.

WHY NOT (D) subtracts the gallons from the miles.

8. **A** 17

1 Miles per gallon = miles ÷ gallons.

2 $255 \div 15 = 17$ miles per gallon.

TIP Check: $17 \times 15 = 255$.

WHY NOT (D) subtracts the gallons from the miles.

9. **C** 8

1 Miles per gallon = miles ÷ gallons.

2 $168 \div 21 = 8$ miles per gallon.

TIP Check: $8 \times 21 = 168$.

WHY NOT (D) subtracts the gallons from the miles.

10. **B** 72

1 Amount = rate × time.

2 $24 \times 3 = 72$ sandbags.

WHY NOT (A) adds the rate and the time instead of multiplying.

11. **C** 4 hours

1 Time = amount ÷ rate.

2 $200 \div 50 = 4$ hours.

TIP Check: $50 \times 4 = 200$ sandwiches.

WHY NOT (D) subtracts the rate from the total.

12. **B** 300

1 Amount = rate × time.

2 $15 \times 20 = 300$ packages.

WHY NOT (A) adds the rate and the time instead of multiplying.

13. **D** 28 miles

1 Change the minutes to hours: 45 min = $\frac{45}{60} = \frac{3}{4}$ hour, so the time is $1\frac{3}{4}$ hours.

2 Distance: $16 \times 1\frac{3}{4} = 16 \times 1 + 16 \times \frac{3}{4} = 16 + 12 = 28$ miles.

WHY NOT (B) leaves out the extra minutes. (C) treats 1 hour 45 minutes as 1.45 hours.

14. **B** 42 mph

1 Change the time to hours: 10 min = $\frac{10}{60} = \frac{1}{6}$ hour.

2 Speed: $7 \div \frac{1}{6} = 7 \times 6 = 42$ miles per hour.

TIP Or scale up to an hour: 10 minutes goes into 60 minutes 6 times, and $7 \times 6 = 42$.

WHY NOT (A) gives miles per minute, not miles per hour. (D) treats 10 minutes as 0.1 hour.

15. **A** 30 minutes

1 Time in hours: $t = d \div r = 30 \div 60 = \frac{30}{60} = \frac{1}{2}$ hour.

2 Change to minutes: $\frac{1}{2} \times 60 = 30$ minutes.

WHY NOT (C) reads 0.5 hour as 50 minutes. (D) divides the speed by the distance before multiplying by 60.

16. **A** 1 hour

1 Moving toward each other, the gap closes at the sum of the speeds: $55 + 60 = 115$ miles per hour.

2 Time to close 115 miles: $115 \div 115 = 1$ hour.

TIP Check: in 1 hour they cover $55 + 60 = 115$ miles.

WHY NOT (B) divides by the average of the two speeds instead of their sum. (D) subtracts the speeds, but the two are closing the gap together.

17. **C** 4 hours

1 Moving toward each other, the gap closes at the sum of the speeds: $11 + 18 = 29$ miles per hour.

2 Time to close 116 miles: $116 \div 29 = 4$ hours.

TIP Check: in 4 hours they cover $44 + 72 = 116$ miles.

WHY NOT (D) divides by the average of the two speeds instead of their sum.

18. **B** 575 miles

1 Moving in opposite directions, the gap grows at the sum of the speeds: $50 + 65 = 115$ miles per hour.

2 After 5 hours: $115 \times 5 = 575$ miles.

WHY NOT (A) subtracts the speeds, but the two are moving apart in opposite directions.

19. **A** $43.55

1 Gallons needed: $182 \div 14 = 13$ gallons.

2 Cost: $13 \times \$3.35 = \$43.55$.

WHY NOT (B) multiplies the miles per gallon by the price. (D) multiplies the miles by the price per gallon.

20. **C** $15.25

1 Gallons needed: $75 \div 15 = 5$ gallons.

2 Cost: $5 \times \$3.05 = \$15.25$.

WHY NOT (A) gives the number of gallons, not the cost. (D) multiplies the miles per gallon by the price.

21. **B** 8 mph

1 Change the time to hours: 30 min = $\frac{30}{60} = \frac{1}{2}$ hour.

2 Speed: $4 \div \frac{1}{2} = 4 \times 2 = 8$ miles per hour.

TIP Or: the pace is $30 \div 4 = 7.5$ minutes per mile, and $60 \div 7.5 = 8$ mph.

WHY NOT (A) gives the pace in minutes per mile, not the speed. (D) multiplies the distance by the time.

22. **B** 40 minutes

1 Time in hours: $5 \div 7.5 = \frac{2}{3}$ hour.

2 Change to minutes: $\frac{2}{3} \times 60 = 40$ minutes.

TIP Or: at 7.5 mph one mile takes $60 \div 7.5 = 8$ minutes, and $5 \times 8 = 40$.

WHY NOT (A) finds the minutes for one mile only. (D) reads 0.66 hour as 66 minutes.

23. **C** 60 minutes

1 Time in hours: $3 \div 3 = 1$ hour.

2 Change to minutes: $1 \times 60 = 60$ minutes.

TIP Or: at 3 mph one mile takes $60 \div 3 = 20$ minutes, and $3 \times 20 = 60$.

WHY NOT (A) divides the speed by the distance. (B) finds the minutes for one mile only.

24. **A** 3.5 hours

1 First part: $10 \div 4 = 2.5$ hours.

2 Second part: $2 \div 2 = 1$ hour.

3 Total: $2.5 + 1 = 3.5$ hours.

WHY NOT (B) divides the total distance by the average of the two speeds. (D) uses the second speed for the whole trip.

25. **C** 4.5 hours

1 First part: $25 \div 10 = 2.5$ hours.

2 Second part: $6 \div 3 = 2$ hours.

3 Total: $2.5 + 2 = 4.5$ hours.

WHY NOT (A) finds the time for the first part only. (B) uses the first speed for the whole trip.

26. **D** 1:35 p.m.

1 Travel time: $195 \div 60 = 3\frac{1}{4}$ hours.

2 Change the fraction of an hour to minutes: $\frac{1}{4} \times 60 =$ 15 minutes, so the trip takes 3 hours 15 minutes.

3 Add to the start time: 10:20 a.m. plus 3 hours 15 minutes is 1:35 p.m.

WHY NOT (C) leaves out the part of an hour.

27. **C** 12:00 p.m.

1 Travel time: $180 \div 48 = 3\frac{3}{4}$ hours.

2 Change the fraction of an hour to minutes: $\frac{3}{4} \times 60 =$ 45 minutes, so the trip takes 3 hours 45 minutes.

3 Add to the start time: 8:15 a.m. plus 3 hours 45 minutes is 12:00 p.m.

WHY NOT (A) leaves out the part of an hour.

28. **D** 500

1 Change the time to hours: 10 min = $\frac{10}{60} = \frac{1}{6}$ hour.

2 Amount: $3{,}000 \times \frac{1}{6} = 500$ gallons.

TIP Or find the rate per minute: $3{,}000 \div 60 = 50$, and $50 \times 10 = 500$.

WHY NOT (A) divides the hourly rate by the minutes.

29. **C** 12 mph

1 Time going: $18 \div 18 = 1$ hour. Time returning: $18 \div 9 = 2$ hours.

2 Total distance: $2 \times 18 = 36$ miles. Total time: $1 + 2 = 3$ hours.

3 Average speed: $36 \div 3 = 12$ miles per hour.

TIP Shortcut for equal distances: $2ab \div (a + b) = 2 \times 18 \times 9 \div 27 = 12$.

WHY NOT (A) uses the one-way distance instead of the round-trip distance. (D) averages the two speeds, but more time is spent at the slower speed.

30. **B** 32 mph

1 Time going: $48 \div 24 = 2$ hours. Time returning: $48 \div 48 = 1$ hour.

2 Total distance: $2 \times 48 = 96$ miles. Total time: $2 + 1 = 3$ hours.

3 Average speed: $96 \div 3 = 32$ miles per hour.

TIP Shortcut for equal distances: $2ab \div (a + b) = 2 \times 24 \times 48 \div 72 = 32$.

WHY NOT (A) uses the one-way distance instead of the round-trip distance. (C) averages the two speeds, but more time is spent at the slower speed. (D) adds the two speeds.

31. **A** 120 mph

1 Time going: $300 \div 150 = 2$ hours. Time returning: $300 \div 100 = 3$ hours.

2 Total distance: $2 \times 300 = 600$ miles. Total time: $2 + 3 = 5$ hours.

3 Average speed: $600 \div 5 = 120$ miles per hour.

TIP Shortcut for equal distances: $2ab \div (a + b) = 2 \times 150 \times 100 \div 250 = 120$.

WHY NOT (B) averages the two speeds, but more time is spent at the slower speed. (D) adds the two speeds.

32. **B** 2 hours

1 Head start: in 1.5 hours the first one goes $12 \times 1.5 = 18$ miles.

2 The gap closes at the difference of the speeds: $21 - 12 = 9$ miles per hour.

3 Time to close it: $18 \div 9 = 2$ hours.

TIP Check: in 2 hours the chaser goes $21 \times 2 = 42$ miles, and the first one has gone $12 \times 3.5 = 42$ miles.

WHY NOT (D) counts the time from when the first one left.

33. **B** 4 hours

1 Head start: in 2 hours the first one goes $16 \times 2 = 32$ miles.

2 The gap closes at the difference of the speeds: $24 - 16 = 8$ miles per hour.

3 Time to close it: $32 \div 8 = 4$ hours.

TIP Check: in 4 hours the chaser goes $24 \times 4 = 96$ miles, and the first one has gone $16 \times 6 = 96$ miles.

WHY NOT (A) adds the speeds, but the gap closes at the difference of the speeds. (C) counts the time from when the first one left.

34. **D** 5 miles

1 Head start: $4 \times 1 = 4$ miles.

2 The gap closes at $20 - 4 = 16$ miles per hour, so it takes $4 \div 16 = 0.25$ hours.

3 Distance from the start: $20 \times 0.25 = 5$ miles.

TIP Check: the slower one has gone $4 \times 1.25 = 5$ miles too.

WHY NOT (A) uses the slower speed for the time the Humvee drives. (C) is the head start, not the meeting point.

35. **C** 110

1 Compare with the known pair: $75 \div 45 = \frac{5}{3}$, so the speed is $\frac{5}{3}$ times 45 mph.

2 Feet per second: $\frac{5}{3} \times 66 = 110$.

TIP Check with 15 mph = 22 feet per second: $75 = 5 \times 15$, and $5 \times 22 = 110$.

WHY NOT (A) adds the difference $66 - 45 = 21$ instead of multiplying by $\frac{5}{3}$.

36. **B** 44

1 Compare with the known pair: $30 \div 90 = \frac{1}{3}$, so the speed is $\frac{1}{3}$ times 90 mph.

2 Feet per second: $\frac{1}{3} \times 132 = 44$.

TIP Check with 15 mph = 22 feet per second: $30 = 2 \times 15$, and $2 \times 22 = 44$.

WHY NOT (C) adds the difference $132 - 90 = 42$ instead of multiplying by $\frac{1}{3}$. (D) divides by $\frac{1}{3}$ instead of multiplying.

37. **C** 13.2 mph

1 First part: $30 \div 15 = 2$ hours.

2 Second part: $3 \div 6 = 0.5$ hours.

3 Total distance: $30 + 3 = 33$ miles. Total time: $2 + 0.5 = 2.5$ hours.

4 Average speed: $33 \div 2.5 = 13.2$ miles per hour.

WHY NOT (B) averages the two speeds, but more time is spent at one of them. (D) adds the two speeds.

38. **D** 52 mph

1 First part: $210 \div 70 = 3$ hours.

2 Second part: $50 \div 25 = 2$ hours.

3 Total distance: $210 + 50 = 260$ miles. Total time: $3 + 2 = 5$ hours.

4 Average speed: $260 \div 5 = 52$ miles per hour.

WHY NOT (C) averages the two speeds, but more time is spent at one of them.

39. **A** $102.00

1 Gas for the truck: $504 \div 12 = 42$ gallons. Gas for the car: $504 \div 28 = 18$ gallons.

2 Gallons saved: $42 - 18 = 24$.

3 Money saved: $24 \times \$4.25 = \$102$.

WHY NOT (D) is the cost for the truck, not the savings.

40. **B** 11:25 a.m.

1 First part: $110 \div 55 = 2$ hours. Second part: $65 \div 65 = 1$ hour.

2 Total time: 2 hours + 1 hour + 45 minutes = 3 hours 45 minutes.

3 Add to the start time: 7:40 a.m. plus 3 hours 45 minutes is 11:25 a.m.

WHY NOT (A) forgets the 45-minute stop.

CHAPTER 25

Work Rates & Mixtures

Tested on: Arithmetic Reasoning

Key Concepts

Work and mixture problems look different, but both are about *rates and totals*: how much of a job gets done each hour, or how much of an ingredient is in each gallon. Set up the totals carefully and the arithmetic stays short.

Work rates

If a job takes t hours, the worker does $\frac{1}{t}$ of the job per hour.

- Working together, **rates add**: $\frac{1}{a} + \frac{1}{b} = \frac{1}{T}$, where T is the time together. Then flip: time = 1 ÷ rate.
- Two-worker shortcut (product over sum): $T = \dfrac{a \times b}{a + b}$. For 3 h and 6 h: $\frac{18}{9} = 2$ h.
- A drain or a leak works against you: **subtract** its rate.
- To find one worker's time from the time together, subtract rates: $\frac{1}{b} = \frac{1}{T} - \frac{1}{a}$.

Equal workers: worker-hours

Total work = number of workers × time. If 6 workers need 8 days, the job is $6 \times 8 = 48$ worker-days, so 4 workers need $48 \div 4 = 12$ days. *More* workers means *less* time (an inverse proportion).

Mixtures

$$\text{amount of pure ingredient} = \text{percent} \times \text{total amount}$$

- When two mixtures are combined, the pure amounts add and the totals add.
- Adding water raises the total but does *not* change the pure amount.
- Price mixtures work the same way: total cost ÷ total pounds = price per pound.

WORKED EXAMPLE

How many liters of a 50% antifreeze solution must be added to 10 liters of a 20% solution to make a 30% solution?

Solution. Let x be the liters of 50% solution. The pure antifreeze going in equals the pure antifreeze in the mixture: $0.20(10) + 0.50x = 0.30(10 + x)$, so $2 + 0.5x = 3 + 0.3x$. Then $0.2x = 1$ and $x = 5$ liters. *Check:* $2 + 2.5 = 4.5$ liters of antifreeze in 15 liters, and $4.5 \div 15 = 0.30 = 30\%$.

No-Calculator Tip Sanity checks save points. Two workers together must be *faster* than the faster one alone (3 h and 6 h together: less than 3 h). A mixture's percent or price must land *between* the two ingredients, closer to the one you use more of.

Common Traps

- Averaging or adding the times: 3 h and 6 h together is 2 h, not 4.5 h or 9 h.
- Treating workers as a direct proportion: fewer workers need *more* time, not less.
- Taking the simple average of two percents or prices when the amounts are different.
- Answering with the final amount when the question asks how much water (or how much of one ingredient) to *add*.

Practice Set 25

Score: _____ / 35 Target time: 42 minutes

Problems 1–10 are warm-ups, 11–24 are test level, and 25–35 are challenge problems.

1. First squad can unload a supply truck in 42 minutes. Second squad can unload the same truck in 21 minutes. How long will it take if both squads work together?

A 10 minutes **B** 14 minutes
C 21 minutes **D** 63 minutes

2. A car's radiator holds 12 quarts of coolant that is 30% antifreeze and the rest water. How many quarts of pure antifreeze are in the radiator?

A 0.36 quarts **B** 3 quarts
C 3.6 quarts **D** 8.4 quarts

3. One snowplow can clear a shopping-center parking lot in 6 hours. A second plow can clear it in 12 hours. How long will the job take if both plows work at the same time?

A 4 hours **B** 4 hours 30 minutes
C 9 hours **D** 18 hours

4. A medic mixes 900 milliliters of a cleaning solution that is 20% bleach and the rest water. How many milliliters of bleach are in the solution?

A 180 milliliters **B** 360 milliliters
C 720 milliliters **D** 1,800 milliliters

5. A crew of 6 movers can load a moving truck in 7 hours. At the same rate, how many hours would it take 3 movers to do the job?

A 14 hours **B** 18 hours
C 20 hours **D** 42 hours

6. Combat engineers mix concrete for a bunker floor using cement, sand, and gravel in the ratio 1 : 2 : 3 by volume. How many cubic feet of cement are in 42 cubic feet of concrete mix?

A 5 cubic feet **B** 7 cubic feet
C 14 cubic feet **D** 35 cubic feet

7. A group of 12 volunteers can pack the boxes for a food drive in 10 hours. At the same rate, how many hours would it take 8 volunteers to do the job?

A 14 hours **B** 15 hours
C 19 hours **D** 120 hours

8. For ruck-march snack packs, a unit mixes peanuts, raisins, and chocolate chips in the ratio 4 : 3 : 1 by weight. How many pounds of raisins are in 24 pounds of trail mix?

A 3 pounds **B** 7 pounds
C 8 pounds **D** 9 pounds

9. A 48-ounce bottle of fruit punch contains 9.6 ounces of real fruit juice. What percent of the punch is real juice?

A 9.6% **B** 18% **C** 20% **D** 80%

10. Working alone, Specialist Santos can fill the sandbags for a fighting position in 4 hours. Specialist Okafor needs 12 hours to do the same job alone. How long will the job take if they work together?

A 2 hours **B** 3 hours
C 8 hours **D** 16 hours

11. One pump can empty a flooded basement in 12 hours. A second pump can empty it in 3 hours. How long will it take to empty the basement if both pumps run at the same time?

A 25 minutes **B** 1 hour 30 minutes
C 2 hours 24 minutes **D** 7 hours 30 minutes

12. Derek and Sofia are stuffing envelopes for a charity mailing. Together they can finish in 4 hours. Alone, Derek would need 12 hours. How many hours would Sofia need alone?

A 6 hours **B** 7 hours
C 8 hours **D** 16 hours

13. A bathtub faucet can fill the tub in 9 minutes. With the plug pulled, the drain empties a full tub in 18 minutes. If the faucet is turned on with the plug pulled, how long will it take to fill the tub?

A 6 minutes **B** 9 minutes
C 18 minutes **D** 27 minutes

14. A team of 20 pickers can harvest an apple orchard in 7 days. How many pickers working at the same rate are needed to finish the job in 4 days?

A 35 **B** 40 **C** 50 **D** 140

15. A cafeteria worker mixes 4 gallons of pure fruit juice with 2 gallons of a punch that is 25% fruit juice. What percent of the mixture is fruit juice?

A 50% **B** 62.5% **C** 75% **D** 125%

16. A base dining facility mixes 69 pounds of 80%-lean ground beef costing $5.50 per pound with 46 pounds of 93%-lean ground beef costing $8.50 per pound. What is the cost per pound of the mixture?

A $5.95 **B** $6.70 **C** $7.00 **D** $7.30

17. A motor pool mechanic mixes 2 gallons of coolant that is 50% antifreeze with 18 gallons of coolant that is 70% antifreeze. What percent of the mixture is antifreeze?

A 68% **B** 72% **C** 74% **D** 120%

18. A coffee shop mixes 20 pounds of Colombian coffee costing $8 per pound with 5 pounds of Kona blend coffee costing $15 per pound. What is the cost per pound of the mixture?

A $9.40 **B** $11.50 **C** $13.60 **D** $235.00

19. A detail of 10 soldiers can unload and stack a shipment of supplies in 12 hours. How many *more* soldiers are needed to finish the job in 6 hours?

A 6 **B** 10 **C** 20 **D** 120

20. A kitchen faucet fills a large sink in 4 minutes. The drain, if left open, empties a full sink in 8 minutes. If the drain is left open, how long will it take the faucet to fill the empty sink?

A 4 minutes **B** 5 minutes
C 6 minutes **D** 8 minutes

21. Working together, Lance Corporal Brooks and Private Nguyen can clean all of the squad's rifles in 4 hours. Lance Corporal Brooks could do the job alone in 6 hours. How long would it take Private Nguyen working alone?

A 2 hours **B** 8 hours
C 12 hours **D** 14 hours

22. Aisha can paint the outside of a house in 13 hours. Luis can paint the same house in 26 hours. If they work together, how long will it take them to paint the house?

A 6 hours 30 minutes **B** 8 hours
C 8 hours 20 minutes **D** 8 hours 40 minutes

23. A chemistry teacher has 3 liters of a 30% acid solution. How many liters of water must be added to dilute it to a 15% solution?

A 3 liters **B** 4 liters **C** 5 liters **D** 6 liters

24. A recipe for lemonade mixes lemonade concentrate and water in the ratio 2 : 7. How many cups of lemonade can be made if 14 cups of lemonade concentrate are used?

A 21 cups **B** 49 cups **C** 58 cups **D** 63 cups

25. A summer camp has enough food to feed 30 campers for 34 days. After 12 days, 15 campers go home. How many more days will the remaining food last?

A 11 days **B** 22 days **C** 44 days **D** 68 days

26. A mechanic has 4 quarts of a 50% antifreeze solution. How many quarts of a 25% antifreeze solution must be added to make a mixture that is 30% antifreeze?

A 1 quart **B** 4 quarts
C 16 quarts **D** 20 quarts

27. A juice plant starts with 160 gallons of juice that is 15% juice solids. How many gallons of water must be removed to make a concentrate that is 50% solids?

A 48 gallons **B** 112 gallons
C 122 gallons **D** 142 gallons

28. A grocery store has 12 pounds of peanuts priced at $5 per pound. How many pounds of cashews priced at $12 per pound must be mixed in to make a blend worth $8 per pound?

A 9 pounds **B** 12 pounds
C 16 pounds **D** 21 pounds

29. A greenhouse worker has 16 gallons of a 10% fertilizer solution. How many gallons of pure fertilizer must be added to make a 20% solution?

A 2 gallons **B** 8 gallons
C 16 gallons **D** 18 gallons

30. The base exchange has 29 pounds of almonds priced at $10 per pound. How many pounds of raisins priced at $2 per pound must be mixed in to make a blend worth $4 per pound?

A 29 pounds **B** 67 pounds
C 77 pounds **D** 87 pounds

31. A medic at an aid station has 200 milliliters of an 80% alcohol solution. How many milliliters of a 50% alcohol solution must be added to make a mixture that is 75% alcohol?

A 40 milliliters **B** 200 milliliters
C 240 milliliters **D** 1,000 milliliters

32. A remote outpost has enough MREs (meals) to feed 75 soldiers for 14 days. After 2 days, 25 more soldiers arrive. How many more days will the remaining MREs last?

A 9 days **B** 11 days **C** 12 days **D** 16 days

33. Working alone, Diego can paint the outside of a house in 10 hours, Rosa in 6 hours, and Jasmine in 30 hours. How long will the job take if all three paint together?

A 2 hours
B 3 hours 20 minutes
C 15 hours 20 minutes
D 46 hours

34. At a supply depot, an experienced forklift driver works twice as fast as a trainee. Together they can unload a cargo container in 7 hours. How long would it take the trainee working alone?

A 3 hours 30 minutes
B 10 hours 30 minutes
C 14 hours
D 21 hours

35. Two pumps fill a water storage tank at a base camp. Working alone, they would take 5 hours and 10 hours. Meanwhile, the camp uses water at a rate that would empty a full tank in 4 hours. Starting with an empty tank, how long will it take to fill it?

A 3 hours 20 minutes **B** 20 hours
C 22 hours **D** 24 hours

Answers & Explanations 25

1 B	**8** D	**15** C	**22** D	**29** A
2 C	**9** C	**16** B	**23** A	**30** D
3 A	**10** B	**17** A	**24** D	**31** A
4 A	**11** C	**18** A	**25** C	**32** A
5 A	**12** A	**19** B	**26** C	**33** B
6 B	**13** C	**20** D	**27** B	**34** D
7 B	**14** A	**21** C	**28** A	**35** B

1. **B** 14 minutes

1 Turn each time into a rate (the part of the job done in one minute): first squad does $\frac{1}{42}$ of the job per minute, and second squad does $\frac{1}{21}$.

2 Add the rates (common denominator 42): $\frac{1}{42}+\frac{1}{21} = \frac{1}{42}+\frac{2}{42} = \frac{3}{42} = \frac{1}{14}$ of the job per minute.

3 Time = 1 job ÷ rate: $1 \div \frac{1}{14} = 14$ minutes.

TIP The answer must be less than the faster time, 21 minutes.

WHY NOT (C) subtracts the two times. (D) adds the two times instead of adding the two rates.

2. **C** 3.6 quarts

1 Amount of antifreeze = percent × total. Change the percent to a decimal: 30% = 0.3.

2 Multiply: $0.3 \times 12 = 3.6$ quarts.

WHY NOT (A) moves the decimal point three places when changing the percent. (D) is the amount of water, not antifreeze.

3. **A** 4 hours

1 Turn each time into a rate (the part of the job done in one hour): the first plow does $\frac{1}{6}$ of the job per hour, and the second plow does $\frac{1}{12}$.

2 Add the rates (common denominator 12): $\frac{1}{6}+\frac{1}{12} = \frac{2}{12}+\frac{1}{12} = \frac{3}{12} = \frac{1}{4}$ of the job per hour.

3 Time = 1 job ÷ rate: $1 \div \frac{1}{4} = 4$ hours.

TIP Product over sum: $\frac{6\times12}{6+12} = \frac{72}{18} = 4$. The answer must be less than the faster time, 6 hours.

WHY NOT (B) halves the average of the two times. (C) averages the two times. (D) adds the two times instead of adding the two rates.

4. **A** 180 milliliters

1 Amount of bleach = percent × total. Change the percent to a decimal: 20% = 0.2.

2 Multiply: $0.2 \times 900 = 180$ milliliters.

WHY NOT (C) is the amount of water, not bleach. (D) moves the decimal point only one place when changing the percent.

5. **A** 14 hours

1 Find the total amount of work in worker-hours: $6 \times 7 = 42$ worker-hours.

2 Share that work among 3 movers: $42 \div 3 = 14$ hours.

TIP Fewer workers means more time, so the answer must be more than 7 hours.

WHY NOT (D) gives the total number of worker-hours, not the time.

6. **B** 7 cubic feet

1 Add the parts: $1 + 2 + 3 = 6$ parts in all.

2 Find the size of one part: $42 \div 6 = 7$ cubic feet.

3 Cement is 1 part, so the answer is 7 cubic feet.

WHY NOT (C) splits the mix into equal amounts. (D) is the amount of everything except the cement.

7. **B** 15 hours

1 Find the total amount of work in worker-hours: $12 \times 10 = 120$ worker-hours.

2 Share that work among 8 volunteers: $120 \div 8 = 15$ hours.

TIP Fewer workers means more time, so the answer must be more than 10 hours.

WHY NOT (A) changes the time by one hour for each worker added or removed. (D) gives the total number of worker-hours, not the time.

8. **D** 9 pounds

1 Add the parts: $4 + 3 + 1 = 8$ parts in all.

2 Find the size of one part: $24 \div 8 = 3$ pounds.

3 Raisins are 3 parts: $3 \times 3 = 9$ pounds.

WHY NOT (A) is the size of just one part. (C) splits the mix into equal amounts.

9. **C** 20%

1 Percent = part ÷ whole: $\frac{9.6}{48}$.

2 Divide and change to a percent: $\frac{9.6}{48} = 0.2 = 20\%$.

WHY NOT (A) gives the amount, not the percent. (D) is the percent that is water and sweetener.

10. **B** 3 hours

1 Turn each time into a rate (the part of the job done in one hour): Specialist Santos does $\frac{1}{4}$ of the job per hour, and Specialist Okafor does $\frac{1}{12}$.

2 Add the rates (common denominator 12): $\frac{1}{4}+\frac{1}{12} = \frac{3}{12}+\frac{1}{12} = \frac{4}{12} = \frac{1}{3}$ of the job per hour.

3 Time = 1 job ÷ rate: $1 \div \frac{1}{3} = 3$ hours.

TIP Product over sum: $\frac{4\times12}{4+12} = \frac{48}{16} = 3$. The answer must be less than the faster time, 4 hours.

WHY NOT (A) halves the faster time, as if both worked as fast as the faster one. (C) averages the two times. (D) adds the two times instead of adding the two rates.

11. **C** 2 hours 24 minutes

1 Turn each time into a rate (the part of the job done in one hour): the first pump does $\frac{1}{12}$ of the job per hour, and the second pump does $\frac{1}{3}$.

2 Add the rates (common denominator 12): $\frac{1}{12}+\frac{1}{3} = \frac{1}{12}+\frac{4}{12} = \frac{5}{12}$ of the job per hour.

3 Time = 1 job ÷ rate: $1 \div \frac{5}{12} = \frac{12}{5} = 2\frac{2}{5}$ hours.

4 Change the fraction of an hour to minutes: $\frac{2}{5}\times60 =$ 24 minutes. So they need 2 hours 24 minutes.

TIP Product over sum: $\frac{3\times12}{3+12} = \frac{36}{15} = 2\frac{2}{5}$. The answer must be less than the faster time, 3 hours.

WHY NOT (A) stops at the combined rate and forgets to flip it into a time. (B) halves the faster time, as if both worked as fast as the faster one. (D) averages the two times.

12. **A** 6 hours

1 Rates: together they do $\frac{1}{4}$ of the job per hour, and Derek alone does $\frac{1}{12}$ of the job per hour.

2 Sofia's rate is the difference: $\frac{1}{4}-\frac{1}{12} = \frac{3}{12}-\frac{1}{12} = \frac{2}{12} = \frac{1}{6}$ of the job per hour.

3 Doing $\frac{1}{6}$ of the job each hour, Sofia needs 6 hours alone.

TIP Check: $\frac{1}{12}+\frac{1}{6} = \frac{1}{4}$, the rate together. ✓

WHY NOT (C) subtracts the times instead of the rates. (D) adds the two times.

13. **C** 18 minutes

1 Rates: the faucet fills $\frac{1}{9}$ of the tub per minute, and the drain empties $\frac{1}{18}$ of it per minute.

2 The drain works against the faucet, so *subtract*: $\frac{1}{9}-\frac{1}{18} = \frac{2}{18}-\frac{1}{18} = \frac{1}{18}$ of the tub per minute.

3 Time = 1 tub ÷ rate = $1 \div \frac{1}{18} = 18$ minutes.

TIP Shortcut for fill and drain: $\frac{9\times18}{18-9} = \frac{162}{9} = 18$.

WHY NOT (A) adds the drain's rate instead of subtracting it. (B) subtracts the two times. (D) adds the two times.

14. **A** 35

1 Total work: $20 \times 7 = 140$ worker-days.

2 To finish in 4 days: $140 \div 4 = 35$ pickers are needed.

WHY NOT (D) gives the total number of worker-days, not the number of workers.

15. **C** 75%

1 Pure fruit juice in each: $1\times4 = 4$ and $0.25\times2 = 0.5$ gallons.

2 Totals: 4+0.5 = 4.5 gallons of fruit juice in 4+2 = 6 gallons of mixture.

3 Percent: $\frac{4.5}{6} = 0.75 = 75\%$.

TIP Check: 75% is between 25% and 100%, closer to the percent of the larger amount.

WHY NOT (A) weights each percent by the wrong amount. (B) averages the two percents without weighting by the amounts. (D) adds the two percents.

16. **B** $6.70

1 Cost of each part: $69 \times 5.50 = 379.50$ and $46 \times 8.50 = 391.00$ dollars.

2 Totals: \$770.50 for 69 + 46 = 115 pounds.

3 Price per pound: $770.50 \div 115 = 6.70$, or \$6.70 per pound.

TIP Check: \$6.70 is between \$5.50 and \$8.50, closer to the price of the item with more pounds.

WHY NOT (C) averages the two prices without weighting by the pounds. (D) weights each price by the wrong number of pounds.

17. **A** 68%

1 Pure antifreeze in each: $0.5\times2 = 1$ and $0.7\times18 = 12.6$ gallons.

2 Totals: 1 + 12.6 = 13.6 gallons of antifreeze in 2 + 18 = 20 gallons of mixture.

3 Percent: $\frac{13.6}{20} = 0.68 = 68\%$.

TIP Check: 68% is between 50% and 70%, closer to the percent of the larger amount.

WHY NOT (D) adds the two percents.

18. **A** $9.40

1 Cost of each part: $20 \times 8 = 160$ and $5 \times 15 = 75$ dollars.

2 Totals: \$235 for 20 + 5 = 25 pounds.

3 Price per pound: $235 \div 25 = 9.40$, or \$9.40 per pound.

TIP Check: \$9.40 is between \$8 and \$15, closer to the price of the item with more pounds.

WHY NOT (B) averages the two prices without weighting by the pounds. (C) weights each price by the wrong number of pounds. (D) is the total cost of the mixture, not the price per pound.

19. **B** 10

1 Total work: $10 \times 12 = 120$ worker-hours.

2 To finish in 6 hours: $120 \div 6 = 20$ soldiers are needed.

3 There are already 10 soldiers, so $20 - 10 = 10$ more are needed.

WHY NOT (A) adds one worker for each hour saved. (C) is the total number of workers needed, not how many more. (D) gives the total number of worker-hours, not the number of workers.

20. **D** 8 minutes

1 Rates: the faucet fills $\frac{1}{4}$ of the sink per minute, and the drain empties $\frac{1}{8}$ of it per minute.

2 The drain works against the faucet, so *subtract*: $\frac{1}{4} - \frac{1}{8} = \frac{2}{8} - \frac{1}{8} = \frac{1}{8}$ of the sink per minute.

3 Time = 1 sink ÷ rate = $1 \div \frac{1}{8} = 8$ minutes.

TIP Shortcut for fill and drain: $\frac{4 \times 8}{8 - 4} = \frac{32}{4} = 8$.

WHY NOT (A) subtracts the two times.

21. **C** 12 hours

1 Rates: together they do $\frac{1}{4}$ of the job per hour, and Lance Corporal Brooks alone does $\frac{1}{6}$ of the job per hour.

2 Private Nguyen's rate is the difference: $\frac{1}{4} - \frac{1}{6} = \frac{3}{12} - \frac{2}{12} = \frac{1}{12}$ of the job per hour.

3 Doing $\frac{1}{12}$ of the job each hour, Private Nguyen needs 12 hours alone.

TIP Check: $\frac{1}{6} + \frac{1}{12} = \frac{1}{4}$, the rate together. ✓

WHY NOT (A) subtracts the times instead of the rates. (B) assumes the two work at the same speed.

22. **D** 8 hours 40 minutes

1 Turn each time into a rate (the part of the job done in one hour): Aisha does $\frac{1}{13}$ of the job per hour, and Luis does $\frac{1}{26}$.

2 Add the rates (common denominator 26): $\frac{1}{13} + \frac{1}{26} = \frac{2}{26} + \frac{1}{26} = \frac{3}{26}$ of the job per hour.

3 Time = 1 job ÷ rate: $1 \div \frac{3}{26} = \frac{26}{3} = 8\frac{2}{3}$ hours.

4 Change the fraction of an hour to minutes: $\frac{2}{3} \times 60 = 40$ minutes. So they need 8 hours 40 minutes.

TIP The answer must be less than the faster time, 13 hours.

WHY NOT (A) halves the faster time, as if both worked as fast as the faster one.

23. **A** 3 liters

1 Pure acid now: $0.3 \times 3 = 0.9$ liters. Adding water does not change this amount.

2 In the new mixture these 0.9 liters must be 15% of the total: $0.9 \div 0.15 = 6$ liters in all.

3 Water to add: $6 - 3 = 3$ liters.

WHY NOT (D) is the final amount of the mixture, not the water added.

24. **D** 63 cups

1 Lemonade concentrate is 2 parts, so one part is $14 \div 2 = 7$ cups.

2 The whole batch is 9 parts: $9 \times 7 = 63$ cups.

WHY NOT (A) adds or subtracts the difference in parts instead of scaling by the size of one part. (B) leaves out the lemonade concentrate.

25. **C** 44 days

1 After 12 days, the original 30 campers still had 22 days' worth of food left: $30 \times 22 = 660$ camper-days.

2 Now there are $30 - 15 = 15$ campers, so the rest takes $660 \div 15 = 44$ days.

WHY NOT (A) sets up a direct proportion instead of an inverse one. (B) ignores the smaller group. (D) forgets the food already eaten in the first 12 days (gives the new group the full supply).

26. **C** 16 quarts

1 Let y be the quarts added. The pure antifreeze going in must equal the pure antifreeze in the mixture: $0.5(4) + 0.25y = 0.3(4 + y)$.

2 Multiply out: $2 + 0.25y = 1.2 + 0.3y$.

3 Collect the y terms: $0.8 = 0.05y$, so $y = 16$ quarts.

TIP Balance shortcut: a 50% antifreeze solution is 20 points from 30% and a 25% antifreeze solution is 5 points away, so $4 \times 20 = y \times 5$, which gives $y = 16$.

WHY NOT (A) swaps the two differences. (B) assumes equal amounts of the two solutions. (D) gives the total amount of the mixture, not the amount added.

27. **B** 112 gallons

1 Pure juice solids: $0.15 \times 160 = 24$ gallons. Removing water does not change this amount.

2 After evaporating, 24 gallons must be 50% of the total: $24 \div 0.5 = 48$ gallons left.

3 Water removed: $160 - 48 = 112$ gallons.

WHY NOT (A) is the amount left after evaporating, not the amount removed.

28. **A** 9 pounds

1 Let y be the pounds of cashews. The money must balance: $5(12) + 12y = 8(12 + y)$.

2 Multiply out: $60 + 12y = 96 + 8y$.

3 Collect terms: $12y - 8y = 96 - 60$, so $4y = 36$ and $y = 9$ pounds.

TIP Balance shortcut: each pound of peanuts is \$3 away from \$8, each pound of cashews is \$4 away; the totals must match: $12 \times 3 = 9 \times 4$.

WHY NOT (B) assumes equal amounts, which only works if the target is halfway between the prices. (C) swaps the two price differences. (D) gives the total weight of the blend.

29. **A** 2 gallons

1 Let y be the gallons of pure fertilizer added (pure means 100%). Pure fertilizer before and after: $0.1(16) + y = 0.2(16 + y)$.

2 Multiply out: $1.6 + y = 3.2 + 0.2y$.

3 Collect terms: $0.8y = 1.6$, so $y = 2$ gallons.

WHY NOT (B) divides by the target percent (20%) instead of by the share that is not fertilizer (80%). (D) gives the total amount of the new mixture.

30. **D** 87 pounds

1 Let y be the pounds of raisins. The money must balance: $10(29) + 2y = 4(29 + y)$.

2 Multiply out: $290 + 2y = 116 + 4y$.

3 Collect terms: $290 - 116 = 4y - 2y$, so $174 = 2y$ and $y = 87$ pounds.

TIP Balance shortcut: each pound of almonds is \$6 away from \$4, each pound of raisins is \$2 away; the totals must match: $29 \times 6 = 87 \times 2$.

WHY NOT (A) assumes equal amounts, which only works if the target is halfway between the prices.

31. **A** 40 milliliters

1 Let y be the milliliters added. The pure alcohol going in must equal the pure alcohol in the mixture: $0.8(200) + 0.5y = 0.75(200 + y)$.

2 Multiply out: $160 + 0.5y = 150 + 0.75y$.

3 Collect the y terms: $10 = 0.25y$, so $y = 40$ milliliters.

TIP Balance shortcut: an 80% alcohol solution is 5 points from 75% and a 50% alcohol solution is 25 points away, so $200 \times 5 = y \times 25$, which gives $y = 40$.

WHY NOT (B) assumes equal amounts of the two solutions. (C) gives the total amount of the mixture, not the amount added. (D) swaps the two differences.

32. **A** 9 days

1 After 2 days, the original 75 soldiers still had 12 days' worth of food left: $75 \times 12 = 900$ soldier-days.

2 Now there are $75 + 25 = 100$ soldiers, so the rest takes $900 \div 100 = 9$ days.

WHY NOT (B) gives the total number of days, not the days remaining. (C) ignores the new arrivals. (D) sets up a direct proportion instead of an inverse one.

33. **B** 3 hours 20 minutes

1 Each rate is 1 job divided by the time alone: $\frac{1}{10}$, $\frac{1}{6}$, and $\frac{1}{30}$ of the job per hour.

2 Add the three rates (common denominator 30): $\frac{1}{10} + \frac{1}{6} + \frac{1}{30} = \frac{3}{30} + \frac{5}{30} + \frac{1}{30} = \frac{9}{30} = \frac{3}{10}$ of the job per hour.

3 Time = 1 job ÷ rate: $1 \div \frac{3}{10} = \frac{10}{3} = 3\frac{1}{3}$ hours.

4 Change the fraction of an hour to minutes: $\frac{1}{3} \times 60 =$ 20 minutes. So they need 3 hours 20 minutes.

TIP Check: the answer must be less than the fastest time alone, 6 hours.

WHY NOT (A) divides the fastest time by 3, as if all three were that fast. (C) averages the three times. (D) adds the three times instead of adding the rates.

34. **D** 21 hours

1 Let the trainee do r of the job per hour. Then the experienced driver does $2r$, and together they do $r + 2r = 3r$ per hour.

2 Together they finish in 7 hours, so $3r = \frac{1}{7}$ and $r = \frac{1}{21}$. So the trainee alone needs 21 hours.

TIP Check: rates $\frac{2}{21} + \frac{1}{21} = \frac{1}{7}$ of the job per hour. ✓

WHY NOT (A) only multiplies or divides the time together by the speed ratio. (B) is the time for the experienced driver, not the one asked about. (C) assumes the two work at the same speed.

35. **B** 20 hours

1 Rates per hour: the pumps add $\frac{1}{5}$ and $\frac{1}{10}$ of the tank; the camp's water use takes away $\frac{1}{4}$.

2 Net rate (common denominator 20): $\frac{1}{5} + \frac{1}{10} - \frac{1}{4} = \frac{4}{20} + \frac{2}{20} - \frac{5}{20} = \frac{1}{20}$ of the tank per hour.

3 Time = 1 tank ÷ rate = $1 \div \frac{1}{20} = 20$ hours.

WHY NOT (A) ignores the camp's water use.

CHAPTER 26

Averages, Statistics & Probability

Tested on: Arithmetic Reasoning

Key Concepts

Averages and probability questions are short, but they hide traps: the wrong count in a division, a list that was never put in order, or two chances that were added when they should have been multiplied. Work with *totals* and *counts*, and these problems become simple arithmetic.

Mean, median, mode, range

- **Mean** (average) = sum of the values ÷ number of values.
- **Median** = the middle value *after* putting the list in order. With an even number of values, average the two middle ones.
- **Mode** = the value that appears most often. **Range** = largest value − smallest value.

Think in totals

$$\text{sum} = \text{mean} \times \text{number of values}$$

To find a missing score, a new average, or the average of two groups, turn every average into a total first, do the adding or subtracting with totals, then divide by the new count. Two groups of different sizes: add the two totals and divide by the total number of people.

Probability and counting

- $P(\text{event}) = \dfrac{\text{number of favorable outcomes}}{\text{total number of outcomes}}$, a number from 0 to 1. $\quad P(\text{not } A) = 1 - P(A)$.
- Independent events ("and"): **multiply**. A coin and a die: $P(\text{heads and } 6) = \frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$.
- Without replacement, the second draw has one fewer item in the bag.
- Counting principle: multiply the number of choices at each step. Ordering n things: $n \times (n-1) \times \cdots \times 1$. Choosing a group of 2 when order doesn't matter: $\frac{n(n-1)}{2}$.
- Expected count = probability × number of tries.

WORKED EXAMPLE

Maria scored 82, 88, and 91 on three tests. What must she score on the fourth test to average 88 on all four?

Solution. For an average of 88 on 4 tests she needs a total of $4 \times 88 = 352$ points. She has $82 + 88 + 91 = 261$. She needs $352 - 261 = 91$ on the fourth test.

No-Calculator Tip Check probability answers for size: a probability can never be more than 1, and "not" events are usually the larger ones. For averages, the answer must lie between the smallest and largest values.

Common Traps

- Finding the median without first putting the numbers in order.
- Averaging two averages when the groups have different sizes.
- Answering with the target average when the question asks what score is *needed*.
- Adding probabilities for "and" events, or forgetting that a drawn item is not put back.

Practice Set 26

Score: _____ / 40 Target time: 48 minutes

Problems 1–12 are warm-ups, 13–28 are test level, and 29–40 are challenge problems.

1. A recruit's times on five six-mile ruck marches were 100, 82, 80, 105, and 98 minutes. What is the mean (average) of these times?

A 93 minutes **B** 98 minutes
C 103 minutes **D** 465 minutes

2. The numbers of recruits who enlisted at a recruiting station in each of five months were 10, 21, 27, 24, 11. What is the range?

A 1 **B** 17 **C** 21 **D** 27

3. Cards numbered 1 through 24 are placed in a box, and one card is drawn at random. What is the probability that the number on the card is greater than 20?

A $\frac{1}{6}$ **B** $\frac{1}{5}$ **C** $\frac{5}{24}$ **D** $\frac{5}{6}$

4. The ages of the seven recruits in a training group are 20, 29, 28, 22, 19, 25, 18. What is the range of their ages?

A 2 **B** 9 **C** 10 **D** 11

5. For a raffle at a school fundraiser, 250 tickets are sold and one winning ticket is drawn. Ryan bought 2 tickets. What is the probability that he wins?

A $\frac{1}{250}$ **B** $\frac{1}{125}$ **C** $\frac{1}{124}$ **D** $\frac{124}{125}$

6. A recruiting station enlisted 20, 19, 11, 20, 8, and 12 recruits in six months. What is the mean (average) number of recruits per month?

A 15 **B** 18 **C** 21 **D** 90

7. A new pickup truck comes in 5 colors, 3 cab styles, and 2 engine sizes. How many different versions of the truck are possible?

A 6 **B** 10 **C** 15 **D** 30

8. Over six weeks, James worked 42, 36, 38, 25, 41, and 22 hours at a part-time job. What is the mean (average) number of hours he worked per week?

A 32 **B** 34 **C** 37 **D** 204

9. The dining facility on a base offers 3 main dishes, 6 side dishes, and 4 desserts. A meal is one main dish, one side, and one dessert. How many different meals are possible?

A 13 **B** 18 **C** 24 **D** 72

10. A jar contains 13 cherry, 6 orange, and 11 lemon hard candies. If one candy is picked at random, what is the probability that it is cherry?

A $\frac{1}{30}$ **B** $\frac{1}{3}$ **C** $\frac{13}{30}$ **D** $\frac{13}{17}$

11. The scores of seven students on a quiz were 72, 85, 69, 93, 64, 63, 96. What is the median of the scores?

A 33 **B** 72 **C** 85 **D** 93

12. On the range, Sergeant Chen hits 70% of the targets on average. Out of 60 targets, how many hits should Chen expect?

A 18 **B** 33 **C** 42 **D** 420

13. On four practice fitness tests, Private Brooks did 51, 64, 38, and 66 push-ups. How many push-ups must Brooks do on the next test to average 57 over all five tests?

A 57 **B** 63 **C** 66 **D** 72

14. The average weight of the 6 soldiers in a squad is 172 pounds. Sergeant Ruiz, who weighs 130 pounds, joins the squad. What is the new average weight of the squad?

A 151 pounds **B** 156 pounds
C 161 pounds **D** 166 pounds

15. Priya scored 88, 92, 73, and 92 on four math tests. What score does she need on the next test to average 89 on all five tests?

A 85 **B** 89 **C** 100 **D** 105

16. The heights, in inches, of the eight players on a basketball team are 64, 67, 78, 74, 77, 77, 66, 65. What is the median of their heights?

A 67 **B** 70 **C** 70.5 **D** 74

17. On a two-mile run, the 15 airmen in A Flight averaged 16 minutes and the 30 airmen in B Flight averaged 19 minutes. What was the average time for all the airmen?

A 17 minutes **B** 18 minutes
C 405 minutes **D** 810 minutes

18. A bag holds 7 yellow, 6 red, and 11 blue marbles. If one marble is picked at random, what is the probability that it is *neither* red nor blue?

A $\frac{7}{24}$ **B** $\frac{7}{17}$ **C** $\frac{11}{24}$ **D** $\frac{17}{24}$

19. Two six-sided dice, one red and one blue, are rolled. What is the probability that the red die shows a number less than 5 and the blue die shows an odd number?

A $\frac{1}{3}$ **B** $\frac{1}{2}$ **C** $\frac{7}{12}$ **D** $\frac{2}{3}$

20. A motor pool has 36 vehicles, and 2 of them are down for maintenance. If one vehicle is picked at random for a convoy, what is the probability that it is *not* down for maintenance?

A $\frac{1}{18}$ **B** $\frac{1}{17}$ **C** $\frac{7}{9}$ **D** $\frac{17}{18}$

21. A patrol carries two radios. Each radio works independently with probability $\frac{5}{6}$. What is the probability that *both* radios work?

A $\frac{5}{18}$ **B** $\frac{11}{36}$ **C** $\frac{25}{36}$ **D** $\frac{5}{6}$

22. In one class, 18 students averaged 87 on a test. In another class, 24 students averaged 94 on the same test. What is the average score of all the students in both classes?

A 91 **B** 101 **C** 1,911 **D** 3,822

23. In a PT test, the soldiers of a squad did these numbers of push-ups: 72, 66, 65, 50, 74, 30, 61, 33. What is the median of the data?

A 63 **B** 63.5 **C** 64 **D** 65

24. A salesperson sold 8, 12, and 25 cars in the first three months of the year. How many cars must the salesperson sell next month to average 16 cars per month over four months?

A 16 **B** 19 **C** 21 **D** 23

25. The average price of the 8 items in Maria's shopping cart is $20. She adds one more item that costs $47. What is the new average price of the items in the cart?

A $23.00 **B** $27.00 **C** $29.00 **D** $33.50

26. A spinner has 5 equal sections, and 2 of them are red. If the spinner is spun 120 times, about how many times should it land on red?

A 24 **B** 48 **C** 60 **D** 72

27. Five runners, including James, finish a race with no ties. How many different orders of finish are possible?

A 15 **B** 120 **C** 135 **D** 3,125

28. Keisha's footlocker has a combination lock with a 3-digit code. Each digit can be any number from 0 to 9, and digits may repeat, but the first digit cannot be 0. How many different codes are possible?

A 29 **B** 600 **C** 729 **D** 900

29. The 10 finalists in a base raffle include 6 soldiers from Alpha Company; the rest are from other companies. Two winners are drawn at random, one after the other (the first name is not put back). What is the probability that both winners are from Alpha Company?

A $\frac{1}{3}$ **B** $\frac{9}{25}$ **C** $\frac{2}{5}$ **D** $\frac{2}{3}$

30. The average high temperature for the first 4 days of a training exercise was 55 degrees. Including the next day, the average for all 5 days is 56 degrees. What was the high temperature on that day?

A 56 **B** 57 **C** 60 **D** 280

31. A box contains 6 glazed donuts and 7 chocolate donuts. Two donuts are taken at random, one after the other (the first is not put back). What is the probability that both donuts are glazed?

A $\frac{5}{52}$ **B** $\frac{2}{13}$ **C** $\frac{30}{169}$ **D** $\frac{5}{26}$

32. In an Army mechanic course, the unit tests count for 80% of the final grade and the final exam counts for 20%. Emily has a test average of 83. What score does she need on the final exam to earn a final grade of 85?

A 83 **B** 85 **C** 87 **D** 93

33. Five soldiers, including Ana, march in single file. Ana, the squad leader, must be first. In how many different orders can the soldiers march?

A 4 **B** 20 **C** 24 **D** 120

34. At a meeting, Kevin and the other 5 people each shake hands once with every other person. How many handshakes take place?

A 15 **B** 21 **C** 30 **D** 36

35. Tasha's club of 8 members will elect a president, a vice president, and a treasurer. No one can hold two offices. How many different results are possible?

A 56 **B** 336 **C** 411 **D** 512

36. Malik wants to take 2 of his 14 favorite books on vacation. How many different pairs of books can he choose?

A 28 **B** 78 **C** 91 **D** 182

37. In a class of 21 students, the 14 boys averaged 89 on a test, and the whole class averaged 85. What was the average score of the girls?

A 77 **B** 81 **C** 85 **D** 539

38. A bag holds 3 green marbles and 4 white marbles. Two marbles are drawn at random, one after the other (the first is not put back). What is the probability of getting one green marble and one white marble, in either order?

A $\frac{12}{49}$ **B** $\frac{2}{7}$ **C** $\frac{24}{49}$ **D** $\frac{4}{7}$

39. Jasmine's average on 10 quizzes is 70. The teacher drops her lowest quiz score, and the average of the other 9 quizzes is 72. What was the score that was dropped?

A 20 **B** 52 **C** 55 **D** 68

40. Mei bowled 190, 123, and 201 in three games. If she bowls the same score in each of the next two games, what must that score be for her to average 152 for all five games?

A 94 **B** 118 **C** 123 **D** 152

Answers & Explanations 26

1 A	**9** D	**17** B	**25** A	**33** C
2 B	**10** C	**18** A	**26** B	**34** A
3 A	**11** B	**19** A	**27** B	**35** B
4 D	**12** C	**20** D	**28** D	**36** C
5 B	**13** C	**21** C	**29** A	**37** A
6 A	**14** D	**22** A	**30** C	**38** D
7 D	**15** C	**23** A	**31** D	**39** B
8 B	**16** C	**24** B	**32** D	**40** C

1. **A** 93 minutes

1 Add the 5 values: 100 + 82 + 80 + 105 + 98 = 465.

2 Divide by the number of values, 5: 465 ÷ 5 = 93.

TIP Check: the mean must be between the smallest value (80) and the largest (105).

WHY NOT (B) is the median (the middle value), not the mean. (D) is the total, not the average.

2. **B** 17

1 Find the largest and smallest values: the largest is 27 and the smallest is 10.

2 Range = largest − smallest = 27 − 10 = 17.

WHY NOT (A) subtracts the first value listed from the last one listed. (C) is the median, not the range. (D) gives the largest value, not the range.

3. **A** $\frac{1}{6}$

1 The numbers greater than 20 are 21 through 24: 24 − 20 = 4 cards.

2 There are 24 cards in all, so the probability is $\frac{4}{24} = \frac{1}{6}$.

WHY NOT (B) compares the favorable cards to the unfavorable ones instead of to all the cards. (C) miscounts the favorable cards by one. (D) is the probability that the card is *not* greater than 20.

4. **D** 11

1 Find the largest and smallest values: the largest is 29 and the smallest is 18.

2 Range = largest − smallest = 29 − 18 = 11.

WHY NOT (A) subtracts the first value listed from the last one listed.

5. **B** $\frac{1}{125}$

1 Favorable outcomes: his 2 tickets. Total outcomes: 250 tickets.

2 Probability = $\frac{2}{250} = \frac{1}{125}$.

WHY NOT (A) uses only one ticket. (C) compares the tickets bought to the tickets not bought instead of to all tickets. (D) is the probability of *not* winning.

6. **A** 15

1 Add the 6 values: 20 + 19 + 11 + 20 + 8 + 12 = 90.

2 Divide by the number of values, 6: 90 ÷ 6 = 15.

TIP Check: the mean must be between the smallest value (8) and the largest (20).

WHY NOT (B) divides by one less than the number of values. (D) is the total, not the average.

7. **D** 30

1 Each choice can be paired with every other choice, so multiply the number of options at each step (the counting principle).

2 $5 \times 3 \times 2 = 30$.

WHY NOT (A) leaves out one of the choices. (B) adds the choices instead of multiplying them. (C) leaves out one of the choices.

8. **B** 34

1 Add the 6 values: 42 + 36 + 38 + 25 + 41 + 22 = 204.

2 Divide by the number of values, 6: 204 ÷ 6 = 34.

TIP Check: the mean must be between the smallest value (22) and the largest (42).

WHY NOT (A) averages only the highest and lowest values. (C) is the median (the middle value), not the mean. (D) is the total, not the average.

9. **D** 72

1 Each choice can be paired with every other choice, so multiply the number of options at each step (the counting principle).

2 $3 \times 6 \times 4 = 72$.

WHY NOT (A) adds the choices instead of multiplying them. (B) leaves out one of the choices. (C) leaves out one of the choices.

10. **C** $\frac{13}{30}$

1 Count all the candies: 13 + 6 + 11 = 30.

2 Favorable outcomes: 13. Probability = $\frac{13}{30}$.

WHY NOT (A) counts only one of the favorable items. (B) assumes each kind is equally likely. (D) compares the favorable outcomes to the unfavorable ones instead of to the total.

11. **B** 72

1 Put the 7 values in order: 63, 64, 69, 72, 85, 93, 96.

2 The middle value is the 4th one (3 values on each side), so the median is 72.

WHY NOT (A) is the range, not the median. (D) takes the middle number of the list without putting it in order first.

12. **C** 42

1 Expected number = percent (as a decimal) × number of targets: $0.7 \times 60 = 42$.

WHY NOT (A) is the expected number of misses. (D) moves the decimal point only one place.

13. **C** 66

1 To average 57 over 5 tests, the total must be $57 \times 5 = 285$.

2 The total so far is 51 + 64 + 38 + 66 = 219.

3 Subtract: 285 – 219 = 66.

TIP Check: 51 + 64 + 38 + 66 + 66 = 285, and 285 ÷ 5 = 57. ✓

WHY NOT (A) is the target average, not the push-ups needed on the next test.

14. **D** 166 pounds

1 Turn the average into a total: $6 \times 172 = 1{,}032$.

2 Add the value: 1,032 + 130 = 1,162.

3 Divide by the new count, 6+1 = 7: 1,162÷7 = 166.

WHY NOT (A) averages the old average with the new value.

15. **C** 100

1 To average 89 over 5 tests, the total must be $89 \times 5 = 445$.

2 The total so far is 88 + 92 + 73 + 92 = 345.

3 Subtract: 445 – 345 = 100.

TIP Check: 88 + 92 + 73 + 92 + 100 = 445, and 445 ÷ 5 = 89. ✓

WHY NOT (B) is the target average, not the score needed on the next test.

16. **C** 70.5

1 Put the 8 values in order: 64, 65, 66, 67, 74, 77, 77, 78.

2 With an even number of values, the median is the average of the two middle ones (the 4th and 5th): (67 + 74) ÷ 2 = 141 ÷ 2 = 70.5.

WHY NOT (A) uses only one of the two middle values. (D) uses only one of the two middle values.

17. **B** 18 minutes

1 Turn each average into a total: $15 \times 16 = 240$ and $30 \times 19 = 570$.

2 Add the totals and the counts: 240 + 570 = 810 for 15 + 30 = 45 airmen.

3 Divide: 810 ÷ 45 = 18.

TIP Check: the answer is closer to the average of the larger group (30 airmen).

WHY NOT (A) weights each average by the other group's size. (C) divides the combined total by 2 instead of by the number of airmen. (D) is the combined total, not the average.

18. **A** $\frac{7}{24}$

1 Total: 7 + 6 + 11 = 24. The ones that are red or blue: 6 + 11 = 17.

2 The rest, 24 – 17 = 7, are favorable. Probability = $\frac{7}{24}$.

TIP Or use $1 - P$: $1 - \frac{17}{24} = \frac{7}{24}$.

WHY NOT (B) compares favorable to unfavorable instead of to the total. (C) leaves out one of the excluded kinds. (D) is the probability of the event happening, not of it *not* happening.

19. **A** $\frac{1}{3}$

1 P(a number less than 5 on the red die) = $\frac{4}{6} = \frac{2}{3}$. P(an odd number on the blue die) = $\frac{3}{6} = \frac{1}{2}$.

2 The two results do not affect each other, so multiply: $\frac{2}{3} \times \frac{1}{2} = \frac{1}{3}$.

WHY NOT (B) ignores the first event. (C) adds the counts instead of multiplying them. (D) ignores the die.

20. **D** $\frac{17}{18}$

1 Vehicles not in maintenance: 36 – 2 = 34.

2 Probability = $\frac{34}{36} = \frac{17}{18}$.

WHY NOT (A) is the probability that the vehicle *is* down for maintenance.

21. **C** $\frac{25}{36}$

1 Each radio has probability $\frac{5}{6}$, and the events are independent, so multiply.

2 $\frac{5}{6} \times \frac{5}{6} = \frac{25}{36}$.

WHY NOT (A) multiplies the denominators but not the numerators. (D) gives the probability for only one radio.

22. **A** 91

1 Turn each average into a total: $18 \times 87 = 1{,}566$ and $24 \times 94 = 2{,}256$.

2 Add the totals and the counts: $1{,}566 + 2{,}256 = 3{,}822$ for $18 + 24 = 42$ students.

3 Divide: $3{,}822 \div 42 = 91$.

TIP Check: the answer is closer to the average of the larger group (24 students).

WHY NOT (C) divides the combined total by 2 instead of by the number of students. (D) is the combined total, not the average.

23. **A** 63

1 Put the 8 values in order: 30, 33, 50, 61, 65, 66, 72, 74.

2 With an even number of values, the median is the average of the two middle ones (the 4th and 5th): $(61 + 65) \div 2 = 126 \div 2 = 63$.

WHY NOT (D) uses only one of the two middle values.

24. **B** 19

1 To average 16 over 4 months, the total must be $16 \times 4 = 64$.

2 The total so far is $8 + 12 + 25 = 45$.

3 Subtract: $64 - 45 = 19$.

TIP Check: $8 + 12 + 25 + 19 = 64$, and $64 \div 4 = 16$. ✓

WHY NOT (A) is the target average, not the cars needed next month.

25. **A** $23.00

1 Turn the average into a total: $8 \times 20 = 160$.

2 Add the value: $160 + 47 = 207$.

3 Divide by the new count, $8 + 1 = 9$: $207 \div 9 = 23$.

WHY NOT (D) averages the old average with the new value.

26. **B** 48

1 P(red) = $\frac{2}{5}$.

2 Expected number: probability × number of spins = $\frac{2}{5} \times 120 = 48$.

WHY NOT (A) counts only one red section. (C) divides by the number of red sections. (D) is the expected number of spins that do *not* land on red.

27. **B** 120

1 First place: 5 choices; second place: 4 (one is used); and so on down to 1.

2 Multiply: $5 \times 4 \times 3 \times 2 \times 1 = 120$.

WHY NOT (A) adds the choices instead of multiplying them. (D) lets the same item fill more than one place.

28. **D** 900

1 Count the choices for each position: 9, 10, 10.

2 Multiply (counting principle): $9 \times 10 \times 10 = 900$.

WHY NOT (A) adds the choices instead of multiplying them. (C) leaves out the digit 0 in every position.

29. **A** $\frac{1}{3}$

1 First pick: $\frac{6}{10}$. After one Alpha Company soldier is taken, 5 of the 9 names left are Alpha Company soldiers: $\frac{5}{9}$.

2 Multiply: $\frac{6}{10} \times \frac{5}{9} = \frac{30}{90} = \frac{1}{3}$.

WHY NOT (B) puts the first one back (uses the same counts for both picks). (C) forgets to remove the first one from the count of Alpha Company soldiers.

30. **C** 60

1 Total for the first 4: $4 \times 55 = 220$.

2 Total for all 5: $5 \times 56 = 280$.

3 The new value is the difference: $280 - 220 = 60$.

WHY NOT (A) is the new average, not the new value. (B) adds the change in the average to the new average. (D) is the total for all 5, not the last value.

31. **D** $\frac{5}{26}$

1 First pick: $\frac{6}{13}$. After one glazed donut is taken, 5 of the 12 donuts left are glazed donuts: $\frac{5}{12}$.

2 Multiply: $\frac{6}{13} \times \frac{5}{12} = \frac{30}{156} = \frac{5}{26}$.

WHY NOT (C) lowers the count of the kind picked but not the total.

32. **D** 93

1 The tests contribute $0.8 \times 83 = 66.4$ points to the grade.

2 The final must contribute $85 - 66.4 = 18.6$ points.

3 The final counts 20%, so the exam score is $18.6 \div 0.2 = 93$.

TIP Check: $0.8(83) + 0.2(93) = 66.4 + 18.6 = 85$. ✓

WHY NOT (B) is the final grade wanted, not the exam score needed. (C) treats the tests and the final as if they counted equally.

33. **C** 24

1 Ana's place is fixed, so only the other 4 people need to be arranged.

2 $4 \times 3 \times 2 \times 1 = 24$.

WHY NOT (A) counts only the choices for one position. (D) ignores the rule about Ana.

34. **A** 15

1 Each of the 6 can be paired with 5 others: $6 \times 5 = 30$.

2 That counts every pair twice (A with B and B with A), so divide by 2: $30 \div 2 = 15$.

WHY NOT (C) counts each pair twice.

35. **B** 336

1 First position: 8 choices; second: 7; third: 6 (no one can repeat).

2 Multiply: $8 \times 7 \times 6 = 336$.

WHY NOT (A) counts groups of three instead of ordered positions. (D) lets the same person fill more than one position.

36. **C** 91

1 Each of the 14 can be paired with 13 others: $14 \times 13 = 182$.

2 That counts every pair twice (A with B and B with A), so divide by 2: $182 \div 2 = 91$.

WHY NOT (B) leaves one person out. (D) counts each pair twice.

37. **A** 77

1 Total for everyone: $21 \times 85 = 1{,}785$.

2 Total for the boys: $14 \times 89 = 1{,}246$.

3 The girls have $1{,}785 - 1{,}246 = 539$ points in all, so their average is $539 \div 7 = 77$.

WHY NOT (B) assumes the two groups are the same size. (C) gives the average of the whole group. (D) is the total for the girls, not their average.

38. **D** $\frac{4}{7}$

1 One order, green marble first and then white marble: $\frac{3}{7} \times \frac{4}{6} = \frac{12}{42}$.

2 The other order, white marble first and then green marble: $\frac{4}{7} \times \frac{3}{6} = \frac{12}{42}$.

3 Add the two orders: $2 \times \frac{12}{42} = \frac{24}{42} = \frac{4}{7}$.

WHY NOT (A) counts only one order and puts the first one back. (B) counts only one order. (C) puts the first one back (uses the same total for both picks).

39. **B** 52

1 Total of all 10: $10 \times 70 = 700$.

2 Total of the other 9: $9 \times 72 = 648$.

3 The dropped score is the difference: $700 - 648 = 52$.

WHY NOT (A) uses 10 scores instead of 9 for the new total. (D) assumes the dropped score was as far below the average as the average went up.

40. **C** 123

1 To average 152 over 5 games, the total must be $152 \times 5 = 760$.

2 The total so far is $190 + 123 + 201 = 514$.

3 The next two must add up to $760 - 514 = 246$, so each one is $246 \div 2 = 123$.

TIP Check: $190 + 123 + 201 + 123 + 123 = 760$, and $760 \div 5 = 152$. ✓

WHY NOT (A) plans for only one more game instead of two. (D) is the target average, not the score needed each game.

CHAPTER 27

Measurement, Units & Multi-Step Problems

Tested on: Arithmetic Reasoning

Key Concepts

Many Arithmetic Reasoning questions are easy arithmetic wrapped in units: feet and inches, quarts and gallons, a 24-hour clock. The rule that saves you: **convert first, then calculate**, and write down each step.

US customary units (memorize these)

Length	Capacity	Weight and time
12 inches = 1 foot	8 fluid ounces = 1 cup	16 ounces = 1 pound
3 feet = 1 yard	2 cups = 1 pint	2,000 pounds = 1 ton
36 inches = 1 yard	2 pints = 1 quart	60 minutes = 1 hour
5,280 feet = 1 mile	4 quarts = 1 gallon	7 days = 1 week

Big unit → small unit: **multiply**. Small unit → big unit: **divide**. (There are *more* inches than feet, so the number gets bigger.)

Metric units

kilo = 1,000 centi = $\frac{1}{100}$ milli = $\frac{1}{1{,}000}$. So 1 km = 1,000 m, 1 m = 100 cm = 1,000 mm, 1 kg = 1,000 g, 1 L = 1,000 mL. Converting is just moving the decimal point: 2.5 km = 2,500 m. US–metric factors (such as 1 mile ≈ 1.6 km) are given in the problem.

Military (24-hour) time and elapsed time

Hours run from 0000 (midnight) to 2359. For p.m. times add 12 to the hour: 3:45 p.m. = 1545. To find elapsed time, subtract hours and minutes separately, and when you borrow an hour, it is worth **60** minutes, not 100. Across midnight: time until 2400, plus time after 0000.

WORKED EXAMPLE

A convoy leaves at 0745 and arrives at 1420. How long is the trip?
Solution. 20 minutes minus 45 minutes won't go, so borrow 1 hour: 14:20 becomes 13 hours 80 minutes. Then 13 – 7 = 6 hours and 80 – 45 = 35 minutes. The trip takes 6 hours 35 minutes.

No-Calculator Tip Decide how to round from the question. "How many trips (boxes, tents) are *needed*?" rounds *up*: 75 soldiers in 6-person tents need 13 tents. "How many *full* pieces (or items you can afford)?" rounds *down*.

Common Traps

- Subtracting clock times like ordinary numbers: 1420 – 0745 = 675 is not 6 hours 75 minutes.

- Fence posts: a 60-foot fence with posts every 6 feet needs 60 ÷ 6 + 1 = 11 posts (one at each end). Cutting a board into 5 pieces takes only 4 cuts.
- "$\frac{1}{3}$ of the *remainder*" is a fraction of what is left, not of the original amount.
- Using the wrong conversion factor (3 feet in a yard, but 12 inches in a foot).

Practice Set 27

Score: _____ / 35 Target time: 42 minutes

Problems 1–10 are warm-ups, 11–24 are test level, and 25–35 are challenge problems.

1. A hallway in the office building where Ethan works is 24 feet long. How many yards of carpet runner are needed to cover its length?

A 2 yards **B** 5 yards
C 8 yards **D** 72 yards

2. Nora's soup recipe calls for 12 quarts of broth. How many cups of broth is that?

A 16 cups **B** 24 cups **C** 48 cups **D** 63 cups

3. A bolt is 32 millimeters long. How many centimeters long is it?

A 0.32 centimeters **B** 2.9 centimeters
C 3.1 centimeters **D** 3.2 centimeters

4. A flight leaves at 11:50 a.m. and lands at 6:25 p.m. (same time zone). How long is the flight?

A 5 hours 25 minutes **B** 5 hours 50 minutes
C 6 hours 5 minutes **D** 6 hours 35 minutes

5. A bag of flour has a mass of 0.5 kilograms. How many grams is that?

A 500 grams **B** 550 grams
C 575 grams **D** 5,000 grams

6. Keisha starts a hike at 9:35 a.m. and hikes for 2 hours 55 minutes. At what time does she finish?

A 11:30 a.m. **B** 12:30 p.m.
C 12:50 p.m. **D** 1:00 p.m.

7. Keisha's party punch recipe makes 32 cups. How many gallons of punch is that?

A 1 gallon **B** 2 gallons
C 16 gallons **D** 512 gallons

8. Ethan needs a piece of trim 24 feet long. How many inches long is the trim?

A 72 inches **B** 288 inches
C 338 inches **D** 363 inches

9. A ruck march has two legs. The first takes 3 hr 29 min and the second takes 4 hr 36 min. What is the total time?

A 7 hr 5 min **B** 8 hr 3 min
C 8 hr 5 min **D** 9 hr 5 min

10. A shelf is 53 inches long. How many binders, each 2 inches thick, can stand on it?

A 26 **B** 27 **C** 28 **D** 30

11. The speed limit on a road overseas is 120 kilometers per hour. About how many miles per hour is that? (1 mile ≈ 1.6 kilometers)

A 75 miles per hour **B** 150 miles per hour
C 192 miles per hour **D** 750 miles per hour

12. A military transport flight takes off at 0735 and lands at 1615 (same time zone). How long is the flight?

A 8 hours 20 minutes **B** 8 hours 40 minutes
C 9 hours 20 minutes **D** 9 hours 40 minutes

13. A roll of fabric holds 10 yd 1 ft. Jamal cuts off 6 yd 2 ft. How much fabric is left on the roll?

A 2 yd 2 ft **B** 3 yd 2 ft **C** 4 yd 1 ft **D** 4 yd 2 ft

14. Maria's pancake recipe uses 3 cups of milk per batch, and 15 batches are needed for a fundraiser breakfast. How many half-gallon cartons of milk must be bought? (1 half gallon = 8 cups)

A 3 **B** 4 **C** 5 **D** 6

15. Hannah has a board that is 11 ft 6 in long and cuts off a piece 4 ft 10 in long. How long is the board that is left?

A 5 ft 8 in **B** 6 ft 6 in **C** 6 ft 8 in **D** 7 ft 4 in

16. In the field, each soldier needs 3 quarts of water per day. How many soldiers can a full 40-gallon water tank supply for one day?

A 53 **B** 54 **C** 58 **D** 63

17. Specialist Okafor begins guard duty at 2:30 p.m. The shift lasts 3 hours 50 minutes. At what time does the shift end, in military time?

A 0620 **B** 1720 **C** 1730 **D** 1820

18. A truck's trip odometer shows 50 miles. About how many kilometers is that? (1 mile $\approx$ 1.6 kilometers)

A 31.25 kilometers **B** 51.6 kilometers
C 80 kilometers **D** 800 kilometers

19. Nora is building shelves and needs ten boards, each 45 inches long. How many feet of board does she need in all?

A 37.5 feet **B** 39 feet
C 45 feet **D** 450 feet

20. Corporal Nguyen's squad has eight full 2-gallon water cans. Each canteen holds 1 quart. How many canteens can the squad fill?

A 16 **B** 64 **C** 69 **D** 128

21. Ana divides a 10-kilogram bag of rice into servings of 125 grams. How many servings does the bag hold?

A 8 **B** 80 **C** 800 **D** 10,000

22. A truck starts with 80 gallons of fuel. It uses $\frac{2}{5}$ of the fuel on the first day and $\frac{1}{6}$ of what is left on the second day. How many gallons are left?

A 8 **B** 30 **C** 32 **D** 40

23. A plumber cuts a 16-foot pipe into equal pieces 2 feet long. How many cuts does the plumber make?

A 5 **B** 6 **C** 7 **D** 8

24. In 5 years, Luis will be three times as old as his daughter will be then. The daughter is 5 now. How old is Luis now?

A 15 **B** 25 **C** 30 **D** 35

25. A saw takes 5 minutes to cut through a steel beam. How long will it take to cut a 12-foot beam into 4 equal pieces?

A 15 minutes **B** 20 minutes
C 25 minutes **D** 60 minutes

26. Nora starts a fitness plan on a Thursday, which counts as day 1. On what day of the week is day 79?

A Friday **B** Wednesday
C Thursday **D** Saturday

27. Sofia is three times as old as her son. Five years ago, Sofia was five times as old as the son. How old is Sofia now?

A 10 **B** 25 **C** 30 **D** 50

28. Of the recruits who started a training course, $\frac{3}{5}$ were sent to other units in the first week, and $\frac{2}{5}$ of the rest were sent away in the second week. 72 recruits remain. How many recruits started the course?

A 144 **B** 180 **C** 300 **D** 350

29. A ship's crew starts loading cargo at 2050, and the job takes 6 hours 50 minutes. At what military time will the loading be finished?

A 0340 **B** 0440 **C** 0510 **D** 1540

30. On the first day a convoy drove $\frac{1}{6}$ of its route, and on the second day it drove $\frac{1}{2}$ of the remaining distance. It still has 30 miles to go. How long is the whole route?

A 36 **B** 50 **C** 57 **D** 72

31. A soldier pulls guard duty from 2125 until 0655 the next morning. How long is the guard shift?

A 6 hours 55 minutes
B 8 hours 30 minutes
C 9 hours 30 minutes
D 14 hours 30 minutes

32. The sum of the ages of Tyrone and his daughter is 44. In 4 years, Tyrone will be three times as old as his daughter. How old is Tyrone now?

A 9 **B** 35 **C** 39 **D** 50

33. Today is Saturday. Priya enlisted exactly 109 days ago. On what day of the week did she enlist?

A Tuesday **B** Friday
C Monday **D** Wednesday

34. Trees are planted on *both* sides of a straight 850-foot road, one every 50 feet, with a tree at each end on each side. How many trees are planted?

A 18 **B** 34 **C** 35 **D** 36

35. Floor tile is sold in boxes that each cover 15 square feet, at $30 per box. How much will enough tile cost to cover an 83-square-foot floor?

A $150 **B** $160 **C** $166 **D** $180

Answers & Explanations 27

1 C	**8** B	**15** C	**22** D	**29** A
2 C	**9** C	**16** A	**23** C	**30** D
3 D	**10** A	**17** D	**24** B	**31** C
4 D	**11** A	**18** C	**25** A	**32** B
5 A	**12** B	**19** A	**26** A	**33** A
6 B	**13** B	**20** B	**27** C	**34** D
7 B	**14** D	**21** B	**28** C	**35** D

1. **C** 8 yards

1 1 yard = 3 feet. Going from a small unit to a big unit, divide: 24 ÷ 3 = 8 yards.

WHY NOT (A) uses 12 instead of 3 feet in a yard. (D) multiplies instead of dividing.

2. **C** 48 cups

1 1 quart = 2 pints = 4 cups. From a big unit to a small unit, multiply: 12 × 4 = 48 cups.

WHY NOT (A) adds 4 instead of multiplying by it. (B) uses 2 cups in a quart instead of 4.

3. **D** 3.2 centimeters

1 1 centimeter = 10 millimeters. Divide by 10 (move the decimal point 1 place to the left): 32 ÷ 10 = 3.2 centimeters.

WHY NOT (A) uses 100 instead of 10.

4. **D** 6 hours 35 minutes

1 From 11:50 a.m. to 12 noon is 10 minutes.

2 From 12 noon to 6:25 p.m. is 6 hours 25 minutes.

3 Add: 10 minutes + 6 hours 25 minutes = 6 hours 35 minutes.

WHY NOT (A) forgets that the clock starts over at 12 noon.

5. **A** 500 grams

1 1 kilogram = 1,000 grams. Multiply by 1,000 (move the decimal point 3 places to the right): 0.5 × 1,000 = 500 grams.

WHY NOT (D) moves the decimal point one place too far.

6. **B** 12:30 p.m.

1 Add the minutes: 35 + 55 = 90 minutes, which is 1 hour 30 minutes. Write down 30 minutes and carry 1 hour.

2 Add the hours: 9 + 2 + 1 = 12.

3 The time is 12:30 p.m.

WHY NOT (A) forgets to carry the extra hour from the minutes.

7. **B** 2 gallons

1 1 gallon = 4 quarts = 16 cups. From a small unit to a big unit, divide: 32 ÷ 16 = 2 gallons.

WHY NOT (C) subtracts 16 instead of dividing by it. (D) multiplies instead of dividing.

8. **B** 288 inches

1 1 foot = 12 inches. Going from a big unit to a small unit, multiply: 24 × 12 = 288 inches.

TIP Check the size: inches are smaller than feet, so there should be more of them.

WHY NOT (A) uses 3 instead of 12 inches in a foot.

9. **C** 8 hr 5 min

1 Add the minutes: 29 + 36 = 65 min. Since 60 min = 1 hr, 65 min = 1 hr 5 min.

2 Add the hours: 3 + 4 + 1 = 8 hr.

3 Total: 8 hr 5 min.

WHY NOT (A) changes 65 min to 5 min but forgets to add 1 to the hours. (D) carries 2 hours instead of 1 when regrouping 65 min.

10. **A** 26

1 Divide: 53 ÷ 2 is 26 with a remainder of 1.

2 The 1 inches left over are not enough for another binder, so the answer is 26 binders.

WHY NOT (B) rounds up, but there is not enough for another binder.

11. **A** 75 miles per hour

1 Fewer miles than kilometers, so divide by 1.6: 120 ÷ 1.6 = 1200 ÷ 16 = 75 miles per hour.

WHY NOT (C) multiplies by 1.6 instead of dividing. (D) moves the decimal point one place too far to the right.

12. **B** 8 hours 40 minutes

1 15 minutes minus 35 minutes won't go, so borrow 1 hour: 1615 becomes 15 hours 75 minutes.

2 Subtract: 15 – 7 = 8 hours and 75 – 35 = 40 minutes.

3 The time is 8 hours 40 minutes.

WHY NOT (A) borrows an hour but then subtracts the minutes backward (smaller from larger) instead of adding 60 to them first. (C) subtracts 1615 – 0735 like ordinary numbers and reads 880 as 8 hours 80 minutes (that is, 9 hours 20 minutes). (D) borrows the hour but forgets to take it away from the hours.

13. **B** 3 yd 2 ft

1 You can't take 2 ft from 1 ft, so borrow 1 yd: 10 yd 1 ft becomes 9 yd 4 ft.

2 Subtract: 4 – 2 = 2 ft and 9 – 6 = 3 yd.

3 What is left: 3 yd 2 ft.

WHY NOT (C) subtracts the smaller number of feet from the larger instead of borrowing. (D) borrows 1 yd but forgets to take it away from the yards.

14. **D** 6

1 Total milk: $15 \times 3 = 45$ cups.

2 Divide: $45 \div 8$ is 5 with a remainder of 5.

3 5 cartons hold only $5 \times 8 = 40$ cups, which is 5 cups short, so buy 6 cartons.

WHY NOT (A) uses 16 cups per carton (that is a full gallon). (C) rounds down, which comes up 5 cups short.

15. **C** 6 ft 8 in

1 You can't take 10 in from 6 in, so borrow 1 ft: 11 ft 6 in becomes 10 ft 18 in.

2 Subtract: 18 – 10 = 8 in and 10 – 4 = 6 ft.

3 What is left: 6 ft 8 in.

WHY NOT (B) borrows 10 inches instead of 12. (D) subtracts the smaller number of inches from the larger instead of borrowing.

16. **A** 53

1 Change gallons to quarts: $40 \times 4 = 160$ quarts.

2 Divide: $160 \div 3$ is 53 with a remainder of 1.

3 The 1 quarts left are not enough for another soldier's day, so 53 soldiers.

WHY NOT (B) rounds up, but there is not enough for another soldier.

17. **D** 1820

1 Change to military time: 2:30 p.m. is 2 + 12 = 14 hours, so 1430.

2 Add the minutes: 30 + 50 = 80 minutes = 1 hour 20 minutes.

3 Add the hours: 14 + 3 + 1 = 18. The shift ends at 1820.

WHY NOT (A) forgets to change the p.m. time to 24-hour time. (B) forgets to carry the extra hour from the minutes. (C) adds only the hours.

18. **C** 80 kilometers

1 Each mile is about 1.6 kilometers, so multiply: $50 \times 1.6 = 80$ kilometers.

WHY NOT (A) divides by 1.6 instead of multiplying. (B) adds 1.6 instead of multiplying. (D) moves the decimal point one place too far to the right.

19. **A** 37.5 feet

1 Total length in inches: $10 \times 45 = 450$ inches.

2 Change inches to feet (12 inches = 1 foot): $450 \div 12 = 37.5$ feet.

WHY NOT (C) divides by 10 instead of 12. (D) gives the total in inches, not feet.

20. **B** 64

1 Total water: $8 \times 2 = 16$ gallons.

2 1 gallon = 4 quarts: $16 \times 4 = 64$ quarts, so 64 canteens.

WHY NOT (A) forgets to change gallons to quarts. (D) changes gallons to pints instead of quarts.

21. **B** 80

1 Change kilograms to grams: $10 \times 1{,}000 = 10{,}000$ grams.

2 Divide by one serving: $10{,}000 \div 125 = 80$ servings.

WHY NOT (A) uses 100 grams in a kilogram instead of 1,000. (C) moves the decimal point one place too far. (D) gives the number of grams, not servings.

22. **D** 40

1 First step: $\frac{2}{5} \times 80 = 32$. Remainder: 80 – 32 = 48.

2 Second step: $\frac{1}{6}$ of the *remainder*: $\frac{1}{6} \times 48 = 8$.

3 Left: 48 – 8 = 40.

WHY NOT (A) gives the fuel used on the second day, not the fuel left. (C) gives the fuel used on the first day, not the fuel left.

23. **C** 7

1 Number of pieces: $16 \div 2 = 8$.

2 The last piece needs no cut of its own, so cuts = pieces −1: $8 - 1 = 7$.

TIP One cut makes 2 pieces, two cuts make 3 pieces: always one fewer cut than pieces.

WHY NOT (D) gives the number of pieces, not the number of cuts.

24. **B** 25

1 In 5 years the daughter will be $5 + 5 = 10$.

2 Then Luis will be $3 \times 10 = 30$.

3 Today Luis is 5 years younger than that: $30 - 5 = 25$.

WHY NOT (A) multiplies the daughter's age today instead of in 5 years. (C) gives Luis's age in 5 years, not now. (D) adds the 5 years instead of subtracting them.

25. **A** 15 minutes

1 Pieces: 4. Cuts = pieces −1 = 3.

2 Time: $3 \times 5 = 15$ minutes.

WHY NOT (B) multiplies by the number of pieces instead of the number of cuts. (D) multiplies the length by the time per cut.

26. **A** Friday

1 Day 79 is $79 - 1 = 78$ days after day 1.

2 $78 \div 7 = 11$ weeks with 1 day left over, so move 1 day past Thursday: Friday.

WHY NOT (B) counts backward instead of forward. (C) counts one day too few. (D) counts 79 days after day 1, but day 79 is only 78 days later.

27. **C** 30

1 Let the son be a years old; then Sofia is $3a$.

2 5 years ago: $3a - 5 = 5(a - 5)$, so $3a - 5 = 5a - 25$.

3 Then $20 = 2a$, so $a = 10$, and Sofia is $3 \times 10 = 30$.

WHY NOT (A) gives the son's age. (B) gives Sofia's age 5 years ago. (D) uses the second ratio for today's ages.

28. **C** 300

1 After the first step, $1 - \frac{3}{5} = \frac{2}{5}$ of the whole is left.

2 The second step uses $\frac{2}{5}$ of that, leaving $1 - \frac{2}{5} = \frac{3}{5}$ of it: $\frac{2}{5} \times \frac{3}{5} = \frac{6}{25}$ of the whole.

3 So $\frac{6}{25}$ of the whole is 72: $72 \div \frac{6}{25} = 300$.

WHY NOT (A) adds the fractions of the amount left. (B) undoes only the first step.

29. **A** 0340

1 Add the minutes: $50 + 50 = 100$ minutes = 1 hour 40 minutes.

2 Add the hours: $20 + 6 + 1 = 27$. Past midnight, subtract 24: $27 - 24 = 3$.

3 The answer is 0340 the next day.

WHY NOT (D) subtracts 12 hours instead of 24 after passing midnight.

30. **D** 72

1 After the first step, $1 - \frac{1}{6} = \frac{5}{6}$ of the whole is left.

2 The second step uses $\frac{1}{2}$ of that, leaving $1 - \frac{1}{2} = \frac{1}{2}$ of it: $\frac{5}{6} \times \frac{1}{2} = \frac{5}{12}$ of the whole.

3 So $\frac{5}{12}$ of the whole is 30: $30 \div \frac{5}{12} = 72$.

WHY NOT (A) undoes only the first step. (B) adds the fractions of the amount left.

31. **C** 9 hours 30 minutes

1 From 2125 to midnight (2400): 2 hours 35 minutes.

2 From midnight to 0655: 6 hours 55 minutes.

3 Add: 2 hours 35 minutes + 6 hours 55 minutes = 9 hours 30 minutes.

WHY NOT (A) counts only the time after midnight. (D) subtracts 0655 from 2125 instead of going past midnight.

32. **B** 35

1 Let the daughter be a; then Tyrone is $44 - a$.

2 In 4 years: $44 - a + 4 = 3(a + 4)$, so $48 - a = 3a + 12$.

3 Then $36 = 4a$, so $a = 9$, and Tyrone is $44 - 9 = 35$.

WHY NOT (A) gives the daughter's age. (C) gives Tyrone's age in 4 years.

33. **A** Tuesday

1 $109 \div 7 = 15$ weeks with 4 days left over. 15 weeks ago was also a Saturday.

2 Go back 4 more days: Friday, Thursday, Wednesday, Tuesday.

WHY NOT (B) moves back 15 days (the number of weeks) instead of 4. (C) counts one day too many. (D) counts forward instead of backward.

34. **D** 36

1 Spaces on one side: $850 \div 50 = 17$, so one side has $17 + 1 = 18$ trees.

2 Two sides: $2 \times 18 = 36$ trees.

WHY NOT (A) counts only one side of the road. (B) forgets the extra tree at the end of each side. (C) adds only one extra tree for both sides.

35. **D** $180

1 Each box covers 15 square feet: 83 ÷ 15 is 5 with a remainder of 8.

2 5 boxes would cover only $5 \times 15 = 75$ square feet, which is not quite enough, so buy 6 boxes.

3 Cost: $6 \times 30 = 180$, so $180.

WHY NOT (A) rounds the number of boxes down, which is not enough. (C) pays for part of a box, which is not sold.

PART V

Practice Tests

Four full-length math sections in the paper-and-pencil format: 30 Arithmetic Reasoning questions in 36 minutes and 25 Mathematics Knowledge questions in 24 minutes.

Practice Test 1
Practice Test 2
Practice Test 3
Practice Test 4

Practice Test 1

Instructions

This test has two parts, timed like the paper-and-pencil ASVAB. **Part 1, Arithmetic Reasoning:** 30 questions, 36 minutes. **Part 2, Mathematics Knowledge:** 25 questions, 24 minutes. No calculator. Use scratch paper. Answer every question—there is no penalty for guessing. Mark your answers on the answer sheet below.

Answer Sheet

Fill in one bubble per question. Photocopy this page to retake the test.

Part 1: Arithmetic Reasoning

1	A	B	C	D
2	A	B	C	D
3	A	B	C	D
4	A	B	C	D
5	A	B	C	D
6	A	B	C	D
7	A	B	C	D
8	A	B	C	D
9	A	B	C	D
10	A	B	C	D
11	A	B	C	D
12	A	B	C	D
13	A	B	C	D
14	A	B	C	D
15	A	B	C	D
16	A	B	C	D
17	A	B	C	D
18	A	B	C	D
19	A	B	C	D
20	A	B	C	D
21	A	B	C	D
22	A	B	C	D
23	A	B	C	D
24	A	B	C	D
25	A	B	C	D
26	A	B	C	D
27	A	B	C	D
28	A	B	C	D
29	A	B	C	D
30	A	B	C	D

Part 2: Mathematics Knowledge

1	A	B	C	D
2	A	B	C	D
3	A	B	C	D
4	A	B	C	D
5	A	B	C	D
6	A	B	C	D
7	A	B	C	D
8	A	B	C	D
9	A	B	C	D
10	A	B	C	D
11	A	B	C	D
12	A	B	C	D
13	A	B	C	D
14	A	B	C	D
15	A	B	C	D
16	A	B	C	D
17	A	B	C	D
18	A	B	C	D
19	A	B	C	D
20	A	B	C	D
21	A	B	C	D
22	A	B	C	D
23	A	B	C	D
24	A	B	C	D
25	A	B	C	D

Part 1: Arithmetic Reasoning

Score: _____ / 30

Target time: 36 minutes

1. A recruiting office processed 244 applications last month. Of these, 75% were incomplete. How many applications were complete?

A 21 **B** 31 **C** 61 **D** 183

2. The sum of two numbers is 86, and their difference is 18. What is the larger number?

A 34 **B** 52 **C** 68 **D** 104

3. Gasoline sold as E10 is 10% ethanol. How many gallons of ethanol are in a full 20-gallon tank of E10 gasoline?

A 0.2 gallons **B** 1.25 gallons
C 2 gallons **D** 18 gallons

4. Mei pays off a $336 phone in 12 equal monthly payments. How much is each payment?

A $2 **B** $8 **C** $26 **D** $28

5. A convoy can take 2 different roads from the base to a checkpoint and 5 different roads from the checkpoint to the training area. How many different routes are there from the base to the training area through the checkpoint?

A 7 **B** 8 **C** 10 **D** 20

6. Ethan deposits $1,200 in a savings account that pays 5% simple interest per year. How much interest will he earn in 15 months?

A $75 **B** $90 **C** $900 **D** $1,275

7. Ryan's rectangular fish tank is 6 feet long, 2 feet wide, and 2 feet deep. How many gallons of water does it hold when it is full? (1 cubic foot holds 7.5 gallons.)

A 24 gal **B** 180 gal **C** 270 gal **D** 420 gal

8. A water trailer at Corporal Lopez's field site holds 275 gallons. How many 1-quart canteens can be filled from a full trailer?

A 275 **B** 550 **C** 800 **D** 1,100

9. At a sporting-goods store, Luis buys a water bottle for $10.25 and a jump rope for $8.95. He pays with a $50 bill. How much change should he receive?

A $19.20 **B** $30.00 **C** $30.80 **D** $31.80

10. Leila marks off a sprint course 10 yards long. How many inches long is the course?

A 120 inches **B** 360 inches
C 385 inches **D** 410 inches

11. The platoon has $260 for a barbecue. After spending $55 on charcoal and paper plates, it will spend the rest on packs of hamburger patties at $18 each. How many packs can it buy?

A 11 **B** 12 **C** 14 **D** 17

12. Engineers need 1 ton of sand for sandbags. The sand comes in 60-pound bags. How many bags must they order?

A 17 **B** 20 **C** 33 **D** 34

13. A motor pool has 30 vehicles, and 3 of them are down for maintenance. If one vehicle is picked at random for a convoy, what is the probability that it is *not* down for maintenance?

A $\frac{1}{10}$ **B** $\frac{1}{9}$ **C** $\frac{7}{10}$ **D** $\frac{9}{10}$

14. Ana buys a certificate of deposit (CD) for $9,000. The CD pays 5% simple interest per year. How much money will be in the account at the end of 5 years?

A $2,250 **B** $11,250 **C** $12,250 **D** $31,500

15. Jasmine is saving for a new laptop, which costs $1,050. She has already saved $90 and can save $50 a week. How many weeks will it take her to have enough money?

A 16 weeks **B** 18 weeks
C 19 weeks **D** 20 weeks

16. A training class has 96 soldiers from Alpha Company and 80 from Bravo Company. The instructor wants to form teams so that every team has the same number of Alpha soldiers and the same number of Bravo soldiers, with no one left out. If the greatest possible number of teams is formed, how many Alpha soldiers are on each team?

A 6 **B** 11 **C** 16 **D** 48

17. Ethan has \$115 in a checking account. He pays \$86 for a phone bill and \$89 for a utility bill. Because the balance went below \$0, the bank also charges a \$30 overdraft fee. How much must Ethan deposit to bring the balance back up to \$0?

A \$90 **B** \$100 **C** \$110 **D** \$120

18. A recruit does 37 push-ups every morning for d days, plus 5 extra push-ups on the last day. Which expression gives the total number of push-ups?

A $37(d + 5)$ **B** $5d + 37$

C $37d + 5$ **D** $42d$

19. A chemistry teacher has 12 liters of a 30% acid solution. How many liters of water must be added to dilute it to a 10% solution?

A 24 liters **B** 26 liters **C** 30 liters **D** 36 liters

20. An 8-pack of soda costs \$4.00. What is the price per can?

A \$0.05 **B** \$0.49 **C** \$0.50 **D** \$5.00

21. A recipe calls for $2\frac{1}{2}$ teaspoons of salt. Leila is making 3 batches. How many teaspoons of salt does she need?

A $6\frac{1}{2}$ teaspoons **B** $7\frac{1}{2}$ teaspoons

C $8\frac{1}{2}$ teaspoons **D** 9 teaspoons

22. Marcus scored 79, 94, 85, and 78 on four math tests. What score does he need on the next test to average 81 on all five tests?

A 69 **B** 72 **C** 78 **D** 81

23. Working together, Carlos and Luis can paint a fence in 6 hours. Working alone, Carlos can paint it in 8 hours. How long would it take Luis to paint the fence alone?

A 2 hours **B** 12 hours

C 24 hours **D** 28 hours

24. A racehorse gallops at 45 miles per hour. How many feet per second is this? (1 mile = 5,280 feet and 1 hour = 3,600 seconds.)

A 58 **B** 66 **C** 3,960 **D** 237,600

25. A supply convoy leaves the base traveling 26 miles per hour. Two hours later, Lieutenant Reyes leaves the base in a repair truck on the same road at 52 miles per hour to catch up. How many hours after the repair truck leaves will it catch up?

A 1 hour **B** 2 hours **C** 3 hours **D** 4 hours

26. At a recruiting event, the ratio of applicants who passed the fitness test to those who did not pass was 8 : 3. The number who passed was 30 more than the number who did not. How many applicants took the test?

A 66 **B** 76 **C** 96 **D** 330

27. A boat travels 36 miles upstream at 6 miles per hour and returns downstream at 18 miles per hour. What is the average speed for the round trip?

A 4.5 mph **B** 9 mph **C** 12 mph **D** 24 mph

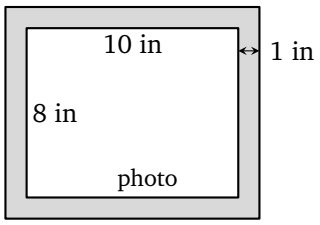
28. Hannah, who is 6 feet tall, casts a shadow 7 feet long. At the same time, a building casts a shadow 126 feet long. How tall is the building?

A 106 ft **B** 108 ft **C** 125 ft **D** 147 ft

29. Petty Officer Kim takes out a loan of \$14,400 to buy furniture for her apartment off base. The loan charges 5% simple interest per year, figured on the full \$14,400 for all 4 years. She will repay the principal plus interest in equal monthly payments over 4 years. How much is each monthly payment?

A \$60 **B** \$300 **C** \$315 **D** \$360

30. A photo 10 inches by 8 inches is surrounded by a frame 1 inch wide on every side. What is the area of the frame alone?



A 19 in^2 **B** 24 in^2 **C** 36 in^2 **D** 40 in^2

Part 2: Mathematics Knowledge

Score: _____ / 25

Target time: 24 minutes

1. What is the greatest common factor of the terms of $16x - 4$?

A 4 **B** 2 **C** $16x$ **D** $4x$

2. The equation $8 \times (9 - 5) = 8 \times 9 - 8 \times 5$ is an example of which property?

A Distributive property
B Commutative property
C Associative property
D Identity property

3. Which inequality represents the statement "A number n decreased by 4 is fewer than 19"?

A $n - 4 \geq 19$ **B** $n - 4 > 19$
C $n - 4 < 19$ **D** $n - 4 \leq 19$

4. What is the value of y in the solution of this system?

$$x + y = 20$$
$$x - y = -4$$

A -12 **B** -4 **C** 8 **D** 12

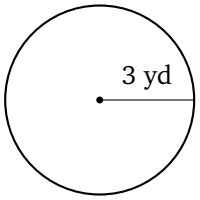
5. What is 25% of 56?

A 14 **B** 42 **C** 70 **D** 140

6. Round 2.839 to the nearest hundredth.

A 2.8 **B** 2.83 **C** 2.84 **D** 2.85

7. The radius of the circle shown is 3 yards. What is the circumference of the circle?



A 3π yd **B** 4π yd **C** 6π yd **D** 9π yd

8. Which expression is equivalent to $5(x + 1) - 2(2x - 5)$?

A $x + 6$ **B** $x - 5$ **C** $9x - 5$ **D** $x + 15$

9. Derek wants to cover a gift box with wrapping paper with no overlap. The box is 10 inches long, 12 inches wide, and 2 inches tall. How many square inches of paper will cover the whole box?

A 328 in^2 **B** 376 in^2 **C** 568 in^2 **D** 720 in^2

10. What is the GCF of 16 and 10?

A 2 **B** 3 **C** 10 **D** 80

11. If y varies directly with x, and $y = 30$ when $x = 10$, what is the value of y when $x = 3$?

A 3 **B** 8 **C** 9 **D** 23

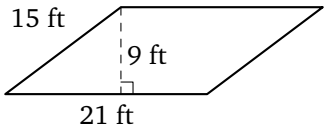
12. What is $(4x^5)^2$?

A $16x^{10}$ **B** $4x^{10}$ **C** $16x^7$ **D** $8x^{10}$

13. To the nearest whole number, what is $\sqrt{116}$?

A 10 **B** 11 **C** 12 **D** 58

14. The parallelogram shown has a base of 21 feet, a height of 9 feet, and slanted sides of 15 feet. What is its area?



A 94.5 ft^2 **B** 189 ft^2 **C** 195 ft^2 **D** 315 ft^2

15. What is the value of $-x^2 + 6x$ when $x = -2$?

A -24 **B** -18 **C** -16 **D** -8

16. A rectangular box has a volume of 140 cubic inches. Its base is 5 inches long and 7 inches wide. How tall is the box?

A 2 in **B** 3 in **C** 4 in **D** 28 in

17. Which of the following is equal to 7.54×10^6?

A 754,000 **B** 7,540,000
C 75,400,000 **D** 754,000,000

18. What is the slope of the line $2x - 6y = 6$?

A $-\frac{1}{3}$ **B** $\frac{1}{3}$ **C** 2 **D** 3

19. Solve the inequality $11 - 5x \geq 1$.

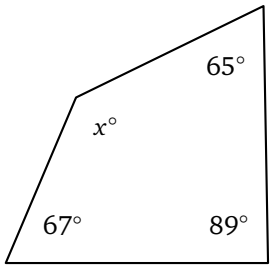
A $x \leq -2$ **B** $x \leq 2$ **C** $x \leq -\frac{12}{5}$ **D** $x \geq 2$

20. Which equation describes the line that passes through $(-3, 3)$ and $(0, 5)$?

A $y = \frac{2}{3}x + 5$ **B** $y = \frac{2}{3}x - 5$

C $y = \frac{2}{3}x + 3$ **D** $y = \frac{3}{2}x + \frac{15}{2}$

21. In the quadrilateral shown, three of the interior angles measure 67°, 89°, and 65°. What is the value of x?



A 90 **B** 129 **C** 134 **D** 139

22. If $3.5x + 0.9 = 32.4$, what is the value of x?

A 0.9 **B** 9 **C** 31.5 **D** 90

23. Final speed is $v = u + at$. Which equation gives a in terms of the other variables?

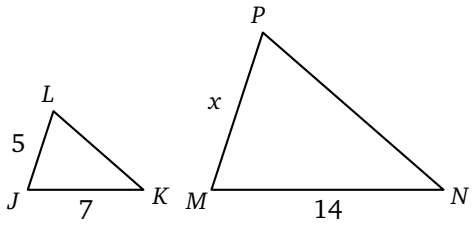
A $a = \frac{v-u}{t}$ **B** $a = t(v - u)$

C $a = \frac{v+u}{t}$ **D** $a = \frac{v}{t} - u$

24. Subtract: $9\frac{1}{2} - 3\frac{2}{3}$

A $2\frac{11}{12}$ **B** $4\frac{5}{6}$ **C** $5\frac{1}{6}$ **D** $5\frac{5}{6}$

25. In the figure, $\triangle JKL$ is similar to $\triangle MNP$, with $JK = 7$, $JL = 5$, and $MN = 14$. What is the value of x?



A 9 **B** 10 **C** 12 **D** 13

Answer Key: Practice Test 1

The small number after each answer is the chapter that teaches the skill.

Part 1: Arithmetic Reasoning

1 C ch. 6	**7** B ch. 20	**13** D ch. 26	**19** A ch. 25	**25** B ch. 24
2 B ch. 13	**8** D ch. 27	**14** B ch. 23	**20** C ch. 5	**26** A ch. 7
3 C ch. 25	**9** C ch. 22	**15** D ch. 22	**21** B ch. 4	**27** B ch. 24
4 D ch. 1	**10** B ch. 27	**16** A ch. 3	**22** A ch. 26	**28** B ch. 17
5 C ch. 26	**11** A ch. 22	**17** A ch. 2	**23** C ch. 25	**29** D ch. 23
6 A ch. 23	**12** D ch. 27	**18** C ch. 10	**24** B ch. 24	**30** D ch. 18

Part 2: Mathematics Knowledge

1 A ch. 15	**6** C ch. 5	**11** C ch. 7	**16** C ch. 20	**21** D ch. 16
2 A ch. 1	**7** C ch. 19	**12** A ch. 14	**17** B ch. 8	**22** B ch. 11
3 C ch. 12	**8** D ch. 10	**13** B ch. 9	**18** B ch. 21	**23** A ch. 11
4 D ch. 13	**9** A ch. 20	**14** B ch. 18	**19** B ch. 12	**24** D ch. 4
5 A ch. 6	**10** A ch. 3	**15** C ch. 2	**20** A ch. 21	**25** B ch. 17

Scoring your practice test

Percent correct	AR (of 30)	MK (of 25)
90% or more — excellent	27–30	23–25
75–89% — good	23–26	19–22
60–74% — keep practicing	18–22	15–18
below 60% — review the chapters	0–17	0–14

These bands measure your progress in this book only. Your real AFQT percentile depends on all four AFQT subtests and is scaled by the Department of Defense.

Explanations: Arithmetic Reasoning

1. **C** 61

1 The question asks about the *other* group: 100% – 75% = 25%.

2 Find that part: $0.25 \times 244 = 61$.

TIP Or find 75% first (183) and subtract: 244 – 183 = 61.

WHY NOT (D) is the number in the group described, not the number asked for.

2. **B** 52

1 Let L be the larger number and s be the smaller number. Then $L + s = 86$ and $L - s = 18$.

2 Add the two equations; s cancels: $2L = 104$, so $L = 52$.

3 Check: the other is 86 – 52 = 34, and 52 – 34 = 18. ✓

WHY NOT (A) gives the smaller number instead. (D) finds $2L = 104$ but forgets to divide by 2.

3. **C** 2 gallons

1 Amount of ethanol = percent × total. Change the percent to a decimal: 10% = 0.1.

2 Multiply: $0.1 \times 20 = 2$ gallons.

WHY NOT (A) moves the decimal point three places when changing the percent. (D) is the amount of gasoline, not ethanol.

4. **D** $28

1 Sharing equally means dividing the total by the number of shares.

2 $336 \div 12 = 28$. Check: $12 \times 28 = 336$.

WHY NOT (A) stops dividing too soon and drops the last digit of the quotient.

5. **C** 10

1 Each choice can be paired with every other choice, so multiply the number of options at each step (the counting principle).

2 $2 \times 5 = 10$.

WHY NOT (A) adds the choices instead of multiplying them.

6. **A** $75

1 Time must be in years: 15 months = $\frac{15}{12} = 1\frac{1}{4} = 1.25$ years.

2 One year's interest: $\$1{,}200 \times 0.05 = \$60$.

3 For $1\frac{1}{4}$ years: $\$60 \times 1\frac{1}{4} = \$75$.

WHY NOT (B) writes 15 months as 1.5 years. (C) uses 15 years instead of 15 months. (D) is the total with interest, not the interest alone.

7. **B** 180 gal

1 Volume: $6 \times 2 \times 2 = 24$ cubic feet.

2 Each cubic foot holds 7.5 gallons: $24 \times 7.5 = 180$ gallons.

TIP Multiplying by 7.5 is the same as multiplying by 15 and halving: $24 \times 15 = 360$, and $360 \div 2 = 180$.

WHY NOT (A) finds the volume in cubic feet but forgets to change it to gallons. (D) uses the surface area instead of the volume.

8. **D** 1,100

1 1 gallon = 4 quarts, so the trailer holds $275 \times 4 = 1{,}100$ quarts.

2 Each canteen takes 1 quart, so 1,100 canteens can be filled.

WHY NOT (A) forgets to convert gallons to quarts. (B) uses 2 quarts in a gallon instead of 4.

9. **C** $30.80

1 Add the prices to find the total cost: $\$10.25 + \$8.95 = \$19.20$.

2 Subtract the total from the amount paid: $\$50.00 - \$19.20 = \$30.80$.

TIP Check by counting up: $\$19.20 + \$30.80 = \$50.00$.

WHY NOT (A) is the total cost, not the change. (B) rounds each price up to a whole dollar instead of using the exact prices. (D) borrows for the cents but forgets to take that dollar away from the dollars.

10. **B** 360 inches

1 1 yard = 36 inches. Going from a big unit to a small unit, multiply: $10 \times 36 = 360$ inches.

TIP Check the size: inches are smaller than feet, so there should be more of them.

WHY NOT (A) uses 12 instead of 36 inches in a yard.

11. **A** 11

1 First take out the fixed cost: $\$260 - \$55 = \$205$ is left to spend.

2 Divide by the price of one: $\$205 \div \$18$ is 11 with some money left over, because $11 \times \$18 = \$198$.

3 One more would cost $216, which is more than $205. Round *down*: 11 packs.

TIP Left over: $\$205 - \$198 = \$7$, not enough for one more.

WHY NOT (B) rounds up, but 12 packs would cost $216. (C) forgets to subtract the $55 first. (D) adds the $55 instead of subtracting it.

12. **D** 34

1 Change tons to pounds: $1 \times 2{,}000 = 2{,}000$ pounds.

2 Divide: $2{,}000 \div 60$ is 33 with a remainder of 20.

3 33 bags hold only $33 \times 60 = 1{,}980$ pounds, which is 20 pounds short, so order 34 bags.

WHY NOT (A) uses 1,000 pounds in a ton instead of 2,000. (B) gives the leftover pounds, not the number of bags. (C) rounds down, which comes up 20 pounds short.

13. **D** $\frac{9}{10}$

1 Vehicles not in maintenance: $30 - 3 = 27$.

2 Probability = $\frac{27}{30} = \frac{9}{10}$.

WHY NOT (A) is the probability that the vehicle *is* down for maintenance.

14. **B** $11,250

1 One year's interest: $\$9{,}000 \times 0.05 = \$450$.

2 Interest for the whole time: $\$450 \times 5 = \$2{,}250$.

3 Add it to the principal: $\$9{,}000 + \$2{,}250 = \$11{,}250$.

WHY NOT (A) is only the interest, not the total. (D) writes 5% as 0.5 instead of 0.05.

15. **D** 20 weeks

1 Money still needed: \$1,050 – \$90 = \$960.

2 Divide by the weekly savings: 960 ÷ 50 is 19 with 10 left over. After 19 weeks she has saved 19 × \$50 = \$950, still \$10 short.

3 So she needs one more week: 19 + 1 = 20 weeks.

WHY NOT (C) rounds down, but after 19 weeks she is still \$10 short.

16. **A** 6

1 First find the greatest number of teams: it must divide both 96 and 80, so it is the GCF.

2 Write each number as a product of primes: $96 = 2^5 \times 3$, $80 = 2^4 \times 5$.

3 Take each prime that both numbers share, with its lowest power: $2^4 = 16$.

4 There are 16 teams, so each gets 96 ÷ 16 = 6 Alpha soldiers.

WHY NOT (B) adds the counts of both items in each group. (C) is the number of teams, not the number of Alpha soldiers in each. (D) splits into only 2 groups, a common factor that is not the greatest.

17. **A** \$90

1 Balance after the payments and the fee: 115 – 86 – 89 – 30 = −90.

2 Deposit needed = target − balance: 0 − (−90) = 0 + 90 = 90.

3 Ethan must deposit \$90.

WHY NOT (D) subtracts the fee twice.

18. **C** $37d + 5$

1 d groups of 37 make $37d$.

2 Add the 5 extra: $37d + 5$.

WHY NOT (A) multiplies the 5 extra by 37 too. (B) swaps the two numbers. (D) multiplies the 5 extra by d too.

19. **A** 24 liters

1 Pure acid now: 0.3 × 12 = 3.6 liters. Adding water does not change this amount.

2 In the new mixture these 3.6 liters must be 10% of the total: 3.6 ÷ 0.1 = 36 liters in all.

3 Water to add: 36 – 12 = 24 liters.

WHY NOT (D) is the final amount of the mixture, not the water added.

20. **C** \$0.50

1 Price per can = total price ÷ number of cans.

2 4.00 ÷ 8 = 0.50. Check: 8 × 0.50 = 4.00.

WHY NOT (A) puts the decimal point one place too far left. (D) puts the decimal point one place too far right.

21. **B** $7\frac{1}{2}$ teaspoons

1 Multiply the whole number and the fraction by 3: $2 \times 3 = 6$ and $\frac{1}{2} \times 3 = \frac{3}{2} = 1\frac{1}{2}$.

2 Add: $6 + 1\frac{1}{2} = 7\frac{1}{2}$.

WHY NOT (A) multiplies only the whole number.

22. **A** 69

1 To average 81 over 5 tests, the total must be $81 \times 5 = 405$.

2 The total so far is 79 + 94 + 85 + 78 = 336.

3 Subtract: 405 – 336 = 69.

TIP Check: 79 + 94 + 85 + 78 + 69 = 405, and 405 ÷ 5 = 81. ✓

WHY NOT (C) makes up the shortfall for only one test, not for all of them. (D) is the target average, not the score needed on the next test.

23. **C** 24 hours

1 Rates: together they do $\frac{1}{6}$ of the job per hour, and Carlos alone does $\frac{1}{8}$ of the job per hour.

2 Luis's rate is the difference: $\frac{1}{6} - \frac{1}{8} = \frac{4}{24} - \frac{3}{24} = \frac{1}{24}$ of the job per hour.

3 Doing $\frac{1}{24}$ of the job each hour, Luis needs 24 hours alone.

TIP Check: $\frac{1}{8} + \frac{1}{24} = \frac{1}{6}$, the rate together. ✓

WHY NOT (A) subtracts the times instead of the rates. (B) assumes the two work at the same speed.

24. **B** 66

1 Change miles to feet: 45 × 5,280 = 237,600 feet per hour.

2 Change hours to seconds: 237,600 ÷ 3,600 = 66 feet per second.

TIP Simplify first: $5{,}280 \div 3{,}600 = \frac{22}{15}$, so $45 \times \frac{22}{15} = 66$.

WHY NOT (C) changes hours to minutes but not to seconds, which gives feet per minute. (D) changes miles to feet but never changes hours to seconds.

25. **B** 2 hours

1 Head start: in 2 hours the first one goes 26×2 = 52 miles.

2 The gap closes at the difference of the speeds: 52 – 26 = 26 miles per hour.

3 Time to close it: 52 ÷ 26 = 2 hours.

TIP Check: in 2 hours the chaser goes 52 × 2 = 104 miles, and the first one has gone 26 × 4 = 104 miles.

WHY NOT (A) divides the head start by the faster speed instead of the difference. (D) counts the time from when the first one left.

26. **A** 66

1 In parts, the applicants who passed are 8 parts and the applicants who did not pass are 3 parts, so the difference is 8 – 3 = 5 parts.

2 Those 5 parts equal 30, so one part is 30 ÷ 5 = 6.

3 The total is 3 + 8 = 11 parts: 11 × 6 = 66.

TIP Check: 48 – 18 = 30, and 48 : 18 simplifies to 8 : 3.

WHY NOT (D) treats the difference of 30 as one part, but it is 5 parts.

27. **B** 9 mph

1 Time going: 36 ÷ 6 = 6 hours. Time returning: 36 ÷ 18 = 2 hours.

2 Total distance: 2 × 36 = 72 miles. Total time: 6 + 2 = 8 hours.

3 Average speed: 72 ÷ 8 = 9 miles per hour.

TIP Shortcut for equal distances: $2ab \div (a + b) = 2 \times 6 \times 18 \div 24 = 9$.

WHY NOT (A) uses the one-way distance instead of the round-trip distance. (C) averages the two speeds, but more time is spent at the slower speed. (D) adds the two speeds.

28. **B** 108 ft

1 Objects and their shadows at the same time of day form similar triangles, so $\frac{\text{height}}{\text{shadow}}$ is the same for both: $\frac{h}{126} = \frac{6}{7}$.

2 Cross-multiply: $7h = 6 \times 126 = 756$.

3 Divide by 7: $h = 756 \div 7 = 108$ feet.

WHY NOT (C) assumes the building is 1 feet shorter than its shadow, just as Hannah is, instead of using a ratio. (D) sets up the proportion upside down.

29. **D** $360

1 Interest: $\$14{,}400 \times 0.05 \times 4 = \$2{,}880$.

2 Total to repay: $\$14{,}400 + \$2{,}880 = \$17{,}280$.

3 Number of payments: $4 \times 12 = 48$ months.

4 Each payment: $\$17{,}280 \div 48 = \$360$.

WHY NOT (A) gives only the interest part of each payment. (B) forgets to add the interest. (C) adds only one year of interest.

30. **D** 40 in^2

1 The frame adds 1 inch on *both* ends of each side, so the outer rectangle is $10+2\times1 = 12$ by $8+2\times1 = 10$ inches.

2 Outer area: 12 × 10 = 120 square inches.

3 Photo area: 10 × 8 = 80 square inches.

4 Frame alone: 120 – 80 = 40 square inches.

WHY NOT (A) adds the frame width to only one end of each side. (C) multiplies the perimeter by the width and misses the four corner squares.

Explanations: Mathematics Knowledge

1. **A** 4

1 Coefficients: the largest number that divides both 16 and 4 is 4.

2 Variables: the second term has no x, so the GCF has no x.

3 So the GCF is 4. (Check: $16x - 4 = 4(4x - 1)$.)

WHY NOT (B) is a common factor, but not the greatest. (C) gives the least common multiple, not the greatest common factor. (D) includes an x, but the second term has no x.

2. **A** Distributive property

1 Compare the two sides of the equation and ask what changed.

2 The number outside the parentheses multiplies *each* number inside. That is the distributive property.

WHY NOT (B) is about changing the order of numbers, which does not happen here. (C) is about moving parentheses among numbers that are all added or all multiplied, which does not happen here. (D) is about adding 0 or multiplying by 1, which does not happen here.

3. **C** $n - 4 < 19$

1 "Fewer than 19" means the quantity is less than 19: it does not include 19 itself, so the inequality symbol is $<$.

2 The quantity is $n-4$, so the inequality is $n-4 < 19$.

WHY NOT (A) points the inequality symbol the wrong way and also gets the endpoint 19 wrong: "fewer than 19" means smaller than 19. (B) points the inequality symbol the wrong way: "fewer than 19" means smaller than 19. (D) uses $\leq$, which includes exactly 19, but 19 is not "fewer than 19".

4. **D** 12

1 Add the equations. The y-terms cancel: $2x = 16$.

2 Divide both sides by 2: $x = \frac{16}{2} = 8$.

3 Substitute $x = 8$ into $x + y = 20$: $8 + y = 20$, so $y = 12$.

WHY NOT (A) has the right size but the wrong sign. (B) gives the value of $x - y$, not y. (C) gives the value of x, not y.

5. **A** 14

1 Change the percent to a decimal: 25% = 0.25.

2 Multiply by the number: $0.25 \times 56 = 14$.

WHY NOT (B) is what is left after removing 25%, not 25% of the number. (C) adds 25% to the number. (D) moves the decimal point only one place.

6. **C** 2.84

1 The hundredths digit is 3. The digit to its right (the thousandths digit) is 9.

2 Since 9 is 5 or more, round up and drop the digits after it: 2.84.

WHY NOT (A) rounds to the nearest tenth instead. (B) drops the extra digits without rounding up (the next digit, 9, is 5 or more).

7. **C** 6π yd

1 The circumference is the distance around: $C = 2\pi r$.

2 Substitute $r = 3$: $C = 2 \times \pi \times 3 = 6\pi$ yards.

WHY NOT (A) uses the radius in $C = \pi d$; the diameter is twice the radius. (D) finds the area, not the circumference.

8. **D** $x + 15$

1 Distribute 5: $5(x + 1) = 5x + 5$.

2 Distribute -2 (the minus sign comes with it): $-2(2x - 5) = -4x + 10$.

3 Combine like terms: $5x - 4x = x$ and $5 + 10 = 15$. The result is $x + 15$.

TIP Check with $x = 1$: the original and $x + 15$ both equal 16.

WHY NOT (A) multiplies only the first term in each set of parentheses. (B) multiplies -5 by 2 instead of -2; the minus sign goes to every term. (C) adds the second product instead of subtracting it.

9. **A** 328 in^2

1 The box has three pairs of matching faces: top and bottom ($10 \times 12 = 120$), front and back ($10 \times 2 = 20$), and the two ends ($12 \times 2 = 24$).

2 Add one face of each pair: $120 + 20 + 24 = 164$.

3 Double it for the matching faces: $2 \times 164 = 328$ square inches.

WHY NOT (D) assumes all six faces are the same size as the bottom.

10. **A** 2

1 List the factors of 10: 1, 2, 5, 10.

2 List the factors of 16: 1, 2, 4, 8, 16.

3 The greatest number in both lists is 2.

WHY NOT (C) is the smallest number, but it is not a factor of 16. (D) is the least common multiple, not the greatest common factor.

11. **C** 9

1 Direct variation means $y = kx$, where k is a constant.

2 Find k by dividing: $k = \frac{y}{x} = \frac{30}{10} = 3$.

3 Use it: $y = 3 \times 3 = 9$.

WHY NOT (A) stops after finding the constant k. (D) adds the change in x instead of multiplying by the constant.

12. **A** $16x^{10}$

1 Raise *every* factor inside the parentheses to the power 2.

2 Coefficient: $4^2 = 16$. Variables: multiply exponents, $(x^5)^2 = x^{10}$.

3 So the result is $16x^{10}$.

WHY NOT (B) forgets to raise the coefficient 4 to the power. (C) adds the exponents instead of multiplying. (D) multiplies the coefficient by 2 instead of raising it to the power 2.

13. **B** 11

1 Trap the root between perfect squares: $10^2 = 100$ and $11^2 = 121$, so $\sqrt{116}$ is between 10 and 11.

2 Compare distances: $116 - 100 = 16$ and $121 - 116 = 5$. 116 is closer to 121.

3 So $\sqrt{116}$ is closest to 11.

WHY NOT (A) is the other neighbor, but 116 is closer to 121. (D) divides 116 by 2 instead of taking the square root.

14. **B** 189 ft^2

1 Area of a parallelogram = base × height, where the height is the dashed line at a right angle to the base (not the slanted side).

2 $21 \times 9 = 189$, so the area is 189 square feet.

TIP Why it works: cut the triangle off one end and slide it to the other end. You get a 21×9 rectangle.

WHY NOT (A) takes $\frac{1}{2}$, as for a triangle. (D) multiplies by the slanted side instead of the height.

15. **C** −16

1 Substitute −2 for x, keeping it in parentheses: $-(-2)^2 + 6(-2)$.

2 Square first: $(-2)^2 = 4$, so $-(-2)^2 = -4$.

3 Multiply: $6(-2) = -12$.

4 Add: $-4 + (-12) = -16$.

WHY NOT (D) squares −2 correctly but drops the minus sign in front of x^2.

16. **C** 4 in

1 Use $V = l \times w \times h$: $140 = 5 \times 7 \times h$.

2 Area of the base: $5 \times 7 = 35$.

3 Divide: $h = 140 \div 35 = 4$ inches.

WHY NOT (D) divides by only one of the two given dimensions.

17. **B** 7,540,000

1 Multiplying by 10^6 moves the decimal point 6 places to the right.

2 7.54 becomes 7,540,000 (fill the empty places with zeros).

WHY NOT (A) moves the decimal point one place too few. (C) moves the decimal point one place too many.

18. **B** $\frac{1}{3}$

1 Solve for y. Move the x-term to the other side: $-6y = -2x + 6$.

2 Divide every term by −6: $y = \frac{1}{3}x - 1$.

3 The slope is the coefficient of x: $\frac{1}{3}$.

WHY NOT (A) forgets to change the sign when moving the x-term. (C) uses the coefficient of x without solving for y. (D) flips the fraction (puts the x-coefficient on the bottom).

19. **B** $x \leq 2$

1 Subtract 11 from both sides (adding or subtracting never changes the direction): $-5x \geq 1 - 11$, so $-5x \geq -10$.

2 Divide both sides by −5. Dividing by a *negative* number reverses the inequality symbol: ≥ becomes ≤: $x \leq \frac{-10}{-5}$, so $x \leq 2$.

3 Check with a number in the answer, $x = 1$: $11 - 5(1) = 6$, and $6 \geq 1$ is true. ✓

WHY NOT (A) gets the number's sign wrong (positive vs. negative) when dividing by −5. (C) moves 11 across without changing its sign. (D) forgets to reverse the inequality symbol when dividing by −5.

20. **A** $y = \frac{2}{3}x + 5$

1 Slope: $m = \frac{5-3}{0-(-3)} = \frac{2}{3}$.

2 Substitute the point $(-3, 3)$ to find b: $3 = \frac{2}{3}(-3) + b = -2 + b$, so $b = 3 - (-2) = 5$.

3 The equation is $y = \frac{2}{3}x + 5$.

4 Check with $(0, 5)$: $\frac{2}{3}(0) + 5 = 5$. ✓

WHY NOT (B) gets the sign of the y-intercept wrong. (C) uses 3, the y-coordinate of a point, as the y-intercept. (D) uses run over rise for the slope.

21. **D** 139

1 A quadrilateral has 4 sides, so its interior angles add up to $(4 - 2) \times 180° = 360°$.

2 Add the known angles: $67 + 89 + 65 = 221$.

3 Subtract: $x = 360 - 221 = 139$.

WHY NOT (A) divides the angle sum by the number of angles, as if all the angles were equal.

22. **B** 9

1 Clear the decimals: multiply *every* term by 10. $35x + 9 = 324$.

2 Undo the addition: subtract 9 from both sides. $35x = 324 - 9 = 315$.

3 Divide both sides by 35: $x = \frac{315}{35} = 9$.

TIP You can also work with the decimals directly: $3.5x = 31.5$, and $31.5 \div 3.5 = 9$.

WHY NOT (A) slips the decimal point. (C) forgets to divide by 3.5. (D) slips the decimal point.

23. **A** $a = \frac{v-u}{t}$

1 Treat every letter except a as if it were a number. Undo what is being done to a, in reverse order, doing the same thing to both sides.

2 Subtract u from both sides: $v - u = at$. Then divide both sides by t: $\frac{v-u}{t} = a$.

3 So $a = \frac{v-u}{t}$.

TIP Check with easy numbers: pick values for the other letters, find a, and see that the original formula holds.

WHY NOT (B) multiplies by t instead of dividing. (C) adds u instead of subtracting it. (D) divides only v by t, not u.

24. **D** $5\frac{5}{6}$

1 Rewrite the fractions with the common denominator 6: $9\frac{3}{6} - 3\frac{4}{6}$.

2 Since $\frac{3}{6}$ is less than $\frac{4}{6}$, regroup: take 1 from 9, so $9\frac{3}{6} = 8\frac{9}{6}$.

3 Subtract: whole numbers $8 - 3 = 5$, fractions $\frac{9}{6} - \frac{4}{6} = \frac{5}{6}$. The answer is $5\frac{5}{6}$.

WHY NOT (C) lowers the whole number by 1 but never adds that 1 to the fraction, then subtracts the smaller fraction from the larger.

25. **B** 10

1 The order of the letters gives the matching vertices: $J \leftrightarrow M$, $K \leftrightarrow N$, $L \leftrightarrow P$. So x is side MP, which matches JL (5), and MN matches JK.

2 Corresponding sides are proportional: $\frac{MP}{JL} = \frac{MN}{JK}$, so $\frac{x}{5} = \frac{14}{7}$.

3 The scale factor is $14 \div 7 = 2$.

4 Multiply: $x = 5 \times 2 = 10$.

WHY NOT (C) adds the difference between corresponding sides instead of multiplying by the scale factor.

Practice Test 2

Instructions

This test has two parts, timed like the paper-and-pencil ASVAB. **Part 1, Arithmetic Reasoning:** 30 questions, 36 minutes. **Part 2, Mathematics Knowledge:** 25 questions, 24 minutes. No calculator. Use scratch paper. Answer every question—there is no penalty for guessing. Mark your answers on the answer sheet below.

Answer Sheet

Fill in one bubble per question. Photocopy this page to retake the test.

Part 1: Arithmetic Reasoning

1	A	B	C	D
2	A	B	C	D
3	A	B	C	D
4	A	B	C	D
5	A	B	C	D
6	A	B	C	D
7	A	B	C	D
8	A	B	C	D
9	A	B	C	D
10	A	B	C	D
11	A	B	C	D
12	A	B	C	D
13	A	B	C	D
14	A	B	C	D
15	A	B	C	D
16	A	B	C	D
17	A	B	C	D
18	A	B	C	D
19	A	B	C	D
20	A	B	C	D
21	A	B	C	D
22	A	B	C	D
23	A	B	C	D
24	A	B	C	D
25	A	B	C	D
26	A	B	C	D
27	A	B	C	D
28	A	B	C	D
29	A	B	C	D
30	A	B	C	D

Part 2: Mathematics Knowledge

1	A	B	C	D
2	A	B	C	D
3	A	B	C	D
4	A	B	C	D
5	A	B	C	D
6	A	B	C	D
7	A	B	C	D
8	A	B	C	D
9	A	B	C	D
10	A	B	C	D
11	A	B	C	D
12	A	B	C	D
13	A	B	C	D
14	A	B	C	D
15	A	B	C	D
16	A	B	C	D
17	A	B	C	D
18	A	B	C	D
19	A	B	C	D
20	A	B	C	D
21	A	B	C	D
22	A	B	C	D
23	A	B	C	D
24	A	B	C	D
25	A	B	C	D

Part 1: Arithmetic Reasoning

Score: _____ / 30

Target time: 36 minutes

1. The base exchange is having a 15%-off sale. Specialist Haddad buys a smartwatch that regularly costs $230. How much does she pay? (There is no sales tax at the exchange.)

A $34.50 **B** $195.50 **C** $215.00 **D** $264.50

2. Marcus sells cars and earns a 5% commission on the price of each car he sells. What is his commission on a car that sells for $26,000?

A $130 **B** $1,300 **C** $13,000 **D** $24,700

3. A survey found that 35% of high school seniors plan to join the military. In a group of 400 seniors, how many would you expect to plan to join?

A 140 **B** 150 **C** 260 **D** 1,400

4. During a field exercise, a Humvee used 12 gallons of diesel to travel 144 miles. How many miles per gallon did it get?

A 12 **B** 13 **C** 15 **D** 132

5. At 5 a.m. the temperature at a cold-weather training site in Alaska was −14°F. By 2 p.m. it had risen 6 degrees. What was the temperature at 2 p.m.?

A −20°F **B** −11°F **C** −8°F **D** 8°F

6. Maria works as a hotel desk clerk and earns $16 per hour, with time-and-a-half for every hour over 40 in a week. Last week she worked 46 hours. How much did she earn last week?

A $144 **B** $688 **C** $736 **D** $784

7. A water park sold 150 day passes on Saturday. Adult passes cost $26 and child passes cost $19. Pass sales were $3,144. How many adult passes were sold?

A 32 **B** 42 **C** 75 **D** 108

8. The total cost of a tent, including a 10% sales tax, was $319. What was the price before tax?

A $31.90 **B** $287.10 **C** $290.00 **D** $309.00

9. Eight years ago, Carlos was three times as old as his son was then. The son is 19 now. How old is Carlos now?

A 25 **B** 33 **C** 36 **D** 41

10. A round field tent has a floor with a diameter of 14 feet. What is the floor area of the tent? Use $\frac{22}{7}$ for π.

A 44 ft^2 **B** 154 ft^2 **C** 308 ft^2 **D** 616 ft^2

11. At a movie theater, adult tickets cost $15 and child tickets cost $9. Which expression gives the total cost, in dollars, of x adult tickets and y child tickets?

A $24(x + y)$ **B** $9x + 15y$
C $15x + 9y$ **D** $24xy$

12. Tasha runs 5 miles in 60 minutes. What is her average speed in miles per hour?

A 3 mph **B** 5 mph
C 12 mph **D** 300 mph

13. On their first weekend pass, 3 Marines eat at a steakhouse. The check comes to $123, and they add an 18% tip. If they split the total evenly, how much does each Marine pay?

A $41.00 **B** $48.38 **C** $63.14 **D** $145.14

14. A warehouse packs cube-shaped boxes that measure 4 inches on each side into a carton that is 28 inches long, 12 inches wide, and 24 inches high. How many boxes fit in one carton?

A 126 **B** 147 **C** 2,016 **D** 8,064

15. A shoe store marks a $150 pair of running shoes down 40%. A week later it cuts the new price by another 25%. What is the final price of the pair of running shoes?

A $37.50 **B** $47.50 **C** $52.50 **D** $67.50

16. A garden center mixes 26 pounds of rye grass seed costing $3.50 per pound with 24 pounds of bluegrass seed costing $6 per pound. What is the cost per pound of the mixture?

A $4.70 **B** $4.75 **C** $4.80 **D** $235.00

17. Lance Corporal Jensen's patrol moves 7 kilometers north, then 24 kilometers east. What is the straight-line distance from the patrol back to its starting point?

A 20 km **B** 24 km **C** 25 km **D** 31 km

18. Diego buys 2 notebooks at \$2.95 each and 6 candy bars at \$1.45 each. What is the total cost?

A \$4.40 **B** \$7.35 **C** \$14.60 **D** \$23.60

19. Tasha is buying carpet for an office that measures 24 feet by 9 feet. The carpet costs \$25 per square yard. What is the cost of the carpet?

A \$24 **B** \$600 **C** \$1,800 **D** \$5,400

20. A platoon of 30 soldiers took a road march, and $\frac{5}{6}$ of them carried the heavy rucksack. Of those soldiers, $\frac{2}{5}$ finished in under three hours. How many soldiers carried the heavy rucksack and finished in under three hours?

A 5 **B** 10 **C** 12 **D** 25

21. Brandon's luggage weighs 88 pounds. About how many kilograms is that? (1 kilogram $\approx$ 2.2 pounds)

A 40 kilograms **B** 85.8 kilograms
C 193.6 kilograms **D** 400 kilograms

22. A hose can fill an empty water trailer in 12 minutes. A loose valve would let a full trailer drain in 15 minutes. If the valve is not fixed, how long will it take the hose to fill the empty trailer?

A 3 minutes **B** 12 minutes
C 27 minutes **D** 60 minutes

23. The average height of the 10 players on a basketball team is 74 inches. A player who is 65 inches tall leaves the team. What is the new average height?

A 66 inches **B** 69 inches
C 74 inches **D** 75 inches

24. At 7:30 a.m., two shuttle buses leave a station together. One shuttle leaves every 9 minutes and the other every 15 minutes. At what time will they next leave together?

A 7:33 a.m. **B** 8:15 a.m.
C 9:00 a.m. **D** 9:45 a.m.

25. Andre won a cash prize. He spent $\frac{3}{8}$ of it on a car repair and saved $\frac{3}{5}$ of the rest. He has \$500 left to spend. How much was the prize?

A \$800 **B** \$1,100 **C** \$2,000 **D** \$20,000

26. In a technical training course, the unit tests count for 75% of the final grade and the final exam counts for 25%. Jamal has a test average of 81. What score does he need on the final exam to earn a final grade of 78?

A 60 **B** 63 **C** 69 **D** 78

27. The sum of three consecutive odd integers is 75. What is the largest of these integers?

A 23 **B** 25 **C** 26 **D** 27

28. A school is taking 485 students on a field trip. Each bus holds 40 students. Each bus costs \$250 to rent for the day. What is the total cost of the buses needed?

A \$3,000.00 **B** \$3,031.25
C \$3,250.00 **D** \$3,500.00

29. A crew of 4 workers can frame a house in 30 days. After 6 days, 2 workers leave for another job. How many more days will it take to finish?

A 24 days **B** 48 days **C** 54 days **D** 60 days

30. A convoy travels 75 miles on paved road at 50 miles per hour, then 10 miles on a dirt road at 20 miles per hour. What is the average speed for the whole trip?

A 42.5 mph **B** 52.5 mph
C 70 mph **D** 170 mph

Part 2: Mathematics Knowledge

Score: _____ / 25

Target time: 24 minutes

1. Which of the following is equal to 0.09?

A 0.9% **B** 9% **C** 13% **D** 90%

2. Evaluate $9(x + 9)$ for $x = 8$.

A 81 **B** 153 **C** 162 **D** 648

3. Which digit is in the hundreds place in 49,628?

A 2 **B** 4 **C** 6 **D** 9

4. How many integer values of x make $-7 \leq x < -1$ true?

A 3 **B** 4 **C** 5 **D** 6

5. If $f(x) = x^2 + 5x + 5$, what is $f(-4)$?

A -31 **B** 1 **C** 2 **D** 41

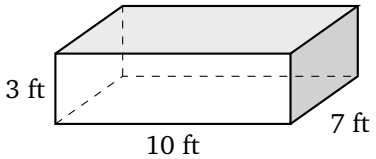
6. Which list shows the numbers -15, 13, -7, 11, -12 in order from least to greatest?

A $13, 11, -7, -12, -15$ **B** $-15, -12, -7, 11, 13$
C $-7, -12, -15, 11, 13$ **D** $-7, 11, -12, 13, -15$

7. Add: 25.1 + 16.61

A 19.12 **B** 41.68 **C** 41.7 **D** 41.71

8. The rectangular box shown is 10 feet long, 7 feet wide, and 3 feet high. What is its volume?



A 20 ft^3 **B** 70 ft^3 **C** 210 ft^3 **D** 242 ft^3

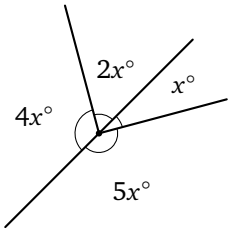
9. What is $6\sqrt{7} - \sqrt{7}$?

A 5 **B** $3\sqrt{7}$ **C** $4\sqrt{7}$ **D** $5\sqrt{7}$

10. A line passes through the points $(-4, -3)$ and $(1, 0)$. What is its slope?

A $-\frac{3}{5}$ **B** $\frac{3}{5}$ **C** $\frac{5}{3}$ **D** 3

11. In the figure, four angles meet at a point and measure $2x°$, $4x°$, $5x°$, and $x°$. What is the value of x?

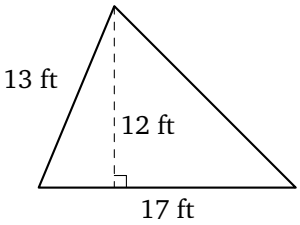


A 30 **B** 35 **C** 90 **D** 150

12. A circle has a circumference of 28π feet. What is the area of the circle?

A 14π ft^2 **B** 196π ft^2 **C** 392π ft^2 **D** 784π ft^2

13. In the triangle shown, the base is 17 feet, the height is 12 feet, and the slanted side is 13 feet. What is the area of the triangle?

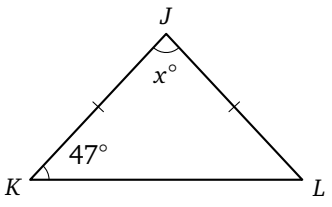


A 102 ft^2 **B** 110.5 ft^2
C 204 ft^2 **D** 221 ft^2

14. Solve for x: $6(x - 1) + 9 = -21$.

A -24 **B** $-\frac{29}{6}$ **C** -4 **D** -3

15. The triangle shown is isosceles, with the two marked sides equal. One base angle measures 47°. What is the value of x?

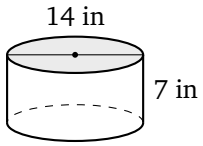


A 86 **B** 96 **C** 101 **D** 133

16. If $\frac{24}{x} = \frac{9}{15}$, what is the value of x?

A 30 **B** 40 **C** 55 **D** 360

17. The cylinder shown has a diameter of 14 inches and a height of 7 inches. What is its volume?



A 343π in^3 **B** 448π in^3
C 686π in^3 **D** $1{,}372\pi$ in^3

18. Simplify: $\frac{6x^3-24x^2-36x}{6x}$

A $x^2 - 4x$ **B** $x^3 - 4x^2 - 6x$
C $x^2 - 10x$ **D** $x^2 - 4x - 6$

19. Divide: $4\frac{2}{3} \div 3\frac{1}{2}$

A $\frac{1}{3}$ **B** $\frac{2}{3}$ **C** $\frac{3}{4}$ **D** $1\frac{1}{3}$

20. If $4x + 5y = 38$ and $5x + 4y = 43$, what is the value of $x - y$?

A 5 **B** 6 **C** 7 **D** 9

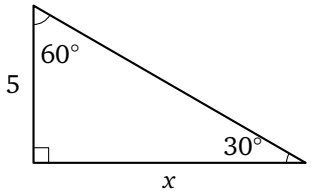
21. What is the smallest positive number that is divisible by each of 4, 10, and 6?

A 20 **B** 30 **C** 60 **D** 120

22. If $(10^2)^n = 10^{10}$, what is the value of n?

A 4 **B** 5 **C** 8 **D** 20

23. In a 30°-60°-90° triangle, the shorter leg is 5 units long. How many units long is the longer leg?



A $\frac{5\sqrt{3}}{3}$ **B** $\frac{5\sqrt{3}}{2}$ **C** $5\sqrt{2}$ **D** $5\sqrt{3}$

24. Multiply: $(6x - 6)^2$

A $36x^2 - 36$ **B** $36x^2 + 36$
C $36x^2 - 72x + 36$ **D** $36x^2 - 72x - 36$

25. A square has an area of 36 square feet. If each side of the square is multiplied by 4, what is the area of the new square?

A 96 ft^2 **B** 100 ft^2 **C** 144 ft^2 **D** 576 ft^2

Answer Key: Practice Test 2

The small number after each answer is the chapter that teaches the skill.

Part 1: Arithmetic Reasoning

1 B ch. 22	**7** B ch. 13	**13** B ch. 22	**19** B ch. 18	**25** C ch. 27
2 B ch. 23	**8** C ch. 23	**14** A ch. 20	**20** B ch. 4	**26** C ch. 26
3 A ch. 26	**9** D ch. 27	**15** D ch. 23	**21** A ch. 27	**27** D ch. 11
4 A ch. 24	**10** B ch. 19	**16** A ch. 25	**22** D ch. 25	**28** C ch. 1
5 C ch. 2	**11** C ch. 10	**17** C ch. 17	**23** D ch. 26	**29** B ch. 25
6 D ch. 22	**12** B ch. 24	**18** C ch. 5	**24** B ch. 3	**30** A ch. 24

Part 2: Mathematics Knowledge

1 B ch. 6	**6** B ch. 2	**11** A ch. 16	**16** B ch. 7	**21** C ch. 3
2 B ch. 10	**7** D ch. 5	**12** B ch. 19	**17** A ch. 20	**22** B ch. 8
3 C ch. 1	**8** C ch. 20	**13** A ch. 18	**18** D ch. 14	**23** D ch. 17
4 D ch. 12	**9** D ch. 9	**14** C ch. 11	**19** D ch. 4	**24** C ch. 14
5 B ch. 15	**10** B ch. 21	**15** A ch. 17	**20** A ch. 13	**25** D ch. 18

Scoring your practice test

Percent correct	AR (of 30)	MK (of 25)
90% or more — excellent	27–30	23–25
75–89% — good	23–26	19–22
60–74% — keep practicing	18–22	15–18
below 60% — review the chapters	0–17	0–14

These bands measure your progress in this book only. Your real AFQT percentile depends on all four AFQT subtests and is scaled by the Department of Defense.

Explanations: Arithmetic Reasoning

1. **B** $195.50

1 Find the discount: 15% of $230 = 0.15 × 230 = $34.50.

2 Subtract it from the regular price: $230 − $34.50 = $195.50.

TIP Shortcut: 15% off means you pay 100% − 15% = 85%, and 0.85 × $230 = $195.50.

WHY NOT (A) is the amount of the discount, not the sale price. (C) subtracts the percent as if it were dollars. (D) adds the discount instead of subtracting it.

2. **B** $1,300

1 Commission = rate × sales. Change the rate to a decimal: 5% = 0.05.

2 Multiply: 0.05 × $26,000 = $1,300.

WHY NOT (A) moves the decimal point three places instead of two. (C) writes 5% as 0.5 instead of 0.05. (D) subtracts the commission from the car's price.

3. **A** 140

1 Expected number = percent (as a decimal) × number of seniors in the group: 0.35 × 400 = 140.

WHY NOT (C) is the expected number of seniors *not* planning to join. (D) moves the decimal point only one place.

4. **A** 12

1 Miles per gallon = miles ÷ gallons.

2 144 ÷ 12 = 12 miles per gallon.

TIP Check: 12 × 12 = 144.

WHY NOT (D) subtracts the gallons from the miles.

5. **C** −8°F

1 A rise means add: −14 + 6.

2 Different signs: subtract the absolute values (14 − 6 = 8) and keep the sign of −14, which has the larger absolute value: −14 + 6 = −8. The temperature was −8°F.

WHY NOT (A) subtracts the rise instead of adding it. (D) has the right size but the wrong sign.

6. **D** $784

1 Only hours over 40 are overtime: 46 − 40 = 6 overtime hours.

2 Regular pay: 40 × $16 = $640. Overtime rate: 1.5 × $16 = $24 per hour.

3 Overtime pay: 6 × $24 = $144.

4 Total: $640 + $144 = $784.

WHY NOT (A) is the overtime pay only. (B) pays the overtime hours only the extra half rate. (C) pays every hour at the regular rate.

7. **B** 42

1 Let a be the number of adult passes. The rest are child passes: $150 - a$.

2 Add up the money: $26a + 19(150 - a) = 3{,}144$.

3 Distribute and combine: $26a + 2{,}850 - 19a = 3{,}144$, so $7a + 2{,}850 = 3{,}144$.

4 Subtract 2,850: $7a = 294$. Divide by 7: $a = 42$.

5 Check: $26(42) + 19(108) = 1{,}092 + 2{,}052 = 3{,}144$. ✓

WHY NOT (C) assumes the two kinds were sold in equal numbers. (D) gives the number of child passes.

8. **C** $290.00

1 The total is 110% of the price: 1.1 × price before tax = $319.

2 Divide: $319 ÷ 1.1 = $290.

TIP Check: 1.1 × $290 = $319.

WHY NOT (A) finds the tax on the total instead of removing it. (B) subtracts 10% of the total, but the tax was 10% of the price before tax. (D) subtracts the percent as if it were dollars.

9. **D** 41

1 8 years ago the son was 19 − 8 = 11.

2 Then Carlos was 3 × 11 = 33.

3 Add the 8 years back: 33 + 8 = 41.

WHY NOT (A) subtracts the 8 years instead of adding them back. (B) gives Carlos's age 8 years ago, not now.

10. **B** 154 ft^2

1 The area formula $A = \pi r^2$ needs the radius: $r = 14 \div 2 = 7$ feet.

2 Square the radius: $7^2 = 49$.

3 $A \approx \frac{22}{7} \times 49 = 22 \times 7 = 154$ square feet.

WHY NOT (A) finds the circumference, not the area. (C) squares the diameter and then halves it; halve first, then square. (D) uses the diameter in place of the radius.

11. **C** $15x + 9y$

1 Each adult ticket costs $15, so x of them cost $15x$ dollars.

2 Each child ticket costs $9, so y of them cost $9y$ dollars.

3 Add the two costs: $15x + 9y$.

WHY NOT (A) charges both prices for every item. (B) matches each price with the wrong item. (D) combines unlike terms.

12. **B** 5 mph

1 Change the time to hours: 60 min = $\frac{60}{60}$ = 1 hour.

2 Speed: 5 ÷ 1 = 5 × 1 = 5 miles per hour.

TIP Or: the pace is 60 ÷ 5 = 12 minutes per mile, and 60 ÷ 12 = 5 mph.

WHY NOT (C) gives the pace in minutes per mile, not the speed. (D) multiplies the distance by the time.

13. **B** $48.38

1 Find the tip: 0.18 × $123 = $22.14.

2 Total with tip: $123 + $22.14 = $145.14.

3 Split it 3 ways: $145.14 ÷ 3 = $48.38.

WHY NOT (A) forgets to add the tip. (C) adds the whole tip to one Marine's share. (D) is the total for the whole group, not one Marine's share.

14. **A** 126

1 Count how many fit along each edge: $28 \div 4 = 7$, $12 \div 4 = 3$, and $24 \div 4 = 6$.

2 Multiply: $7 \times 3 \times 6 = 126$.

TIP Check with volumes: the container holds $28 \times 12 \times 24 = 8{,}064$ cubic inches, and one cube is $4^3 = 64$, so $8{,}064 \div 64 = 126$.

WHY NOT (C) divides the volume by the edge length instead of by the volume of one cube. (D) finds the volume of the container but forgets to divide.

15. **D** $67.50

1 First discount: $\$150 - 0.4 \times \$150 = \$150 - \$60 = \$90$.

2 The second discount is taken from the *new* price: $\$90 - 0.25 \times \$90 = \$90 - \$22.50 = \$67.50$.

TIP In one line: $\$150 \times 0.6 \times 0.75 = \$67.50$.

WHY NOT (C) adds the discounts to get 65% off.

16. **A** $4.70

1 Cost of each part: $26 \times 3.50 = 91.00$ and $24 \times 6.00 = 144.00$ dollars.

2 Totals: \$235 for $26 + 24 = 50$ pounds.

3 Price per pound: $235.00 \div 50 = 4.70$, or \$4.70 per pound.

TIP Check: \$4.70 is between \$3.50 and \$6, closer to the price of the item with more pounds.

WHY NOT (B) averages the two prices without weighting by the pounds. (C) weights each price by the wrong number of pounds. (D) is the total cost of the mixture, not the price per pound.

17. **C** 25 km

1 Traveling north and then east makes a right angle, so the two legs of the trip and the straight-line distance form a right triangle.

2 $7^2 + 24^2 = 49 + 576 = 625$.

3 $\sqrt{625} = 25$, so the answer is 25 kilometers.

WHY NOT (D) adds the two distances instead of using the Pythagorean theorem.

18. **C** $14.60

1 First item: $2 \times 2.95 = 5.90$.

2 Second item: $6 \times 1.45 = 8.70$.

3 Total: $5.90 + 8.70 = 14.60$, so \$14.60.

WHY NOT (A) adds one of each price and ignores the quantities. (B) forgets to multiply the second price by 6. (D) charges every item at the higher price.

19. **B** $600

1 Change each length to yards: $24 \div 3 = 8$ yd and $9 \div 3 = 3$ yd.

2 Area in square yards: $8 \times 3 = 24$ square yards.

3 Cost: $24 \times 25 = 600$, so the carpet costs \$600.

TIP Or find square feet first ($24 \times 9 = 216$) and divide by 9: $216 \div 9 = 24$ square yards.

WHY NOT (A) finds the number of square yards but forgets to multiply by the price. (C) divides by 3 instead of 9 (a square yard is 3 ft by 3 ft). (D) forgets to change square feet to square yards.

20. **B** 10

1 First group: $\frac{5}{6} \times 30 = 25$.

2 Then take $\frac{2}{5}$ of that group: $\frac{2}{5} \times 25 = 10$.

WHY NOT (C) takes $\frac{2}{5}$ of the whole group instead of the smaller group. (D) is the size of the first group only.

21. **A** 40 kilograms

1 A kilogram is heavier than a pound, so there are fewer kilograms. Divide by 2.2: $88 \div 2.2 = 880 \div 22 = 40$ kilograms.

WHY NOT (B) subtracts 2.2 instead of dividing. (C) multiplies by 2.2 instead of dividing. (D) moves the decimal point one place too far to the right.

22. **D** 60 minutes

1 Rates: the hose fills $\frac{1}{12}$ of the trailer per minute, and the valve empties $\frac{1}{15}$ of it per minute.

2 The valve works against the hose, so *subtract*: $\frac{1}{12} - \frac{1}{15} = \frac{5}{60} - \frac{4}{60} = \frac{1}{60}$ of the trailer per minute.

3 Time = 1 trailer ÷ rate = $1 \div \frac{1}{60} = 60$ minutes.

TIP Shortcut for fill and drain: $\frac{12 \times 15}{15 - 12} = \frac{180}{3} = 60$.

WHY NOT (A) subtracts the two times. (B) ignores the valve. (C) adds the two times.

23. **D** 75 inches

1 Turn the average into a total: $10 \times 74 = 740$.

2 Subtract the value: $740 - 65 = 675$.

3 Divide by the new count, $10 - 1 = 9$: $675 \div 9 = 75$.

WHY NOT (C) assumes the average does not change.

24. **B** 8:15 a.m.

1 Write each number as a product of primes: $9 = 3^2$, $15 = 3 \times 5$.

2 Take every prime that appears, with its highest power: $3^2 \times 5 = 45$.

3 They leave together again after 45 minutes: 45 minutes after 7:30 a.m. is 8:15 a.m.

WHY NOT (A) uses the greatest common factor, 3 minutes. (C) is the second time they leave together, not the first. (D) uses $9 \times 15 = 135$ minutes, a common multiple but not the least one.

25. **C** $2,000

1 After the first step, $1 - \frac{3}{8} = \frac{5}{8}$ of the whole is left.

2 The second step uses $\frac{3}{5}$ of that, leaving $1 - \frac{3}{5} = \frac{2}{5}$ of it: $\frac{5}{8} \times \frac{2}{5} = \frac{1}{4}$ of the whole.

3 So $\frac{1}{4}$ of the whole is \$500: $500 \div \frac{1}{4} = 2{,}000$.

WHY NOT (A) undoes only the first step. (B) increases the amount by each fraction instead of dividing. (D) subtracts both fractions from the whole amount.

26. **C** 69

1 The tests contribute $0.75 \times 81 = 60.75$ points to the grade.

2 The final must contribute $78 - 60.75 = 17.25$ points.

3 The final counts 25%, so the exam score is $17.25 \div 0.25 = 69$.

TIP Check: $0.75(81)+0.25(69) = 60.75+17.25 = 78$. ✓

WHY NOT (D) is the final grade wanted, not the exam score needed.

27. **D** 27

1 Let n be the smallest. Consecutive odd integers go up by 2, so the numbers are $n, n+2, n+4$.

2 Add them: $3n + 6 = 75$.

3 Subtract 6: $3n = 69$. Divide by 3: $n = 23$.

4 The numbers are 23, 25, 27. The largest is 27.

WHY NOT (A) is the smallest number, not the largest. (B) is the middle number, not the largest. (C) counts by 1 instead of by 2.

28. **C** $3,250.00

1 Divide: $485 \div 40 = 12$ remainder 5.

2 The 5 leftover students need one more bus, so 13 buses are needed.

3 Cost: $13 \times 250 = 3{,}250$ dollars.

WHY NOT (A) rounds the number of buses down. (B) pays for a fraction of a bus.

29. **B** 48 days

1 After 6 days, the original 4 workers still had 24 days' worth of work left: $4 \times 24 = 96$ worker-days.

2 Now there are $4 - 2 = 2$ workers, so the rest takes $96 \div 2 = 48$ days.

WHY NOT (A) ignores the smaller group. (C) gives the total number of days, not the days remaining. (D) forgets the work already done in the first 6 days (gives the new crew the whole job).

30. **A** 42.5 mph

1 First part: $75 \div 50 = 1.5$ hours.

2 Second part: $10 \div 20 = 0.5$ hours.

3 Total distance: $75 + 10 = 85$ miles. Total time: $1.5 + 0.5 = 2$ hours.

4 Average speed: $85 \div 2 = 42.5$ miles per hour.

WHY NOT (C) adds the two speeds. (D) divides the total distance by the second part's time only.

Explanations: Mathematics Knowledge

1. **B** 9%

1 To change a decimal to a percent, multiply by 100 (move the decimal point two places to the right).

2 $0.09 \times 100 = 9$, so $0.09 = 9\%$.

WHY NOT (A) moves the decimal point only one place. (D) moves the decimal point three places.

2. **B** 153

1 Substitute $x = 8$: $9(8 + 9)$.

2 Parentheses first: $8 + 9 = 17$. Then multiply: $9 \times 17 = 153$.

WHY NOT (A) multiplies only x by 9, not the whole sum.

3. **C** 6

1 Read the digits from the right, naming each place: 8 (ones), 2 (tens), 6 (hundreds).

2 So the hundreds digit is 6.

WHY NOT (A) is the digit in the tens place. (B) is the digit in the ten-thousands place. (D) is the digit in the thousands place.

4. **D** 6

1 −7 is included (≤), and −1 is not included (<). So the integers run from −7 to −2.

2 List them: −7, −6, −5, −4, −3, −2. That is 6 integers.

TIP Count from −7 to −2 with last − first +1: −2 − (−7) + 1 = 6.

WHY NOT (C) leaves out both endpoints.

5. **B** 1

1 Replace every x with (−4): $f(-4) = (-4)^2+5(-4)+5$.

2 Square first (order of operations): $(-4)^2 = 16$. So $f(-4) = 16 - 20 + 5$.

3 Combine: $f(-4) = 1$.

WHY NOT (A) treats $(-4)^2$ as −16. (D) makes a sign error with $5x$.

6. **B** −15, −12, −7, 11, 13

1 Negative numbers come first. Among them, the one farthest from 0 is the least: −15 < −12 < −7.

2 Then the positive numbers: 11 < 13.

3 So the order is −15, −12, −7, 11, 13.

WHY NOT (A) lists the numbers from greatest to least. (C) puts the negative numbers in the wrong order; for negatives, the larger absolute value is the smaller number. (D) orders the numbers by their distance from 0, ignoring the signs.

7. **D** 41.71

1 Line up the decimal points and fill in zeros so both numbers have 2 decimal places: 25.10 + 16.61.

2 Add column by column, keeping the point in place: 25.10 + 16.61 = 41.71.

WHY NOT (A) lines up the last digits instead of the decimal points.

8. **C** 210 ft^3

1 The volume of a rectangular box is $V = l \times w \times h$.

2 Multiply in a convenient order: $10 \times 3 = 30$, then $30 \times 7 = 210$.

3 The volume is 210 cubic feet.

WHY NOT (A) adds the three dimensions instead of multiplying them. (B) finds the area of the base only and forgets the height. (D) finds the surface area, not the volume.

9. **D** $5\sqrt{7}$

1 Both terms have $\sqrt{7}$, so they are like radicals: combine the numbers in front, just like $6x - x$.

2 6 − 1 = 5, so the answer is $5\sqrt{7}$. The $\sqrt{7}$ stays the same.

WHY NOT (A) drops the $\sqrt{7}$ after subtracting.

10. **B** $\frac{3}{5}$

1 Use $m = \frac{y_2-y_1}{x_2-x_1}$ with $(-4, -3)$ and $(1, 0)$.

2 Rise: 0 − (−3) = 3. Run: 1 − (−4) = 5.

3 Slope: $\frac{3}{5}$.

TIP Quick check: as x increases, y increases, so the slope must be positive.

WHY NOT (A) subtracts the coordinates in opposite orders on the top and bottom. (C) divides the change in x by the change in y (run over rise). (D) finds the rise but forgets to divide by the run.

11. **A** 30

1 The angles go all the way around a point, so they add up to 360°: $2x + 4x + 5x + x = 360$.

2 Combine like terms: $12x = 360$.

3 Divide both sides by 12: $x = 360 \div 12 = 30$.

WHY NOT (C) divides 360° by the number of angles, as if they were all equal. (D) gives the largest angle instead of the value of x.

12. **B** 196π ft^2

1 First find the radius: $2\pi r = 28\pi$, so $2r = 28$ and $r = 14$.

2 Then $A = \pi r^2 = \pi \times 14^2 = 196\pi$ square feet.

WHY NOT (A) forgets to square the radius. (C) mixes up the formulas and uses $2\pi r^2$. (D) uses 28 (the diameter) as the radius.

13. **A** 102 ft^2

1 Use the base and the height. The height is the dashed line (12 feet), which meets the base at a right angle; the slanted side (13 feet) is not the height.

2 Area of a triangle = $\frac{1}{2} \times$ base $\times$ height.

3 $\frac{1}{2} \times 17 \times 12 = \frac{1}{2} \times 204 = 102$, so the area is 102 square feet.

WHY NOT (B) uses the slanted side instead of the height. (C) forgets to multiply by $\frac{1}{2}$. (D) uses the slanted side and forgets the $\frac{1}{2}$.

14. **C** −4

1 Distribute 6 to *both* terms in the parentheses: $6 \cdot x = 6x$ and $6 \cdot (-1) = -6$. The equation becomes $6x - 6 + 9 = -21$.

2 Combine the plain numbers: −6 + 9 = 3, so $6x + 3 = -21$.

3 Undo the addition: subtract 3 from both sides. $6x = -21 - 3 = -24$.

4 Undo the multiplication: divide both sides by 6. $x = \frac{-24}{6} = -4$.

WHY NOT (A) forgets to divide by 6. (B) multiplies only the x by 6, not the −1.

15. **A** 86

1 The two base angles of an isosceles triangle are equal, so both measure 47°.

2 The angles add up to 180: $x = 180 - 47 - 47 = 180 - 94 = 86$.

WHY NOT (D) subtracts only one base angle from 180°.

16. **B** 40

1 Cross-multiply: $9 \times x = 24 \times 15$, so $9x = 360$.

2 Divide both sides by 9: $x = 360 \div 9 = 40$.

3 Check: $\frac{24}{40}$ and $\frac{9}{15}$ both simplify to $\frac{3}{5}$. ✓

WHY NOT (A) uses a constant difference: 24 is 15 more than 9, so it adds 15 to 15 instead of scaling. (D) multiplies 24 by 15 but forgets to divide by 9.

17. **A** 343π in^3

1 The radius is half the diameter: $r = 14 \div 2 = 7$ inches.

2 Base: $\pi r^2 = \pi \times 7^2 = 49\pi$ square inches.

3 Volume: $V = \pi r^2 h = 49\pi \times 7 = 343\pi$ cubic inches.

WHY NOT (C) squares the diameter and then halves it; halve first. (D) uses the diameter in place of the radius.

18. **D** $x^2 - 4x - 6$

1 Divide *each* term of the top by $6x$.

2 $\frac{6x^3}{6x} = x^2$, $\frac{-24x^2}{6x} = -4x$, and $\frac{-36x}{6x} = -6$.

3 So the result is $x^2 - 4x - 6$.

WHY NOT (A) drops the last term, but $\frac{-36x}{6x}$ is -6, not 0. (B) divides the coefficients but forgets to subtract the exponents. (C) leaves an x on the last term.

19. **D** $1\frac{1}{3}$

1 Change each mixed number to an improper fraction: $4\frac{2}{3} = \frac{4\times3+2}{3} = \frac{14}{3}$ and $3\frac{1}{2} = \frac{7}{2}$.

2 Dividing means multiplying by the reciprocal: $\frac{14}{3} \div \frac{7}{2} = \frac{14}{3} \times \frac{2}{7}$.

3 Cancel before multiplying: 14 and 7 share 7, so the problem becomes $\frac{2}{3} \times 2$.

4 Multiply top times top and bottom times bottom: $\frac{2}{3} \times 2 = \frac{4}{3} = 1\frac{1}{3}$.

WHY NOT (C) divides in the wrong order.

20. **A** 5

1 Look at the coefficients: they are swapped. Subtract the second equation from the first: $-x + y = -5$.

2 Divide by -1: $x - y = \frac{-5}{-1} = 5$.

TIP No need to find x and y separately (they are 7 and 2).

WHY NOT (C) gives the value of x. (D) adds the equations, which gives $x + y$.

21. **C** 60

1 Write each number as a product of primes: $4 = 2^2$, $6 = 2 \times 3$, $10 = 2 \times 5$.

2 Take every prime that appears, with its highest power: $2^2 \times 3 \times 5 = 60$.

3 Check: 60 divided by each of the numbers gives a whole number.

WHY NOT (A) is not divisible by 6. (B) is not divisible by 4. (D) is divisible by all of them, but it is not the smallest such number.

22. **B** 5

1 A power of a power multiplies the exponents: $2 \cdot n = 10$.

2 Divide by 2: $n = 10 \div 2 = 5$.

WHY NOT (C) subtracts the exponents; a power of a power multiplies them. (D) multiplies instead of dividing.

23. **D** $5\sqrt{3}$

1 In a 30°-60°-90° triangle the sides are in the ratio $1 : \sqrt{3} : 2$ (short leg : long leg : hypotenuse). The short leg is across from the 30° angle.

2 Long leg = short leg $\times\sqrt{3} = 5\sqrt{3}$.

WHY NOT (A) divides the short leg by $\sqrt{3}$ instead of multiplying. (C) multiplies the short leg by $\sqrt{2}$ (the 45°-45°-90° rule) instead of by $\sqrt{3}$.

24. **C** $36x^2 - 72x + 36$

1 Use the pattern $(A - B)^2 = A^2 - 2AB + B^2$ with $A = 6x$ and $B = 6$.

2 $A^2 = (6x)^2 = 36x^2$, $-2AB = -2(6x)(6) = -72x$, and $B^2 = 6^2 = 36$.

3 So $(6x - 6)^2 = 36x^2 - 72x + 36$.

TIP Check with $x = 1$: the original and your answer must give the same number.

WHY NOT (A) squares each term, leaves out the middle term, and makes the last term negative. (B) squares each term but leaves out the middle term $-72x$. (D) makes the last term negative, but $(-6)^2 = 36$.

25. **D** 576 ft^2

1 The original side is 6 ft, because $6 \times 6 = 36$.

2 The new side is $6 \times 4 = 24$ ft.

3 New area: $24 \times 24 = 576$ square feet.

WHY NOT (A) gives the new perimeter instead of the new area. (B) adds 4 to the side instead of multiplying. (C) multiplies the area by 4 instead of 16.

Practice Test 3

Instructions

This test has two parts, timed like the paper-and-pencil ASVAB. **Part 1, Arithmetic Reasoning:** 30 questions, 36 minutes. **Part 2, Mathematics Knowledge:** 25 questions, 24 minutes. No calculator. Use scratch paper. Answer every question—there is no penalty for guessing. Mark your answers on the answer sheet below.

Answer Sheet

Fill in one bubble per question. Photocopy this page to retake the test.

Part 1: Arithmetic Reasoning

1	A	B	C	D
2	A	B	C	D
3	A	B	C	D
4	A	B	C	D
5	A	B	C	D
6	A	B	C	D
7	A	B	C	D
8	A	B	C	D
9	A	B	C	D
10	A	B	C	D
11	A	B	C	D
12	A	B	C	D
13	A	B	C	D
14	A	B	C	D
15	A	B	C	D
16	A	B	C	D
17	A	B	C	D
18	A	B	C	D
19	A	B	C	D
20	A	B	C	D
21	A	B	C	D
22	A	B	C	D
23	A	B	C	D
24	A	B	C	D
25	A	B	C	D
26	A	B	C	D
27	A	B	C	D
28	A	B	C	D
29	A	B	C	D
30	A	B	C	D

Part 2: Mathematics Knowledge

1	A	B	C	D
2	A	B	C	D
3	A	B	C	D
4	A	B	C	D
5	A	B	C	D
6	A	B	C	D
7	A	B	C	D
8	A	B	C	D
9	A	B	C	D
10	A	B	C	D
11	A	B	C	D
12	A	B	C	D
13	A	B	C	D
14	A	B	C	D
15	A	B	C	D
16	A	B	C	D
17	A	B	C	D
18	A	B	C	D
19	A	B	C	D
20	A	B	C	D
21	A	B	C	D
22	A	B	C	D
23	A	B	C	D
24	A	B	C	D
25	A	B	C	D

Part 1: Arithmetic Reasoning

Score: _____ / 30

Target time: 36 minutes

1. At 6 p.m. the temperature at an Army post in upstate New York was 6°F. By midnight it had dropped 8 degrees. What was the temperature at midnight?

A −14°F **B** −2°F **C** 2°F **D** 14°F

2. Carlos reads about 25 pages per hour. How many pages can he read in 3 hours?

A 28 **B** 75 **C** 85 **D** 105

3. Private Kim hits a target $\frac{9}{10}$ of the time. If each shot is independent, what is the probability that Kim hits the target on both of the next 2 shots?

A $\frac{81}{100}$ **B** $\frac{83}{100}$ **C** $\frac{9}{10}$ **D** $\frac{99}{100}$

4. A battalion's goal for its charity drive is $2,000. It has reached 25% of the goal. How much more money does it need to raise?

A $500 **B** $1,500 **C** $1,900 **D** $1,975

5. A cafeteria worker mixes 8 gallons of a punch that is 40% fruit juice with 4 gallons of a punch that is 25% fruit juice. What percent of the mixture is fruit juice?

A 30% **B** 32.5% **C** 35% **D** 65%

6. Lance Corporal Brooks's take-home pay is $3,240 a month. She pays $\frac{1}{8}$ of it toward the car loan. How much of the month's pay is left?

A $405 **B** $1,335 **C** $2,835 **D** $3,835

7. Lance Corporal Chen's yearly base pay is $31,200. Service members are paid twice a month, on the 1st and the 15th. How much is each paycheck before taxes and other deductions?

A $600 **B** $900 **C** $1,200 **D** $1,300

8. Grace's work shift starts at 8:35 a.m. and ends at 12:30 p.m. How long is the shift?

A 3 hours 55 minutes **B** 4 hours
C 4 hours 35 minutes **D** 8 hours 5 minutes

9. A bathroom floor is a rectangle 10 feet by 16 feet. It will be covered with square tiles that measure 2 feet on each side. How many tiles are needed?

A 26 **B** 40 **C** 80 **D** 160

10. Hannah buys a lawn mower at a local store for $180. The sales tax rate is 4%. What is the total cost, including tax?

A $7.20 **B** $172.80 **C** $184.00 **D** $187.20

11. The daily high temperatures (in degrees Fahrenheit) for eight days were 86, 57, 68, 66, 63, 63, 93, 70. What is the median of the temperatures?

A 66 **B** 67 **C** 68 **D** 68.5

12. A board is $9\frac{1}{2}$ feet long. A carpenter cuts off a piece $4\frac{5}{8}$ feet long. How long is the rest of the board? (Ignore the width of the saw cut.)

A $4\frac{7}{8}$ feet **B** $5\frac{1}{8}$ feet **C** $5\frac{7}{8}$ feet **D** $14\frac{1}{8}$ feet

13. A parking garage charges $8 for the first hour and $3 for each additional hour. Ryan paid $20 to park. For how many hours did he park?

A 4 hours **B** 5 hours **C** 6 hours **D** 7 hours

14. Omar has $250 saved and earns $16 per hour at a part-time job. What is the least number of whole hours he must work to have at least $600?

A 20 hours **B** 21 hours
C 22 hours **D** 38 hours

15. A boat's outboard motor runs on gasoline mixed with oil in the ratio 32 : 1 (gasoline to oil). How many fluid ounces of oil should be mixed with 1 gallon of gasoline? (1 gallon = 128 fluid ounces)

A 4 fluid ounces **B** 6 fluid ounces
C 8 fluid ounces **D** 32 fluid ounces

16. Priya buys 3 packs of socks at $8 each and 2 pairs of jeans at $43 each and pays with two $100 bills. How much change should Priya get back?

A $90 **B** $110 **C** $133 **D** $149

17. A box of supplies weighs 13 lb 1 oz. After Andre removes 12 lb 4 oz of supplies, how much does the box weigh?

A 7 oz **B** 13 oz
C 1 lb 3 oz **D** 1 lb 13 oz

18. After a promotion ceremony, Staff Sergeant Walker takes her team out to lunch. The food costs $144. There is a 6% sales tax, and she adds a 20% tip figured on the food cost before tax. How much does she pay in all?

A $152.64 **B** $172.80 **C** $181.44 **D** $221.44

19. Two warning lights blink together. One blinks every 10 seconds and the other every 14 seconds. After how many seconds will they next blink at the same time?

A 2 seconds **B** 24 seconds
C 50 seconds **D** 70 seconds

20. Emily borrows $4,000 from a credit union to buy a used car. The loan charges 8% simple interest per year. After how many years will the interest on the loan add up to $1,280?

A 0.04 years **B** 4 years
C 5 years **D** 48 years

21. A radar station covers a circular area with a diameter of 20 miles. How many square miles does it cover? Use 3.14 for π.

A 62.8 mi^2 **B** 300 mi^2
C 314 mi^2 **D** 1,256 mi^2

22. A train leaves the first station at 10:15 a.m. and travels 75 miles at 50 miles per hour. It waits 20 minutes at a second station, then travels another 90 miles at 60 miles per hour to the end of the line. At what time does it reach the end of the line?

A 1:05 p.m. **B** 1:15 p.m.
C 1:20 p.m. **D** 1:35 p.m.

23. A rental truck costs $43 per day plus $0.25 per mile. Which expression gives the cost, in dollars, of renting the truck for d days and driving it m miles?

A $43d + 0.25m$ **B** $43m + 0.25d$
C $43 + 0.25m$ **D** $(43 + 0.25)dm$

24. Hannah buys 3 pens at $1.12 each and 2 notebooks at $3.83 each. She pays with a $20 bill. How much change should she get back?

A $8.98 **B** $11.02 **C** $15.05 **D** $16.64

25. A platoon starts a road march at 4 miles per hour. Two hours later, Private Okafor follows the same route in a Humvee carrying water at 20 miles per hour. How many hours after the Humvee leaves will it catch up?

A 0.4 hours **B** 0.5 hours
C 2 hours **D** 2.5 hours

26. A convoy leaves at 2240 for a drive that takes 10 hours 25 minutes. At what military time will it arrive?

A 0805 **B** 0905 **C** 1005 **D** 2105

27. Five friends, including Priya, line up for a photo. Priya must stand at the left end. How many different line-ups are possible?

A 4 **B** 20 **C** 22 **D** 24

28. Jamal is filling a sandbox that is 9 feet long and 7 feet wide with sand 8 inches deep. Sand is sold in bags that each hold half a cubic foot. How many bags does he need to buy?

A 21 **B** 42 **C** 82 **D** 84

29. Three squads are filling sandbags for a checkpoint. Working alone, first squad would finish in 10 hours, second squad in 3 hours, and third squad in 15 hours. How long will it take if all three squads work together?

A 30 minutes **B** 1 hour
C 2 hours **D** 9 hours 20 minutes

30. The dining facility on base mixes regular coffee that costs $5 per pound with dark-roast coffee that costs $11 per pound to make 15 pounds of a coffee blend that is worth $7 per pound. How many pounds of regular coffee are in the mix?

A 5 pounds **B** 7 pounds
C 8 pounds **D** 10 pounds

Part 2: Mathematics Knowledge

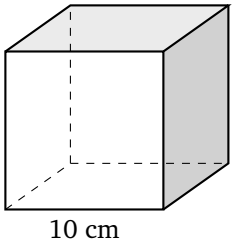
Score: _____ / 25

Target time: 24 minutes

1. Round 7,092 to the nearest hundred.

A 7,000 **B** 7,100 **C** 7,192 **D** 7,200

2. What is the volume of a cube with an edge length of 10 centimeters?



A 30 cm^3 **B** 100 cm^3
C 1,000 cm^3 **D** 1,200 cm^3

3. What is 25% of 116?

A 29 **B** 87 **C** 145 **D** 290

4. A cylinder has a diameter of 10 meters and a volume of 75π cubic meters. What is its height?

A 0.75 m **B** 1 m **C** 3 m **D** 15 m

5. What is the ratio of 10 quarts to 3 gallons, in simplest form?

A 5 : 6 **B** 6 : 5 **C** 10 : 3 **D** 40 : 3

6. What is the x-intercept of the line $y = 2x - 14$?

A -14 **B** -7 **C** $-\frac{1}{7}$ **D** 7

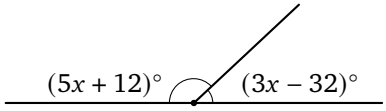
7. Divide: $34.8 \div 6$

A 0.58 **B** 2.9 **C** 5.4 **D** 5.8

8. Evaluate $\sqrt{\frac{9}{121}}$.

A $\frac{3}{121}$ **B** $\frac{9}{242}$ **C** $\frac{3}{11}$ **D** $\frac{9}{11}$

9. In the figure, a ray meets a straight line, forming angles of $(3x - 32)°$ and $(5x + 12)°$. What is the value of x?



A 20 **B** 22 **C** 25 **D** 43

10. Which ordered pair (x, y) is the solution of this system?

$$3x + y = -9$$
$$x - y = -3$$

A $(-3, 0)$ **B** $(0, -3)$ **C** $(-2, -3)$ **D** $(-2, 1)$

11. Simplify $\frac{15x^8}{3x^3}$. Assume $x \neq 0$.

A $5x^6$ **B** $5x^5$ **C** $5x^{11}$ **D** $12x^5$

12. What is 4,100,000 written in scientific notation?

A 4.1×10^5 **B** 4.1×10^6
C 4.1×10^7 **D** 41×10^5

13. Which one of 477, 484, 504, or 976 is divisible by 4 *and* by 9?

A 477 **B** 484 **C** 504 **D** 976

14. Multiply: $(x - 9)(x + 1)$

A $x^2 - 8x - 9$ **B** $x^2 - 8x + 9$
C $x^2 + 8x - 9$ **D** $x^2 - 9$

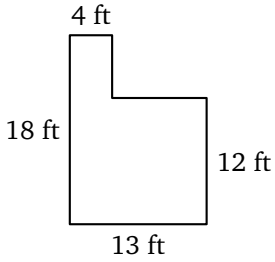
15. A tabletop is shaped like a quarter circle with a radius of 30 inches. What is the area of the tabletop?

A 15π in^2 **B** 30π in^2
C 225π in^2 **D** 900π in^2

16. Which is the factored form of $x^2 - 9$?

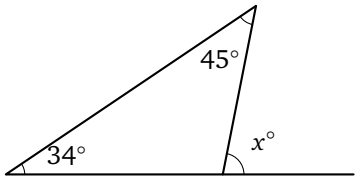
A $(x - 3)^2$ **B** $(x - 1)(x + 9)$
C $(x + 3)^2$ **D** $(x - 3)(x + 3)$

17. The figure shows the floor of a storage building. All corners are right angles. The sides shown measure 13 feet, 18 feet, 4 feet, and 12 feet. What is the perimeter?



A 42 ft **B** 47 ft **C** 54 ft **D** 62 ft

18. In the figure, one side of the triangle is extended. The two interior angles shown measure 34° and 45°. What is the value of x?



A 11 **B** 74 **C** 79 **D** 101

19. If $4x - 13 = 35$, what is the value of x?

A 12 **B** 13 **C** $\frac{87}{4}$ **D** 48

20. Which inequality describes all values of x that make $-10 < -8 - 2x < 4$ true?

A $-1 < x < 6$ **B** $-6 < x < 1$

C $x \leq -6$ or $x \geq 1$ **D** $2 < x < 9$

21. Subtract: $6\frac{1}{2} - 4\frac{7}{10}$

A $\frac{5}{9}$ **B** $\frac{4}{5}$ **C** $1\frac{1}{5}$ **D** $1\frac{4}{5}$

22. The sides of a triangle have lengths $4x-7$, $2x-7$, and $2x + 7$. Which expression represents the perimeter of the triangle?

A $8x - 7$ **B** x

C $16x - 14$ **D** $8x + 21$

23. What is the slope of a line that is perpendicular to the line $4x + y = 9$?

A -4 **B** $-\frac{1}{4}$ **C** $\frac{1}{4}$ **D** 4

24. Evaluate: $-3 \times (6 - 8) + (-24) \div (-3)$

A 2 **B** 14 **C** 15 **D** 24

25. What are the solutions of $5x^2 - 3x - 14 = 0$?

A $x = -\frac{14}{5}$ or $x = 4$ **B** $x = -2$ or $x = \frac{7}{5}$

C $x = 2$ **D** $x = -\frac{7}{5}$ or $x = 2$

Answer Key: Practice Test 3

The small number after each answer is the chapter that teaches the skill.

Part 1: Arithmetic Reasoning

1 B ch. 2	**7** D ch. 22	**13** B ch. 11	**19** D ch. 3	**25** B ch. 24
2 B ch. 24	**8** A ch. 27	**14** C ch. 12	**20** B ch. 23	**26** B ch. 27
3 A ch. 26	**9** B ch. 18	**15** A ch. 25	**21** C ch. 19	**27** D ch. 26
4 B ch. 23	**10** D ch. 22	**16** A ch. 1	**22** D ch. 24	**28** D ch. 20
5 C ch. 25	**11** B ch. 26	**17** B ch. 27	**23** A ch. 10	**29** C ch. 25
6 C ch. 22	**12** A ch. 4	**18** C ch. 23	**24** A ch. 5	**30** D ch. 13

Part 2: Mathematics Knowledge

1 B ch. 1	**6** D ch. 21	**11** B ch. 14	**16** D ch. 15	**21** D ch. 4
2 C ch. 20	**7** D ch. 5	**12** B ch. 8	**17** D ch. 18	**22** A ch. 10
3 A ch. 6	**8** C ch. 9	**13** C ch. 3	**18** C ch. 17	**23** C ch. 21
4 C ch. 20	**9** C ch. 16	**14** A ch. 14	**19** A ch. 11	**24** B ch. 2
5 A ch. 7	**10** A ch. 13	**15** C ch. 19	**20** B ch. 12	**25** D ch. 15

Scoring your practice test

Percent correct	AR (of 30)	MK (of 25)
90% or more — excellent	27–30	23–25
75–89% — good	23–26	19–22
60–74% — keep practicing	18–22	15–18
below 60% — review the chapters	0–17	0–14

These bands measure your progress in this book only. Your real AFQT percentile depends on all four AFQT subtests and is scaled by the Department of Defense.

Explanations: Arithmetic Reasoning

1. **B** $-2°$F

1 A drop means subtract: $6 - 8$.

2 $6 - 8 = 6 + (-8)$. Different signs: subtract the absolute values ($8 - 6 = 2$) and keep the sign of -8, which has the larger absolute value: $6 + (-8) = -2$. The temperature was $-2°$F.

WHY NOT (C) subtracts the smaller number from the larger and drops the negative sign. (D) adds the drop instead of subtracting it.

2. **B** 75

1 Amount = rate × time.

2 $25 \times 3 = 75$ pages.

WHY NOT (A) adds the rate and the time instead of multiplying.

3. **A** $\frac{81}{100}$

1 Each shot has probability $\frac{9}{10}$, and the events are independent, so multiply.

2 $\frac{9}{10} \times \frac{9}{10} = \frac{81}{100}$.

WHY NOT (C) gives the probability for only one shot. (D) finds the probability of at least one, not all.

4. **B** \$1,500

1 Amount done: $0.25 \times \$2{,}000 = \$500$.

2 Amount still needed: $\$2{,}000 - \$500 = \$1{,}500$.

TIP Or: $100\% - 25\% = 75\%$ is left, and $0.75 \times \$2{,}000 = \$1{,}500$.

WHY NOT (A) is the amount already done, not the amount still needed. (D) subtracts the percent as if it were an amount.

5. **C** 35%

1 Pure fruit juice in each: $0.4 \times 8 = 3.2$ and $0.25 \times 4 = 1$ gallons.

2 Totals: $3.2 + 1 = 4.2$ gallons of fruit juice in $8 + 4 = 12$ gallons of mixture.

3 Percent: $\frac{4.2}{12} = 0.35 = 35\%$.

TIP Check: 35% is between 25% and 40%, closer to the percent of the larger amount.

WHY NOT (A) weights each percent by the wrong amount. (B) averages the two percents without weighting by the amounts. (D) adds the two percents.

6. **C** \$2,835

1 Find $\frac{1}{8}$ of the pay: \$3,240 ÷ 8 = \$405. That is the amount paid on the car loan.

2 Subtract it from the pay: \$3,240 – \$405 = \$2,835.

TIP Or: $1 - \frac{1}{8} = \frac{7}{8}$ of the pay is left, and $\frac{7}{8} \times$ \$3,240 = \$2,835.

WHY NOT (A) is the amount paid on the car loan, not the amount left.

7. **D** \$1,300

1 Twice a month means $12 \times 2 = 24$ paychecks a year.

2 Divide the yearly pay by the number of paychecks: \$31,200 ÷ 24 = \$1,300.

TIP Check: $24 \times$ \$1,300 = \$31,200.

WHY NOT (A) divides by 52, the number of weeks, not the number of paychecks. (C) uses 26 paychecks, but twice a month gives 24 paychecks a year.

8. **A** 3 hours 55 minutes

1 From 8:35 a.m. to 12 noon is 3 hours 25 minutes.

2 From 12 noon to 12:30 p.m. is 30 minutes.

3 Add: 3 hours 25 minutes + 30 minutes = 3 hours 55 minutes.

WHY NOT (B) counts only the change in the hour. (C) writes the times as 0835 and 1230, subtracts them like ordinary numbers, and reads 395 as 3 hours 95 minutes (that is, 4 hours 35 minutes). (D) forgets that the clock starts over at 12 noon.

9. **B** 40

1 Area of the floor: $10 \times 16 = 160$ square feet.

2 Each tile covers $2 \times 2 = 4$ square feet.

3 Number of tiles: 160 ÷ 4 = 40.

TIP Or count tiles along each side: $10 \div 2 = 5$ by $16 \div 2 = 8$, and $5 \times 8 = 40$.

WHY NOT (A) works with the perimeter instead of the area. (C) divides by the side of a tile (2) instead of its area (4). (D) forgets that each tile covers 4 square feet.

10. **D** \$187.20

1 Find the tax: 4% of \$180 = $0.04 \times 180 =$ \$7.20.

2 Add the tax to the price: \$180 + \$7.20 = \$187.20.

TIP In one step: with tax you pay 104%, so multiply by 1.04.

WHY NOT (A) is the tax alone, not the total cost. (B) subtracts the tax instead of adding it. (C) adds the tax rate as if it were dollars.

11. **B** 67

1 Put the 8 values in order: 57, 63, 63, 66, 68, 70, 86, 93.

2 With an even number of values, the median is the average of the two middle ones (the 4th and 5th): $(66 + 68) \div 2 = 134 \div 2 = 67$.

WHY NOT (A) uses only one of the two middle values. (C) uses only one of the two middle values.

12. **A** $4\frac{7}{8}$ feet

1 Common denominator 8: $9\frac{4}{8} - 4\frac{5}{8}$.

2 Regroup: $9\frac{4}{8} = 8\frac{12}{8}$.

3 Subtract: $8\frac{12}{8} - 4\frac{5}{8} = 4\frac{7}{8}$ feet.

WHY NOT (B) subtracts the smaller fraction from the larger instead of regrouping. (C) regroups but forgets to take 1 from the whole number. (D) adds instead of subtracting.

13. **B** 5 hours

1 Let h be the total number of hours. The first hour costs \$8; the other $h - 1$ hours cost \$3 each: $8 + 3(h - 1) = 20$.

2 Subtract the first-hour charge: $3(h - 1) = 20 - 8 = 12$.

3 Divide by 3: $h - 1 = 4$, so $h = 5$ hours.

WHY NOT (A) finds only the additional hours and forgets to count the first hour.

14. **C** 22 hours

1 Let h be the number of hours. The total must be at least the goal: $250 + 16h \geq 600$.

2 Subtract 250: $16h \geq 350$. Divide by 16: $h \geq \frac{350}{16} = 21\frac{7}{8}$.

3 h must be a whole number at least that big, so round *up*: 22. Check: $250 + 16(21) = 586$ is too little, but $250 + 16(22) = 602$ is enough.

WHY NOT (B) rounds down, which falls just short of the goal. (D) ignores the \$250 already saved.

15. **A** 4 fluid ounces

1 Change gallons to fluid ounces: $1 \times 128 = 128$ fluid ounces of gasoline.

2 The oil is $\frac{1}{32}$ of the gasoline amount: $128 \div 32 = 4$ fluid ounces.

WHY NOT (D) makes two slips: multiplies the gallons by the ratio instead of dividing, and skips the change to ounces.

16. **A** $90

1 Find the cost of each kind of item: $3 \times 8 = 24$ and $2 \times 43 = 86$.

2 Total cost: $24 + 86 = 110$ dollars.

3 Change: $200 - 110 = 90$ dollars.

WHY NOT (B) is the total cost, not the change. (C) forgets to multiply the second price by 2. (D) subtracts one of each price and ignores the quantities.

17. **B** 13 oz

1 You can't take 4 oz from 1 oz, so borrow 1 lb: 13 lb 1 oz becomes 12 lb 17 oz.

2 Subtract: $17 - 4 = 13$ oz and $12 - 12 = 0$ lb.

3 What is left: 13 oz.

WHY NOT (A) borrows 10 ounces instead of 16. (C) subtracts the smaller number of ounces from the larger instead of borrowing. (D) borrows 1 lb but forgets to take it away from the pounds.

18. **C** $181.44

1 Tax: $0.06 \times \$144 = \$8.64$.

2 Tip: $0.2 \times \$144 = \$28.80$.

3 Total: $\$144 + \$8.64 + \$28.80 = \$181.44$.

TIP Both percents are of the same amount, so add them first: $6\% + 20\% = 26\%$, and $1.26 \times \$144 = \$181.44$.

WHY NOT (A) forgets the tip. (B) forgets the tax.

19. **D** 70 seconds

1 The events line up again after a common multiple of 10 and 14; the first time is the least common multiple.

2 Multiples of 14: 14, 28, 42, 56, 70. The first one that is also a multiple of 10 is 70.

WHY NOT (A) is the greatest common factor, not the least common multiple. (B) adds the two numbers.

20. **B** 4 years

1 One year's interest: $\$4{,}000 \times 0.08 = \$320$.

2 Number of years: $\$1{,}280 \div \$320 = 4$.

TIP Check: $\$320 \times 4 = \$1{,}280$.

WHY NOT (A) forgets to change the percent to a decimal. (D) gives the number of months, not years.

21. **C** 314 mi^2

1 The area formula $A = \pi r^2$ needs the radius: $r = 20 \div 2 = 10$ miles.

2 Square the radius: $10^2 = 100$.

3 $A \approx 3.14 \times 100 = 314$ square miles.

TIP Multiply by 3 and by 0.14, then add: $3 \times 100 = 300$ and $0.14 \times 100 = 14$, so the total is 314.

WHY NOT (A) finds the circumference, not the area. (B) uses 3 for π instead of 3.14. (D) uses the diameter in place of the radius.

22. **D** 1:35 p.m.

1 First part: $75 \div 50 = 1.5$ hours (1 hour 30 minutes). Second part: $90 \div 60 = 1.5$ hours (1 hour 30 minutes).

2 Total time: 1 hour 30 minutes + 1 hour 30 minutes + 20 minutes = 3 hours 20 minutes.

3 Add to the start time: 10:15 a.m. plus 3 hours 20 minutes is 1:35 p.m.

WHY NOT (B) forgets the 20-minute stop.

23. **A** $43d + 0.25m$

1 Daily charges: $43d$ dollars.

2 Mileage charges: $0.25m$ dollars.

3 Total: $43d + 0.25m$.

WHY NOT (B) matches each rate with the wrong quantity. (C) charges the daily rate only once instead of d times. (D) multiplies everything together.

24. **A** $8.98

1 Cost of each kind: $3 \times 1.12 = 3.36$ and $2 \times 3.83 = 7.66$.

2 Total: $3.36 + 7.66 = 11.02$.

3 Change: $20.00 - 11.02 = 8.98$, so $8.98.

TIP Count up from $11.02: to the next dollar, then to $20.

WHY NOT (B) is the total cost, not the change. (C) subtracts one of each price and ignores the quantities. (D) subtracts only the first item.

25. **B** 0.5 hours

1 Head start: in 2 hours the first one goes $4 \times 2 = 8$ miles.

2 The gap closes at the difference of the speeds: $20 - 4 = 16$ miles per hour.

3 Time to close it: $8 \div 16 = 0.5$ hours.

TIP Check: in 0.5 hours the chaser goes $20 \times 0.5 = 10$ miles, and the first one has gone $4 \times 2.5 = 10$ miles.

WHY NOT (A) divides the head start by the faster speed instead of the difference. (D) counts the time from when the first one left.

26. **B** 0905

1 Add the minutes: 40 + 25 = 65 minutes = 1 hour 5 minutes.

2 Add the hours: 22 + 10 + 1 = 33. Past midnight, subtract 24: 33 − 24 = 9.

3 The answer is 0905 the next day.

WHY NOT (A) forgets to carry the extra hour from the minutes. (D) subtracts 12 hours instead of 24 after passing midnight.

27. **D** 24

1 Priya's place is fixed, so only the other 4 people need to be arranged.

2 $4 \times 3 \times 2 \times 1 = 24$.

WHY NOT (A) counts only the choices for one position.

28. **D** 84

1 Change the depth to feet: 8 in = $\frac{8}{12}$ ft = $\frac{2}{3}$ ft.

2 Volume: $9 \times 7 \times \frac{2}{3} = 42$ cubic feet.

3 Each bag holds $\frac{1}{2}$ cubic foot, so it takes 2 bags for every cubic foot: $42 \times 2 = 84$ bags.

WHY NOT (A) multiplies by the bag size instead of dividing by it. (B) finds the number of cubic feet but forgets to divide by the bag size.

29. **C** 2 hours

1 Each rate is 1 job divided by the time alone: $\frac{1}{10}$, $\frac{1}{3}$, and $\frac{1}{15}$ of the job per hour.

2 Add the three rates (common denominator 30): $\frac{1}{10} + \frac{1}{3} + \frac{1}{15} = \frac{3}{30} + \frac{10}{30} + \frac{2}{30} = \frac{15}{30} = \frac{1}{2}$ of the job per hour.

3 Time = 1 job ÷ rate: $1 \div \frac{1}{2} = 2$ hours.

TIP Check: the answer must be less than the fastest time alone, 3 hours.

WHY NOT (A) stops at the combined rate and forgets to flip it into a time. (B) divides the fastest time by 3, as if all three were that fast. (D) averages the three times.

30. **D** 10 pounds

1 Let r be the pounds of regular coffee; then $15 - r$ pounds are dark-roast coffee.

2 The value of the mix is $15 \times 7 = 105$ dollars, so $5r + 11(15 - r) = 105$.

3 Distribute and combine: $5r + 165 - 11r = 105$, so $-6r + 165 = 105$.

4 Subtract 165: $-6r = -60$. Divide by −6: $r = 10$ pounds of regular coffee.

TIP Sense check: $7 per pound is closer to the price of regular coffee, so the mix must contain more regular coffee.

WHY NOT (A) gives the pounds of dark-roast coffee.

Explanations: Mathematics Knowledge

1. **B** 7,100

1 The hundreds digit is 0. Look at the digit just to its right (the tens digit): 9.

2 Because 9 is 5 or more, round up: the hundreds digit 0 becomes 1.

3 Change every digit after the hundreds place to zero: 7,100.

WHY NOT (A) rounds down even though the next digit, 9, is 5 or more. (C) raises the hundreds digit but forgets to change the digits after it to zeros.

2. **C** 1,000 cm^3

1 All three dimensions of a cube equal the edge: $V = s^3 = s \times s \times s$.

2 $10 \times 10 = 100$, and $100 \times 10 = 1{,}000$.

3 The volume is 1,000 cubic centimeters.

WHY NOT (A) multiplies the edge by 3 instead of cubing it. (B) squares the edge instead of cubing it.

3. **A** 29

1 Change the percent to a decimal: 25% = 0.25.

2 Multiply by the number: $0.25 \times 116 = 29$.

WHY NOT (B) is what is left after removing 25%, not 25% of the number. (C) adds 25% to the number. (D) moves the decimal point only one place.

4. **C** 3 m

1 The radius is $10 \div 2 = 5$ meters.

2 Use $V = \pi r^2 h$: $75\pi = \pi \times 5^2 \times h = 25\pi h$.

3 Divide both sides by 25π: $h = 75 \div 25 = 3$ meters.

WHY NOT (A) uses the diameter in place of the radius. (D) forgets to square the radius.

5. **A** 5 : 6

1 Both amounts must be in the same unit. Change gallons to quarts: $3 \times 4 = 12$ quarts.

2 Now compare quarts to quarts, in the order asked: 10 : 12.

3 Divide both terms by their greatest common factor, 2: $10 \div 2 = 5$ and $12 \div 2 = 6$. The ratio is 5 : 6.

WHY NOT (B) reverses the order of the ratio. (C) forgets to change gallons to quarts before comparing. (D) multiplies the quarts by 4 instead of the gallons.

6. **D** 7

1 The x-intercept is where the line crosses the x-axis, so set $y = 0$: $0 = 2x - 14$.

2 $2x = 14$, so $x = 14 \div 2 = 7$.

WHY NOT (A) gives the y-intercept instead of the x-intercept. (B) makes a sign error when solving for x. (C) divides the slope by the y-intercept instead of dividing $-b$ by the slope.

7. **D** 5.8

1 Divide as with whole numbers and put the point in the answer directly above the point in 34.8.

2 $34.8 \div 6 = 5.8$. Check: $6 \times 5.8 = 34.8$.

WHY NOT (A) puts the decimal point one place too far left.

8. **C** $\frac{3}{11}$

1 Take the square root of the top and of the bottom: $\sqrt{\frac{9}{121}} = \frac{\sqrt{9}}{\sqrt{121}}$.

2 $\sqrt{9} = 3$ and $\sqrt{121} = 11$, so the answer is $\frac{3}{11}$.

WHY NOT (A) takes the square root of the numerator only. (B) divides by 2 instead of taking the square root. (D) takes the square root of the denominator only.

9. **C** 25

1 Angles that form a straight line add up to 180°: $(3x - 32) + (5x + 12) = 180$.

2 Combine like terms: $8x - 20 = 180$.

3 Add 20 to both sides: $8x = 180 + 20 = 200$.

4 Divide both sides by 8: $x = 200 \div 8 = 25$.

TIP Check: with $x = 25$ the angles are 43° and 137°, and $43 + 137 = 180$.

WHY NOT (A) makes a sign error when moving the constant term. (D) is the measure of an angle, not the value of x.

10. **A** $(-3, 0)$

1 A solution must make *both* equations true. Test $(-3, 0)$, that is, $x = -3$ and $y = 0$.

2 First equation: $3(-3) + 0 = -9$. ✓

3 Second equation: $-3 - 0 = -3$. ✓

TIP You can also solve it directly: adding the equations cancels y.

WHY NOT (B) swaps the values of x and y. (C) works in the first equation but not the second. (D) works in the second equation but not the first.

11. **B** $5x^5$

1 Divide the coefficients: $15 \div 3 = 5$.

2 Divide like bases by *subtracting* exponents: $\frac{x^8}{x^3} = x^{8-3} = x^5$.

3 So the result is $5x^5$.

WHY NOT (C) adds the exponents instead of subtracting. (D) subtracts the coefficients instead of dividing.

12. **B** 4.1×10^6

1 Move the decimal point to get a number that is at least 1 but less than 10: 4.1.

2 The decimal point moved 6 places to the left, which made the number smaller, so multiply by 10^6 to keep the same value.

3 So $4{,}100{,}000 = 4.1 \times 10^6$.

WHY NOT (A) counts one place too few. (C) counts one place too many. (D) is equal to 4,100,000 but is not scientific notation: the first factor must be less than 10.

13. **C** 504

1 Divisible by 4: the number formed by its last two digits is divisible by 4. Divisible by 9: its digit sum is divisible by 9.

2 The number 504 passes both tests (last two digits 04; digit sum 9): $504 \div 36 = 14$.

3 Each of the other choices fails one of the tests.

WHY NOT (A) is divisible by 9, but not by 4: its last two digits, 77, are not divisible by 4. (B) is divisible by 4, but not by 9: its digit sum, 16, is not divisible by 9. (D) is divisible by 4, but not by 9: its digit sum, 22, is not divisible by 9.

14. **A** $x^2 - 8x - 9$

1 Use FOIL. First: $x \cdot x = x^2$. Outer: $x \cdot 1 = x$. Inner: $(-9) \cdot x = -9x$. Last: $(-9) \cdot 1 = -9$.

2 Add them and combine the two middle terms: $x^2 + x - 9x - 9 = x^2 - 8x - 9$.

WHY NOT (B) gets the sign of the last term wrong. (C) gets the sign of the middle term wrong. (D) multiplies only the First and Last terms and leaves out the middle.

15. **C** 225π in^2

1 A whole circle with radius 30 has area $\pi \times 30^2 = 900\pi$.

2 A quarter circle is one fourth of it: $900\pi \div 4 = 225\pi$ square inches.

WHY NOT (A) finds the length of the curved edge, not the area. (B) forgets to square the radius. (D) finds the area of the whole circle.

16. **D** $(x-3)(x+3)$

1 $x^2 - 9$ is a difference of squares: $(x)^2 - 3^2$.

2 Use $A^2 - B^2 = (A+B)(A-B)$: $x^2 - 9 = (x-3)(x+3)$.

WHY NOT (A) is a perfect square, not a difference of squares. (B) uses 9 and −1, which multiply to −9 but do not add to 0. (C) is a perfect square, not a difference of squares.

17. **D** 62 ft

1 Find the missing corner: it is 13 − 4 = 9 ft wide and 18 − 12 = 6 ft tall.

2 Go around the shape and add all six sides: 13 + 12 + 9 + 6 + 4 + 18 = 62 feet.

TIP Shortcut: the two inside edges together make up for the missing pieces of the top and side, so the perimeter equals that of a 13 × 18 rectangle: $2 \times (13 + 18) = 62$.

WHY NOT (B) adds only the four labeled sides and leaves out the two edges of the cut-out corner.

18. **C** 79

1 An exterior angle of a triangle equals the sum of the two *remote* (non-adjacent) interior angles.

2 $x = 34 + 45 = 79$.

3 Check: the interior angle next to it is 180 − 79 = 101, and 34 + 45 + 101 = 180. ✓

WHY NOT (A) subtracts the two remote angles instead of adding them. (D) finds the interior angle next to the exterior angle, not the exterior angle.

19. **A** 12

1 Undo the subtraction: add 13 to both sides. $4x = 35 + 13 = 48$.

2 Undo the multiplication: divide both sides by 4. $x = \frac{48}{4} = 12$.

3 Check: $4(12) - 13 = 48 - 13 = 35$. ✓

WHY NOT (C) divides only 35 by 4, not the −13: $x - 13 = \frac{35}{4}$. (D) forgets to divide by 4.

20. **B** $-6 < x < 1$

1 Do the same thing to all three parts. Add 8 to each part: $-2 < -2x < 12$.

2 Divide each part by −2. Dividing by a negative number reverses *both* inequality symbols: $1 > x > -6$.

3 Rewrite it with the smaller number on the left: $-6 < x < 1$.

4 Check with $x = -5$: the middle is $-8 - 2(-5) = 2$, and $-10 < 2 < 4$ is true. ✓

WHY NOT (A) forgets that dividing by −2 changes the signs of the numbers. (C) gives the numbers *outside* the answer. (D) subtracts 8 instead of adding it.

21. **D** $1\frac{4}{5}$

1 Rewrite the fractions with the common denominator 10: $6\frac{5}{10} - 4\frac{7}{10}$.

2 Since $\frac{5}{10}$ is less than $\frac{7}{10}$, regroup: take 1 from 6, so $6\frac{5}{10} = 5\frac{15}{10}$.

3 Subtract: whole numbers 5 − 4 = 1, fractions $\frac{15}{10} - \frac{7}{10} = \frac{8}{10} = \frac{4}{5}$. The answer is $1\frac{4}{5}$.

WHY NOT (C) lowers the whole number by 1 but never adds that 1 to the fraction, then subtracts the smaller fraction from the larger.

22. **A** $8x - 7$

1 The perimeter is the sum of the three sides.

2 Add the x-terms: $4 + 2 + 2 = 8$. Add the constants: $-7 - 7 + 7 = -7$.

3 The perimeter is $8x - 7$.

WHY NOT (B) combines unlike terms. (C) doubles the sum, as if the figure were a rectangle. (D) adds every constant, ignoring the minus signs.

23. **C** $\frac{1}{4}$

1 Solve for y to find the slope: $y = -4x + 9$, so $y = -4x + 9$ and the slope is −4.

2 A perpendicular slope is the negative reciprocal: flip −4 and change its sign to get $\frac{1}{4}$.

3 Check: $(-4) \times \frac{1}{4} = -1$. ✓

WHY NOT (A) gives the slope of a parallel line. (B) flips the fraction but forgets to change the sign. (D) changes the sign but forgets to flip the fraction.

24. **B** 14

1 Parentheses first: 6 − 8 = −2.

2 Multiply: $-3 \times (-2) = 6$ (same signs, positive).

3 Divide: $(-24) \div (-3) = 8$ (same signs, positive).

4 Add: 6 + 8 = 14.

WHY NOT (A) gets the sign of $-3 \times (-2)$ wrong.

25. **D** $x = -\frac{7}{5}$ or $x = 2$

1 Here $a = 5$, $b = -3$, $c = -14$.

2 Compute $b^2 - 4ac$: $(-3)^2 - 4(5)(-14) = 9 + 280 = 289$, and $\sqrt{289} = 17$.

3 $x = \frac{3 \pm 17}{10}$: either $x = \frac{3+17}{10} = 2$ or $x = \frac{3-17}{10} = -\frac{7}{5}$.

TIP You can check by factoring: $5x^2 - 3x - 14 = (5x + 7)(x - 2)$.

WHY NOT (A) divides by 5 instead of by $2a = 10$. (B) uses -3 instead of 3 in the formula. (C) uses only the + sign of $\pm$.

Practice Test 4

Instructions

This test has two parts, timed like the paper-and-pencil ASVAB. **Part 1, Arithmetic Reasoning:** 30 questions, 36 minutes. **Part 2, Mathematics Knowledge:** 25 questions, 24 minutes. No calculator. Use scratch paper. Answer every question—there is no penalty for guessing. Mark your answers on the answer sheet below.

Answer Sheet

Fill in one bubble per question. Photocopy this page to retake the test.

Part 1: Arithmetic Reasoning

1	A	B	C	D
2	A	B	C	D
3	A	B	C	D
4	A	B	C	D
5	A	B	C	D
6	A	B	C	D
7	A	B	C	D
8	A	B	C	D
9	A	B	C	D
10	A	B	C	D
11	A	B	C	D
12	A	B	C	D
13	A	B	C	D
14	A	B	C	D
15	A	B	C	D
16	A	B	C	D
17	A	B	C	D
18	A	B	C	D
19	A	B	C	D
20	A	B	C	D
21	A	B	C	D
22	A	B	C	D
23	A	B	C	D
24	A	B	C	D
25	A	B	C	D
26	A	B	C	D
27	A	B	C	D
28	A	B	C	D
29	A	B	C	D
30	A	B	C	D

Part 2: Mathematics Knowledge

1	A	B	C	D
2	A	B	C	D
3	A	B	C	D
4	A	B	C	D
5	A	B	C	D
6	A	B	C	D
7	A	B	C	D
8	A	B	C	D
9	A	B	C	D
10	A	B	C	D
11	A	B	C	D
12	A	B	C	D
13	A	B	C	D
14	A	B	C	D
15	A	B	C	D
16	A	B	C	D
17	A	B	C	D
18	A	B	C	D
19	A	B	C	D
20	A	B	C	D
21	A	B	C	D
22	A	B	C	D
23	A	B	C	D
24	A	B	C	D
25	A	B	C	D

Part 1: Arithmetic Reasoning

Score: _____ / 30

Target time: 36 minutes

1. At sunset the temperature at a town in northern Minnesota was 5°F. By midnight it had dropped 25 degrees. What was the temperature at midnight?

A −30°F **B** −20°F **C** 20°F **D** 30°F

2. Ryan reads about 20 pages per hour. How many hours will it take him to read an 80-page book?

A 4 hours **B** 6 hours
C 7 hours **D** 60 hours

3. During a PT session, five soldiers did the following numbers of push-ups in two minutes: 39, 68, 39, 73, and 56. What is the mean (average) number of push-ups?

A 55 **B** 56 **C** 61 **D** 275

4. Jamal has $75 saved and earns $12 per hour at a part-time job. What is the least number of whole hours he must work to have at least $400?

A 25 hours **B** 26 hours
C 27 hours **D** 28 hours

5. Diego wants to buy a pair of running shoes that normally costs $95. The store is taking 15% off the price this weekend. How much will the pair of running shoes cost on sale?

A $14.25 **B** $80.00 **C** $80.75 **D** $109.25

6. Tyrone's hiking club has 10 members. One person will be chosen as president and a different person as vice president. In how many ways can the two positions be filled?

A 90 **B** 95 **C** 100 **D** 105

7. A pump can fill an empty fuel bladder at a forward operating base in 6 hours. A damaged valve would let a full bladder drain in 15 hours. If the valve is not fixed, how long will it take the pump to fill the empty bladder?

A 6 minutes **B** 6 hours
C 9 hours **D** 10 hours

8. A tractor-trailer travels 156 miles in 3 hours. What is the truck's average speed?

A 47 mph **B** 52 mph
C 153 mph **D** 468 mph

9. A crew of 5 movers can load a moving truck in 9 hours. At the same rate, how many hours would it take 3 movers to do the job?

A 9 hours **B** 11 hours
C 13 hours **D** 15 hours

10. Corporal Walker's monthly base pay is $3,350. This year's military pay raise is 5%. What will her new monthly base pay be?

A $167.50 **B** $3,355.00
C $3,517.50 **D** $5,025.00

11. Yuki bought a circular rug with a diameter of 12 feet. What is the area of the rug? Use 3.14 for π.

A 37.68 ft^2 **B** 113.04 ft^2
C 226.08 ft^2 **D** 452.16 ft^2

12. Sergeant Brooks has 32 chemical light sticks and 56 batteries. Every field kit must be identical, with nothing left over. What is the greatest number of field kits that can be made?

A 2 **B** 4 **C** 8 **D** 224

13. A hardware store takes 40% off a $120 power drill, and a rewards card takes 20% off the sale price. What is the final price of the power drill?

A $57.60 **B** $72.00 **C** $96.00 **D** $110.40

14. A hardware store buys hammers from a supplier for $11 each and marks up the cost by 50%. What is the selling price of each hammer?

A $5.50 **B** $16.50 **C** $22.50 **D** $61.00

15. Jasmine rides a train that travels at 36 miles per hour for 3 hours 15 minutes. How many miles does the train travel?

A 107 miles **B** 108 miles
C 113.4 miles **D** 117 miles

16. Grace has a roll of wire 51 meters long. How many pieces 60 centimeters long can be cut from it?

A 85 **B** 90 **C** 850 **D** 5,100

17. The average age of the 11 players on a softball team is 26 years. A new player who is 50 years old joins the team. What is the new average age of the players?

A 28 **B** 30 **C** 32 **D** 38

18. Last year the cost of a yearly gym membership was $260. This year it is $221. What is the percent decrease?

A 13% **B** 15% **C** 39% **D** 85%

19. Sergeant Nguyen is leaving the Army after four years and has two civilian job offers. Job A pays $18 per hour, and Job B pays a salary of $34,840 a year. At either job he would work 40 hours a week, 52 weeks a year. How much more per year does the better-paying job pay?

A $280 **B** $1,160 **C** $2,600 **D** $37,440

20. A guy wire runs from the top of a 24-foot pole to a stake in the ground 10 feet from the base of the pole. How long is the wire?

A 26 ft **B** 31 ft **C** 34 ft **D** 676 ft

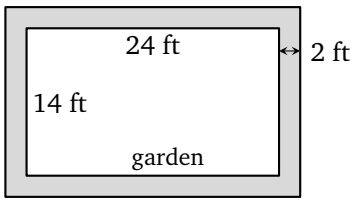
21. At a grocery store, coffee comes in three sizes: a 12-ounce bag of coffee for $5.04, a 24-ounce bag of coffee for $8.64, and a 32-ounce bag of coffee for $12.80. Which size is the best buy?

A The 32-ounce bag
B The 24-ounce bag
C The 12-ounce bag
D All three cost the same per ounce

22. Trees are planted every 10 feet along one side of a straight 210-foot sidewalk, with a tree at each end. How many trees are planted?

A 21 **B** 22 **C** 23 **D** 28

23. Derek's rectangular garden is 24 feet by 14 feet. He puts a gravel border 2 feet wide around the outside of the garden. What is the area of the border alone?



A 80 ft^2 **B** 152 ft^2 **C** 156 ft^2 **D** 168 ft^2

24. A water storage tank at a forward operating base is 14 feet long, 8 feet wide, and 5 feet deep. How many gallons of water does it hold when it is full? (1 cubic foot holds 7.5 gallons.)

A 560 gal **B** 3,330 gal
C 4,200 gal **D** 5,040 gal

25. A floor plan is drawn to a scale of 1 inch = 2 feet. A wall measures $7\frac{3}{4}$ inches on the plan. How long is the actual wall?

A 15.5 feet **B** 17 feet
C 31 feet **D** 155 feet

26. The sum of four consecutive odd integers is 256. What is the smallest of these integers?

A 53 **B** 61 **C** 63 **D** 65

27. Nora earned $2,300 last month. She spent $\frac{1}{4}$ of it on rent and then $\frac{1}{3}$ of what was left on groceries. How much money is left?

A $575 **B** $950 **C** $1,150 **D** $1,725

28. A base dining facility has 75 pounds of 93%-lean ground beef priced at $6 per pound. How many pounds of 80%-lean ground beef priced at $3 per pound must be mixed in to make a blend worth $4 per pound?

A 75 pounds **B** 120 pounds
C 150 pounds **D** 225 pounds

29. Grace works at a restaurant and earns $12.50 per hour for the first 40 hours each week. Every hour over 40 is overtime and pays 1.5 times the regular rate. Last week she worked 43 hours. How much did she earn?

A $537.50 **B** $556.25 **C** $557.25 **D** $806.25

30. Jasmine starts a 152-day savings challenge on a Saturday. The challenge ends 152 days later. On what day of the week does it end?

A Friday
B Wednesday
C Monday
D Thursday

Part 2: Mathematics Knowledge

Score: _____ / 25

Target time: 24 minutes

1. The point $(4, -8)$ is plotted on a coordinate plane. In which quadrant is it?

A Quadrant I **B** Quadrant II
C Quadrant III **D** Quadrant IV

2. Which power is equal to $\frac{1}{125}$?

A 5^{-3} **B** 3^{-5} **C** 5^3 **D** -5^3

3. In a survey, everyone chose either coffee or tea, and the ratio of coffee drinkers to tea drinkers was 3 : 4. What fraction of the people surveyed were tea drinkers?

A $\frac{1}{7}$ **B** $\frac{3}{7}$ **C** $\frac{4}{7}$ **D** $\frac{4}{3}$

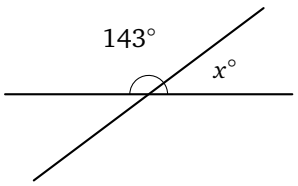
4. Which of the following is equal to 0.75?

A 7.5% **B** 75% **C** 85% **D** 750%

5. Subtract: $\frac{11}{12} - \frac{1}{3}$

A $\frac{7}{12}$ **B** $\frac{5}{6}$ **C** $1\frac{1}{9}$ **D** $1\frac{1}{4}$

6. Two lines cross, and one of the four angles they form measures 143°, as shown. Find x.



A 17 **B** 22 **C** 37 **D** 143

7. Which of the following is a solution of $15 < 4x + 3 < 43$?

A 3 **B** 6 **C** 10 **D** 11

8. What is the sum of $5x^2 + 3x - 3$ and $3x^2 + 3x + 4$?

A $8x^2 + 6x + 1$ **B** $8x^2 + 1$
C $14x^2 + 1$ **D** $8x^4 + 6x^2 + 1$

9. What is the value of $(2\sqrt{6})^2$?

A 12 **B** 24 **C** 48 **D** 144

10. What is the product of $-4x^4y^3$ and $-9xy^4$?

A $36x^4y^{12}$ **B** $36x^4y^7$
C $-13x^5y^7$ **D** $36x^5y^7$

11. Divide: $1{,}440 \div 4$

A 36 **B** 360 **C** 370 **D** 3,600

12. Multiply: 0.25×3.8

A 0.95 **B** 4.05 **C** 9.5 **D** 950

13. What number is halfway between 13 and −3 on a number line?

A −5 **B** 2 **C** 4 **D** 5

14. A rectangle that is 6 centimeters by 12 centimeters has a perimeter of 36 centimeters. If its length and width are both multiplied by 4, what is the perimeter of the new rectangle?

A 36 cm **B** 52 cm **C** 144 cm **D** 576 cm

15. What is the area of a circle with a diameter of 14 meters?

A 49π m^2 **B** 64π m^2
C 98π m^2 **D** 196π m^2

16. What is the value of $a^2 - 2ab$ when $a = 2$ and $b = -5$?

A −16 **B** 24 **C** 28 **D** 64

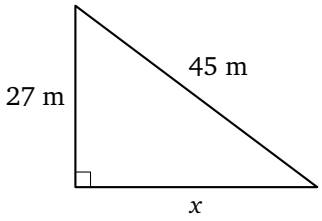
17. If the radius of a circle is divided by 3, the area of the new circle is what fraction of the area of the original circle?

A $\frac{1}{27}$ **B** $\frac{1}{18}$ **C** $\frac{1}{9}$ **D** $\frac{1}{3}$

18. If $-6x - 1 = 11$, what is the value of $3x - 5$?

A −13 **B** −12 **C** −11 **D** −2

19. In the right triangle shown, the hypotenuse is 45 meters and one leg is 27 meters. What is the value of x?



A 36 m **B** 38 m **C** 39 m **D** 1,296 m

20. A cube has a total surface area of 294 square inches. What is its volume?

A 49 in^3 **B** 294 in^3 **C** 343 in^3 **D** 512 in^3

21. Which of the following is the prime factorization of 648?

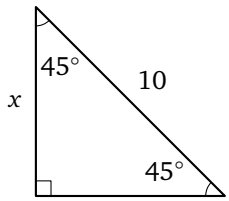
A $4 \times 2 \times 3^4$ **B** $2^3 \times 3^3$

C 2×3 **D** $2^3 \times 3^4$

22. The volume of a box is $V = lwh$. If $V = 72$, $l = 6$, and $w = 4$, what is h?

A 3 **B** 7.2 **C** 48 **D** 62

23. The figure shows a 45°-45°-90° triangle in which the hypotenuse is 10 units long. What is the value of x?



A 5 **B** $5\sqrt{2}$ **C** $5\sqrt{3}$ **D** $10\sqrt{2}$

24. If $5x + 4y = 68$ and $4x + 5y = 58$, what is the value of $x - y$?

A 2 **B** 8 **C** 9 **D** 10

25. Solve for x: $2x^2 - 19x + 42 = 0$.

A $x = -6$ or $x = -\frac{7}{2}$ **B** $x = \frac{7}{2}$ or $x = 6$

C $x = 6$ or $x = 7$ **D** $x = 3$ or $x = 7$

Answer Key: Practice Test 4

The small number after each answer is the chapter that teaches the skill.

Part 1: Arithmetic Reasoning

1 B ch. 2	**7** D ch. 25	**13** A ch. 23	**19** C ch. 22	**25** A ch. 7
2 A ch. 24	**8** B ch. 24	**14** B ch. 23	**20** A ch. 17	**26** B ch. 11
3 A ch. 26	**9** D ch. 25	**15** D ch. 24	**21** B ch. 22	**27** C ch. 4
4 D ch. 12	**10** C ch. 23	**16** A ch. 27	**22** B ch. 27	**28** C ch. 25
5 C ch. 22	**11** B ch. 19	**17** A ch. 26	**23** D ch. 18	**29** B ch. 5
6 A ch. 26	**12** C ch. 3	**18** B ch. 6	**24** C ch. 20	**30** D ch. 27

Part 2: Mathematics Knowledge

1 D ch. 21	**6** C ch. 16	**11** B ch. 1	**16** B ch. 10	**21** D ch. 3
2 A ch. 8	**7** B ch. 12	**12** A ch. 5	**17** C ch. 19	**22** A ch. 10
3 C ch. 7	**8** A ch. 14	**13** D ch. 2	**18** C ch. 11	**23** B ch. 17
4 B ch. 6	**9** B ch. 9	**14** C ch. 18	**19** A ch. 17	**24** D ch. 13
5 A ch. 4	**10** D ch. 14	**15** A ch. 19	**20** C ch. 20	**25** B ch. 15

Scoring your practice test

Percent correct	AR (of 30)	MK (of 25)
90% or more — excellent	27–30	23–25
75–89% — good	23–26	19–22
60–74% — keep practicing	18–22	15–18
below 60% — review the chapters	0–17	0–14

These bands measure your progress in this book only. Your real AFQT percentile depends on all four AFQT subtests and is scaled by the Department of Defense.

Explanations: Arithmetic Reasoning

1. **B** −20°F

1 A drop means subtract: 5 − 25.

2 5 − 25 = 5 + (−25). Different signs: subtract the absolute values (25 − 5 = 20) and keep the sign of −25, which has the larger absolute value: 5 + (−25) = −20. The temperature was −20°F.

WHY NOT (C) subtracts the smaller number from the larger and drops the negative sign. (D) adds the drop instead of subtracting it.

2. **A** 4 hours

1 Time = amount ÷ rate.

2 80 ÷ 20 = 4 hours.

TIP Check: 20 × 4 = 80 pages.

WHY NOT (D) subtracts the rate from the total.

3. **A** 55

1 Add the 5 values: 39 + 68 + 39 + 73 + 56 = 275.

2 Divide by the number of values, 5: 275 ÷ 5 = 55.

TIP Check: the mean must be between the smallest value (39) and the largest (73).

WHY NOT (B) is the median (the middle value), not the mean. (D) is the total, not the average.

4. **D** 28 hours

1 Let h be the number of hours. The total must be at least the goal: $75 + 12h \geq 400$.

2 Subtract 75: $12h \geq 325$. Divide by 12: $h \geq \frac{325}{12} = 27\frac{1}{12}$.

3 h must be a whole number at least that big, so round *up*: 28. Check: $75 + 12(27) = 399$ is too little, but $75 + 12(28) = 411$ is enough.

WHY NOT (C) rounds down, which falls just short of the goal.

5. **C** \$80.75

1 Find the discount: 15% of \$95 = 0.15×95 = \$14.25.

2 Subtract it from the regular price: \$95 – \$14.25 = \$80.75.

TIP Shortcut: 15% off means you pay 100% – 15% = 85%, and $0.85 \times \$95 = \$80.75$.

WHY NOT (A) is the amount of the discount, not the sale price. (B) subtracts the percent as if it were dollars. (D) adds the discount instead of subtracting it.

6. **A** 90

1 President: 10 choices. Vice president: 9 choices (anyone except the president).

2 Multiply: $10 \times 9 = 90$.

WHY NOT (C) lets the same person hold both positions.

7. **D** 10 hours

1 Rates: the pump fills $\frac{1}{6}$ of the bladder per hour, and the valve empties $\frac{1}{15}$ of it per hour.

2 The valve works against the pump, so *subtract*: $\frac{1}{6} - \frac{1}{15} = \frac{5}{30} - \frac{2}{30} = \frac{3}{30} = \frac{1}{10}$ of the bladder per hour.

3 Time = 1 bladder ÷ rate = $1 \div \frac{1}{10} = 10$ hours.

TIP Shortcut for fill and drain: $\frac{6 \times 15}{15 - 6} = \frac{90}{9} = 10$.

WHY NOT (A) stops at the net rate and forgets to flip it into a time. (B) ignores the valve. (C) subtracts the two times.

8. **B** 52 mph

1 Use $r = d \div t$: speed = distance ÷ time.

2 $r = 156 \div 3 = 52$ miles per hour.

WHY NOT (C) subtracts the time from the distance. (D) multiplies the distance by the time instead of dividing.

9. **D** 15 hours

1 Find the total amount of work in worker-hours: $5 \times 9 = 45$ worker-hours.

2 Share that work among 3 movers: $45 \div 3 = 15$ hours.

TIP Fewer workers means more time, so the answer must be more than 9 hours.

WHY NOT (A) assumes the time does not change. (B) changes the time by one hour for each worker added or removed.

10. **C** \$3,517.50

1 Find the raise: $0.05 \times \$3{,}350 = \$167.50$.

2 Add it to the old pay: \$3,350 + \$167.50 = \$3,517.50.

TIP In one step: $1.05 \times \$3{,}350 = \$3{,}517.50$.

WHY NOT (A) is the amount of the raise, not the new monthly base pay. (B) adds the percent as if it were dollars. (D) moves the decimal point only one place.

11. **B** 113.04 ft^2

1 The area formula $A = \pi r^2$ needs the radius: $r = 12 \div 2 = 6$ feet.

2 Square the radius: $6^2 = 36$.

3 $A \approx 3.14 \times 36 = 113.04$ square feet.

TIP Multiply by 3 and by 0.14, then add: $3 \times 36 = 108$ and $0.14 \times 36 = 5.04$, so the total is 113.04.

WHY NOT (A) finds the circumference, not the area. (C) squares the diameter and then halves it; halve first, then square. (D) uses the diameter in place of the radius.

12. **C** 8

1 Every field kit gets the same number of chemical light sticks and the same number of batteries, so the number of field kits must divide both 32 and 56. The greatest such number is the GCF.

2 Write each number as a product of primes: $32 = 2^5$, $56 = 2^3 \times 7$.

3 Take each prime that both numbers share, with its lowest power: $2^3 = 8$.

WHY NOT (A) is a common factor, but not the greatest one. (D) is the least common multiple, not the greatest common factor.

13. **A** $57.60

1 First discount: $\$120 - 0.4 \times \$120 = \$120 - \$48 = \$72$.

2 The second discount is taken from the *new* price: $\$72 - 0.2 \times \$72 = \$72 - \$14.40 = \$57.60$.

TIP In one line: $\$120 \times 0.6 \times 0.8 = \$57.60$.

WHY NOT (B) stops after the first discount. (C) applies only the second discount. (D) multiplies the two rates ($40\% \times 20\% = 8\%$) and takes only that much off.

14. **B** $16.50

1 Markup: $0.5 \times \$11 = \$5.50$.

2 Selling price: $\$11 + \$5.50 = \$16.50$.

TIP In one step: $1.5 \times \$11 = \$16.50$.

WHY NOT (A) is the markup, not the selling price. (D) adds the percent as if it were dollars.

15. **D** 117 miles

1 Change the minutes to hours: 15 min = $\frac{15}{60} = \frac{1}{4}$ hour, so the time is $3\frac{1}{4}$ hours.

2 Distance: $36 \times 3\frac{1}{4} = 36 \times 3 + 36 \times \frac{1}{4} = 108 + 9 = 117$ miles.

WHY NOT (B) leaves out the extra minutes. (C) treats 3 hours 15 minutes as 3.15 hours.

16. **A** 85

1 Change meters to centimeters: $51 \times 100 = 5{,}100$ centimeters.

2 Divide by one piece: $5{,}100 \div 60 = 85$ pieces.

WHY NOT (C) uses 1,000 centimeters in a meter instead of 100. (D) gives the length in centimeters, not the number of pieces.

17. **A** 28

1 Turn the average into a total: $11 \times 26 = 286$.

2 Add the value: $286 + 50 = 336$.

3 Divide by the new count, $11 + 1 = 12$: $336 \div 12 = 28$.

WHY NOT (D) averages the old average with the new value.

18. **B** 15%

1 Find the amount of decrease: $260 - 221 = 39$ dollars.

2 Divide by the *original* amount: $\frac{39}{260} = 0.15$.

3 Convert to a percent: $0.15 = 15\%$ decrease.

TIP Hard division? Test the choices instead: 15% of $260 is $39, exactly the change.

WHY NOT (C) gives the dollar change instead of the percent change. (D) gives the new amount as a percent of the old amount.

19. **C** $2,600

1 Hours in a year: $40 \times 52 = 2{,}080$ hours.

2 Job A per year: $2{,}080 \times \$18 = \$37{,}440$.

3 Compare with Job B's $34,840: $\$37{,}440 - \$34{,}840 = \$2{,}600$. Job A pays $2,600 more.

WHY NOT (A) assumes 4 weeks in every month (48 weeks), but a year has 52 weeks. (B) uses 2,000 hours a year instead of 2,080. (D) is Job A's yearly pay, not the difference.

20. **A** 26 ft

1 The pole, the ground, and the wire form a right triangle with the wire as the hypotenuse.

2 $10^2 + 24^2 = 100 + 576 = 676$.

3 $\sqrt{676} = 26$, so the answer is 26 feet.

TIP Shortcut: divide by 2 and you get the 5-12-13 triple, so the hypotenuse is $13 \times 2 = 26$.

WHY NOT (C) adds the two distances instead of using the Pythagorean theorem. (D) forgets to take the square root.

21. **B** The 24-ounce bag

1 Find each price per ounce (price ÷ number of ounces): $\$5.04 \div 12 = \$0.42$; $\$8.64 \div 24 = \$0.36$; $\$12.80 \div 32 = \$0.40$.

2 The lowest price per ounce is $0.36. The 24-ounce bag is the best buy.

TIP The biggest package is not always the best deal, so always compare unit prices.

WHY NOT (A) is the largest size, but it costs $0.40 per ounce, more than $0.36. (C) has the lowest price tag, but it costs $0.42 per ounce, more than $0.36. (D) is not true: the prices per ounce are $0.42, $0.36, and $0.40.

22. **B** 22

1 Number of spaces: $210 \div 10 = 21$.

2 A straight line with a post at each end has one more post than spaces: $21 + 1 = 22$.

TIP Picture a short case: a 10-foot fence with posts every 5 feet has posts at 0, 5, and 10, which is 3 posts for 2 spaces.

WHY NOT (A) counts the spaces between posts, not the posts.

23. **D** 168 ft^2

1 The border adds 2 feet on *both* ends of each side, so the outer rectangle is $24 + 2 \times 2 = 28$ by $14 + 2 \times 2 = 18$ feet.

2 Outer area: $28 \times 18 = 504$ square feet.

3 Garden area: $24 \times 14 = 336$ square feet.

4 Border alone: $504 - 336 = 168$ square feet.

WHY NOT (A) adds the border width to only one end of each side. (B) multiplies the perimeter by the width and misses the four corner squares.

24. **C** 4,200 gal

1 Volume: $14 \times 8 \times 5 = 560$ cubic feet.

2 Each cubic foot holds 7.5 gallons: $560 \times 7.5 = 4{,}200$ gallons.

TIP Multiplying by 7.5 is the same as multiplying by 15 and halving: $560 \times 15 = 8{,}400$, and $8{,}400 \div 2 = 4{,}200$.

WHY NOT (A) finds the volume in cubic feet but forgets to change it to gallons. (B) uses the surface area instead of the volume.

25. **A** 15.5 feet

1 Each inch on the plan stands for 2 feet, so multiply the measurement by 2.

2 Split it up: $7 \times 2 = 14$ and $\frac{3}{4} \times 2 = 1.5$, so the total is $14 + 1.5 = 15.5$ feet.

WHY NOT (D) misplaces the decimal point in the product.

26. **B** 61

1 Let n be the smallest. Consecutive odd integers go up by 2, so the numbers are $n, n+2, n+4, n+6$.

2 Add them: $4n + 12 = 256$.

3 Subtract 12: $4n = 244$. Divide by 4: $n = 61$.

4 Check: $61 + 63 + 65 + 67 = 256$. ✓

WHY NOT (C) is the second number, not the smallest. (D) is the third number, not the smallest.

27. **C** $1,150

1 Rent: $\frac{1}{4} \times 2{,}300 = 575$, which leaves $2{,}300 - 575 = 1{,}725$ dollars.

2 Groceries: $\frac{1}{3}$ of what was left is $\frac{1}{3} \times 1{,}725 = 575$.

3 What remains: $1{,}725 - 575 = 1{,}150$ dollars.

TIP Shortcut: keep $\frac{3}{4}$ and then $\frac{2}{3}$ of that: $\frac{3}{4} \times \frac{2}{3} \times 2{,}300 = 1{,}150$.

WHY NOT (A) is the amount for groceries, not what is left. (D) stops after rent.

28. **C** 150 pounds

1 Let y be the pounds of 80%-lean ground beef. The money must balance: $6(75) + 3y = 4(75 + y)$.

2 Multiply out: $450 + 3y = 300 + 4y$.

3 Collect terms: $450 - 300 = 4y - 3y$, so $y = 150$ pounds.

TIP Balance shortcut: each pound of 93%-lean ground beef is \$2 away from \$4, each pound of 80%-lean ground beef is \$1 away; the totals must match: $75 \times 2 = 150 \times 1$.

WHY NOT (A) assumes equal amounts, which only works if the target is halfway between the prices. (D) gives the total weight of the blend.

29. **B** $556.25

1 Regular pay: $40 \times 12.50 = 500.00$.

2 Overtime rate: $1.5 \times 12.50 = 18.75$ per hour.

3 Overtime pay: $3 \times 18.75 = 56.25$.

4 Total: $500.00 + 56.25 = 556.25$, so \$556.25.

WHY NOT (A) pays the overtime hours at the regular rate. (D) pays every hour at the overtime rate.

30. **D** Thursday

1 Every 7 days the day of the week repeats. $152 \div 7 = 21$ weeks with 5 days left over.

2 After 21 full weeks it is Saturday again. Count 5 more days: Sunday, Monday, Tuesday, Wednesday, Thursday.

WHY NOT (A) counts one day too many. (B) counts one day too few. (C) counts backward instead of forward.

Explanations: Mathematics Knowledge

1. **D** Quadrant IV

1 The x-coordinate 4 is positive, so the point is right of the y-axis. The y-coordinate -8 is negative, so it is below the x-axis.

2 That is the lower right region. The quadrants are numbered counterclockwise from the upper right (I, II, III, IV), so this is Quadrant IV.

WHY NOT (A) reads the sign of the y-coordinate backwards. (B) swaps the x- and y-coordinates. (C) reads the sign of the x-coordinate backwards.

2. **A** 5^{-3}

1 Write the denominator as a power: $125 = 5 \times 5 \times 5 = 5^3$.

2 A reciprocal of a power is the same power with a negative exponent: $\frac{1}{5^3} = 5^{-3}$.

WHY NOT (B) switches the base and the exponent. (C) is the number itself; the reciprocal needs a negative exponent. (D) confuses a negative exponent with a negative number.

3. **C** $\frac{4}{7}$

1 Think in parts: the whole group is 3 + 4 = 7 equal parts.

2 The tea drinkers are 4 of those 7 parts, so the fraction is $\frac{4}{7}$.

WHY NOT (A) is the size of just one part, not the whole group's share. (B) is the fraction for the other group. (D) uses the ratio 4 : 3 as if it were the fraction of the total.

4. **B** 75%

1 To change a decimal to a percent, multiply by 100 (move the decimal point two places to the right).

2 $0.75 \times 100 = 75$, so 0.75 = 75%.

WHY NOT (A) moves the decimal point only one place. (D) moves the decimal point three places.

5. **A** $\frac{7}{12}$

1 12 is a multiple of 3, so use 12 as the common denominator.

2 Rewrite $\frac{1}{3}$ with denominator 12: $\frac{1}{3} = \frac{4}{12}$.

3 Subtract the numerators: $\frac{11}{12} - \frac{4}{12} = \frac{7}{12}$.

WHY NOT (B) uses the common denominator without rewriting the numerators. (C) subtracts the numerators and subtracts the denominators. (D) adds instead of subtracting.

6. **C** 37

1 The $x°$ angle and the 143° angle sit side by side along one straight line, so together they make 180°.

2 $x + 143 = 180$, so $x = 180 - 143 = 37$.

WHY NOT (D) treats the angles as vertical angles, but these two angles together form a straight line.

7. **B** 6

1 Solve first. Subtract 3 from all three parts: $12 < 4x < 40$, then divide all three by 4: $3 < x < 10$.

2 Only 6 is in this range. Check: $4(6) + 3 = 27$, and $15 < 27 < 43$. ✓

WHY NOT (A) makes the middle equal to 15, but that end is not included. (C) makes the middle equal to 43, but that end is not included. (D) gives $4(11) + 3 = 47$, which is outside the range.

8. **A** $8x^2 + 6x + 1$

1 Group like terms: $(5 + 3)x^2 + (3 + 3)x + (-3 + 4)$.

2 Add the coefficients and keep each exponent: $8x^2 + 6x + 1$.

WHY NOT (B) subtracts the x-terms instead of adding them. (C) combines unlike terms (x^2-terms with x-terms). (D) adds the exponents as well as the coefficients.

9. **B** 24

1 Square each factor: $(2\sqrt{6})^2 = 2^2 \times (\sqrt{6})^2$.

2 $2^2 = 4$ and $(\sqrt{6})^2 = 6$, so the answer is $4 \times 6 = 24$.

WHY NOT (A) squares the $\sqrt{6}$ but forgets to square the 2. (D) squares 6 instead of $\sqrt{6}$.

10. **D** $36x^5y^7$

1 Multiply the coefficients: $(-4) \cdot (-9) = 36$.

2 Multiply like bases by *adding* exponents: $x^4 \cdot x^1 = x^{4+1} = x^5$ and $y^3 \cdot y^4 = y^{3+4} = y^7$ (a letter with no exponent has exponent 1).

3 So the product is $36x^5y^7$.

WHY NOT (A) multiplies the exponents instead of adding them. (B) treats an x with no exponent as x^0 instead of x^1. (C) adds the coefficients instead of multiplying them.

11. **B** 360

1 Since 4 does not go into 1, start with 14: 14÷4 = 3, remainder 2.

2 Bring down the 4 to make 24: 24 ÷ 4 = 6.

3 Bring down the 0: 4 does not go into 0, so write 0 in the quotient.

4 The quotient is 360. Check: $4 \times 360 = 1{,}440$.

WHY NOT (A) skips the zero in the quotient. (D) adds an extra zero.

12. **A** 0.95

1 Ignore the decimal points and multiply: $25 \times 38 = 950$.

2 Count the decimal places: 2 in 0.25 and 1 in 3.8, 3 in all.

3 Put 3 decimal places in the product: 0.950, which is 0.95.

WHY NOT (B) adds instead of multiplying. (C) counts only the decimal places in 0.25 and ignores the one in 3.8. (D) ignores the decimal points completely.

13. **D** 5

1 The halfway point is the average of the two numbers: $(-3 + 13) \div 2$.

2 $-3 + 13 = 10$, and $10 \div 2 = 5$.

3 Check: 5 is 8 units from each number.

WHY NOT (A) has the right size but the wrong sign.

14. **C** 144 cm

1 Perimeter is a length (a sum of sides), so when every side is multiplied by 4, the perimeter is multiplied by 4 too.

2 $36 \times 4 = 144$, so the new perimeter is 144 centimeters.

WHY NOT (A) gives the original perimeter. (B) adds 4 to the length and to the width instead of multiplying. (D) multiplies by 16, which is how the area changes.

15. **A** 49π m^2

1 The formula $A = \pi r^2$ needs the radius, which is half the diameter: $r = 14 \div 2 = 7$ meters.

2 Square the radius: $7^2 = 49$.

3 So $A = \pi \times 49 = 49\pi$ square meters.

WHY NOT (C) squares the diameter and then halves it; halve first, then square. (D) uses the diameter in place of the radius.

16. **B** 24

1 Substitute, putting each value in parentheses: $(2)^2 - 2(2)(-5)$.

2 Powers first: $(2)^2 = 4$.

3 Multiply each term (watch the signs): $4 + 20$.

4 Add and subtract from left to right: $4 + 20 = 24$.

WHY NOT (A) forgets that a negative times a negative is positive.

17. **C** $\frac{1}{9}$

1 The new radius is $\frac{1}{3}r$.

2 New area: $\pi\left(\frac{1}{3}r\right)^2 = \frac{1}{9}\pi r^2$.

3 So the new area is $\frac{1}{9}$ of the original.

WHY NOT (A) cubes the factor; that is how volume changes. (D) assumes the area shrinks by the same factor as the radius.

18. **C** -11

1 First solve for x. Undo the subtraction: add 1 to both sides. $-6x = 11 + 1 = 12$.

2 Undo the multiplication: divide both sides by -6. $x = \frac{12}{-6} = -2$.

3 The question asks for $3x - 5$, not x. Substitute: $3(-2) - 5 = -6 - 5 = -11$.

TIP Always reread the question before choosing: finding x is only part of the job.

WHY NOT (D) gives the value of x, not of $3x - 5$.

19. **A** 36 m

1 The hypotenuse is the longest side, so use $a^2 + b^2 = c^2$ and solve for the missing leg: leg^2 = c^2 – (other leg)^2.

2 $45^2 - 27^2 = 2{,}025 - 729 = 1{,}296$.

3 Take the square root: $\sqrt{1{,}296} = 36$ m.

TIP Shortcut: divide by 9 and you get the 3-4-5 triple, so the missing side is $4 \times 9 = 36$.

WHY NOT (D) forgets to take the square root.

20. **C** 343 in^3

1 Each of the 6 faces has area $294 \div 6 = 49$ square inches.

2 A square with area 49 has side 7, so the edge is 7 inches.

3 Volume: $7^3 = 343$ cubic inches.

WHY NOT (A) stops at the area of one face. (B) gives the surface area, not the volume.

21. **D** $2^3 \times 3^4$

1 Divide by 2 as many times as you can: $648 \rightarrow 324 \rightarrow 162 \rightarrow 81$ (3 factors of 2).

2 Divide by 3 as many times as you can: $81 \rightarrow 27 \rightarrow 9 \rightarrow 3 \rightarrow 1$ (4 factors of 3).

3 So $648 = 2^3 \times 3^4$.

WHY NOT (A) uses 4, which is not prime. (B) multiplies out to 216, not 648. (C) lists each prime only once; that product is only 6.

22. **A** 3

1 Substitute: $72 = 6 \times 4 \times h$, so $72 = 24h$.

2 Divide by 24: $h = 72 \div 24 = 3$.

WHY NOT (B) divides by $l + w$ instead of $l \times w$. (C) subtracts instead of dividing. (D) subtracts l and w instead of dividing.

23. **B** $5\sqrt{2}$

1 In a 45°-45°-90° triangle the sides are in the ratio $1 : 1 : \sqrt{2}$ (leg : leg : hypotenuse).

2 Leg = hypotenuse $\div\sqrt{2} = \frac{10}{\sqrt{2}}$.

3 Multiply top and bottom by $\sqrt{2}$: $\frac{10\sqrt{2}}{2} = 5\sqrt{2}$.

WHY NOT (A) halves the hypotenuse, which is the 30°-60°-90° rule for the short leg. (C) halves the hypotenuse and multiplies by $\sqrt{3}$, the 30°-60°-90° rule for the long leg. (D) multiplies the hypotenuse by $\sqrt{2}$ instead of dividing.

24. **D** 10

1 Look at the coefficients: they are swapped. Subtract the second equation from the first: $x - y = 10$.

2 That is $x - y = 10$ already.

TIP No need to find x and y separately (they are 12 and 2).

WHY NOT (A) gives the value of y.

25. **B** $x = \frac{7}{2}$ or $x = 6$

1 Factor: $2x^2 - 19x + 42 = (2x - 7)(x - 6)$ (check the middle term: $-12x - 7x = -19x$).

2 Set each factor equal to 0: $2x - 7 = 0$ gives $x = \frac{7}{2}$; $x - 6 = 0$ gives $x = 6$.

3 The solutions are $x = \frac{7}{2}$ or $x = 6$.

WHY NOT (A) uses the numbers in the factors instead of their opposites. (C) forgets to divide by 2 when solving $2x - 7 = 0$. (D) puts -7 and -6 in the wrong factors.

Formula Sheet

The ASVAB does not give you formulas. Everything on this page is fair game—learn it until you can write it from memory.

Percents & money

- part = percent × whole
- percent change = $\dfrac{\text{new} - \text{original}}{\text{original}} \times 100\%$
- sale price = price $\times(1 - \text{discount rate})$
- total with tax = price $\times(1 + \text{tax rate})$
- simple interest: $I = Prt$ (t in years)

Rates

- distance = rate × time, $d = rt$
- average speed = $\dfrac{\text{total distance}}{\text{total time}}$
- work together: $\dfrac{1}{a} + \dfrac{1}{b} = \dfrac{1}{t}$
- unit rate = $\dfrac{\text{amount}}{\text{one unit}}$

Statistics & probability

- mean = $\dfrac{\text{sum of values}}{\text{number of values}}$
- median: middle value of the *sorted* list
- mode: most frequent value; range = max − min
- $P(\text{event}) = \dfrac{\text{favorable outcomes}}{\text{total outcomes}}$
- $P(\text{not } A) = 1 - P(A)$; independent: $P(A \text{ and } B) = P(A)\,P(B)$

Exponents & roots

- $a^m \cdot a^n = a^{m+n}$, $\dfrac{a^m}{a^n} = a^{m-n}$, $(a^m)^n = a^{mn}$
- $a^0 = 1$, $a^{-n} = \dfrac{1}{a^n}$, $(ab)^n = a^n b^n$
- $\sqrt{ab} = \sqrt{a}\,\sqrt{b}$, $\sqrt{\dfrac{a}{b}} = \dfrac{\sqrt{a}}{\sqrt{b}}$

Algebra

- $(a+b)^2 = a^2 + 2ab + b^2$
- $(a-b)^2 = a^2 - 2ab + b^2$
- $(a+b)(a-b) = a^2 - b^2$
- quadratic formula: $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

- slope $m = \dfrac{y_2 - y_1}{x_2 - x_1}$; line: $y = mx + b$
- distance = $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
- midpoint = $\left(\dfrac{x_1 + x_2}{2}, \dfrac{y_1 + y_2}{2}\right)$

Angles & triangles

- complementary: sum 90°; supplementary: sum 180°
- angles of a triangle add to 180°
- polygon with n sides: angle sum $(n-2) \times 180°$
- Pythagorean theorem: $a^2 + b^2 = c^2$
- common triples: 3-4-5, 5-12-13, 8-15-17, 7-24-25
- 45-45-90: sides $x,\ x,\ x\sqrt{2}$; 30-60-90: $x,\ x\sqrt{3},\ 2x$

Perimeter & area

- rectangle: $P = 2l + 2w$, $A = lw$
- square: $P = 4s$, $A = s^2$
- triangle: $A = \frac{1}{2}bh$
- parallelogram: $A = bh$ (height, not slanted side)
- trapezoid: $A = \frac{1}{2}(b_1 + b_2)h$

Circles & solids

- $d = 2r$, circumference $C = 2\pi r = \pi d$
- area $A = \pi r^2$; $\pi \approx 3.14 \approx \frac{22}{7}$
- box: $V = lwh$, $SA = 2lw + 2lh + 2wh$
- cube: $V = s^3$, $SA = 6s^2$
- cylinder: $V = \pi r^2 h$
- cone: $V = \frac{1}{3}\pi r^2 h$; sphere: $V = \frac{4}{3}\pi r^3$

Quick Reference Tables

Squares and cubes

$1^2 = 1$	$14^2 = 196$	$1^3 = 1$
$2^2 = 4$	$15^2 = 225$	$2^3 = 8$
$3^2 = 9$	$16^2 = 256$	$3^3 = 27$
$4^2 = 16$	$17^2 = 289$	$4^3 = 64$
$5^2 = 25$	$18^2 = 324$	$5^3 = 125$
$6^2 = 36$	$19^2 = 361$	$6^3 = 216$
$7^2 = 49$	$20^2 = 400$	$7^3 = 343$
$8^2 = 64$	$21^2 = 441$	$8^3 = 512$
$9^2 = 81$	$22^2 = 484$	$9^3 = 729$
$10^2 = 100$	$23^2 = 529$	$10^3 = 1{,}000$
$11^2 = 121$	$24^2 = 576$	
$12^2 = 144$	$25^2 = 625$	
$13^2 = 169$		

Fractions, decimals, percents

Fraction	Decimal	Percent
$\frac{1}{2}$	0.5	50%
$\frac{1}{3}$	$0.333\ldots$	$33\frac{1}{3}\%$
$\frac{2}{3}$	$0.666\ldots$	$66\frac{2}{3}\%$
$\frac{1}{4}$	0.25	25%
$\frac{3}{4}$	0.75	75%
$\frac{1}{5}$	0.2	20%
$\frac{2}{5}$	0.4	40%
$\frac{3}{5}$	0.6	60%
$\frac{4}{5}$	0.8	80%
$\frac{1}{6}$	$0.1666\ldots$	$16\frac{2}{3}\%$
$\frac{1}{8}$	0.125	12.5%
$\frac{3}{8}$	0.375	37.5%
$\frac{5}{8}$	0.625	62.5%
$\frac{7}{8}$	0.875	87.5%
$\frac{1}{10}$	0.1	10%
$\frac{1}{20}$	0.05	5%
$\frac{1}{25}$	0.04	4%
$\frac{1}{50}$	0.02	2%
$\frac{1}{100}$	0.01	1%

Prime numbers less than 100

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97

Divisibility rules

2	last digit is even
3	sum of digits divisible by 3
4	last two digits divisible by 4
5	last digit 0 or 5
6	divisible by 2 and by 3
9	sum of digits divisible by 9
10	last digit 0

Units of measure

12 inches = 1 foot; 3 feet = 1 yard
5,280 feet = 1 mile; 1,760 yards = 1 mile
8 fluid ounces = 1 cup; 2 cups = 1 pint
2 pints = 1 quart; 4 quarts = 1 gallon
16 ounces = 1 pound; 2,000 pounds = 1 ton
60 seconds = 1 minute; 60 minutes = 1 hour
1 yard^2 = 9 feet^2; 1 foot^2 = 144 inches^2
1 yard^3 = 27 feet^3

Metric: kilo = 1,000; centi = $\frac{1}{100}$; milli = $\frac{1}{1000}$
1 km = 1,000 m; 1 m = 100 cm = 1,000 mm
1 kg = 1,000 g; 1 L = 1,000 mL

24-hour (military) time

Hours after noon: add 12. 1:00 p.m. = 1300, 6:45 p.m. = 1845, midnight = 0000, 12:30 a.m. = 0030.

Glossary of Math Terms

Mathematics Knowledge questions often test vocabulary directly ("Which of the following is a prime number?"). Make sure every term below is familiar.

Absolute value. The distance of a number from 0 on the number line; never negative. $|-7| = 7$.

Acute angle. An angle greater than $0°$ and less than $90°$.

Area. The amount of surface a flat figure covers, measured in square units.

Average (mean). The sum of a set of values divided by the number of values.

Binomial. A polynomial with two terms, such as $x + 5$.

Circumference. The distance around a circle: $C = 2\pi r = \pi d$.

Coefficient. The number multiplying a variable. In $7x^2$, the coefficient is 7.

Complementary angles. Two angles whose measures add up to $90°$.

Composite number. A whole number greater than 1 that has more than two factors, such as 9 or 12.

Congruent. Having exactly the same size and shape (equal sides, equal angles).

Consecutive integers. Integers that follow one another in order, such as 7, 8, 9.

Constant. A term with no variable, such as the 5 in $3x+5$.

Denominator. The bottom number of a fraction; it tells how many equal parts make a whole.

Diameter. A segment through the center of a circle from side to side; twice the radius.

Difference. The result of a subtraction.

Digit. Any of the symbols 0–9 used to write numbers.

Divisor. The number you divide by. In $20 \div 4$, the divisor is 4.

Equation. A statement that two expressions are equal, such as $2x + 3 = 11$.

Equilateral triangle. A triangle with three equal sides and three $60°$ angles.

Even number. An integer divisible by 2.

Exponent. The small raised number that tells how many times the base is used as a factor: $2^3 = 2 \cdot 2 \cdot 2$.

Factor. A number that divides another number evenly. The factors of 12 are 1, 2, 3, 4, 6, 12.

Greatest common factor (GCF). The largest number that divides each of two or more numbers.

Hypotenuse. The longest side of a right triangle, opposite the right angle.

Improper fraction. A fraction whose numerator is greater than or equal to its denominator, such as $\frac{7}{4}$.

Inequality. A statement using $<$, $>$, $\leq$, or $\geq$ to compare two expressions.

Integer. A whole number or its opposite: $\ldots, -2, -1, 0, 1, 2, \ldots$

Irrational number. A number that cannot be written as a fraction of integers, such as $\sqrt{2}$ or π.

Isosceles triangle. A triangle with at least two equal sides; the angles opposite them are equal.

Least common multiple (LCM). The smallest positive number that is a multiple of each of two or more numbers.

Like terms. Terms with the same variables raised to the same powers, such as $3x^2$ and $-5x^2$.

Linear equation. An equation whose graph is a straight line; the variable has no exponent above 1.

Median. The middle value of a sorted list (or the average of the two middle values).

Mixed number. A whole number and a fraction together, such as $2\frac{1}{3}$.

Mode. The value that occurs most often in a data set.

Monomial. A polynomial with one term, such as $-4x^3y$.

Multiple. The product of a number and a whole number. Multiples of 6: 6, 12, 18, …

Numerator. The top number of a fraction.

Obtuse angle. An angle greater than $90°$ and less than $180°$.

Odd number. An integer that is not divisible by 2.

Parallel lines. Lines in the same plane that never meet; they have equal slopes.

Percent. Per hundred: $35\% = \frac{35}{100} = 0.35$.

Perimeter. The distance around a figure.

Perpendicular lines. Lines that meet at a right angle; their slopes are negative reciprocals.

Polygon. A closed figure made of straight sides (triangle, quadrilateral, pentagon, …).

Polynomial. A sum of terms of the form ax^n, such as $3x^2 - 5x + 2$.

Prime number. A whole number greater than 1 whose only factors are 1 and itself: 2, 3, 5, 7, 11, ...

Probability. The chance that an event occurs, from 0 (impossible) to 1 (certain).

Product. The result of a multiplication.

Proportion. An equation stating that two ratios are equal, such as $\frac{3}{4} = \frac{9}{12}$.

Quadrant. One of the four regions of the coordinate plane, numbered I–IV counterclockwise from the upper right.

Quadratic equation. An equation of the form $ax^2 + bx + c = 0$ with $a \neq 0$.

Quotient. The result of a division.

Radius. The distance from the center of a circle to any point on it.

Range. The difference between the largest and smallest values in a data set.

Ratio. A comparison of two quantities by division, written $3 : 4$ or $\frac{3}{4}$.

Rational number. Any number that can be written as a fraction of two integers.

Reciprocal. The number that multiplies with a given number to give 1. The reciprocal of $\frac{2}{3}$ is $\frac{3}{2}$.

Right angle. An angle of exactly 90°.

Scientific notation. A number written as $a \times 10^n$ with $1 \leq a < 10$, such as 4.2×10^5.

Slope. The steepness of a line: rise over run, $\frac{y_2 - y_1}{x_2 - x_1}$.

Square root. A number that, multiplied by itself, gives the original number: $\sqrt{49} = 7$.

Sum. The result of an addition.

Supplementary angles. Two angles whose measures add up to 180°.

Trinomial. A polynomial with three terms, such as $x^2 + 5x + 6$.

Variable. A letter that stands for an unknown or changing number.

Vertex. A corner point of an angle, polygon, or solid (plural: vertices).

Vertical angles. The opposite angles formed by two intersecting lines; they are equal.

Volume. The amount of space a solid occupies, measured in cubic units.

Whole numbers. The numbers 0, 1, 2, 3, ...

Progress Tracker

Record your score the first time you complete each practice set, then redo the problems you missed about a week later and record that score too.

Ch.	Practice set	Score	Date	Redo
1	Whole Numbers & Order of Operations	/ 25		/ 25
2	Negative Numbers & Absolute Value	/ 25		/ 25
3	Factors, Multiples, Primes, GCF & LCM	/ 25		/ 25
4	Fractions & Mixed Numbers	/ 25		/ 25
5	Decimals	/ 25		/ 25
6	Percents	/ 25		/ 25
7	Ratios & Proportions	/ 25		/ 25
8	Exponents & Scientific Notation	/ 25		/ 25
9	Square Roots & Radicals	/ 25		/ 25
10	Algebraic Expressions	/ 25		/ 25
11	Linear Equations	/ 25		/ 25
12	Inequalities	/ 25		/ 25
13	Systems of Equations	/ 25		/ 25
14	Polynomials & Exponent Rules	/ 25		/ 25
15	Factoring & Quadratic Equations	/ 25		/ 25
16	Lines & Angles	/ 25		/ 25
17	Triangles & the Pythagorean Theorem	/ 25		/ 25
18	Perimeter & Area	/ 25		/ 25
19	Circles	/ 25		/ 25
20	Volume & Surface Area	/ 25		/ 25
21	Coordinate Geometry	/ 25		/ 25
22	Money: Shopping, Discounts, Tax & Wages	/ 40		/ 40
23	Interest, Commission & Percent Change	/ 35		/ 35
24	Rate, Distance & Time	/ 40		/ 40
25	Work Rates & Mixtures	/ 35		/ 35
26	Averages, Statistics & Probability	/ 40		/ 40
27	Measurement, Units & Multi-Step Problems	/ 35		/ 35
–	Diagnostic Test	/ 30		
–	Practice Test 1	/ 55		
–	Practice Test 2	/ 55		
–	Practice Test 3	/ 55		
–	Practice Test 4	/ 55		

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